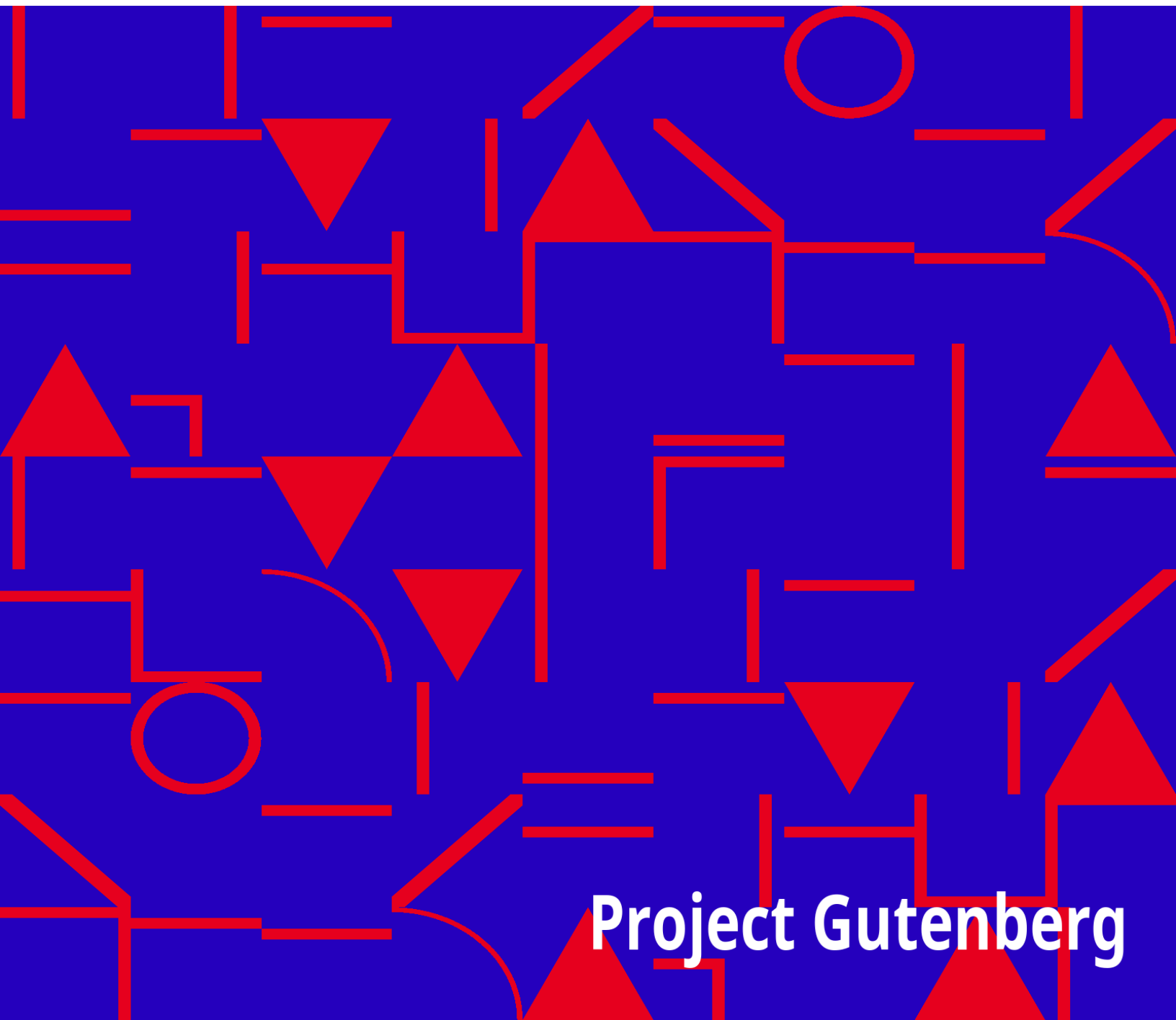


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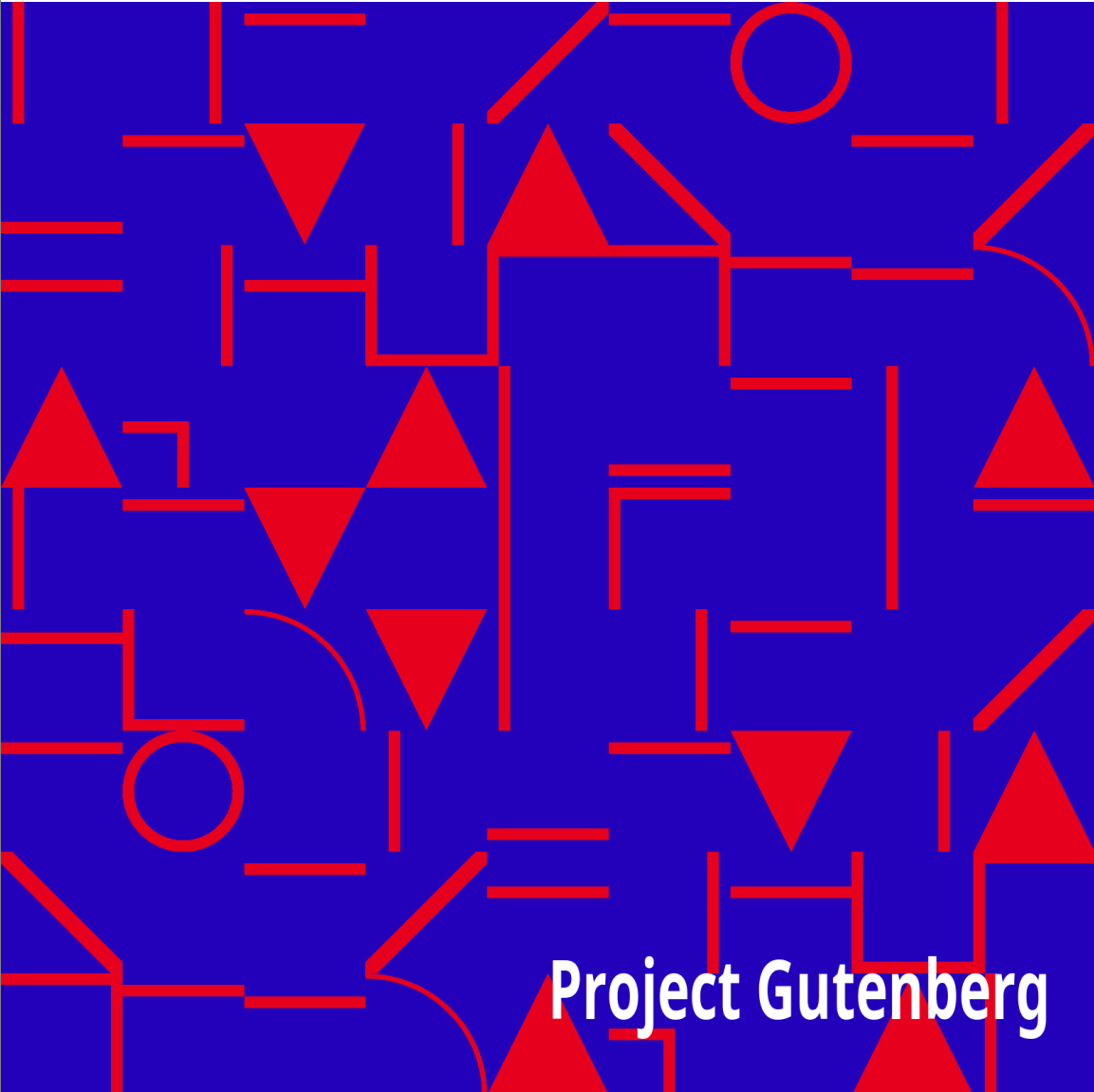
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Transcriber's
note:

A few typographical errors have been corrected. They appear in the text like this, and the explanation will appear when the mouse pointer is moved over the

marked passage.

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KEY TO PRONUNCIATION

The method of marking pronunciations here employed is either (1) by marking the syllable on which the accent falls, or (2) by a simple system of transliteration, to which the following is the Key:—

VOWELS

ā, as in *fate*, or in *bare*.

ä, as in *alms*, Fr. *âme*, Ger. *Bahn* = á of Indian names.

a', the same sound short or medium, as in Fr. *bal*, Ger. *Mann*.

a, as in *fat*.

a", as in *fall*.

a, obscure, as in *rural*, similar to *u* in *but*, è in *her*: common in Indian names.

ē, as in *me* = *i* in *machine*.

e, as in *met*.

è, as in *her*.

ī, as in *pine*, or as *ei* in Ger. *mein*.

i, as in *pin*, also used for the short sound corresponding to ē, as in French and Italian words.

eu, a long sound as in Fr. *jeûne* = Ger. long *ö*, as in *Söhne*, *Goethe* (Goethe).

eu, corresponding sound short or medium, as in Fr. *peu* = Ger. *ö* short.

ō, as in *note*, *moan*.

o, as in *not*, *soft*—that is, short or medium.

ö, as in *move*, *two*.

ū as in *tube*.

u, as in *tub*: similar to *è* and also to *a*.

u“, as in *bull*.

ü, as in Sc. *abune* = Fr. *û* as in *dû*, Ger. *ü* long as in *grün*, *Bühne*.

u‘, the corresponding short or medium sound, as in Fr. *but*, Ger. *Müller*.

oi, as in *oil*.

ou, as in *pound*; or as *au* in Ger. *Haus*.

CONSONANTS

Of the *consonants*, **b, d, f, h, j, k, l, m, n, ng, p, sh, t, v, z**, always have their common English sounds, when used to transliterate foreign words. The letter **c** is not used by itself in re-writing for pronunciation, **s** or **k** being used instead. The only consonantal symbols, therefore, that require explanation are the following:—

ch is always as in *rich*.

d, nearly as *th* in *this* = Sp. *d* in *Madrid*, &c.

g is always hard, as in *go*.

h represents the guttural in Scotch *loch*, Ger. *nach*, also other similar gutturals.

n‘, Fr. nasal *n* as in *bon*.

r represents both English *r*, and *r* in foreign words, which is generally much more strongly trilled.

s, always as in so.

th, as *th* in *thin*.

th, as *th* in *this*.

w always consonantal, as in we.

x = ks, which are used instead.

y always consonantal, as in yea (Fr. *ligne* would be re-written *lēny*).

zh, as s in pleasure = Fr. *j*.

Atreb'ates, ancient inhabitants of that part of Gallia Belgica, afterwards called *Artois*. A colony of them settled in Britain, in a part of Berkshire and Oxfordshire.

At'rek, a river of Asia, forming the boundary between Persia and the Russian Transcaspian territory, and flowing into the Caspian; length 250 miles.

Atreus (at'rūs), in Greek mythology, a son of Pelops and Hippodamīa, and grandson of Tantālus. Atreus was the father of Agamemnon, according to Homer; other writers call him Agamemnon's grandfather. He succeeded Eurystheus, his father-in-law, as King of Mycēnæ, and in revenge for the seduction of his wife by his brother Thyestes gave a banquet at which the latter partook of the flesh of his own sons. Atreus was killed by Ægisthus, a son of Thyestes. The tragic events connected with this family furnished materials to some of the great Greek dramatists.

Atri (ancient, **Hadria**), an episcopal city in the province of Teramo, Italy, 8 miles from the Adriatic. It has an old (thirteenth century) Gothic cathedral, ruins of ancient Roman walls and buildings, and a palace of the Agraviva family, who were Dukes of Atri from 1398 to 1775. Pop. 14,043.

At´riplex. See *Orache*.

A´trium, the entrance-hall and most important apartment of a Roman house, generally ornamented with statues, pictures, and *imagines* or ancestral likenesses, which were portrait masks in wax kept in cases. The *atrium* formed the reception-room for visitors and clients. It was lighted by the *compluvium*, an opening in the roof, towards which the roof sloped so as to throw the rainwater into a cistern in the floor called the *impluvium*.

In zoology the term is applied to the large chamber or 'cloaca' into which the intestine opens in the Tunicata.

At´ropa, the nightshade genus of plants. See *Belladonna*.

At´rophy, a wasting of the flesh due to some interference with the nutritive processes. It may arise from a variety of causes, such as permanent, oppressive, and exhausting passions, organic disease, a want of proper food or of pure air, suppurations in important organs, copious evacuations of blood, saliva, semen, &c., and it is also sometimes produced by poisons, for example arsenic, mercury, lead, in miners, painters, gilders, &c. In old age the whole frame except the heart undergoes atrophic change, and it is of frequent occurrence in infancy as a consequence of improper, unwholesome food, exposure to cold, damp, or impure air, &c. Single organs or parts of the body may be affected irrespective of the general state of nutrition; thus local atrophy may be superinduced by palsies, the pressure of tumours upon the nerves of the limbs, or by artificial pressure, as in the feet of Chinese ladies.

At´ropin, or **At´ropine**, a crystalline alkaloid obtained from the deadly nightshade (*Atrōpa Belladonna*). It is very poisonous, and produces persistent dilation of the pupil.

At´ropos, the eldest of the three Fates (the others being Clotho and Lachēsis), who cuts the thread of life with her shears.

Attaché (at´a-shā), a junior member of the diplomatic services attached to an embassy or legation.

Attach´ment, in English law, a taking of the person, goods, or estate by virtue of a writ or precept. It is distinguished from an *arrest* by proceeding

out of a higher court by precept or writ, whereas the latter proceeds out of an inferior court by precept only. An arrest lies only against the body of a man, whereas an attachment lies often against the goods only, and sometimes against the body and goods. It differs from a *distress* in that an attachment does not extend to lands, while a distress cannot touch the body. —*Foreign attachment* answers to what in Scotland is termed arrestment, by means of which a creditor may obtain the security of the goods or other personal property of his debtor in the hands of a third person for the purpose of enforcing the appearance of the debtor to answer to an action, and afterwards, upon his continued default, of obtaining the property absolutely in satisfaction of the demand.

Attack', the opening act of hostility by a force seeking to dislodge an enemy from its position. It is considered more advantageous to offer than to await attack, even in a defensive war. The historic forms of attack are: (1) the parallel; (2) the form in which both wings attack and the centre is kept back; (3) the form in which the centre is pushed forward and the wings kept back; (4) the famous oblique mode, dating at least from Epaminondas, and employed by Frederick the Great, where one wing advances to engage, whilst the other is kept back, and occupies the attention of the enemy by pretending an attack. Napoleon preferred to mass heavy columns against an enemy's centre. The forms of attack have changed with the weapons used. In the days of the pike heavy masses were the rule, but the use of the musket led to an extended battle-front to give effect to the fire. The advance in long and slender lines which grew out of this has been not less famous in the annals of British attack than the square formation in those of defence. In the European War (1914-18) the Germans often attacked in mass-formation; but British attacks were usually carried out by successive waves; one wave secured its objective and consolidated it while another wave passed through to attack a more advanced objective. Artillery preparation became of increasingly great importance; it broke down the enemy's wire, counteracted his artillery-fire, and made his infantry keep under cover. In trench-to-trench attacks machine-guns and trench-mortars were of great value, and many casualties were avoided by the skilful use of tanks in the attack. But it is still a fundamental principle of tactics that the infantry is the chief factor in the attack, and that no attack can be considered overwhelmingly successful without the use of the bayonet.

Attain' der, the legal consequences of a sentence of death or outlawry pronounced against a person for treason or felony, the person being said to be *attainted*. It resulted in forfeiture of estate and 'corruption of blood', rendering the party incapable of inheriting property or transmitting it to heirs; but these results now no longer follow. Formerly persons were often subjected to attainder by a special Bill or Act passed in Parliament called *Bills of Attainder*, the last being passed in 1798, in the case of Lord Edward Fitzgerald, one of the Irish rebel leaders.

Attaint', a writ at common law against a jury for a false verdict, finally abolished in England in 1825.

Attale'a, a genus of American palms, comprising the piassava palm, which produces coquilla-nuts.

Att'alus, the names of three kings of ancient Pergamus (241-133 B.C.), the last of whom bequeathed his kingdom to the Romans. They were all patrons of art and literature.

At'tar, in the East Indies, a general term for a perfume from flowers; in Europe generally used only of the *attar* or *otto of roses*, an essential oil made from *Rosa centifolia*, the hundred-leaved or cabbage-rose, *R. damascēna*, or damask-rose, *R. moschāta*, or musk-rose, &c., 100,000 roses yielding only 180 grains of attar. Cashmere, Shiraz, and Damascus are celebrated for its manufacture, and there are extensive rose farms in the valley of Kezanlik in Roumelia and at Ghazipur in Benares. The oil is at first greenish, but afterwards it presents various tints of green, yellow, and red. It is concrete at all ordinary temperatures, but becomes liquid about 84° F. It consists of two substances, a hydrocarbon and an oxygenated oil, and is frequently adulterated with the oils of rhodium, sandal-wood, and geranium, with the addition of camphor or spermaceti.

At'terbury, Francis, an English prelate, born in 1662, and educated at Westminster and Oxford. In 1687 he took his degree of M.A., and appeared as a controversialist in a defence of the character of Luther, entitled, *Considerations on the Spirit of Martin Luther*, &c. He also assisted his pupil, the Hon. Charles Boyle, in his famous controversy with Bentley on the *Epistles of Phalaris*. Having taken orders in 1691 he settled in London, became chaplain to William and Mary, preacher of Bridewell, and lecturer

of St. Bride's. Controversy was congenial to him, and in 1706 he commenced one with Dr. Wake, which lasted four years, on the rights, privileges, and powers of convocations. For this service he received the thanks of the lower house of convocation and the degree of Doctor of Divinity from Oxford. Soon after the accession of Queen Anne he was made Dean of Carlisle, aided in the defence of the famous Sacheverell, and wrote *A Representation of the Present State of Religion*. In 1712 he was made Dean of Christ Church, and in 1713 Bishop of Rochester and Dean of Westminster. After the death of the queen in 1714 he distinguished himself by his opposition to George I; and, having entered into a correspondence with the Pretender's party, was apprehended in Aug., 1722, and committed to the Tower. Being banished the kingdom, he settled in Paris, where he chiefly occupied himself in study and in correspondence with men of letters. But even here, in 1725, he was actively engaged in fomenting discontent in the Scottish Highlands. He died in 1731, and his body was privately interred in Westminster Abbey. His sermons and letters are marked by ease and grace; but as a critic and a controversialist he is rather dexterous and popular than accurate and profound.

Attercliffe, a parliamentary division of the borough of Sheffield.

Attic, an architectural term variously used. An *Attic base* is a peculiar kind of base, used by the ancient architects in the Ionic order and by Palladio and some others in the Doric. An *Attic story* is a low story in the upper part of a house rising above the main portion of the building. In ordinary language an attic is an apartment lighted by a window in the roof.

At'tica, a State of ancient Greece, the capital of which is Athens. The territory was triangular in shape, with Cape Sunium (Colonna) as its apex and the ranges of Mounts Cithæron and Parnes as its base. On the north these ranges separated it from Bœotia; on the west it was bounded by Megaris and the Saronic Gulf; on the east by the Ægean. Its most marked physical divisions consisted of the highlands, midland district, and coast district, with the two famous plains of Eleusis and of Athens. The Cephissus and Ilissus, though small, were its chief streams; its principal hills, Cithæron, Parnes, Hymettus, Pentelicus, and Laurium. Its soil has probably undergone considerable deterioration, but produced good fruit, especially olives and figs. These are still cultivated, as well as the vine and cereals, but

Attica is better suited for pasture than tillage. According to tradition the earliest inhabitants of Attica lived in a savage manner until the time of Cecrops, who came, 1550 B.C., with a colony from Egypt, taught them all the essentials of civilization, and founded Athens. One of Cecrops' descendants founded eleven other cities in the regions round, and there followed a period of mutual hostility. To Theseus is assigned the honour of uniting these cities in a confederacy, with Athens as the capital, thus forming the Attic State. After the death of Codrus, 1068 B.C., the monarchy was abolished, and the government vested in archons elected by the nobility, at first for life, in 752 B.C. for ten years, and in 683 B.C. for one year only. The severe constitution of Draco was succeeded in 594 by the milder code of Solon, the democratic elements of which, after the brief tyranny of the Pisistratids, were emphasized and developed by Clisthenes. He divided the people into ten classes, and made the Senate consist of 500 persons, establishing as the Government an oligarchy modified by popular control. Then came the splendid era of the Persian War, which elevated Athens to the summit of fame. Miltiades at Marathon, and Themistocles at Salamis, conquered the Persians by land and by sea. The chief external danger being removed, the rights of the people were enlarged; the archons and other magistrates were chosen from all classes without distinction. The period from the Persian War to the time of Alexander (500 to 336 B.C.) was most remarkable for the development of the Athenian constitution. Attica appears to have contained a territory of nearly 850 sq. miles, with some 500,000 inhabitants, 360,000 of whom were slaves, while the inhabitants of the city numbered 180,000. Cimon and Pericles (444 B.C.) raised Athens to its point of greatest splendour, though under the latter began the Peloponnesian War, which ended with the conquest of Athens by the Lacedæmonians. The succeeding tyranny of the Thirty, under the protection of a Spartan garrison, was overthrown by Thrasybulus, with a temporary partial restoration of the power of Athens; but the battle of Cheronæa (338 B.C.) made Attica, in common with the rest of Greece, a dependency of Macedon. The attempts at revolt after the death of Alexander were crushed, and in 260 B.C. Attica was still under the sway of Antigonos Gonatas, the Macedonian king. A period of freedom under the shelter of the Achæan League then ensued, but their support of Mithridates led, in 146 B.C., to the subjugation of the Grecian States by Rome. After the division of the Roman Empire, Attica belonged to the Empire of the East until, in A.D. 396, it was conquered by

Alaric the Goth and the country devastated. Attica and Bœotia now form a nome or province of the kingdom of Greece, with a population of 407,063.—BIBLIOGRAPHY: Sir J. G. Frazer, *Pausanias's Description of Greece*, vols. ii and v; C. Wordsworth, *Athens and Attica*.

At'ticus, Titus Pomponius, a Roman of great wealth and culture, born 109 B.C., and died 32 B.C. On the death of his father he removed to Athens to avoid participation in the civil war, to which his brother Sulpicius had fallen a victim. There he so identified himself with Greek life and literature as to receive the surname Atticus. It was his principle never to mix in politics, and he lived undisturbed amid the strife of factions. Sulla and the Marian party, Cæsar and Pompey, Brutus and Antony, were alike friendly to him, and he was in favour with Augustus. Of his close friendship with Cicero proof is given in the series of letters addressed to him by Cicero. He married at the age of fifty-three, and had one daughter, Pomponia, named by Cicero Atticula and Attica. He reached the age of seventy-seven years without sickness, but, being then attacked by an incurable disease, ended his life by voluntary starvation. He was a type of the refined Epicurean, and an author of some contemporary repute, though none of his works have reached us.—The name Atticus was given to Addison by Pope, in a well-known passage (Prologue to the *Satires*, addressed to Dr. Arbuthnot).

At'tila (in Ger. *Etzel*), the famous leader of the Huns, was the son of Mundzuk, and the successor, in conjunction with his brother Bleda, of his uncle Roua. The rule of the two leaders extended over a great part of Northern Asia and Europe, and they threatened the Eastern Empire, and twice compelled the weak Theodosius II to purchase an inglorious peace. Attila caused his brother Bleda to be murdered (444), and in a short time extended his dominion over all the peoples of Germany and exacted tribute from the Eastern and Western emperors. The Vandals, the Ostrogoths, the Gepidæ, and a part of the Franks united under his banners, and he speedily formed a pretext for leading them against the Empire of the East. He laid waste all the countries from the Black to the Adriatic Sea, and in three encounters defeated the Emperor Theodosius, but could not take Constantinople. Thrace, Macedonia, and Greece all submitted to the invader, who destroyed seventy flourishing cities; and Theodosius was obliged to purchase a peace. Turning to the west, the 'scourge of God', as his defeated enemies termed him, crossed with an immense army the Rhine, the

Moselle, and the Seine, came to the Loire, and laid siege to Orleans. The inhabitants of this city repelled the first attack, and the united forces of the Romans under Aetius, and of the Visigoths under their king, Theodoric, compelled Attila to raise the siege. He retreated to Champagne, and waited for the enemy in the plains of Châlons. In apparent opposition to the prophecies of the soothsayers the ranks of the Romans and Goths were broken; but when the victory of Attila seemed assured, the Gothic prince Thorismond, the son of Theodoric, poured down from the neighbouring height upon the Huns, who were defeated with great slaughter. Rather irritated than discouraged, he sought in the following year a new opportunity to seize upon Italy, and demanded Honoria, the sister of Valentinian III, in marriage, with half the kingdom as a dowry. When this demand was refused he conquered and destroyed Aquileia, Padua, Vicenza, Verona, and Bergamo, laid waste the plains of Lombardy, and was marching on Rome when Pope Leo I went with the Roman ambassadors to his camp and succeeded in obtaining a peace. Attila went back to Hungary, and died on the night of his marriage with Hilda or Ildico (453), either from the bursting of a blood-vessel or by her hand. The description that Jordanès (or Jornandes) has left us of him is in keeping with his Kalmuck-Tartar origin. He had a large head, a flat nose, broad shoulders, and a short and ill-formed body; but his eyes were brilliant, his walk stately, and his voice strong and well-toned.—BIBLIOGRAPHY: Thierry, *Koenig Attila und seine Zeit*; E. Hutton, *Attila and the Huns*.

Attilly, a village in France. See *Scarpe*, *Battle of the*.

Attleborough, a manufacturing town of the United States, in Massachusetts. Pop. 19,731.

At'tock, a town and fort in Rawal Pindi district, Punjab, overhanging the Indus at the point where it is joined by the Kabul River. It is at the head of the steam navigation of the Indus, and is connected with Lahore by railway. It is an important post on the military road to the frontier. Pop. 2822.

Attor'ney, a person appointed to do something for and in the stead and name of another. An attorney may have *general* powers to act for another; or his power may be *special*, and limited to a particular act or acts. A special attorney is appointed by a deed called a *power* or *letter of attorney*,

specifying the acts which he is authorized to do. An *attorney at law* is a person qualified to appear for another before a court of law to prosecute or defend any action on behalf of his client. The term in England was formerly applied especially to those practising before the supreme courts of common law at Westminster, and corresponded to the term *solicitor* used in courts of Chancery; but this distinction was abolished in 1873, and *solicitor* is now the regular term for all such legal agents. In the United States the term is in common use, and is wide enough to include what in England would be called barristers (or counsel), in Scotland advocates, having indeed the general sense of lawyer. In America women are admitted as attorneys.

Attorney-General, in England and Ireland, the first law-officer and legal adviser of the Crown, acting on its behalf in its revenue and criminal proceedings, carrying on prosecutions in crimes that have a public character, guarding the interests of charitable endowments, and granting patents. He is *ex officio* the leader of the bar, and, as a member of Parliament, has charge of all Government measures on legal questions. The Solicitor-General holds a similar position, and may act in his place. In Scotland the Attorney-General is called Lord-Advocate. There are also Attorneys-General in the colonies. In the United States he is head of the Department of Justice. The individual States have also an Attorney-General.

Attrac'tion, the tendency of all material bodies, whether masses or particles, to approach each other, to unite, and to remain united. Newton was the first to adopt the theory of a universal attractive force, and to determine its laws. When bodies tend to come together from sensible distances the tendency is termed either the attraction of *gravitation*, *magnetism*, or *electricity*, according to circumstances; when the attraction operates at insensible distances it is known as *adhesion* with respect to surfaces, as *cohesion* with respect to the particles of a body, and as *affinity* when the particles of different bodies tend together. It is by the attraction of gravitation that all bodies fall to the earth when unsupported.—
BIBLIOGRAPHY: Newton, *Principia*; Thomson and Tait, *Natural Philosophy*; Laplace, *Mécanique Céleste*; Poynting, *The Mean Density of the Earth*.

Attrek. See *Atrek*.

At´tribute, in philosophy, a quality or property of a substance, as whiteness or hardness. A substance is known to us only as a congeries of attributes.

In the fine arts an attribute is a symbol regularly accompanying and marking out some personage. Thus the caduceus, purse, winged hat, and sandals are attributes of Mercury, the trampled dragon that of St. George.

Attwood, George, F.R.S., an English mathematician, born 1746, died 1807, best known by his invention, called after him *Attwood's Machine*, for verifying the laws of falling bodies. It consists essentially of a freely-moving pulley over which runs a fine cord with two equal weights suspended from the ends. A small additional weight is laid upon one of them, causing it to descend with uniform acceleration. Means are provided by which the added weight can be removed at any point of the descent, thus allowing the motion to continue from this point onward with uniform velocity. See *Gravitation*.

Atys, or **Attis** (at'is), in classical mythology, the shepherd lover of Cybēle, who, having broken the vow of chastity which he made her, castrated himself. In Asia Minor Atys seems to have been a deity, with somewhat of the same character as Adonis. Catullus has written a celebrated poem (Carmen 63) on the subject of Attis.

Aubagne (ō-ba´n-yè), a town in France, department of Bouches-du-Rhône, with manufactures of cottons, pottery, cloth. Pop. 8800.

Aubaine, **Droit d'** (drwä dō-bān). See *Droit d'Aubaine*.

Aube (ōb), a north-eastern French department; area 2326 sq. miles; pop. 227,745. The surface is undulating, and watered by the Aube, &c. The N. and N.W. districts are bleak and infertile, the southern districts remarkably fertile. A large extent of ground is under forests and vineyards, and the soil is admirable for grain, pulse, and hemp. The chief manufactures are worsted and hosiery. Troyes is the capital.—The River Aube, which gives name to the department, rises in Haute-Marne, flows N.W., and after a course of 113 miles joins the Seine.

Aubenais (ōb-nä), a town of France, department Ardèche, with a trade in coal, silk, &c. Pop. 7206.

Auber (ō-bār), Daniel François Esprit, a French operatic composer, born 1782, at Caen, in Normandy, died at Paris 1871. He was originally intended for a mercantile career, but devoted himself to music, studying under Cherubini. His first great success was his opera *La Bergère Châtelaine*, produced in 1820. In 1822 he had associated himself with Scribe as librettist, and other operas now followed in quick succession. Chief among them were *Masaniello* or *La Muette de Portici* (1828), *Fra Diavolo* (1830), *Lestocq* (1834), *L'Ambassadrice* (1836), *Le Domino Noir* (1837), *Les Diamants de la Couronne* (1841), *Marco Spada* (1853), *La Fiancée du Roi de Garbe* (1864). Despite his success in *Masaniello*, his peculiar field was comic opera, in which his charming melodies, bearing strongly the stamp of the French national character, his uniform grace and piquancy, won him a high place.

Aubergine (ō'bèr-zhēn), the fruit of the eggplant (q.v.).

Aubervilliers (ō-bār-vēl-yā), a suburban locality of Paris, with a fort belonging to the defensive works of the city. Pop. 37,558.

Aubigné, Merle d'. See *Merle d'Aubigné*.

Aubin (ō-ban'), a town of Southern France, department of Aveyron, 20 miles N.E. of Villefranche; mining district: coal, sulphur, alum, and iron. Pop. 9574.

Au'brey, John, F.R.S., an English antiquary, born in Wiltshire in 1625 or 1626, died about 1700. He was educated at Oxford; collected materials for the *Monasticon Anglicanum*, and afforded important assistance to Wood, the antiquary. He left large collections of manuscripts, which have been used by subsequent writers. His *Miscellanies* (London, 1696) contain much curious information, but display credulity and superstition. His *Survey of Surrey* was incorporated in Rawlinson's *Natural History and Antiquities of the County of Surrey*, which was published in 1719.

Au'burn, the name of many places in America, the chief being a handsome city of New York State, at the north end of Owasco Lake. It is chiefly famous for its State prison, large enough to receive 1000 prisoners. In the town or vicinity various manufactures are carried on. Pop. 36,142.—

Another Auburn is in Maine, on the Androscoggin River, a manufacturing town. Pop. (1920), 16,985.

Aubusson (ō-bu'-sōn), a town of the interior of France, department Creuse, celebrated for its carpets. Pop. 7211.

Aubusson (ō-bu'-sōn), Pierre d', grand-master of the knights of St. John of Jerusalem, born in 1423 of a noble French family, served in early life against the Turks, then entered the order of St. John, obtained a commandery, was made grand-prior, and in 1476 succeeded the Grand-master Orsini. In 1480 the Island of Rhodes, the head-quarters of the order, was invaded by a Turkish army of 100,000 men. The town was besieged for two months and then assaulted, but the Turks were obliged to retire with great loss. He died at Rhodes in 1503.

Auch (ōsh), a town in S.W. France, capital of department Gers; the seat of an archbishop, with one of the finest Gothic cathedrals in France; manufactures linens, leather, &c. Pop. 13,638.

Auchenia (a'-kē'ni-a). See *Llama*.

Auchterar'der, a town, Perthshire, Scotland, with manufactures of tweeds, tartans, &c. The opposition to the presentee to the church of Auchterarder (1839) originated the struggle which ended in the formation of the Free Church of Scotland. Pop. (1921), 3151.

Auck'land, a town of New Zealand, in the North Island, founded in 1840, and situated on Waitemata Harbour, one of the finest harbours of New Zealand, where the island is only 6 miles across, there being another harbour (Manukau) on the opposite side of the isthmus. At dead low water there is sufficient depth in the harbour for the largest steamers. The working ship channel has an average depth of 36 feet, and varies in width from 1 to 2 miles. The harbour has two good entrances, with a lighthouse; and is defended by batteries. There are numerous wharves and jetties, and two graving-docks, one of which—the Calliope Dock, opened in 1887—is one of the largest in the whole of the Southern Seas. Its site is picturesque, the streets are spacious, and the public buildings—churches, educational establishments, including a university college—are numerous and handsome. It has a large and increasing trade, there being connection with

the chief places on the island by rail, and regular communication with the other ports of the colony, Australia, and Fiji by steam. It was formerly the capital of the colony. Pop. (including suburbs) 157,750.—The provincial district of Auckland forms the northern part of North Island, with an area of 25,364 sq. miles. Pop. 308,766. The surface is very diversified; volcanic phenomena are common, including geysers, hot lakes, &c.; rivers are numerous; wool, timber, kauri-gum, &c., are exported. Much gold has been obtained in the Thames Valley and elsewhere.

Auckland, William Eden, Lord, an English statesman, born 1744; educated at Eton and Oxford, called to the bar 1768, Under-Secretary of State 1772, and in 1776 a lord of the Board of Trade. In 1778 he was nominated in conjunction with Lord Howe and others to act as mediator between Britain and the insurgent American colonies. He was afterwards Secretary of State for Ireland, Ambassador Extraordinary to France, Ambassador Extraordinary to the Netherlands, &c. He was raised to the peerage in 1788, and died in 1814.

Auckland Islands, a group of islands about 180 miles S. of New Zealand, discovered in 1806, and belonging to Britain. They are of volcanic origin and fertile; and the largest, which is 30 miles by 15, has two good harbours. There are no settled inhabitants.

Auction is a public sale to the party offering the best price where the buyers bid against each other, or to the bidder who first accepts the terms offered by the vendor where he sells by reducing his terms until someone accepts them. The latter form is known as a *Dutch Auction*. A sale by auction must be conducted in the most open and public manner possible; and there must be no collusion on the part of the buyers. Puffing, or mock bidding, to raise the value by apparent competition, is illegal.

Auctioneer, a person who conducts sales by auction. It is his duty to state the conditions of sale, to declare the respective biddings, and to terminate the sale by *knocking down* the thing sold to the highest bidder. In Britain an auctioneer must have a licence (for which he pays £10), renewable annually. Verbal declarations by an auctioneer are not suffered to control the printed conditions of sale. The Auctioneers' Institute of the United Kingdom was founded in 1886.

Au´cuba, a genus of plants, ord. Cornaceæ, one species of which, *A. japoníca*, a laurel-like shrub with spotted leaves, a native of Japan and China, is now common in ornamental grounds in Europe. The flowers are diœcious and inconspicuous. For a long time only the female plant, introduced into Britain from Japan in 1783, was cultivated, but in 1850 the male was introduced, and the fruit, which consists of beautiful coral-red berries, is now frequently developed, and adds greatly to the attractiveness of the plant. *A. himalaica*, also brought to Europe, is less hardy.

Aude (ōd), a maritime department in the S. of France; area 2448 sq. miles, mainly covered by hills belonging to the Pyrenees or the Cevennes, and traversed w. to E. by a valley drained by the Aude. The loftier districts are bleak and unproductive; the others tolerably fertile, yielding good crops of grain. The wines, especially the white wines, are famous; olives and other fruits are also cultivated. The manufactures are varied; the trade is facilitated by the Canal du Midi. Carcassonne is the capital; other towns are Narbonne and Castelnaudary. Pop. 286,552. The River Aude rises in the Eastern Pyrenees, and, flowing nearly parallel to the Canal du Midi, falls into the Mediterranean after a course of 130 miles.

Audebert (ōd-bār), Jean Baptiste, French engraver and naturalist, born in 1759, died in 1800; published *Histoire Naturelle des Singes, des Makis, et des Galéopithèques*; *Histoire des Colibris*, &c.; and began *Histoire des Grimpereaux et des Oiseaux de Paradis*, finished by Desray—all finely-illustrated works.

Au´denshaw, a town of England, in Lancashire, 4 miles E. of Manchester, with cotton-mills, engineering-works, &c. Pop. (1921), 7878.

Audiometer, an instrument for the measurement of hearing, invented by Professor D. E. Hughes, of London, in 1879.

Au´diphone, an acoustic instrument by means of which deaf persons are enabled to hear. It consists essentially of a fan-shaped plate of hardened caoutchouc, which is bent to a greater or less degree by strings, and is very sensitive to sound-waves. When used, the up edge is pressed against the upper front teeth, with the convexity outward, and the sounds being collected are conveyed from the teeth to the auditory nerve without passing through the external ear.

Au´dit, an examination into accounts or dealings with money or property, along with vouchers or other documents connected therewith, especially by proper officers, or persons appointed for the purpose. Also the occasion of receiving the rents from the tenants on an estate.

Au´ditor, a person appointed to examine accounts, public or private, to see whether they are correct and in accordance with vouchers. In Britain the public accounts are audited by the Exchequer and Audit Department, Somerset House, at the head of it being a comptroller and auditor-general, and an assistant-comptroller and auditor, with a large staff of clerks. In Scotland there is an auditor attached to the Court of Session appointed to tax costs in litigation.

Auditory Nerves. See *Ear*.

Audley, a town (urban district) of England, in Staffordshire, to the north-west of the district of The Potteries, with coal and iron mines. Pop. (1921), 14,751.

Audran (ō-drän), Gerard, a celebrated French engraver, born 1640; studied at Rome; was appointed engraver to Louis XIV; died at Paris 1703. He engraved Le Brun's *Battles of Alexander*, two of Raphael's cartoons, Poussin's *Coriolanus*, &c., and takes a first place among historical engravers. Other members of the family were successful in the same profession: Benoît, 1661-1721; Claude *père*, 1592-1677; Claude *fils*, 1640-84; Germain, 1631-1710; Jean, 1667-1756.

Au´dubon, John James, an American naturalist of French extraction, born near New Orleans in 1775, was educated in France, and studied painting under David. In 1798 he settled in Pennsylvania, but having a great love for ornithology he set out in 1810 with his wife and child, descended the Ohio, and for many years roamed the forests in every direction, drawing the birds which he shot. In 1826 he came to England, exhibited his drawings in Liverpool, Manchester, and Edinburgh, and finally published them in an unrivalled work of double-folio size, with 435 coloured plates of birds the size of life (*The Birds of America*, 4 vols., 1827-39), with an accompanying text (*Ornithological Biography*, 5 vols., 8vo, partly written by Professor Macgillivray). On his final return to America he laboured with Dr. Bachman

on a finely-illustrated work entitled *The Quadrupeds of America* (1843-50, 3 vols.). He died at New York in 1851.

Auerbach, a manufacturing town of Germany, in Saxony. Pop. 2200.

Auerbach (ou'èr-bäh), Berthold, a distinguished German author of Jewish extraction, born 1812, died 1882. He abandoned the study of Jewish theology in favour of philosophy, publishing in 1836 his *Judaism and Modern Literature*, and a translation of the works of Spinoza with critical biography (5 vols., 1841). His later works were tales or novels, and his *Village Tales of the Black Forest* (*Schwarzwälder Dorfgeschichten*), as well as others of his writings, have been translated into several languages. Other works: *Barfüssele*, *Joseph im Schnee*, *Edelweiss*, *Auf der Höhe*, *Das Landhaus am Rhein*, *Waldfried*, *Brigitta*.

Auerstädt (ou'èr-stet), battle at, 14th Oct., 1806. See *Jena*.

Augeas (a"-jē'as), a fabulous king of Elis, in Greece, whose stable contained 3000 oxen, and had not been cleaned for thirty years. Hercules undertook to clear away the filth in one day in return for a tenth part of the cattle, and executed the task by turning the River Alphēus through it. Augeas, having broken the bargain, was deposed and slain by Hercules.

Auger (a"-gèr), an instrument for boring holes considerably larger than those bored by a gimlet, used by carpenters and joiners, ship-wrights, &c.

Augereau (ōzh-rō), Pierre François Charles, Duke of Castiglione, Marshal of France, son of a mason, born at Paris 1757. He adopted the life of a soldier, and by 1796 had reached the rank of general of division in the army of Italy. At Casale, Lodi, Castiglione, and Arcole he highly distinguished himself. In 1797 he was at Paris, and was the instrument of the *coup d'état* of the 18th of Fructidor (4th Sept.). In 1799 he was chosen a member of the Council of Five Hundred. He then obtained the command of the army in Holland, and fought till the end of the campaign. In 1803 he was appointed to lead the army collected at Bayonne against Portugal. In 1804 he was named marshal of the empire, and grand officer of the Legion of Honour. He subsequently took part in the battles of Jena and Eylau, held a command in Spain, and in July, 1813, led the army in Bavaria against Saxony, taking

part in the battle of Leipzig. On Napoleon's abdication he submitted to Louis XVIII, who named him a peer. He died 1816.

Aughrim, a village in Co. Galway, Ireland, memorable for the decisive victory gained in the neighbourhood, 12th July, 1691, by the forces of William of Orange, under Ginkel, over the Irish and French troops, under St. Ruth. The total English casualties were about 1700, while the Irish lost at least 7000 men as well as all their war material. This battle caused the complete submission of the country.

Augier (ō-zhi-ā), Emile, noted French dramatist, born 1820, came young to Paris, entered a lawyer's office, but relinquished law for literature; elected an academician in 1857, in 1868 a commander of the Legion of Honour. His first and one of his best dramas was the comedy *La Ciguë* (1844); among his other works are *L'Aventurière*, *Gabrielle*, *Paul Forestier*, *Le Mariage d'Olympe*, *Le Gendre de M. Poirier*, *Les Effrontés*, *Le Fils de Giboyer*, *Les Lions et les Renards*, *Maître Guérin*, *Les Fourchambault*, &c. He died in 1889.

Augite (a''jīt) is the commonest member of the pyroxene group of minerals; a constituent of many igneous rocks, such as basalt, gabbro, &c. It crystallizes in short almost rectangular prisms of the monoclinic system, modifying planes causing cross-sections to be eight-sided. Its specific gravity is about 3.2; lustre vitreous; hardness sufficient to scratch glass; colour usually dark. It is a silicate of calcium, magnesium, iron, and aluminium, and alters in geological time by slow recrystallization into hornblende, possibly with some loss of calcium. It may be imitated by the artificial fusion of its constituents. A transparent, green, non-aluminous variety found at Zillertal, in Tyrol, is used in jewellery.

Augsburg (ougz'bu`rh; Lat. *Augusta Vindelicorum*), a city of Bavaria, at the junction of the Wertach and Lech, antique in appearance, but with some fine streets, squares, and handsome or interesting buildings, including a splendid town hall, a lofty belfry (Perlach Tower), cathedral, with paintings by Domenichino, Holbein, &c.; St. Ulrich's Church; the bishop's palace, where the Augsburg Confession was presented to the Diet, afterwards a royal residence; the Fugger Palace, or mansion of the celebrated Fugger family; the public library; the theatre; the Academy of Arts; and the Fuggerei, a

separate quarter of the city, consisting of fifty-three small houses, tenanted at a merely nominal rent by indigent Roman Catholics. Augsburg was a renowned commercial centre in the Middle Ages, and is still an important emporium of South German and Italian trade; industries: cotton spinning and weaving, dyeing, woollen manufacture, machinery and metal goods, books and printing, chemicals, &c. The Emperor Augustus established a colony here about 12 B.C. In 1276 it became a free city, and, besides being a great mart for the commerce between the north and south of Europe, it was a great centre of German art in the Middle Ages. It early took a conspicuous part in the Reformation. (See next article.) In 1806 it was incorporated in Bavaria. Pop. (1919), 154,555.

Augsburg Confession, a document which was presented by the Protestants at the Diet of Augsburg, 1530, to the Emperor Charles V and the Diet, and being signed by the Protestant States was adopted as their creed. Luther made the original draft; but as its style appeared too violent it was given to Melanchthon for amendment. The original is to be found in the imperial Austrian archives. Afterwards Melanchthon arbitrarily altered some of the articles, and there arose a division between those who held the original and those who held the altered Augsburg Confession. The former is received by the Lutherans, the latter by the German Reformed.

Au'gurs, a board or college of diviners who, amongst the Romans, predicted future events and announced the will of the gods from the occurrence of certain signs. These consisted of signs in the sky, especially thunder and lightning; signs from the flight and cries of birds; from the feeding of the sacred chickens; from the course taken or sounds uttered by various quadrupeds or by serpents; from accidents or occurrences, such as spilling the salt, sneezing, &c. The answers of the augurs as well as the signs by which they were governed were called *auguries*, but bird-predictions were properly termed *auspices*. Nothing of consequence could be undertaken without consulting the augurs, and by the mere utterance of the words *alio die* ('meet on another day') they could dissolve the assembly of the people and annul all decrees passed at the meeting.

Au'gust, the eighth month from January. It was the sixth of the Roman year, and hence was called *Sextilis* till the Emperor Augustus affixed to it his own name.

Augus'ta, the name of many ancient places, as Augusta Trevirorum, now *Trèves*; Augusta Taurinorum, now *Turin*; Augusta Vindelicorum, now *Augsburg*; &c.

Augusta (ou-gu's'ta'), or **Agos'ta**, a seaport in the south-east of Sicily, 12 miles north of Syracuse. It exports salt, oil, honey, &c. Pop. 17,250.

Augus'ta, capital of Maine, United States, on the River Kennebec, which is crossed by a bridge and is navigable for small vessels 43 miles from its mouth, while a dam enables steamboats to ply for 20 miles farther up and furnishes immense water-power. Pop. (1920), 14,144.

Augusta, the capital of Richmond county, Georgia, United States, on the left bank of the Savannah River, 231 miles from its mouth; well built, and connected with the river by high-level canals; an important manufacturing centre, having cotton-mills, machine-shops, and railroad works, &c. Pop. (1920), 52,548.

Au'gustine (Aurelius Augustinus), **St.**, a renowned father of the Christian Church, was born at Tagaste, in Africa, in 354, his mother Monica being a Christian, his father Patricius a pagan. His parents sent him to Carthage to complete his education, but he disappointed their expectations by his neglect of serious study and his devotion to pleasure. A lost book of Cicero's, called *Hortensius*, led him to the study of philosophy; but dissatisfied with this he went over to the Manichæans. He was one of their disciples for nine years, but left them, went to Rome, and thence to Milan, where he announced himself as a teacher of rhetoric. St. Ambrose, the bishop of this city, converted him to the faith of his boyhood, and the reading of Paul's *Epistles* wrought an entire change in his life and character. He retired into solitude, and prepared himself for baptism, which he received in his thirty-third year from the hands of Ambrose. Returning to Africa, he sold his estate and gave the proceeds to the poor, retaining only enough to support him. At the desire of the people of Hippo, Augustine became the assistant of the bishop of that town, preached with extraordinary success, and in 395 succeeded to the see. He entered into a warm controversy with Pelagius concerning the doctrines of free-will, grace, and predestination, and wrote treatises concerning them, but of his various works his *Confessions* is most secure of immortality. He died 28th Aug.,

430, while Hippo was besieged by the Vandals. He was a man of great enthusiasm, self-devotion, zeal for truth, and powerful intellect, and though there have been fathers of the Church more learned, none have wielded a more powerful influence. His doctrine of grace, which was an important contribution to Christian thought, triumphed at last in the Reformation and evangelical religion. His writings are partly autobiographical, as the *Confessions*, partly polemical, homiletic, or exegetical. The greatest is the *City of God (De Civitate Dei)*, a vindication of Christianity.—BIBLIOGRAPHY: Joseph McCabe, *St. Augustine and his Age*; Nourrisson, *La Philosophie de St. Augustin*.

Au'gustine, or **Austin, St.**, the *Apostle of the English*, first Archbishop of Canterbury, flourished at the close of the sixth century, was sent, in 596, with forty monks by Pope Gregory I to introduce Christianity into Saxon England, and was kindly received by Ethelbert, King of Kent, whom he converted, baptising 10,000 of his subjects in one day. In acknowledgment of his tact and success Augustine received the archiepiscopal pall from the Pope, with instructions to establish twelve sees in his province, but he could not persuade the British bishops in Wales to unite with the new English Church. He died in 604 or 605. Cf. Sir H. H. Howorth, *St. Augustine of Canterbury*.

Au'gustins, or **Augustines**, members of several monastic fraternities who follow rules framed by the great St. Augustine, or deduced from his writings, of which the chief are the Canons Regular of St. Augustine, or Austin Canons, and the Begging Hermits or Austin Friars. The Austin Canons were introduced into Britain about 1100, and had about 170 houses in England and about twenty-five in Scotland. They took the vows of chastity and poverty, and their habit was a long black cassock with a white rochet over it, having over that a black cloak and hood. The Austin Friars, originally hermits, were a much more austere body, went barefooted, and formed one of the four orders of mendicants. An order of nuns had also the name of Augustines. Their garments, at first black, were afterwards violet.

Augusto'vo, a town of Poland, formerly in Russia, in the government of Suwalki, founded in 1557 by Sigismund II. Pop. 11,797. The battle of Augustovo was fought between the Russians and the Germans between 14th Sept and 3rd Oct., 1914.

August'ulus, Romulus, the last of the Western Roman emperors; reigned for one year (475-6), when he was overthrown by Odoacer and banished.

August'us, Gaius Julius Cæsar Octavianus (originally called Gaius Octavius), Roman emperor, was the son of Gaius Octavius and Atia, a daughter of Julia, the sister of Julius Cæsar. He was born 63 B.C., and died A.D. 14. Octavius was at Apollonia, in Epirus, when he received news of the death of his uncle (44 B.C.), who had previously adopted him as his son. He returned to Rome to claim Cæsar's property and avenge his death, and now took, according to usage, his uncle's name with the surname Octavianus. He was aiming secretly at the chief power, but at first he joined the republican party, and assisted at the defeat of Antony at Mutina. He got himself chosen consul in 43. Soon after the second triumvirate was formed between him and Antony and Lepidus, and this was followed by the conscription and assassination of three hundred Senators and two thousand knights of the party opposed to the triumvirate. Next year Octavianus and Antony defeated the republican army under Brutus and Cassius at Philippi. The victors now divided the Roman world between them, Octavianus getting the West, Antony the East, and Lepidus Africa. Sextus Pompeius, who had made himself formidable at sea, had now to be put down; and Lepidus, who had hitherto retained an appearance of power, was deprived of all authority (36 B.C.) and retired into private life. Antony and Octavianus now shared the Empire between them; but while the former, in the East, gave himself up to a life of luxury, and alienated the Romans by his alliance with Cleopatra and his adoption of Oriental manners, Octavianus skilfully cultivated popularity, and soon declared war ostensibly against the Queen of Egypt. The naval victory of Actium, in which the fleet of Antony and Cleopatra was defeated, made Octavianus master of the world, 31 B.C. He returned to Rome, 29 B.C., celebrated a splendid triumph, and caused the temple of Janus to be closed in token of peace being restored. Gradually all the highest offices of State, civil and religious, were united in his hands, and the new title of Augustus was also assumed by him, being formally conferred by the Senate in 27 B.C. Great as was the power given to him, he exercised it with wise moderation, and kept up the show of a republican form of government. Under him successful wars were carried on in Africa and Asia (against the Parthians), in Gaul and Spain, in Pannonia, Dalmatia, &c.; but the defeat of Varus by the Germans under Arminius with the loss of three

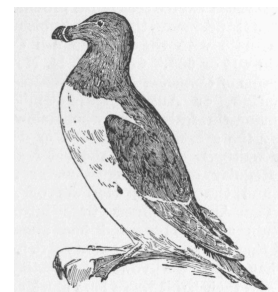
legions, A.D. 9, was a great blow to him in his old age. Many useful decrees proceeded from him, and various abuses were abolished. He gave a new form to the Senate, employed himself in improving the morals of the people, enacted laws for the suppression of luxury, introduced discipline into the armies, and order into the games of the circus. He adorned Rome in such a manner that it was said: "He found it of brick, and left it of marble". The people erected altars to him, and, by a decree of the Senate, the month Sextilis was called *Augustus* (our August). He was a patron of literature; Virgil and Horace were befriended by him, and their works and those of their contemporaries are the glory of the *Augustan Age*. His death, which took place at Nola, plunged the Empire into the greatest grief. He was thrice married, but had no son, and was succeeded by his stepson Tiberius, whose mother Livia he had married after prevailing on her husband to divorce her.—BIBLIOGRAPHY: J. B. Firth, *Augustus Cæsar* (in *Heroes of the Nations* series); E. S. Shuckburgh, *Augustus*.

Augustus II (or **Frederick Augustus I**), Elector of Saxony and King of Poland, second son of John George III, Elector of Saxony, was born at Dresden in 1670, died at Warsaw 1733. He succeeded his brother in the Electorate in 1694, and the Polish throne having become vacant, in 1696, by the death of John Sobieski, Augustus presented himself as a candidate for it and was successful. He joined with Peter the Great in the war against Charles XII of Sweden, invaded Livonia, but was defeated by Charles near Riga, and at Clissow, between Warsaw and Cracow. In 1704 he was deposed, and two years later formally resigned the crown to Stanislaus I, now devoting himself to his Saxon dominions. In 1709, after the defeat of Charles at Poltava, the Poles recalled Augustus, who united himself anew with Peter. The two monarchs, in alliance with Denmark, sent troops into Pomerania, but the Swedish general Steinbock defeated the allies at Gadebusch, 20th Dec., 1712. The death of Charles XII put an end to the war, and Augustus concluded a peace with Sweden. A confederation was now formed in Poland against the Saxon troops, but through the mediation of Peter an arrangement was concluded by which the Saxon troops were removed from the kingdom. Augustus now gave himself up to voluptuousness and a life of pleasure. His Court was one of the most splendid and polished in Europe. The Poles yielded but too readily to the example of their king, and the last years of his reign were characterized by

boundless luxury and corruption of manners. His wife left him one son. The Countess of Königsmark bore him the celebrated commander Marshal Saxe (Maurice of Saxony).

Augustus III (or **Frederick Augustus II**), Elector of Saxony and King of Poland, son of Augustus II, born at Dresden 1696, succeeded his father as Elector in 1733, and was chosen King of Poland through the influence of Austria and Russia. He closely followed the example of his father, distinguishing himself by the splendour of his feasts and the extravagance of his Court. He preferred Dresden to Warsaw, and through his long absence from Poland the government sank into entire inactivity. During the first Silesian war he formed a secret alliance with Austria. The consequence was that during the second Silesian war Frederick the Great of Prussia pushed on into Saxony, and occupied the capital, from which Augustus fled. By the peace of Dresden, 25th Dec., 1745, he was reinstated in the possession of Saxony. In 1756 he was involved anew in a war against Prussia. When Frederick declined his proposal of neutrality he left Dresden, and entered the camp at Pirna, where 17,000 Saxon troops were assembled. Frederick surrounded the Saxons, who were obliged to surrender, and Augustus fled to Poland. On the threat of invasion by Russia he returned to Dresden, where he died in 1763. His son, Frederick Christian, succeeded him as Elector of Saxony, and Stanislaus Poniatowski as King of Poland.

Auk, a name of certain swimming birds, family Alcidae, including the great auk, the little auk, the puffin, &c. The genus *Alca*, or auks proper, contains only two species, the great auk (*Alca impennis*), and the razor-bill (*Alca torda*). The great auk or gair-fowl, a bird about 3 feet in length, used to be plentiful in northerly regions, and also visited the British shores, but has become extinct. Some seventy skins, about as many eggs, with bones representing perhaps a hundred individuals, are preserved in various museums. Though the largest species of the family, the wings were only 6 inches from the carpal joint to the tip, totally useless for flight, but employed as fins in swimming, especially under water. The tail was about 3 inches long; the beak was high, short, and compressed; the head, neck, and upper parts were blackish; a large spot under each eye, and most of the under parts white. Its



Razor-bill (*Alca torda*)

legs were placed so far back as to cause it to sit nearly upright. The razor-bill is about 15 inches in length, and its wings are sufficiently developed to be used for flight. It is found in numbers on some parts of the British shores, as the Isle of Man.

Aulap'olay, or **Alleppi**, a seaport on the south-west coast of Hindustan, Travancore, between the sea and a lagoon, with a safe roadstead all the year round; exports timber, coir, coconuts, &c. Pop. 24,918.

Auld Lights. See *Presbyterianism*.

Aulic (Lat. *aula*, a court or hall), an epithet given to a council (the *Reichshofrath*) in the old German Empire, one of the two supreme courts of the German Empire, the other being the court of the imperial chamber (*Reichskammergericht*). It had not only concurrent jurisdiction with the latter court, but in many cases exclusive jurisdiction, in all feudal processes, and in criminal affairs, over the immediate feudatories of the Emperor and in affairs which concerned the imperial Government.

Au'lis, in ancient Greece, a seaport in Bœotia, on the strait called Euripus, between Bœotia and Eubœa. See *Iphigenia*.

Aullagas (ou-lyä'gäs), a salt lake of Bolivia, which receives the surplus waters of Lake Titicaca through the Rio Desaguadero, and has only one perceptible insignificant outlet, so that what becomes of its superfluous water is still a matter of uncertainty.

Aulnoy (ō-nwä), Countess d', French writer, born 1650, died 1705, was the author of *Contes des Fées* (*Fairy Tales*), many of which, such as *The White Cat*, *The Yellow Dwarf*, &c., have been translated into English. She also wrote a number of novels, historical memoirs, &c.

Aumale (ō-mäl), a small French town, department of Seine Inférieure, 35 miles N.E. of Rouen, which has given titles to several notables in French history.—Jean d'Arcourt, Eighth Count d'Aumale, fought at Agincourt, and defeated the English at Gravelle (1423).—Claude II, Duc d'Aumale, one of the chief instigators of the massacre of St. Bartholomew, was killed 1573.—Charles de Lorraine, Duc d'Aumale, was an ardent partisan of the League in the politico-religious French wars of the sixteenth century.—Henri-Eugene-Philippe Louis d'Orleans, Duc d'Aumale, son of Louis Philippe, king of the

French, was born in 1822. In 1847 he succeeded Marshal Bugeaud as Governor-General of Algeria, where he had distinguished himself in the war against Abd-el-Kader. After the revolution of 1848 he retired to England; but he returned to France in 1871, and was elected a member of the Assembly; became Inspector-General of the army in 1879, and was expelled along with the other royal princes in 1886, but was allowed to return. Author of a *History of the House of Condé*, &c. He died in 1897.

Aun'gerville, Richard, known as Richard de Bury (from his birthplace Bury St. Edmund's), English statesman, bibliographer, and correspondent of Petrarch, born 1281, died 1345. He entered the order of Benedictine monks, and became tutor to the Prince of Wales, afterwards Edward III. Promoted to several offices of dignity, he ultimately became Bishop of Durham, and Lord Chancellor of England. During his frequent embassies to the Continent he made the acquaintance of many of the eminent men of the day. He was a diligent collector of books, and formed a library at Oxford. Author of *Philobiblon*, printed at Cologne in 1473; *Epistolæ Familiarium*, including letters to Petrarch, &c.

Aurangabad', a town of India, in the territory of the Nizam of Hyderabad, 175 miles from Bombay. It contains a ruined palace of Aurangzib and a mausoleum erected to the memory of his favourite wife. It was formerly a considerable trading centre, but its commercial importance decreased when Hyderabad became the capital of the Nizam. Pop. 34,000.

Aurangzib ('ornament of the throne'), one of the greatest of the Mogul emperors of Hindustan, born in Oct. 1618 or 1619. When he was nine years old his weak and unfortunate father, Shah Jehan, succeeded to the throne. Aurangzib was distinguished, when a youth, for his serious look, his frequent prayers, his love of solitude, his profound hypocrisy, and his deep plans. In his twentieth year he raised a body of troops by his address and good fortune, and obtained the government of the Deccan. He stirred up dissensions between his brothers, made use of the assistance of one against the other, and finally shut his father up in his harem, where he kept him prisoner. He then murdered his relatives one after the other, and in 1659 ascended the throne. Notwithstanding the means by which he had got possession of power, he governed with much wisdom. Two of his sons, who endeavoured to form a party in their own favour, he caused to be arrested

and put to death by slow poison. He carried on many wars, conquered Golconda and Bijapur, and drove out, by degrees, the Mahrattas from their country. After his death, on 4th March, 1707, the Mogul Empire declined.

Aurantia´ceæ, the orange tribe, a nat. ord. of plants, polypetalous dicotyledons, with leaves containing a fragrant essential oil in transparent dots, and a superior pulpy fruit, originally natives of India; examples comprise the orange, lemon, lime, citron, and shaddock.

Auray (ō-rā), a seaport of North-West France, department Morbihan, with a deaf and dumb institute, and within 2 miles of St. Anne of Auray, a famous place of pilgrimage. Pop. 6653.

Aure´lian, Lucius Domitius Aurelianus, Emperor of Rome, of humble origin, was born about A.D. 212, rose to the highest rank in the army, and on the death of Claudius II (270) was chosen emperor. He delivered Italy from the barbarians (Alemanni and Marcomanni), and conquered the famous Zenobia, Queen of Palmyra. He followed up his victories by the reformation of abuses, and the restoration throughout the Empire of order and regularity. He lost his life, A.D. 275, by assassination, when heading an expedition against the Persians.

Aure´lius Antoni´nus, Marcus, often called simply **Marcus Aurelius**, Roman emperor and philosopher, son-in-law, adopted son, and successor of Antoninus Pius, born A.D. 121, succeeded to the throne 161, died 180. His name originally was Marcus Annius Verus. He voluntarily shared the government with Lucius Verus, whom Antoninus Pius had also adopted. Brought up and instructed by Plutarch's nephew, Sextus, the orator Herodes Atticus, and L. Volusius Mecianus, the jurist, he had become acquainted with learned men, and formed a particular love for the Stoic philosophy. A war with Parthia broke out in the year of his accession, and did not terminate till 166. A confederacy of the northern tribes now threatened Italy, while a frightful pestilence, brought from the East with the army, raged in Rome itself. Both emperors set out in person against the rebellious tribes. In 169 Verus died, and the sole command of the war devolved on Marcus Aurelius, who prosecuted it with the utmost rigour, and nearly exterminated the Marcomanni. His victory over the Quadi (174) is connected with a famous legend. Dion Cassius tells us that the twelfth legion of the Roman

army was shut up in a defile, and reduced to great straits for want of water, when a body of Christians enrolled in the legion prayed for relief. Not only was rain sent, which enabled the Romans to quench their thirst, but a fierce storm of hail beat upon the enemy, accompanied by thunder and lightning, which so terrified them that a complete victory was obtained, and the legion was ever after called *The Thundering Legion* (*Legio Fulminatrix*). After this victory the Marcomanni, the Quadi, as well as the rest of the barbarians, sued for peace. The sedition of the Syrian governor Avidius Cassius, with whom Faustina, the empress, was in treasonable communication, called off the emperor from his conquests, but before he reached Asia the rebel was assassinated. Aurelius returned to Rome, after visiting Egypt and Greece, but soon new incursions of the Marcomanni compelled him once more to take the field. He defeated the enemy several times, but was taken sick at Sirmium, and died at Vindobona (Vienna) in 180. His only extant work is the *Meditations*, written in Greek. It has been translated into most modern languages (into English first by George Long in 1862, and by J. Jackson in 1906). This may be regarded as a manual of practical morality, in which wisdom, gentleness, and benevolence are combined in the most fascinating manner. Many believe it to have been intended for the instruction of his son Commodus. Aurelius was one of the best emperors Rome ever saw, although his philosophy and the magnanimity of his character did not restrain him from the persecution of the Christians, whose religious doctrines he was led to believe were subversive of good government, and whom he charged, therefore, with obstinacy, the greatest social crime in the eyes of Roman authority. Marcus Aurelius was not so much a philosopher as a seeker after righteousness.—BIBLIOGRAPHY: P. B. Watson, *M. Aurelius Antoninus*; Sir Samuel Dill, *Roman Society from Nero to Marcus Aurelius*; Translations of the *Meditations* by G. H. Kendall, and J. Jackson.

Aurengzebe. See *Aurangzib*.

Aure'ola, or **Au'reole**, in paintings, an illumination surrounding the whole figure of a holy person, as Christ, a saint, or a martyr, intended to represent a luminous cloud or haze emanating from him. It is generally of an oval shape, or may be nearly or quite circular, and is of similar character with the nimbus surrounding the heads of sacred personages.

Au´rēus, the first gold coin which was coined at Rome, 207 B.C. Its value varied at different times, from about 12s. to £1, 4s. See *Numismatics*.

Aurich (ou´rēh), a German town, province of Hanover. Pop. 6070.

Au´ricle. See *Heart*.

Auric´ula, a garden flower derived from the yellow *Primūla Auricūla*, found native in the Swiss Alps, and sometimes called bear's-ear from the shape of its leaves. It has for over three centuries been an object of cultivation by florists, who have succeeded in raising from seed a great number of beautiful varieties. Its leaves are obovate, entire or serrated, and fleshy, varying, however, in form in the numerous varieties. The flowers are borne on an erect umbel and central scape with involucre. The original colours of the corolla are yellow, purple, and variegated, and there is a mealy covering on the surface. There are auricula clubs and societies in the north of England.

Auricular Confession. See *Confession*.

Au´rifaber, the Latinized name of Johann Goldschmidt, one of Luther's companions, born 1519, became pastor at Erfurt in 1566, died there in 1579. He collected the unpublished MSS. of Luther, and edited the *Epistolæ* and the *Table-talk*.

Auriflamme. See *Oriiflamme*.

Auri´ga, in astronomy, the *Waggoner*, a constellation of the northern hemisphere, containing Capella, a star of the first magnitude. *Nova Aurigæ*, a temporary star, appeared in the constellation in 1892.

Aurillac (ō-rē-ya´k), a town of France, capital of the department Cantal, in a valley watered by the Jordanne, about 270 miles S. of Paris; well built, with wide streets; copper-works, paper-works, manufactures of lace, tapestry, leather, &c. Pop. 18,036.

Aurochs (a´roks), a species of wild bull or buffalo, the *urus* of Cæsar, *bison* of Pliny, the European bison, *Bos* or *Bonassus* *Bison* of modern naturalists. The animal was once abundant in Europe, but were it not for the

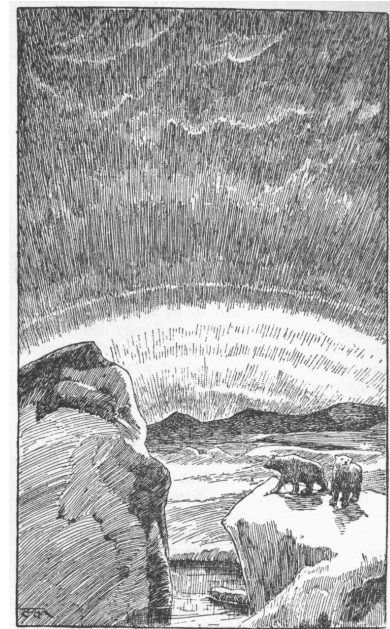
protection afforded by the late Emperor of Russia to a few herds which inhabit the forests of Lithuania it would before this have been extinct.

Auro´ra, an American city, of Kane county, Illinois, on Fox River, 40 miles W. by S. of Chicago; it has flourishing manufactures, railway-works, and a considerable trade. Pop. (1920), 36,265.

Auro´ra (Gr. *Eōs*), in classical mythology, the goddess of the dawn, daughter of Hyperion and Theia, and sister of Helios and Selēnē (Sun and Moon). She was represented as a charming figure, 'rosy-fingered', clad in a yellow robe, rising at dawn from the ocean and driving her chariot through the heavens. Among the mortals whose beauty captivated the goddess, poets mentioned Orion, Tithōnus, and Cephālus.

Auro´ra, one of the New Hebrides Islands, S. Pacific Ocean, about 30 miles long by 5 wide. It rises to a considerable elevation, and is covered with a luxuriant vegetation.

Auro´ra Borea´lis, a luminous meteoric phenomenon appearing in the north, most frequently in high latitudes, the corresponding phenomenon in the southern hemisphere being called *Aurora Australis*, and both being also called *Polar Light*, *Streamers*, &c. The northern aurora has been far the most observed and studied. It usually manifests itself by streams of light ascending towards the zenith from a dusky line of cloud or haze a few degrees above the horizon, and stretching from the north towards the west and east, so as to form an arc with its ends on the horizon, and its different parts and rays are constantly in motion. Sometimes it appears in detached places; at other times it almost covers the whole sky. It assumes many shapes and a variety of colours, from a pale red or yellow to a deep red or blood colour; and in far northern latitudes serves to illuminate the earth and cheer the gloom of the long winter nights. The appearance of the aurora borealis so exactly resembles the effects of experimental electrical



Aurora Borealis

phenomena that there is every reason to believe that their causes are similar. When electricity passes through rarefied air it exhibits a diffused luminous stream which has all the characteristic appearances of the aurora, and hence it is highly probable that this natural phenomenon is occasioned by the passage of electricity through the upper regions of the atmosphere. The synchronism of auroral display with disturbances of the magnetic needle is an ascertained fact, and the connection between auroræ and magnetism is further evident from the fact that the beams or coruscations issuing from a point in the horizon west of north are frequently observed to run in the magnetic meridian. What are known as magnetic storms are invariably connected with exhibitions of the aurora, and with spontaneous galvanic currents in the ordinary telegraph wires; and this connection is found to be so certain that, upon remarking the display of one of the three classes of phenomena, we can at once assert that the other two are also present. In recent years it has been established that auroræ wax and wane in frequency *pari passu* with sun-spots in an 11-year cycle, and that they often manifest themselves about the time of transit of a conspicuous spot across the sun's central meridian. Also they frequently recur at successive intervals of about 27 days, which is the period of a solar rotation relative to the earth. It is therefore inferred that auroræ are largely excited by influences proceeding from the sun, and it is suggested that they are the result of the impinging upon our upper atmosphere of streams of electric corpuscles expelled from the solar orb, these streams when approaching our planet being mainly directed to its higher latitudes as a consequence of its magnetic polarity. The aurora borealis is said to be frequently accompanied by sound, which is variously described as resembling the rustling of pieces of silk against each other, or the sound of wind against the flame of a candle. The aurora of the southern hemisphere is quite a similar phenomenon to that of the north.—
BIBLIOGRAPHY: A. Angot, *Les Aurores Polaires*; Captain H. P. Dawson, *Observations of the International Polar Expeditions, 1882-3, Fort Rae*.

Aurungabad. See *Aurangabad*.

Aurungzebe. See *Aurangzib*.

Auscultation, a method of distinguishing the state of the internal parts of the body, particularly of the thorax and abdomen, by observing the sounds arising in the part either through the immediate application of the ear to its

surface (immediate auscultation), or by applying the stethoscope to the part, and listening through it (mediate auscultation). Auscultation may be used with more or less advantage in all cases where morbid sounds are produced, but its general applications are: the *auscultation* of respiration, the *auscultation* of the voice; *auscultation* of coughs; *auscultation* of sounds foreign to all these, but sometimes accompanying them; *auscultation* of the actions of the heart; obstetric *auscultation*. The parts when struck also give different sounds in health and disease.

Auso´nia, an ancient poetical name of Italy.

Auso´nius, Decimus Magnus (c. A.D. 310-395), Roman poet, born at Burdigala (Bordeaux). Valentinian entrusted to him the education of his son Gratian, and appointed him afterwards quæstor and pretorian prefect. Gratian appointed him consul in Gaul, and after this emperor's death he lived upon an estate at Bordeaux, devoted to literary pursuits. He wrote epigrams, idyls, eclogues, letters in verse, &c., still extant, and was probably a Christian. He was rather a man of letters than a poet, and his poems are devoid of inspiration.

Aus´pices, among the ancient Romans strictly omens or auguries derived from birds, though the term was also used in a wider sense. Nothing of importance was done without taking the auspices, which, however, simply showed whether the enterprise was likely to result successfully or not, without supplying any further information. Magistrates possessed the right of taking the auspices, in which they were usually assisted by an augur. Before a war or campaign a Roman general always took the auspices, and hence the operations were said to be carried out 'under his auspices'. See *Augur*.

Aus´sig, a town in Bohemia, in the republic of Czecho-Slovakia, formerly in Austria, near the junction of the Bila with the Elbe, 42 miles N.N.W. of Prague; has large manufactures of woollens, chemicals, &c. The town is now known as Ousti nad Labem. Pop. 40,000.

Aus´ten, Jane, English novelist, born 1775, at Steventon, in Hants, of which parish her father was rector. Her principal novels are, *Sense and Sensibility*; *Pride and Prejudice*, which Disraeli is said to have read seventeen times; *Mansfield Park*; and *Emma*. Two more were published after her death,

entitled *Northanger Abbey* and *Persuasion*, which were, however, her most early attempts. Her novels are marked by ease, humour, and a complete knowledge of the domestic life of the English middle classes of her time. She died in 1817.

Austenite, a constituent of high-carbon steel (q.v.).

Aus'terlitz, a town with 3703 inhabitants, in Moravia, 10 miles E. of Brünn, famous for the battle of 2nd Dec., 1805, fought between the French (70,000 in number) and the allied Austrian and Russian armies (95,000). The decisive victory of the French led to the Peace of Pressburg between France and Austria.

Aus'tin, capital of the State of Texas, on the Colorado, about 200 miles from its mouth, and accessible to steamboats during certain seasons. There is a State university and other institutions, and a splendid capitol built of red granite. Pop. (1920), 34,876.

Austin, Alfred, English poet, born at Hedingley, near Leeds, in 1835, educated at Stonyhurst and St. Mary's College, Oscott; took the degree of B.A. at London in 1853, was called to the bar and practised, but gave up law for literature in 1861. He published, in 1861, a satire called *The Season*, followed by many poems, including *The Human Tragedy*, *The Golden Age*, *Savonarola* (a tragedy), *English Lyrics*, *Fortunatus the Pessimist*, *Lyrical Poems*, *Narrative Poems*, *Prince Lucifer*, *Alfred the Great*, *A Tale of True Love*, *Flodden Field* (a tragedy), &c. His works in prose include *The Garden that I love*, *In Veronica's Garden*, *Spring and Autumn in Ireland*, *Haunts of Ancient Peace*, *The Bridling of Pegasus*, &c. He was made Poet Laureate in 1896, about four years after the death of Tennyson. He died in 1913.

Aus'tin, John, an English writer on jurisprudence, born 1790, died 1859. From 1826 to 1835 he filled the chair of jurisprudence at London University. He served on several royal commissions, one of which took him to Malta; lived for some years on the Continent, and finally settled at Weybridge, in Surrey. His fame rests solely on his great works: *The Province of Jurisprudence Determined*, published in 1832; and his *Lectures on Jurisprudence*, published by his widow between 1861 and 1863.—His wife, Sarah, one of the Taylors of Norwich, produced translations of

German works, and other books bearing on Germany or its literature; also, *Considerations on National Education*, &c. Born 1793, died 1867. Her daughter, Lady Duff Gordon, translated Meinhold's *Mary Schweidler, the Amber Witch*, and other German works.

Austin, St. See *Augustine*.

Austin Friars. See *Augustins*.

Australasia, a division of the globe usually regarded as comprehending the Islands of Australia, Tasmania, New Zealand, New Caledonia, the New Hebrides, the Solomon Islands, New Ireland, New Britain, the Admiralty Islands, New Guinea, and the Arru Islands, besides numerous other islands and island groups; estimated area, 3,400,000 sq. miles; pop. 6,000,000. It forms one of the three portions into which some geographers have divided Oceania, the other two being Malaysia and Polynesia. The British territories in Australasia comprise the Commonwealth of Australia, the Australian dependencies of Papua and Northern Territory, New Zealand and adjacent islands, and the crown colony of Fiji.



Australia (older name, **New Holland**), the largest island in the world, a sea-girt continent, lying between the Indian and Pacific Oceans, s.e. of Asia; between lat. 10° 39' and 39° 11' s.; long. 113° 5' and 153° 16' e.; greatest length, from w. to e., 2400 miles; greatest breadth, from n. to s., 1700 to 1900 miles. It is separated from New Guinea on the north by Torres Strait, from Tasmania on the south by Bass Strait. It is divided into two unequal parts by the Tropic of Capricorn, and consequently belongs partly to the South Temperate, partly to the Torrid Zone. It is occupied by five British colonies, namely, New South Wales, Victoria, and Queensland in the east; South Australia in the middle, stretching from sea to sea; and Western Australia in the west, which, with the Island of Tasmania, form the Commonwealth of Australia. Their area and population are as follows:—

	Area in sq. m.	Pop. in 1920
New South Wales	309,432	2,091,115
Victoria	87,884	1,528,151
Queensland	670,500	752,245
South Australia	380,070	491,177
Western Australia	975,920	330,819
Tasmania	26,215	212,847
Northern Territory	523,620	3,992
Federal Territory	940	1,972

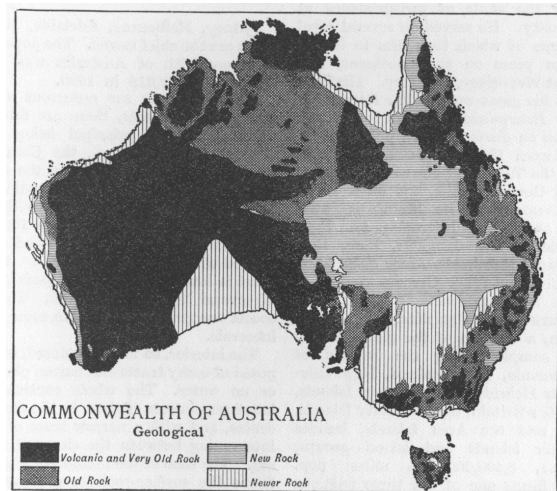
Sydney, Melbourne, Adelaide, Brisbane, and Perth are the chief towns. The population of the Commonwealth of Australia was 4,895,894 in 1917 and 5,412,318 in 1920.

Although there are numerous spacious harbours on the coasts, there are few remarkable indentations, the principal being the Gulf of Carpentaria, on the n., the Great Australian Bight, and Spencer's Gulf, on the s. The chief projections are Cape York Peninsula and Arnhem Land in the north. Parallel to the n.e. coast runs the Great Barrier Reef for 1000 miles. In great part the e. coast is bold and rocky, and is fringed with many small islands. Part of the s. coast is low and sandy, and part presents cliffs of several hundred feet high. The n. and w. coasts are generally low, with some elevations at intervals.

The interior, so far as explored, is largely composed of rocky tracts and barren plains with little or no water. The whole continent forms an immense plateau, highest in the east, low in the centre, and with a narrow tract of land usually intervening between the elevated area and the sea. The base of the table-land is granite, which forms the surface-rock in a great part of the south-west, and is common in the higher grounds along the east side. Secondary (cretaceous) and tertiary rocks are largely developed in the interior. Silurian rocks occupy a large area in South Australia, on both sides of Spencer Gulf. The mountainous region in the south-east and east is mainly composed of volcanic, Silurian, carbonaceous, and carboniferous rocks yielding good coal. No active volcano is known to exist, but in the south-east there are some craters only recently extinct. The highest and most extensive mountain-system is a belt about 150 miles wide skirting the whole eastern and south-eastern border of the continent, and often called, in whole or in part, the Great Dividing Range, from forming the great watershed of Australia. A part of it, called the Australian Alps, in the south-east, contains the highest summits in Australia, Mount Kosciusko (7175 feet), Mount Clarke (7256 feet), and Mount Townshend (7353 feet). West of the Dividing Range are extensive plains or downs admirably adapted for pastoral purposes. The deserts and scrubs, which occupy large areas of the interior, are a characteristic feature of Australia. The former are destitute of vegetation, or are covered only with a coarse, spiny grass that affords no sustenance to cattle or horses; the latter are composed of a dense growth of shrubs and low trees, often impenetrable till the traveller has cleared a track with his axe.

The rivers of Australia are nearly all subject to great irregularities in volume, many of them at one time showing a channel in which there is merely a series of pools, while at another they inundate the whole adjacent country. The chief is the Murray, which, with its affluents the Murrumbidgee, Lachlan, and Darling, drains a great part of the interior west of the Dividing Range, and falls into the sea on the south coast (after entering Lake Alexandrina). Its greatest tributary is the Darling, which may even be regarded as the main stream. On the east coast are the Hunter, Clarence, Brisbane, Fitzroy, and Burdekin; on the west, the Swan, Murchison, Gascoyne, Ashburton, and De Grey; on the north, the Fitzroy, Victoria, Flinders, and Mitchell. The Australian rivers are of little service in

facilitating internal communication. Many of them lose themselves in swamps or sandy wastes of the interior. A considerable river of the interior is Cooper's Creek, or the Barcoo, which falls into Lake Eyre, one of a group of lakes on the south side of the continent having no outlet, and accordingly salt. The principal of these are Lakes Eyre, Torrens, and Gairdner, all of which vary in size and saltiness according to the season. Another large salt lake of little depth, Lake Amadeus, lies a little west of the centre of Australia. Various others of less magnitude are scattered over the interior.



COMMONWEALTH OF AUSTRALIA
Geological

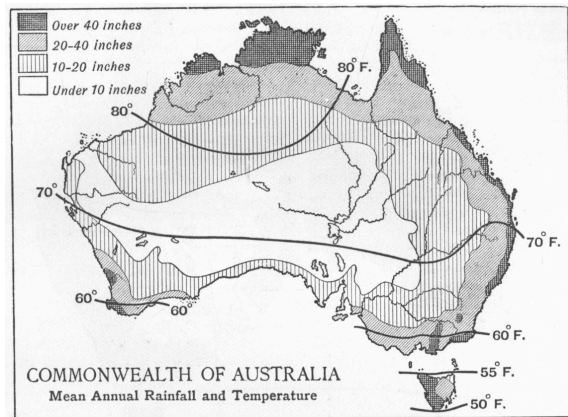
The climate of Australia is generally hot and dry, but very healthy. In the tropical portions there are heavy rains, and in most of the coast districts there is a sufficiency of moisture, but in the interior the heat and drought are extreme. Considerable portions now devoted to pasturage are liable at times to suffer from drought. At Melbourne the mean temperature is about 56°, at Sydney about 63°. The south-eastern settled districts are at times subject to excessively hot winds from the interior, which cause great discomfort, and are often followed by a violent cold wind from the south ('southerly bursters'). In the mountainous and more temperate parts snow-storms are common in winter (June, July, and August).

Australia is a region containing a vast quantity of mineral wealth. Foremost come its rich and extensive deposits of gold, which, since the precious metal was first discovered in 1851, have produced a total of nearly £600,000,000. The greatest quantity has been obtained in Victoria, but New South Wales and Queensland have also yielded a considerable amount, and now Western Australia stands first in respect of annual output. Australia also possesses silver, copper, tin, lead, zinc, antimony, mercury, plumbago, &c., besides coal (now worked to a considerable extent in New South Wales) and iron. Various precious stones are found, as the garnet, ruby,

topaz, sapphire, and even the diamond. Of building-stone there are granite, limestone, marble, and sandstone.

The Australian flora presents peculiarities which mark it off by itself in a very decided manner. Many of its most striking features have an unmistakable relation to the general dryness of the climate. The trees and bushes have for the most part a scanty foliage, presenting little surface for evaporation, or thick leathery leaves well fitted to retain moisture. The most widely-spread types of Australian vegetation are the various kinds of

gum tree (*Eucalyptus*), the shea-oak (*Casuarina*), the acacia or wattle, the grass tree (*Xanthorrhæa*), many varieties of Proteaceæ, and a great number of ferns and tree-ferns. Of the gum tree there are found upwards of 150 species, many of which are of great value. Individual specimens of the 'peppermint' (*E. amygdalina*) have been found to measure from 480 to 500 feet in height. As timber trees the most valuable members of this genus are the *E. rostrata* (or red gum), *E. leucoxylon*, and *E. marginata* (jarrah), the timber of which is hard, and almost indestructible. A number of the gum trees have deciduous bark. The wattle or acacia includes about 300 species, some of them of considerable economic value, yielding good timber or bark for tanning. The most beautiful and most useful is that known as the golden wattle (*A. dealbata*) which in spring is adorned with rich masses of fragrant yellow blossom. Palms—of which there are twenty-four species, all except the coco-palm peculiar to Australia—are confined to the north and east coasts. In the 'scrubs' already mentioned hosts of densely-intertwisted bushes occupy extensive areas. The *mallee* scrub is formed by a species of dwarf eucalyptus, the *mulga* scrub by a species of thorny acacia. A plant which covers large areas in the arid regions is the *spinifex* or porcupine grass, a hard, coarse, and excessively spiny plant, which renders travelling difficult, wounds the feet of horses, and is utterly uneatable by any animal. Other large tracts are occupied by herbs or bushes of a more valuable kind,



COMMONWEALTH OF AUSTRALIA Mean Annual Rainfall and Temperature

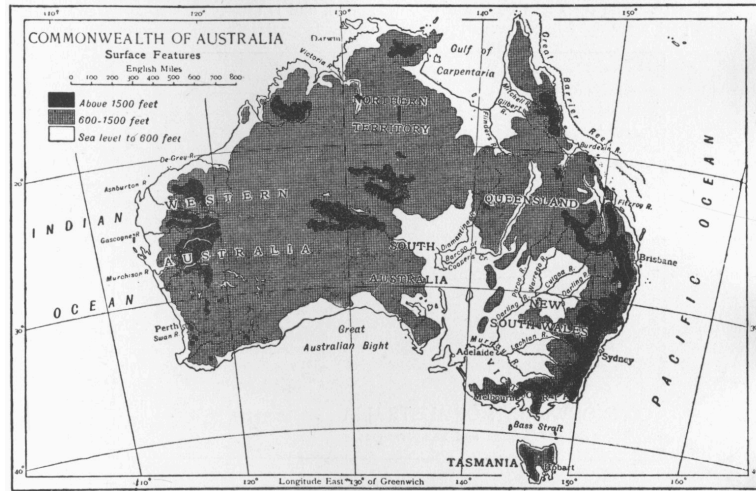
from their affording fodder. Foremost among those stands the salt-bush (*Atriplex nummularia*, ord. Chenopodiaceæ). Beautiful flowering plants are numerous. Australia also possesses great numbers of turf-forming grasses, such as the kangaroo-grass (*Anthistiria austrālis*), which survives even a tolerably-protracted drought. The native fruit trees are few and unimportant, and the same may be said of the plants yielding roots used as food; but exotic fruits and vegetables may now be had in the different colonies in great abundance and of excellent quality. The vine, the olive, and mulberry thrive well, and quantities of wine are now produced. The cereals of Europe and maize are extensively cultivated, and large tracts of country, particularly in Queensland, are under the sugar-cane.

The Australian fauna is almost unique in its character. Its great feature is the nearly total absence of all the forms of mammalia which abound in the rest of the world, their place being supplied by a great variety of marsupials—these animals being nowhere else found, except in the opossums of America. There are about 110 kinds of marsupials (of which the kangaroo, wombat, bandicoot, and phalangers or opossums, are the best-known varieties), over twenty kinds of bats, a wild dog (the dingo), and a number of rats and mice. Two extraordinary animals, the platypus, or water-mole of the colonist (*Ornithorhynchus*), and the porcupine ant-eater (*Echidna*) constitute the lowest order of mammals (*Monotremata*), and are confined to Australia. Their young are produced from eggs. Australia now possesses a large stock of the domestic animals of Britain, which thrive there remarkably well. The breed of horses is excellent. Horned cattle and sheep are largely bred, the first attaining a great size, while the sheep improve in fleece and their flesh in flavour. There are upwards of 650 different species of birds, the largest being the emu, or Australian ostrich, and a species of cassowary. Peculiar to the country are the black swan, the honey-sucker, the lyre-bird, the brush-turkey, and other mound-building birds, the bower-birds, &c. The parrot tribe preponderates over most other groups of birds in the continent. There are many reptiles, the largest being the alligator, found in some of the northern rivers. There are upwards of sixty different species of snakes, some of which are very venomous. Lizards, frogs, and insects are also numerous in various parts. The seas, rivers, and lagoons abound in fish of numerous varieties, and other aquatic animals, many of them peculiar. Whales and seals frequent the coasts. On the northern coasts are extensive

fisheries of trepang, much visited by native traders from the Indian Archipelago. Some animals of European origin, such as the rabbit and the sparrow, have developed into real pests in several of the colonies.

The natives belong to the Australian negro stock, and are sometimes considered the lowest as regards intelligence in the whole human family, though this is doubtful. At the census of 1891 they were believed to number about 60,000, exclusive of those in the unexplored parts. They are of a dark-brown or

black colour, with jet-black curly but not woolly hair, of medium size, but inferior muscular development. In the settled parts of the continent they are inoffensive, and rapidly dying out. They have no fixed habitations; in the summer they live almost entirely in the open air, and in the more inclement weather they shelter themselves with bark erections of the rudest construction. They have no cultivation and no domestic animals. Their food consists of such animals as they can kill, and no kind of living creature seems to be rejected, snakes, lizards, frogs, and even insects being eaten, often half raw. They are ignorant of the potter's art. In their natural condition they wear little or no clothing. They speak a number of different languages or dialects. The women are regarded merely as slaves, and are frequently maltreated. They have no religion; they practise polygamy, and are said sometimes to resort to cannibalism, but only in exceptional circumstances. They are occasionally employed by the settlers in light kinds of work, and as horse-breakers; but they dislike continuous occupation, and soon give it up. The weapons of all the tribes are generally similar, consisting of spears, shields, boomerangs, wooden axes, clubs, and stone hatchets. Of these the boomerang is the most singular, being an invention confined to the Australians.



COMMONWEALTH OF AUSTRALIA Surface Features

The five colonies, independently of each other, having declared their desire for a federal union, the Commonwealth of Australia was proclaimed. A Convention, which sat at Adelaide in 1897-8, drafted a Constitution Bill, and on 9th July, 1900, the British Parliament passed the Act to constitute the Commonwealth. There is now a Governor-General and Central or Federal Parliament, consisting of a Senate and a House of Representatives, while each State has also (as before) a Governor, an administration, and a Parliament of its own. Each Parliament consists of two Houses corresponding to the British House of Lords and House of Commons, but both Houses are elected by popular vote. Altogether the machinery of government very much resembles that of the home country. The aggregate annual revenue of the colonies is over £30,000,000. The public debt on 30th June, 1917, was £372,517,623. The former militia and volunteer units have been gradually merged into the new Citizen Army started by the Australian Government in 1911. The Government also agreed in the same year to furnish an Australian fleet unit, upon which King George conferred the title of 'Royal Australian Navy'. There is no established Church in the colonies. The denomination which numbers most adherents is the English or Anglican Church, next to which come the Roman Catholics, Presbyterians, and Methodists. Education is well provided for, instruction in the primary schools being in some cases free and compulsory, and the higher education being more and more attended to. There are flourishing universities in Melbourne, Sydney, and Adelaide. Newspapers are exceedingly numerous, and periodicals of all kinds are abundant. There is as yet no native literature of any distinctive type, but names of Australian writers of ability both in prose and poetry are beginning to be known beyond their own country.

Pastoral and agricultural pursuits and mining are the chief occupations of the people, though manufactures and handicrafts also employ large numbers. The total land area under cultivation was calculated at 13,298,576 acres in 1919-20. For sheep-rearing and the growth of wool the Australian colonies are unrivalled, and while the production of gold has considerably decreased, that of wool is constantly on the increase. The great bulk of the wool exported goes to Britain, which receives over 300,000,000 pounds from the Australian colonies annually (£37,256,915 in 1919-20). The commerce is rapidly extending, and becoming every year more important to Britain, whence the colonists derive their chief supplies of manufactured

goods in return for wool, gold, and other produce. Next to wool come gold, tin, copper, wheat, preserved meat, and tallow, hides and skins, cotton, tobacco, sugar, and wine as the most important items of export. The chief imports consist of textile fabrics, haberdashery, and clothing, machinery and metal goods. The aggregate imports in 1920 amounted to about £97,456,899 in value, the exports to £148,564,523. There are upwards of 23,000 miles of railway in actual use or in course of construction, and about 133,000 miles of telegraph. In 1912 the building of the Trans-Australian railway from Port Augusta to Kalgoorlie was commenced. The longest telegraph line is that running northwards across the continent from Adelaide. The two chief routes for mails between Britain and the Australian colonies are by way of the Suez Canal, and by San Francisco across the American continent. The coinage is the same as in the mother country. Banks and banking offices are numerous, including post office or other savings banks for the reception of small sums.

It is doubtful when Australia was first discovered by Europeans. Between 1531 and 1542 the Portuguese published the existence of a land which they called Great Java, and which corresponded to Australia, and probably the first discovery of the country was made by them early in the sixteenth century. The first authenticated discovery is said to have been made in 1601 by a Portuguese named Manoel Godinho de Eredia. In 1606 Torres, a Spaniard, passed through the strait that now bears his name, between New Guinea and Australia. Between this period and 1628 a large portion of the coastline of Australia had been surveyed by various Dutch navigators. In 1664 the continent was named New Holland by the Dutch Government. In 1688 Dampier coasted along part of Australia, and about 1700 explored a part of the w. and N.W. coasts. In 1770 Cook carefully surveyed the E. coast, named a number of localities, and took possession of the country for Britain. He was followed by Bligh in 1789, who carried on a series of observations on the N.E. coast, adding largely to the knowledge already obtained of this new world. Colonists had now arrived on the soil, and a penal settlement was formed (1788) at Port Jackson. In this way was laid the foundation of the future colony of New South Wales. The Moreton Bay district (Queensland) was settled in 1825; in 1835 the Port Phillip district. In 1851 the latter district was erected into a separate colony under the name of Victoria. Previous to this time the colonies both of Western Australia and

of South Australia had been founded—the former in 1829, the latter in 1836. The latest of the colonies is Queensland, which dates from 1859. The discovery of gold in abundance took place in 1851 and caused an immense excitement and great influx of immigrants. The population was then only about 350,000, and was slowly increasing; but the discovery of the precious metal started the country on that career of prosperity which has since been almost uninterrupted. Convicts were long sent to Australia from the mother country, but transportation to New South Wales practically ceased in 1840, and the last convict vessel to W. Australia arrived in 1868. Altogether about 70,000 convicts were landed in Australia (besides almost as many in Tasmania).

The record of interior exploration forms an interesting part of Australian history. This has been going on since early last century, and is as yet far from complete. There is still a large area of the continent of which little or nothing is known, comprising especially a vast territory belonging to Western Australia, and a portion of South Australia. Among the men who have won fame in the field of Australian exploration are Oxley (1817-23), who partly explored the Lachlan and Macquarie, discovered the Brisbane, &c.; Hume and Hovell (1824), who crossed what is now the colony of Victoria from north to south; Cunningham (1827), who discovered the Darling Downs; Sturt (1828-9), who examined the Macquarie, part of the Darling, and the Murrumbidgee, which he traced to the Murray, sailing down the latter to Lake Alexandrina; in 1844 he penetrated to near the middle of the continent from the south; Mitchell (1831-6) made extensive explorations in New South Wales and Victoria; M'Millan (1839) explored and traversed Gippsland; Eyre (1840) travelled by the coast from Adelaide to King George's Sound; Leichhardt in 1844-5 travelled from Brisbane to Port Essington, discovering fine tracts of territory and the numerous rivers flowing into the Gulf of Carpentaria; in 1848 he was lost in the northern interior, in attempting to cross Australia from east to west, and nothing further regarding his fate has been discovered; Kennedy (1848) was killed in exploring Cape York Peninsula; A. C. Gregory (1855-6) explored part of North-Western Australia, and crossed from that to the Brisbane district, an important exploring journey; M'Douall Stuart (1859-60-2) crossed the continent from south to north and back again nearly in the line of the present overland telegraph; Burke, Wills, Gray, and King (1860-1) crossed

from Melbourne to the Gulf of Carpentaria, but Burke, Wills, and Gray perished on the return journey; F. T. Gregory (1861) explored the region of the Ashburton, Fortescue, and other rivers of North-West Australia; Warburton (1873) travelled with camels from the centre of the continent to the north-west coast; J. Forrest (1874) made an important journey in Western Australia; Giles (1874-6) explored Central Western Australia; Favenc (1878-9) travelled from Brisbane to Port Darwin; A. Forrest (1879) explored part of Northern Australia; Mills (1883) traversed with camels a considerable stretch of new ground in Western Australia; Winnicke (1883-4), also with camels, explored and mapped about 40,000 sq. miles of the unknown interior; Lindsay (1885-6) travelled north-west from Lake Eyre, and then north-east to the Gulf of Carpentaria. Among subsequent explorations mention may be made of Carnegie's eight months' journeyings in W. Australia (1896-7). (See articles on the separate colonies.)—BIBLIOGRAPHY: Wall, *Physical Geography of Australia*; Aflalo, *Natural History of Australia*; T. A. Coghlan, *A Statistical Account of the Seven Colonies of Australasia* (Sydney); G. W. Rusden, *History of Australia*.

Austra'lioids, one of the five groups into which Professor Huxley classifies man, comprising the indigenous non-Aryan inhabitants of Central and Southern India, the ancient Egyptians and their descendants, and the modern Fellahs.

Austria, or **German Austria**, a republic formed out of the fragments of the old Austro-Hungarian Empire. The republic was proclaimed on 12th Nov., 1918, and the government was taken in hand by the National Constitutional Assembly. The new republic, the frontiers of which were provisionally defined by the treaty of St. Germain (q.v.), includes Upper Austria, Lower Austria, Salzburg, and parts of Styria, Carinthia, Tyrol, and Vorarlberg, some districts having been assigned to Yugo-Slavia and Italy. German Austria now has an area of 30,716 sq. miles, and a pop. of 6,130,197.

Austria-Hungary (in Ger. *Oesterreich*, that is, Eastern Empire), previous to the changes resulting from the European War (1914-18), a Central European monarchy, inhabited by several nationalities, and consisting of the Austrian Empire, the Hungarian monarchy, and the territory of Bosnia and Herzegovina annexed by Austria on 5th Oct., 1908. The States had together a total area of about 261,242 sq. miles, and were bounded s. by Turkey, the

Adriatic, and Italy; w. by Switzerland, Bavaria, and Saxony; n. by Prussia and Russian Poland; and e. by Russia and Roumania. On 12th Nov., 1918, the German-Austrian Republic was proclaimed, and the government was taken in hand by the National Constitutional Assembly. The account given in the following pages applies to the countries only as they existed before the end of the European War.

The population of Austria proper in 1913 was estimated at 29,193,293.

The prevailing character of the Austrian dominions is mountainous or hilly, the plains not occupying more than a fifth part of the whole surface. The loftiest ranges belong to the Alps, and are found in Styria, Salzburg, and Carinthia, the highest summits being the Ortlerspitzen (12,814 feet) on the western boundary of Tyrol, and the Grossglockner (12,300 feet) on the borders of Salzburg, Tyrol, and Carinthia. Another great range is that of the Carpathians. The chief rivers are the Danube, with its tributaries, and the Dniester. The Danube is navigable for pretty large vessels; the tributaries also are largely navigable. The lakes are numerous and often picturesque, the chief being Lake Balaton or the Plattensee. The climate is exceedingly varied, but generally good. The principal products of the north are wheat, barley, oats, and rye; in the centre vines and maize are added; and in the south olives and various fruits. The cereals grow to perfection, other crops being hops, tobacco, flax, and hemp. Sheep and cattle are largely reared.—Wild deer, wild swine, chamois, foxes, lynxes, and a species of small black bear are found in many districts, the fox and lynx being particularly abundant.—In mineral productions Austria is very rich, possessing, with the exception of platinum, all the useful metals, the total annual value of the mineral products of Austria being estimated at upwards of £15,000,000, the principal being coal, salt, and iron.

Before the European War, manufactures were in the most flourishing condition in Bohemia, Moravia, Silesia, and Lower Austria; less so in the eastern provinces, and insignificant in Dalmatia, Bukowina, Herzegovina, &c. Among the most important manufactures were those of machinery and metal goods, Austria holding a high place for the manufacture of musical and scientific instruments, gold and silver plate and jewellery; of stone and china-ware and of glass, which is one of the oldest and most highly-developed industries in Austria; of chemicals; of sugar from beet; of beer,

spirits, &c.; and especially the manufactures of wool, cotton, hemp, and flax.

In addition to the general import and export trade, Austria carried on a very considerable amount of business in the transit of goods through her territory. In 1914 the total value of imports into Austria-Hungary was £114,716,000, of exports £83,996,000; the value of imports in 1913 was £141,433,000, the exports £115,129,000. Among imports were cotton and other fibres, textile goods and yarn, metals, machinery, drugs, chemicals, oils, fats, hides, skins, &c. The chief exports were cereals, animals, metallic goods, woven fabrics, pottery and glass manufactures. Nearly two-thirds of the commerce are with Germany, next in importance being the trade with Roumania, Italy, and Russia. The exports direct to the United Kingdom in 1913 were £7,705,949; the imports of British produce direct, £4,480,760; but these amounts do not include indirect exports and imports through other countries. The staple exports to the United Kingdom were corn and flour. The chief imports from it were cotton manufactures, machinery and metals, woollen goods, fish, &c. In 1913 the mercantile navy of Austria had a total burden of about 471,252 tons. The principal ports were Trieste, in Austria, and Fiume, in Hungary.

None of the European States, except Russia, exhibited such a diversity of race and language as the former dual monarchy. The Slavs—who differ greatly, however, amongst themselves in language and civilization—amounted to above 20,000,000, or 46 per cent of the total population, and formed the great mass of the population of Bohemia, Moravia, Carniola, Galicia, Dalmatia, Croatia, and Slavonia, and Northern Hungary, and half the population of Silesia and Bukowina. The Germans, about 9,950,266, formed almost the sole population of the archduchy of Austria, Salzburg, the greatest portion of Styria and Carinthia, almost the whole of Tyrol and Vorarlberg, large portions of Bohemia and Moravia, the whole of West Silesia, &c.; and they were also numerous in Hungary and Transylvania. The Magyars or Hungarians (10,061,549) formed nearly half of the inhabitants of the former Kingdom of Hungary and the connected provinces.

The State religion of the former dual monarchy was the Roman Catholic. In 1910 there were in the Austrian portion of the monarchy 22,530,000 Roman

Catholics, 3,417,000 Greek Catholics united to the Roman Church, 689,000 non-united, 589,000 Protestants, and 1,314,000 Jews.

The intellectual culture of the people was highest in the German provinces, but in some of the other provinces the percentage of uneducated was very high. Attendance at the elementary schools was compulsory on all children from their sixth to the end of their fourteenth year in most of the Austrian provinces; to at least the twelfth year in the whole empire. There were numerous gymnasia and 'real schools', the gymnasia being intended chiefly to prepare pupils for the universities, while in the real schools a more practical end was kept in view, and modern languages and physical science formed the groundwork of the educational course. The technical high schools in Austria were very important institutions, and there are many other special schools where the students received training in mining, agriculture, industries of all kinds, art, music, commerce, &c. In 1917 there were eight universities maintained by the State, viz. in Vienna, Prague (2), Gratz, Cracow, Lemberg, Innsbruck, and Czernowitz. Most of these have four faculties—Catholic theology, law and politics, medicine, and philosophy.

Before the revolution of 1918 the ruler of the dual monarchy had the title of emperor in his Austrian dominions, but he was only king of Hungary. The Parliament of the Austrian division of the empire was known as the Reichsrath, or council of the realm, consisting of an Upper House (Herrenhaus), composed of princes of the imperial family, nobles with the hereditary right to sit, archbishops, and life-members nominated by the Emperor; and a Lower House (Abgeordnetenhaus) of 516 deputies elected on the basis of universal suffrage. There were seventeen provincial Diets or Assemblies, each provincial division having one. In the Hungarian division of the Empire the legislature was a Parliament consisting of an Upper House or House of Magnates and of a Lower House or House of Representatives, the latter elected by all citizens of full age paying a small amount in taxes, or otherwise qualified. Its powers corresponded to those of the Austrian Parliament or Reichsrath. All matters affecting the joint interests of the two divisions of the monarchy, such as foreign affairs, war, and finance, were dealt with by the Parliaments of the two States. There were three Budgets, viz. one for common affairs, one for Austria, and one for Hungary. The Budget estimates for the former Austrian kingdom in

1913 were £130,716,773 (the revenue balancing the expenditure). In 1918 the estimates were: expenditure 24,321,140,000 crowns, revenue 4,854,789 crowns. Austria's special debt in Feb., 1918, was placed at 54,081,765,681 crowns, of which 29,274,603,300 crowns were war debt. On 20th Nov., 1918, the Provisional Government of the Austrian Republic passed a law adopting the Budget of 1918-9, and empowering the Minister of Finance to make good the difference between revenue and expenditure.

Military service was compulsory and universal throughout the dual monarchy. The old Austro-Hungarian army numbered about 820,000 men on a peace footing.

History.—In 791 Charlemagne drove the Avars from the territory between the Ens and the Raab, and united it to his empire under the name of *the Eastern Mark* (that is, March or boundary land); and from the establishment by him of a margraviate in this new province the former Austrian Empire took its rise. On the invasion of Germany by the Hungarians it became subject to them from 900 till 955, when Otho I, by the victory of Augsburg, reunited a great part of this province to the German Empire, which by 1043 had extended its limits to the Leitha. The margraviate of Austria was hereditary in the family of the Counts of Babenberg (Bamberg) from 982 till 1156, in which year the boundaries of Austria were extended so as to include the territory above the Ens, and the whole was created a duchy. The territory was still further increased in 1192 by the gift of the duchy of Styria as a fief from the Emperor Henry VI, Vienna being by this time the capital. The male line of the House of Bamberg became extinct in 1246, and the Emperor Frederick II declared Austria and Styria a vacant fief, the hereditary property of the German emperors. In 1282 the Emperor Rudolph granted Austria, Styria, and Carinthia to his two sons, Albert and Rudolph. The former became sole ruler (duke), and since then, until 12th Nov., 1918, Austria remained under the House of Habsburg. Albert, who was an energetic ruler, was elected emperor in 1298, but was assassinated in 1308. The first of his successors whom we need specially mention was Albert V, son-in-law of the Emperor Sigismund. He assisted Sigismund in the Hussite wars, and was elected, after his death, King of Hungary and of Bohemia, and German emperor (1438). Ladislaus, his posthumous son, was the last of the Austrian line proper, and its possessions devolved upon the collateral

Styrian line in 1457; since which time the House of Austria furnished an unbroken succession of German emperors.

In 1453 the Emperor Frederick III, a member of this House, had conferred upon the country the rank of an archduchy before he himself became ruler of all Austria. His son Maximilian I, by his marriage with Mary, the surviving daughter of Charles the Bold, united the Netherlands to the Austrian dominions. After the death of his father, in 1493, Maximilian was made Emperor of Germany, and transferred to his son Philip the government of the Netherlands. He also added to his paternal inheritance Tyrol, with several other territories, particularly some belonging to Bavaria, and acquired for his family new claims to Hungary and Bohemia. The marriage of his son Philip to Joanna of Spain raised the House of Habsburg to the throne of Spain. Philip, however, died in 1506, and the death of Maximilian, in 1519, was followed by the union of Spain and Austria, his grandson (the eldest son of Philip), Charles I, King of Spain, being elected Emperor of Germany as Charles V. Charles thus became the greatest monarch in Europe, but in 1521 he ceded to his brother Ferdinand all his dominions in Germany. Ferdinand I, by his marriage with Anna, the sister of Louis II, King of Hungary, acquired the kingdoms of Hungary and Bohemia, with Moravia, Silesia, and Lusatia, the appendages of Bohemia. To oppose him the woywode of Transylvania, John Zapolya, sought the help of the Sultan, Soliman II, who appeared in 1529 at the gates of Vienna, but was compelled to retreat. In 1535 a treaty was made by which John Zapolya was allowed to retain the royal title and half of Hungary, but after his death new disputes arose, and Ferdinand maintained the possession of Lower Hungary only by paying Soliman the sum of 30,000 ducats annually (1562). In 1556 Ferdinand obtained the imperial crown, when his brother Charles laid by the sceptre for a cowl. He died in 1564, leaving his territories to be divided amongst his three sons.

Maximilian II, the eldest, succeeded his father as emperor, obtaining Austria, Hungary, and Bohemia; Ferdinand, the second son, received Tyrol and Lower Austria; and Charles, the youngest, obtained Styria, Carinthia, Carniola, and Görz. Maximilian died in 1576, and was succeeded on the imperial throne by his eldest son Rudolph II, who had already been crowned King of Hungary in 1572, and King of Bohemia in 1575. Rudolph's reign was distinguished by the war against Turkey and

Transylvania; the persecutions of the Protestants, who were driven from his dominions; the cession of Hungary in 1608; and in 1611 of Bohemia and his hereditary estates in Austria to his brother Matthias. Matthias, who succeeded Maximilian on the imperial throne, concluded a peace with the Turks, but was disturbed by the Protestant Bohemians, who took up arms in defence of their religious rights, thus commencing the Thirty Years' War. After his death in 1619 the Bohemians refused to acknowledge his successor, Ferdinand II, until after the battle of Prague in 1620, when Bohemia had to submit, and was deprived of the right of choosing her king. Lutheranism was strictly forbidden in all the Austrian dominions. Hungary, which revolted under Bethlen Gabor, Prince of Transylvania, was, after a long struggle, subdued. During the reign of Ferdinand III (1637-57), successor of Ferdinand II, Austria was continually the theatre of war; Lusatia was ceded to Saxony in 1635; and Alsace to France in 1648, when peace was restored in Germany by the Treaty of Westphalia.

The Emperor Leopold I, son and successor of Ferdinand III, was victorious through the talents of Prince Eugene in two wars with Turkey; and Vienna was delivered by Sobieski and the Germans from the attacks of Kara Mustapha in 1683. In 1687 he united Hungary to Transylvania, and in 1699 restored to Hungary the country lying between the Danube and the Theiss. It was the chief aim of Leopold to secure to Charles, his second son, the inheritance of the Spanish monarchy, and in 1701, upon the victory of French diplomacy in the appointment of the grandson of Louis XIV, the War of the Spanish Succession commenced. Leopold died in 1705, but Joseph I, his eldest son, continued the war. As he died without children in 1711, his brother Charles was elected emperor, but was obliged to accede in 1714 to the Peace of Utrecht, by which Austria received the Netherlands, Milan, Mantua, Naples, and Sardinia. In 1720 Sicily was given to Austria in exchange for Sardinia. This monarchy now embraced over 190,000 sq. miles; but its power was weakened by new wars with Spain and France. In the peace concluded at Vienna (1735 and 1738) Charles VI was forced to cede Naples and Sicily to Spain and part of Milan to the King of Sardinia; and in 1739, by the Peace of Belgrade, he was obliged to transfer to the Porte Belgrade, Serbia, &c., partly in order to secure the succession to his daughter, Maria Theresa, by the Pragmatic Sanction. He died in 1740.

On Maria Theresa's marriage to Francis Stephen, Duke of Lorraine (the dynasty henceforth being that of Habsburg-Lorraine), and her accession to the Austrian throne, the Empire was threatened with dismemberment. Frederick II of Prussia subdued Silesia; the Elector of Bavaria was crowned in Linz and Prague, and in 1742 chosen emperor under the name of Charles VII; Hungary alone supported the heroic and beautiful queen. Charles, however, died in 1745, and the husband of Theresa was crowned Emperor of Germany as Francis I; but a treaty concluded in 1745 confirmed to Frederick the possession of Silesia, and by the Peace of Aix-la-Chapelle, 1748, Austria was obliged to cede the duchies of Parma, Piacenza, and Guastalla to Philip, Infant of Spain, and several districts of Milan and Sardinia. To recover Silesia, Maria Theresa formed an alliance with France, Russia, Saxony, and Sweden, and entered upon the Seven Years' War; but by the Peace of Hubertsherg, 1763, Silesia was recognized as Prussian territory. On the death of Francis I, in 1765, Joseph II, his eldest son, was appointed to assist his mother in the government and elected Emperor of Germany. The partition of Poland (1772) gave Galicia and Lodomeria to Austria, which also obtained Bukowina from the Porte in 1777. At the death of the Empress in 1780 Austria contained 235,000 sq. miles, with a pop. estimated at 24,000,000.

The liberal home administration of the Empress was continued and extended by her successor, Joseph II, who did much to further the spread of religious tolerance, education, and the industrial arts. The Low Countries, however, revolted, and he was unsuccessful in the war of 1788 against the Porte. His death took place in 1790. He was succeeded by his eldest brother, Leopold II, under whom peace was restored in the Netherlands, and in Hungary, and also with the Porte. On the death of his sister and her husband Louis XVI of France he formed an alliance with Prussia, but died in 1792, before the French revolutionary war broke out.

His son, Francis II, succeeded, and was elected German Emperor, by which time France had declared war against him as King of Hungary and Bohemia. In 1795, in the third partition of Poland, West Galicia fell to Austria, and by the Peace of Campo-Formio (1797) she received the largest part of the Venetian territory as compensation for her loss of Lombardy and the Netherlands. In 1799 Francis, in alliance with Russia, renewed the war with France until 1801, when the Peace of Lunéville was concluded. In

1804 Francis declared himself hereditary Emperor of Austria as Francis I, and united all his States under the name of the Empire of Austria, immediately taking up arms once more with his allies Russia and Great Britain against France. The war of 1805 was terminated by the Peace of Pressburg (26th Dec.), by which Francis had to cede to France the remaining provinces of Italy, as well as to give up portions of territory to Bavaria, Würtemberg, and Baden, receiving in return Salzburg and Berchtesgaden. After the formation of the Confederation of the Rhine (12th July, 1806) Francis was forced to resign his dignity as Emperor of Germany, which had been in his family more than 500 years. A new war with France in 1809 cost the monarchy 42,380 sq. miles of territory and 3,500,000 subjects. Napoleon married Maria Louisa, daughter of the Emperor, and in 1812 concluded an alliance with him against Russia. But in 1813 Francis again declared war against France, and formed an alliance with Britain, Prussia, and Sweden against his son-in-law. By the Congress of Vienna (1815) Austria gained Lombardy and Venetia, and recovered, together with Dalmatia, the hereditary territories which it had been obliged to cede.

In the troubled period following the French revolution of 1830 insurrections took place in Modena, Parma, and the Papal States (1831-2), but were suppressed without much difficulty; and though professedly neutral during the Polish insurrections Austria clearly showed herself on the side of Russia, with whom her relations became more intimate as those between Great Britain and France grew more cordial. The death of Francis I (1835) and accession of his son, Ferdinand I, made little change in the Austrian system of government, and much discontent was the consequence. In 1846 the failure of the Polish insurrection led to the incorporation of Cracow with Austria. In Italy the declarations of Pio Nono in favour of reform increased the difficulties of Austria, and in Hungary the opposition under Kossuth and others assumed the form of a great constitutional movement. In 1848, when the expulsion of Louis Philippe shook all Europe, Metternich found it impossible any longer to guide the helm of the State, and the Government was compelled to admit a free press and the right of citizens to arms. Apart from the popular attitude in Italy and in Hungary, where the Diet declared itself permanent under the presidency of Kossuth, the insurrection made equal progress in Vienna itself, and the royal family, no longer in safety, removed to Innsbruck. After various ministerial changes the Emperor

abdicated in favour of his nephew, Francis Joseph; more vigorous measures were adopted; and Austria, aided by Russia, reduced Hungary to submission.

The year 1855 is memorable for the Concordat with the Pope, which put the educational and ecclesiastical affairs of the Empire entirely into the hands of the Papal see. In 1859 the hostile intentions of France and Sardinia against the possessions of Austria in Italy became so evident that she declared war by sending an army across the Ticino; but after disastrous defeats at Magenta and Solferino she was compelled to cede Milan and the north-west portion of Lombardy to Sardinia. In 1864 she joined with the German States in the war against Denmark, but a dispute about Schleswig-Holstein involved her in a war with her allies (1866), while at the same time Italy renewed her attempts for the recovery of Venice. The Italians were defeated at Custozza and driven back across the Mincio; but the Prussians, victorious at Königgrätz (or Sadowa), threatened Vienna. Peace was concluded with Prussia on 23rd Aug. and with Italy on 3rd Oct., the result of the war being the cession of Venetia through France to Italy and the withdrawal of Austria from all interference in the affairs of Germany.

For nearly half a century (from 1866 to 1914) the internal affairs of Austria-Hungary gave much occupation to statesmen of the dual monarchy. Hungarian demands for self-government were finally agreed to, and the Empire was divided into the two parts already mentioned—Austria and Hungary. This settlement was consummated by the coronation of the Emperor Francis Joseph I, at Budapest, as King of Hungary, on 8th June, 1867. In the same year the Concordat of 1855 came up for discussion, and measures were passed for the re-establishment of civil marriage, the emancipation of schools from the domination of the Church, and the placing of different creeds on a footing of equality. The fact of the Austro-Hungarian dominions comprising so many different nationalities gave the central Government much trouble, both in regard to internal and to external affairs. In regard to the 'Eastern Question', for instance, the action of Austria was hampered by the sympathies shown by the Magyars for their blood relations, the Turks, while the Slavs were naturally more favourable to Russia. During the war between Russia and Turkey in 1877-8 Austria remained neutral; but at its close it was decided at the Congress of Berlin that Bosnia and Herzegovina should in future be administered by Austria

instead of Turkey. In 1908, taking advantage of the condition of Turkey, Austria formally annexed these provinces. By the middle of Oct., 1918, the Austro-Hungarian monarchy began to break up into independent national States. Four States of some size laid claim to independence, viz. German-Austria, Czecho-Slovakia (recognized by the Allies), Hungary, and Yugo-Slavia. On 13th Oct., 1921, German Austria signed with Hungary the Burgenland pact, agreeing to a plebiscite in several districts. For recent history of Austria, see *Francis Joseph; European War*.—Cf. S. Whitman, *Austria* (in *Story of the Nations Series*); H. W. Steed, *The Hapsburg Monarchy*.

Auteuil (ō-tè-yè), formerly a suburban village of Paris, but now enclosed within the fortifications.

Autobiog'raphy, a department of literature of which, as a definite branch or form at least, we have no very ancient examples, though many autobiographical details are contained, for instance, in Cicero's and Pliny's letters, while St. Augustine has left a fragmentary autobiography in his *Confessions*. Various writers of note, without claiming or perhaps intending to write a formal autobiography, have left similar materials in the form of their own memoirs or reminiscences, or, for a portion of their lives, in diaries. The chief English diarists are Pepys and Evelyn, while Crabb Robinson is an admirable modern example. Brief autobiographies were written by Gibbon, Hume, and Sir Walter Scott (of his early life only), and among other writers who have given us autobiographies or kindred works are Fanny Burney, Cobbett, Haydon, Miss Mitford, Harriet Martineau, Leigh Hunt, Hugh Miller (in *My Schools and Schoolmasters*), John Stuart Mill, Anthony Trollope, Sir Henry Taylor, Miss F. P. Cobbe, Carlyle, Ruskin, Herbert Spencer, Alexander Bain, and Alfred Russel Wallace. Lord Roberts's *Forty-one Years in India* and General Grant's *Personal Memoirs* are most interesting autobiographical works by men of action. One of the most famous of autobiographies is that of the Italian Benvenuto Cellini. See *Biography*.

Autochrome. See *Photography*.

Autochthones (a'-tok'tho-nēz), the Greek name for the aboriginal inhabitants of a country. See *Aborigines*.

Au'toclave (derived from the Gr. *autos*, self, and Lat. *clavis*, key) is a strong steam-tight vessel in which hydrolysis of liquids can be carried out, under pressure, at temperatures considerably higher than their normal boiling-points. The vessels may be of copper, iron, or other suitable material, are sometimes enamelled within, or may have renewable refractory linings when used for corrosive liquids. Some types have mechanical agitators. Used in the manufacture of candles, coal-tar colours, &c. May be heated by high-pressure steam or in baths of oil or molten lead. See *Papin*.

Au'tocrat (Gr. *autos*, self, *kratos*, rule), an absolute or uncontrolled ruler; the head of a State who is not controlled by any constitutional limitations, such as the Tsars of Russia till 1917.

Auto-de-fe (Sp.); **Auto-da-fe** (Pg.), literally, 'act of faith'. See *Inquisition*.

Au'tograph, a person's own handwriting; an original manuscript or signature, as opposed to a copy. The practice of collecting autographs or signatures originated in Germany and the Low Countries, chiefly among members of the universities, and dates at least from the sixteenth century. Among the earliest collections known are those of John Cotton and Hans Sloane in the British Museum, and of Philippe de Bethune, brother of Sully, and Loménie de Brienne at the Bibliothèque Nationale. The most celebrated collection formed in England in recent years is that of Alfred Morrison.

Automatic Writing. See *Psychical Research*; *Spiritualism*.

Autom'atism, the confinement of activity in men or animals within a purely mechanical limit, resulting from injury to or partial removal of the brain.

Autom'aton (Gr. *automatos*, spontaneous), a self-moving machine performing actions like those of a living being, and often shaped like one. The walking statues of Dædalus, the flying dove of Archytas, the brazen head of Friar Bacon, the iron fly of Regiomontanus, the door-opening figure of Albertus Magnus, the parading knights of the clock presented to Charlemagne by Harun al Rashid, the toy carriage and attendants constructed by Camus for Louis XIV, the flute-player, tambour-player, and

duck of Vaucanson, and the writing child of the brothers Droz are among the more noteworthy of traditional automata. See *Conjuring*.

Automobile. See *Motor Vehicles*.

Auton´omy, the power of a State, institution, &c., to legislate for itself.

Autoph´agi (-jī), birds which feed themselves as soon as hatched.

Au´toplasty, the operation by which wounds and diseased parts are repaired with healthy tissues taken from other parts of the same person's body.

Autoplate. See *Printing*.

Autop´sy, literally, personal observation or inspection, commonly restricted to post-mortem examination. See *Coroner*.

Au´totype, a species of photographic print. A thin sheet of gelatine on paper is rendered sensitive to light by treatment with bichromate of potash, and then exposed under an ordinary photographic negative. The portions of gelatine affected by the light become insoluble, the remainder of the gelatine is then washed away, and the picture remains reproduced in the gelatine, there being slight elevations and depressions corresponding with the distribution of light and shade. This may be printed from, but it is more often made use of to obtain electrotypes or other reverses, from which impressions can more easily be taken.

Autumn, the season between summer and winter, in the northern hemisphere often regarded as embracing August, September, and October, or three months about that time. The beginning of the astronomical autumn is 22nd Sept., the autumnal equinox; and the end is 21st Dec., the shortest day. The autumn of the southern hemisphere takes place at the time of the northern spring.

Autun (ō-tu´n; ancient, **Bibracte**, later, **Augustodunum**), a town, South-Eastern France, department of Saône-et-Loire. It has two Roman gates of exquisite workmanship, the ruins of an amphitheatre and of several temples, the cathedral of St. Lazare, a fine Gothic structure of the twelfth century

with chapels added in the fifteenth; manufactures of carpets, woollens, cotton, velvet, hosiery, &c. Pop. 15,498.

Autunite, a hydrous phosphate of uranium and calcium, crystallizing in the rhombic system in yellow plates. Like other minerals of the uranium series it shows radio-activity.

Auvergne (ō-vār-nyè), a province, Central France, now merged into departments Cantal and Puy-de-Dôme, and an arrondissement of Haute-Loire. The Auvergne Mountains, separating the basins of the Allier, Cher, and Creuse from those of the Lot and Dordogne, contain the highest points of Central France: Mount Dor, 6188 feet; Cantal, 6093 feet, and Puy-de-Dôme, 4806 feet. The number of extinct volcanoes and general geologic formation make the district one of great scientific interest. The minerals include iron, copper, and lead, and there are warm and cold mineral springs. Auvergne contributes a large supply to the labour markets of Paris and Belgium, there being in Paris alone some 50,000 Auvergnats.

Auxerre (ō-sār), a town, France, department of Yonne, 110 miles S.E. of Paris. Principal edifices: a fine Gothic cathedral, unfinished; the abbey of St. Germain, with curious crypts; and an old episcopal palace, now the Hôtel de Prefecture; it manufactures woollens, hats, casks, leather, earthenware, violin strings, &c.; trade, chiefly in wood and wines, of which the best known is white Chablis. Pop. 21,930.

Auxom'eter, an instrument to measure the magnifying powers of an optical apparatus.

Auxonne (ō-son; ancient, **Aussona**), a town, France, department of Côte-d'Or (Burgundy), on the Saône; a fortified place, with some manufactures. Pop. 6300.

A'va, a town in Asia, formerly the capital of Burma, on the Irawadi, now almost wholly in ruins.

Ava-Ava, **Arva**, **Kava**, or **Yava** (*Macropiper methysticum*), a plant of the nat. ord. Piperaceæ (pepper family), so called by the inhabitants of Polynesia, who make an intoxicating drink out of it. Its leaves are chewed with betel, in South-Eastern Asia.

Avad´avat. See *Amadavat*.

Av´alanches, large masses of snow or ice precipitated from the mountains, and distinguished as *wind* or *dust avalanches* when they consist of fresh-fallen snow whirled like a dust-storm into the valleys; as *sliding avalanches* when they consist of great masses of snow sliding down a slope by their own weight; and as *glacier* or *summer avalanches* when ice-masses are detached by heat from the high glaciers.

Avâl Islands. Same as *Bahrein Islands*.

Avallon (a´-va´-lōn), a town of Central France, department Yonne. Pop. 5900.

Av´alon, a sort of fairyland or elysium mentioned in connection with the legends of King Arthur, being his abode after disappearing from the haunts of men: called also *Avilion*. The name is also identified with Glastonbury; and has been given to a peninsula of Newfoundland.

Avan´turine, or **Aven´turine**, a variety of quartz containing glittering spangles of mica through it; also a sort of artificial gem of similar appearance.

Av´ars, a nation, probably of Turanian origin, who at an early period may have migrated from the region east of the Tobol in Siberia to that about the Don, the Caspian Sea, and the Volga. A part advanced to the Danube in A.D. 555, and settled in Dacia. They served in Justinian's army, aided the Lombards in destroying the kingdom of the Gepidæ, and in the sixth century conquered under their Khan Bajan the region of Pannonia. They then won Dalmatia, pressed into Thuringia and Italy against the Franks and Lombards, and subdued the Slavs dwelling on the Danube, as well as the Bulgarians on the Black Sea. But they were ultimately limited to Pannonia, where they were overcome by Charlemagne, and nearly extirpated by the Slavs of Moravia. After 827 they disappear from history. Traces of their fortified settlements are found, and known as Avarian rings. See *Lesghians*.

Avatar´, more properly **Avatara**, in Hindu mythology, an incarnation of the Deity. Of the innumerable avatars the chief are the ten incarnations of Vishnu, who appeared successively as a fish, a tortoise, a boar, a man-lion, a dwarf, &c.

Avatch´a, a volcano and bay in Kamchatka. The volcano, which is 9000 feet high, was last active in 1855. The town of Petropavlovsk lies in the bay.

Avebury (āv´be-ri), a village of England, in Wiltshire, occupying the site of a so-called Druidical temple, which originally consisted of a large outer circle of 100 stones, from 15 to 17 feet in height, and about 40 feet in circumference, surrounded by a broad ditch and lofty rampart, and enclosing two smaller circles. Few traces now remain of the structure. On the neighbouring downs are numerous barrows or tumuli, one of which, called Silbury Hill, rises to the height of 130 feet, with a circumference of 2027 feet at the base, covering fully 5 acres.

Avebury, John Lubbock, first Baron. See *Lubbock*.

Aveiro (a´-vā´i-ru´), a coast town in Portugal, province of Douro, with a cathedral, an active fishery, and a thriving trade. Pop. 11,523.

Avellino (a´-vel-lē´nō), a town in Southern Italy, capital of the province of Avellino, 29 miles east of Naples, the seat of a bishop. Avellino nuts were celebrated under the Romans. Pop. 24,620. Area of the province, 1165 sq. miles; pop. 411,813.

A´ve Mari´a ('Hail, Mary'), the first two words of the angel Gabriel's salutation (*Luke*, i, 28), and the beginning of the very common Latin prayer to the Virgin in the Roman Catholic Church. Its lay use was sanctioned at the end of the twelfth century, and a papal edict of 1326 ordains the repetition of the prayer thrice each morning, noon, and evening, the hour being indicated by sound of bells called the Ave Maria or Angelus Domini. The prayers are counted upon the small beads of the rosary, as the pater-nosters are upon the large ones.

Ave´na. See *Oat*.

Av´ens, the English name of two rosaceous plants of the genus *Geum*. Common avens, or herb-bennet, *G. urbānum*, possesses astringent properties, and was formerly used in medicine. The mountain avens, *Dryas octopetala*, belongs to the same nat. ord.

Av´entaile, the movable face-guard of the helmet, through which the wearer breathed.

Av´entine, one of the seven hills of Rome, east of the Tiber. It was included in the city by Servius Tullius.

Aven´turine. See *Avanturine*.

Av´erage, in maritime law, any charge or expense over and above the freight of goods, and payable by their owner.—*General average* is the sum falling to be paid by the owners of ship, cargo, and freight, in proportion to their several interests, to make good any loss or expense intentionally incurred for the general safety of ship and cargo, e.g. throwing goods overboard, cutting away masts, port dues in cases of distress, &c.—*Particular average* is the sum falling to be paid for unavoidable loss when the general safety is not in question, and therefore chargeable on the individual owner of the property lost. A policy of insurance generally covers both general and particular average, unless specially excepted.—*Averaging*, in Stock Exchange language, denotes the operation of a speculator in increasing transactions at a higher or lower price when the price is moving against him, so that the average price of the whole will be higher or lower than his original purchase or sale.

Aver´nus, a lake, now called **Lago d'Averno**, in Campania, Italy, between the ancient Cumæ and Puteoli, about 8 miles from Naples. It is the crater of an old volcano, and is in some places 180 feet deep. Formerly the gloom of its forest surroundings and its mephitic exhalations caused it to be regarded as the entrance to the infernal regions. It was the fabled abode of the Cimmerians, and especially dedicated to Proserpine.

Averroes (a-ver´o-es; corrupted from **Ibn Roshd**), the most renowned Arabian philosopher, born at Cordova, in Spain, 1126, died at Morocco 1198. His ability procured him the succession to his father's office of chief magistrate, and the King of Morocco appointed him at the same time *cadi* in the province of Mauretania. Accused of being an infidel, he was, however, deprived of his offices, and banished to Spain; but, being persecuted there also, he fled to Fez, where he was condemned to recant and undergo public penance. Upon this he went back to his own country, where the Caliph Almansur finally restored him to his dignities. Averroes regarded Aristotle as the greatest of all philosophers, and devoted himself so largely to the exposition of his works as to be called among the Arabians *The Interpreter*.

He wrote a compendium of medicine, and treatises on theology, philosophy, jurisprudence, &c. His commentaries upon Aristotle appeared before 1250 in a Latin translation attributed to Michael Scott, the reputed wizard (1194-1250), and others. Averroes was at once a philosopher, a theologian, and a theosophist.

Averrunca'tor, a garden implement for pruning trees without a ladder, consisting of two blades similar to stout shears, one fixed rigidly to a long handle, and the other moved by a lever to which a cord passing over a pulley is attached.

Aver'sa, a well-built town of Southern Italy, 7 miles N. of Naples, in a beautiful vine and orange district, the seat of a bishop, with a cathedral and various religious institutions, and a large lunatic asylum. Andreas of Hungary, husband of Queen Joanna I, was strangled in a convent here, 18th Sept., 1345. Pop. 23,203.

Avesnes (a'-vān), a town of France, department Nord. Pop. 5829.

Avesta. See *Zendavesta*.

Aveyron (a'-vā-rōn'), a department occupying the southern extremity of the central plateau of France, traversed by mountains belonging to the Cevennes and the Cantal ranges; principal rivers: Aveyron, Lot, and Tarn, the Lot alone being navigable. The climate is cold, and agriculture is in a backward state, but considerable attention is paid to sheep-breeding. It is noted for its 'Roquefort cheese'. It has coal, iron, and copper mines, besides other minerals. Area, 3385 sq. miles; capital, Rhodéz. Pop. (1921), 332,940.

Avezzano (a'-vet-zä'nō), a town of S. Italy, province Aquila. Pop. 11,279.

Av'iary, a building or enclosure for keeping, breeding, and rearing birds. Aviaries appear to have been used by the Persians, Greeks, and Romans, and are highly prized in China. In England they were in use at least as early as 1577, when William Harrison refers to "our costlie and curious aviaries". An aviary may be simply a kind of very large cage, or a series of enclosures.

Aviatik, a German type of biplane. It was originally built at Mulhausen in Alsace.

Aviation. See *Aeronautics*.

Avicen'na, or **Ebn-Sina**, an Arabian philosopher and physician, born at Kharmaithen, in the province of Bokhara, A.D. 980. After practising as a physician he quitted Bokhara at the age of twenty-two, and for a number of years led a wandering life, settling at last at Hamadan, subsequently as vizier of the Emir. On the death of his patron he lived in retirement at Hamadan, but having secretly offered his services to the Sultan of Ispahan he was imprisoned by the new Emir. Escaping, he fled to Ispahan, was received with great honour by the Sultan, and passed there in quietness the last fourteen years of his life, writing upon medicine, logic, metaphysics, astronomy, and geometry. His philosophy was Aristotelianism mingled with neo-Platonism, and his influence is most marked in Dante and the Mystics. He died at Hamadan, in Northern Persia, 1037, leaving many writings, mostly commentaries on Aristotle. Of his 100 treatises the best known is the *Canon Medicinæ*, which was still in use as a textbook at Louvain and Montpellier in the middle of the seventeenth century. His *Philosophia Orientalis*, mentioned by Roger Bacon, is lost.

Avie'nus, Rufus Festus, a Latin descriptive poet, who flourished about the end of the fourth century A.D., and wrote *Descriptio Orbis Terræ*, a general description of the earth; *Ora Maritima*, an account of the Mediterranean coasts, &c.

Avifau'na, a collective term for the birds of any region.

Avigliano (a'-vêl-yä'nō), a town of S. Italy, province Potenza. Pop. 17,413.

Avignon (a'-vê-nyōn; ancient, **Avenio**), an old town of S.E. France, capital of department Vaucluse, on the left bank of the Rhone; enclosed by lofty battlemented and turreted walls, well built, but with rather narrow streets. It is an archbishop's see since 1475, and has a large and ancient cathedral on a rock overlooking the town, the immense palace in which the Popes resided (used as a barracks and prison for a long time), and other old buildings. The industries of the city are numerous and varied, the principal being connected with silk. The silk manufacture and the breeding of silk-worms are the principal employments in the district. Here Petrarch lived several years, and made the acquaintance of Laura, whose tomb is in the Franciscan church. From 1309 to 1377 seven Popes in succession, from Clement V to

Gregory XI, resided in this city. After its purchase by Pope Clement VI, in 1348, Avignon and its district continued, with a few interruptions, under the rule of a vice-legate of the Pope's till 1791, when it was formally united to the French Republic. Pop. 49,304.

Avignon Berries. See *French Berries*.

Avila (ä'vē-la´), a town of Spain, capital of province of Avila, a modern division of Old Castile. It is the see of the Bishop suffragan of Santiago, with fine cathedral. Once one of the richest towns of Spain. Principal employment in the town, spinning; in the province, breeding sheep and cattle. Pop., town, 12,060; province, 214,008.

Avila, Gil Gonzalez d', a Spanish antiquary and biographer, 1577-1658; made historiographer of Castile in 1612, and of the Indies in 1641. Most valuable works: *Teatro de las Grandezas de Madrid*, 1623, and *Teatro Ecclesiastico*, 1645-53.

Avila y Zuñiga (ä'vē-la´ ē thö-nyē'ga´), Don Luis d', Spanish general, diplomatist, and historian; a favourite of Charles V; born about 1490, died about 1560. His chief work, printed in 1548 and translated into five languages, was on the war of Charles V in Germany.

Aviles (a´-vē'les), a town of Northern Spain, province Oviedo, with a good harbour. Pop. 13,660.

Aviz, an order of knighthood in Portugal, instituted by Sancho, its first king, and having as its original object the subjection of the Moors.

Avizan´dum, in Scots law, private consideration. To make *avizandum* is to remove a cause from the public court to the private consideration of the judge.

Avlo´na (Ital. *Valona*), a seaport of Albania on the Adriatic, with a considerable trade. It was occupied and made a naval and military base by the Italians during the European War. Pop. 6500.

Avo´ca, or **Ovoca**, a river and valley in Ireland, County Wicklow, celebrated as the scene and subject of one of Moore's *Irish Melodies*. The river is formed by the union of the Avonbeg and Avonmore, and below the

junction receives the Aughrim, the name in Celtic signifying 'the meeting of the waters'. After a course of about 9 miles it falls into the sea about a mile below Arklow. The scenery here is singularly beautiful, and attracts many visitors.

Avoca´do-pear. See *Alligator-pear*.

Avogad´ro's Law, in physics, asserts that equal volumes of different gases at the same pressure and temperature contain an equal number of molecules.

Avoirdupois (a-vèr'du¨-pois; from old French, literally, 'goods of weight'), a system of weights used for all goods except precious metals, gems, and medicines, and in which a pound contains 16 ounces, or 7000 grains, while a pound troy contains 12 ounces, or 5760 grains. A hundredweight contains 112 pounds avoirdupois; a *cental* of 100 pounds is common in America, and is a legal British weight. See *Weights and Measures*.

Av´ola, a seaport on the east of Sicily, with a trade in almonds, sugar, &c. Pop. 17,711.

A´von, the name of several rivers in England, of which the principal are: 1. The Upper Avon, rising in Leicestershire, and flowing s.w. into the Severn at Tewkesbury. Stratford-on-Avon lies on this river. 2. The Lower Avon, rising in Gloucestershire, and falling into the Severn N.W. of Bristol; navigable as far as Bath. 3. In Monmouthshire. 4. In Wiltshire and Hampshire, entering the English Channel at Christchurch Bay. There are also streams of this name in Wales and Scotland.

Avonmouth, a place at the mouth of the River Avon, 6 miles from Bristol, and connected with it by rail and river, with large docks belonging to the Bristol corporation, and forming part of the shipping accommodation of that city.

Av'oset, a bird about the size of a lapwing, of the genus *Recurvirostra* (*R. avosetta*), family Scolopacidae (snipes), ord. Grallatores. The bill is long, slender, elastic, and bent upward toward the tip, the legs long, the feet webbed, and the plumage variegated with black and white. The bird feeds on worms and other small animals, which it scoops up from the mud of the marshes and fens that it frequents. It is found in Europe, Asia, Africa, and America; but the American species is slightly different from the others.



Avoset
(*Recurvirostra*
avosetta)

Avranches (a' -vrän' sh; ancient, **Abrincæ**), a town, France, department La Manche, about 3 miles from the Atlantic. It formerly had a fine cathedral, destroyed during the first French revolution. Manufactures: lace, thread, and candles. Pop. 7174.

Awe (a''), a Scottish loch in Argyllshire, about 28 miles long and 2 broad, and communicating by the Awe with Loch Etive. Ben Cruachan stands at its northern extremity. It has many islands and beautiful scenery, and abounds in trout, salmon, &c.

Axe, or **Ax**, a well-known tool for cutting or chipping wood, consisting of an iron head with an arched cutting edge of steel, which is in line with the wooden handle of the tool, and not at right angles to it as in the adze.

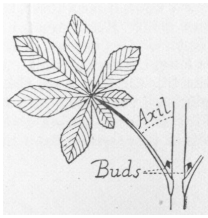
Axel. See *Absalon*.

Axe-stone. See *Jade*.

Axholme Isle (aks'ōm), a sort of island in England formed by the Rivers Trent, Idle, and Don, in the north-west angle of Lincolnshire, 17 miles long, 4½ broad.

Axil, or **Axilla**, in botany, the angle between the upper side of a leaf and the stem or branch from which it springs. Buds usually appear in the axils, and flowers or flower-stalks growing in this way are called *axillary*.

Ax'im, a town of W. Africa, on the Gold Coast.



Leaf and stem of
Horse-chestnut,
showing axil

Axinite, a mineral, silicate of alumina, lime, &c., with boracic acid, deriving its name from the form of the crystals, the edges of which bear some resemblance to the edge of an axe.

Axinomancy, an ancient method of divination by the movements of an axe (Gr. *axinē*) balanced on a stake, or of an agate placed on a red-hot axe. The names of suspected persons being uttered, the movements at a particular name indicated the criminal.

Axiom, a universal proposition, which the understanding must perceive to be true as soon as it perceives the meaning of the words, and therefore called a *self-evident truth*: e.g. *A is A*. In mathematics, axioms are those propositions which are assumed without proof, as being in themselves independent of proof, and which are made the basis of all the subsequent reasoning; as, 'The whole is greater than its part'; 'Things that are equal to the same thing are equal to one another'. See *Geometry*.

Axis, the straight line, real or imaginary, passing through a body or magnitude, on which it revolves, or may be supposed to revolve; especially a straight line with regard to which the different parts of a magnitude, or several magnitudes, are symmetrically arranged; e.g. the *axis of the world*, the imaginary line drawn through its two poles.

In *botany* the word is also used, the stem being termed the *ascending axis*, the root the *descending axis*.

In *anatomy* the name is given to the second vertebra from the head, that on which the *atlas* moves. See *Atlas*.

Axis (*Cervus axis*), a species of Indian deer, also known as the Spotted Hog-deer, of a rich fawn colour, nearly black along the back, with white spots, and under parts white. Breeds freely in many parks in Europe.

Axminster, a market town, England, county Devon, on the Axe, at one time celebrated for its woollen cloth and carpet manufactures, and giving name to an expensive variety of carpet having a thick soft pile, and also to a cheaper variety. Pop. (1921), 2049.

Ax'olotl (*Amblystōma maculātum*), a curious Mexican amphibian, not unlike a newt, from 8 to 10 inches in length, with gills formed of three long ramified or branch-like processes floating on each side of the neck. It reproduces by laying eggs, and was for some time regarded as a perfect animal with permanent gills. It is said, however, that they frequently lose their gills like the other members of the genus, though some authorities maintain that the true axolotl never loses its gills, and that merely confusion with *A. tigrīnum* has led to the belief, as this species sometimes retains its branchiæ, though usually it loses them. The axolotl is esteemed a luxury by the Mexicans. There are a number of species of *Amblystoma* in North America.

Ax'um, or **Aksum**, a town in Tigré, a division of Abyssinia, once the capital of an important kingdom, and at one time the great depot of the ivory trade in the Red Sea. It is the seat of an *abuna* or Patriarch, and has a pop. of 5000. The site of the town still exhibits many remains of its former greatness.

Ayacucho (ä-ya' -kö'chō), the name of a department of Peru, and of its capital. The department has an area of 18,185 sq. miles; a pop. of 302,469. The town (formerly Guamanga or Huamanga) has a cathedral and a university, and a pop. of 20,000.

Ayala (a' -yā'la'), Pedro Lopez de, Spanish historian and poet, Chancellor of Castile in the second half of the fourteenth century, and the author of a history of Castile from 1350 to 1396. He took an active part in the struggle between Henry II and Pedro the Cruel, and was taken prisoner by the English in 1367. During his English captivity he wrote part of his chief poetical work, *A Book in Rhyme concerning Court Life*. Died 1407.

Ayamonte (a' -ya' -mon'tā), a seaport town, Spain, province of Huelva, 2 miles from the mouth of the Guadiana. Pop. 6000.

Ayass'oluk. See *Ephesus*.

Aye-aye (ī-ī; *Cheirōmys madagascariensis*), an animal of Madagascar, so called from its cry; now referred to the lemur family. It is about the size of a hare, has large flat ears and a bushy tail; large eyes; long sprawling fingers, the third so slender as to appear shrivelled; colour, musk-brown, mixed with

black and grey ash; feeds on grubs, fruits, &c.; habits, nocturnal. See *Primates*, *Lemurs*.

Ayesha (a-yesh'a), daughter of Abu-Bekr and favourite wife of Mahomet, though she bore him no child, born in 610 or 611. After his death she opposed the succession of Ali, but was defeated and taken prisoner. She died at Medina in 677 or 678 (A.H. 58).



Aye-aye (*Cheirōmys
madagascariensis*)

Aylesbury (ālz'be-ri), county town of Buckinghamshire, England, with a fine old parish church; chief industries, silk-throwing, printing, making condensed milk, and poultry-rearing for the London market. The Aylesbury duck, also called 'white English', is famous. Previous to 1885 it and its hundred sent two members to Parliament, and it still gives its name to a parliamentary division. Pop. (1921), 12,114.

Ay'loffe, Sir Joseph, an English antiquary, born about 1708, died 1781; member of the first council of the Society of Antiquaries, a commissioner for the preservation of State papers, the author and editor of several works, of which the best known is his *Calendars of the Auntient Charters*, &c.

Aymaras (ī'ma' -ra' z), an Indian race of Bolivia and Peru, speaking a language akin to the Quichua.

Ay'mon, the surname of four brothers, Alard, Richard, Guiscard, and Renaud, who hold a first place among the heroes of the Charlemagne cycle of romance. Their exploits were the subject of a romance, *Les Quatre Fils d'Aymon*, by Huon de Villeneuve, a trouvère of the thirteenth century, and Renaud is a leading figure in Ariosto's *Orlando*.

Ayr (ār), a town of Scotland, a royal and parliamentary burgh, and capital of Ayrshire, at the mouth of the River Ayr, on the Firth of Clyde. It was the site of a Roman station. William the Lion built a castle there in 1197 and constituted it a royal burgh in 1202; and the Parliament which confirmed Robert Bruce's title to the crown sat in Ayr. It is picturesquely situated, and has excellent public and other buildings. Two bridges connect Ayr proper with the suburbs of Newton and Wallacetown, a third stands farther up the river. The 'New Brig' of Burns's 'Brigs of Ayr' (built in 1788) had to be

replaced by a new structure in 1879, while the 'Auld Brig' (built in 1252) still stands, after being repaired. Carpets, lace goods, leather, iron goods, &c., are manufactured. The harbour accommodation is good, and in particular coals are largely exported. The house in which Burns was born stands within 1½ miles of the town, and near 'Alloway's auld haunted kirk', and a monument to him is near. Ayr unites with Ardrossan, Irvine, Prestwick, Saltcoats, and Troon in sending a member to Parliament. Pop. 32,986.—The county has an area of 724,523 acres. It is divided into the districts of Carrick in the south, Kyle in the middle, and Cunningham in the north. The surface is irregular, and a large portion of it hilly, but much of it is fertile. The principal streams are the Ayr, Stinchar, Girvan, Doon, Irvine, and Garnock. Coal and iron are abundant; and there are numerous collieries and ironworks. Limestone and freestone are also abundant. Agriculture is in a flourishing state, the principal crops being oats, turnips, and early potatoes, while dairy husbandry is extensively practised; the Ayrshire cows, referred to in the eighteenth century as the Cunningham, are celebrated as milkers, and Ayrshire cheese ('Dunlop' and 'Cheddar') has a high reputation. Woollen manufactures are extensive, particularly carpets and bonnets; lace goods are largely made; Kilmarnock is a centre of engineering and of dairying education. Goods made include explosives, boots and shoes, chemicals, and leather. Chief towns, Ayr, Kilmarnock, Irvine, Ardrossan, Saltcoats, Stevenston, Maybole. Ayr and Buteshire send three members to Parliament. Pop. 267,600 (1918).

Ayrer (Ī'rer), Jacob, a German dramatist of the sixteenth century, who almost rivalled Hans Sachs in copiousness and importance. He was a citizen and legal official of Nuremberg, and died in 1605. His works, published at Nuremberg in 1618, under the title *Opus Theatricum*, include thirty comedies and tragedies and thirty-six humorous pieces.

Ayrton, William Edward, English physicist and electrical engineer, born in 1847. From 1873 to 1879 he was professor of natural philosophy and telegraphy in the Imperial College of Engineering at Tokio. In 1879 he was appointed professor of applied physics in the City and Guilds of London Institute. He invented an ammeter, voltmeter, &c., in conjunction with Professor Perry. He died in 1908. He published *Practical Electricity*, and many papers on scientific subjects.

Aytoun (ā'tun), Sir Robert, poet, born in Fifeshire, Scotland, 1570, died 1638. After studying at St. Andrews he lived for some time in France, whence, in 1603, he addressed a panegyric in Latin verse to King James on his accession to the crown of England. By the grateful monarch he was appointed one of the gentlemen of the bedchamber, and private secretary to the queen, receiving also the honour of knighthood. At a later period of his life he was secretary to Henrietta Maria, queen of Charles I. His poems are few in number, but are distinguished by elegance of diction. Several of his Latin poems are preserved in the work called *Delitiæ Poetarum Scotorum*, published at Amsterdam in 1637 at the expense of Sir John Scot of Scotstarvet.

Aytoun, William Edmonstoune, poet and prose writer, born at Edinburgh in 1813, died at Blackhills, Elgin, 1865. He studied at the University of Edinburgh, became a writer to the signet in 1835, and passed as advocate in 1840. He issued a volume of poems in 1832, by 1836 was a contributor to *Blackwood's Magazine*, and in 1840 he published the *Life and Times of Richard 1*. In 1848 he published a collection of ballads entitled *Lays of the Scottish Cavaliers*, which has proved the most popular of all his works. It was followed in 1854 by *Firmilian, a Spasmodic Tragedy* (intended to ridicule certain popular writers); the *Bon Gaultier Ballads* (parodies and other humorous pieces, in conjunction with Theodore Martin), 1855; in 1856 the poem *Bothwell*; and in subsequent years by *Norman Sinclair*, *The Glenmutchkin Railway*, and other stories. In 1858 he edited a critical and annotated collection of the *Ballads of Scotland*. A translation of the poems and ballads of Goethe was executed by him in conjunction with Theodore Martin. In 1845 he became professor of rhetoric and English literature in the University of Edinburgh—a position which he held till his death. In 1852 he was appointed Sheriff of Orkney and Shetland.

Ayuntamiento (a' -yu'n-ta' -mē-en'tō), the name given to the town and village councils in Spain and Spanish America.

Ayu'thia, the ancient capital of Siam, on the Menam, now a scene of splendid ruin.

Aza'lea, a genus of plants, nat. ord. Ericaceæ, or heaths, remarkable for the beauty and fragrance of their flowers, and distinguished from the

rhododendrons chiefly by the flowers having five stamens instead of ten. Many beautiful rhododendrons with deciduous leaves are known under the name of *azalea* in gardens. The azaleas are common in North America, and two species of these—*A. viscōsa* and *A. nudiflōra*—are well known in Britain. An Asiatic species, *A. pontica*, famous for the stupefying effect which its honey is said to have produced on Xenophon's army (cf. *Anabasis*, book iv, chap. 8), is also common in British gardens and shrubberies; and another, *A. indica*, is a brilliant greenhouse plant.

Azamgarh, or **Azimgarh**, a town of India, United Provinces, capital of district of same name. Pop. 18,835.—The district has an area of 2418 sq. miles; a pop. of 1,530,000.

Azeglio (a 'd-zel'yō), Massimo Taparelli, Marquis d', an Italian 'admirable Crichton', artist, novelist, publicist, statesman, and soldier, born at Turin in 1798, died 1866. After gaining some reputation in Rome as a painter, he married the daughter of Manzoni, and achieved success in literature by his novels *Ettore Fieramosca* (1833) and *Niccolo dei Lapi* (1841). These embodied much of the patriotic spirit, and in a short time he devoted himself exclusively to fostering the national sentiment by personal action and by his writings. Many of the reforms of Pius IX were due to him. He commanded a legion in the Italian struggle of 1848, and was severely wounded at Vincenza. Chosen a member of the Sardinian Chamber of Deputies, he was, after the battle of Novara, made president of the cabinet, and in 1859 appointed to the military post of general and commissioner-extraordinary for the Roman States.

Azerbaijan (a ' -zer-bī-jän'), a province of North-Western Persia; area, 40,000 sq. miles; pop. estimated at 2,000,000. It consists generally of lofty mountain ranges, some of which rise to a height of between 12,000 and 13,000 feet. Principal rivers: the Aras or Araxes, and the Kizil-Uzen, which enter the Caspian; smaller streams discharge themselves within the province into the great salt lake of Urumiyah. Agricultural products: wheat, barley, maize, fruit, cotton, tobacco, and grapes. Horses, cattle, sheep, and camels are reared in considerable numbers. Chief minerals: iron, lead, copper, salt, saltpetre, and marble. Tabriz is the capital; pop. 200,000.

Azerbaijan, a new republic proclaimed in 1918, after the Russian revolution. It consists of the former Russian province of Baku on the coast of the Caspian Sea. In 1920 the Bolshevik party overthrew the Government and broke off relations with the Entente. See *Russia*.

Azimuth of a heavenly body, the arc of the horizon comprehended between the meridian of the observer and a vertical circle passing through the centre of the body. The azimuth and altitude give the exact position of the body with reference to the horizon.

Azincourt. See *Agincourt*.

Azo Compounds, a class of substances which belong to the benzene series and contain carbon, hydrogen, and nitrogen. All azo compounds are highly coloured, and in some cases they are direct cotton dyes. The simplest azo compound is *azobenzene*, the chemical formula of which is $\text{C}_6\text{H}_5\text{—N}=\text{N—C}_6\text{H}_5$. It is prepared from nitrobenzene by reducing it with stannous chloride and sodium hydroxide. Azobenzene is a highly-coloured crystalline substance, but it is not a dye; it forms, however, the basis of the dyes known as *azodyes*. The azodyes form one of the largest groups of dyes, and derive their name from the presence of the characteristic chemical group $\text{—N}=\text{N—}$, known as the *azo* group. Azo compounds may become exceedingly complicated in structure. They are divided into mon-azo, di-azo, tri-azo compounds according to the number of azo groups present. The well-known dyes Congo Red and Bismarck Brown are azo compounds of complicated structure. Azo dyes were discovered by Griess in 1856, but as a class did not come on the market until about 1876.

Azof. See *Azov*.

Azoic. See *Pre-Cambrian*.

Azoimide, or **Hydrazoic Acid**, N_3H , is an acid liquid of disagreeable odour, producing headache if inhaled. The acid itself and its salts, with the exception of the alkali salts, are exceedingly explosive and dangerous to handle. Lead hydrazoate, PbN_6 , is one of the most stable of these salts. It is a more powerful explosive than mercury fulminate, and has been used instead, in the manufacture of detonators.

Azores (a-zōrz' or a-zō'res), or **Western Islands**, a group belonging to and 900 miles west of Portugal, in the North Atlantic Ocean. They are nine in number, and form three distinct groups—a N.W., consisting of Flores and Corvo; a central, consisting of Terceira, São Jorge, Pico, Fayal, and Graciosa; and a S.E., consisting of São Miguel (or St. Michael) and Santa Maria. The total area is 922 sq. miles; São Miguel (containing the capital Ponta Delgada), Pico, and Terceira are the largest. The islands, which are volcanic and subject to earthquakes, are of comparatively recent origin, and are conical, lofty, precipitous, and picturesque. The most remarkable summit is the peak of Pico, about 7600 feet high. There are numerous hot springs. They are covered with luxuriant vegetation, and diversified with woods, corn-fields, vineyards, lemon and orange groves, and rich, open pastures. The mild and somewhat humid climate, combined with the natural fertility of the soil, brings all kinds of vegetable products rapidly to perfection, among the most important being grain, oranges, pine-apples, bananas, potatoes, yams, beans, coffee, and tobacco. The inhabitants are mainly of Portuguese descent, indolent and devoid of enterprise. Principal exports: wine and brandy, oranges, maize, beans, pineapples, cattle. The climate is recommended as suitable for consumptive patients. The Azores were discovered by Cabral about 1431, shortly after which date they were taken possession of and colonized by the Portuguese. When first visited they were uninhabited, and had scarcely any other animals except birds, particularly hawks, to which, called in Portuguese *açores*, the islands owe their name. Pop. 242,613 (1911).

Az'ote, a name formerly given to nitrogen; hence substances containing nitrogen and forming part of the structure of plants and animals are known as *azotized* bodies. Such are albumen, fibrine, caseine, gelatine, urea, kreatine, &c. See *Nitrogen*.

A'zov, or **Asov**, a town in the Russian government of Ekaterinoslav, upon an island at the mouth of the Don, where it flows into the Sea of Azov; formerly a place of extensive trade, but its harbour has become almost sanded up. Pop. 31,111.

Azov, Sea of (ancient, **Palus Moeōtis**), an arm of the Black Sea, with which it is united by the Straits of Kertch or Kaffa; length about 170 miles, breadth about 80 miles; greatest depth not more than 8 fathoms. The W.

part, called the Putrid Sea, is separated from the main expanse by a long sandy belt called Arabat, along which runs a military road. The sea teems with fish. The Don and other rivers enter it, and its waters are very fresh.

Azpeitia (a`th-pā'i-ti-a`), a town of N.E. Spain, province Guipuzcoa. Near it is the convent of Loyola, a large edifice, now a museum. Pop. 6692.

Az'rael. See *As'rael*.

Az'tecs (from *Aztatl* (heron), and *Flacatl* (man), 'people of the heron'), a race of people who settled in Mexico early in the fourteenth century, ultimately extended their dominion over a large territory, and were still extending their supremacy at the time of the arrival of the Spaniards, by whom they were speedily subjugated. Their political organization, termed by the Spanish writers an absolute monarchy, consisted of a military chief exercising important, but not unlimited, power in civil affairs, in which the council of chiefs and periodic assemblies of the judges had also a voice. Their most celebrated ruler was Montezuma, who was reigning when the Spaniards arrived, about the middle of the fifteenth century. It is inferred that considerable numbers of them lived in large communal residences, and that land was held and cultivated upon the communal principle. Slavery and polygamy were both legitimate, but the children of slaves were regarded as free. Although ignorant of the horse, ox, &c., they had a considerable knowledge of agriculture, maize and the agave being the chief produce. Silver, lead, tin, and copper were obtained from mines, and gold from the surface and river beds, but iron was unknown to them, their tools being of bronze and obsidian. Metal coins were not in use. In metal-work, feather-work, weaving, and pottery they possessed a high degree of skill. To record events they used picture-writing, and their lunar calendars were of unusual accuracy. Two special deities claimed their reverence: Huitzilo-pochtli, the god of war, propitiated with human sacrifices; and Quetzalcoatl, the beneficent god of light and air, with whom at first the Aztecs were disposed to identify Cortez. Their temples, with large terraced pyramidal bases, were in the charge of an exceedingly large priesthood, to whom the education of the young was entrusted. As a civilization of apparently independent origin, yet closely resembling in many features the archaic Oriental civilizations, the Aztec civilization is of the first interest, but in most accounts of it a large speculative element has to be discounted.

Az'uline, or **Az'urine**, blue dyes belonging to the coal-tar class.

A'zure, the heraldic term for the colour blue, represented in engraving by horizontal lines.

Az'urine (*Leuciscus cærulëus*), a fresh-water fish of the same genus as the roach, chub, and minnow, found in some parts of Europe, but rare in Britain; called also *Blue Roach*.

Az'urite, a blue mineral, a carbonate of copper, often occurring in crystals. Its formula is $2\text{CuCO}_3.\text{Cu}(\text{OH})_2$, and it is also called *Blue Malachite*. Also a name of lazulite.

Azymite. See *Eucharist*.

B

B is the second letter and the first consonant in the English and most other alphabets. It is a mute and labial, pronounced solely by the lips, and is distinguished from *p* by being sonant, that is, produced by the utterance of voice as distinguished from breath.

B, in *music*, the seventh note of the model diatonic scale or scale of C. It is called the leading note, as there is always a feeling of suspense when it is sounded until the keynote is heard.

Baader, Franz Xaver von (fra'nts-zä'fer fon bä'der), German philosopher and theologian, born in Munich, 1765, died 1841. He studied engineering, became superintendent of mines, and was ennobled for services. He was deeply interested in the religious speculations of Eckhart, St. Martin, and Böhme, and in 1826 was appointed professor of philosophy and speculative theology in the University of Munich. During the last three years of his life he was interdicted from lecturing for opposing the interference in civil matters of the Roman Catholic Church.

Ba'al, or **Bel**, a Semitic word, which primarily signified lord or proprietor, and was afterwards applied to many different divinities, or, with qualifying epithets, to the same divinity regarded in different aspects, describing him as an occupier of some physical object or locality, or as a possessor of some attribute. Thus in *Hos.* ii, 16 it is applied to Jehovah himself, while *Baal-berith* (the Covenant-lord) was the god of the Shechemites, and *Baal-zebub* (the Fly-god) the idol of the Philistines at Ekron. Baal was the sacred title applied to the Sun as the principal male deity of the Phœnicians and their descendants, the Carthaginians, as well as of the ancient Canaanitish nations, and was worshipped as the supreme ruler and vivifier of nature. The word enters into the composition of many Hebrew, Phœnician, and Carthaginian names of persons and places; thus, *Jerubaal*, *Hasdrubal* (help of Baal), *Hannibal* (grace of Baal), and *Baal-Hammon*, *Baal-Thamar*, &c.

Baalbek' (ancient, **Heliopölis**, 'city of the sun'), a place in Syria, in a fertile valley at the foot of Antilibanus, 40 miles from Damascus, famous for its

magnificent ruins. Of these the chief is the Temple of the Sun, built either by Antoninus Pius or by Septimius Severus. Some of the blocks used in its construction are 60 feet long by 12 thick; and its fifty-four columns, of which six are still standing, were 72 feet high and 22 in circumference. Near it is a temple of Jupiter, of smaller size though larger than the Parthenon at Athens, and there are other structures of an elaborately ornate type. Originally a centre of Sun-worship, it became a Roman colony under Julius Cæsar, was garrisoned by Augustus, and acquired increasing renown under Trajan as the seat of an oracle. Under Constantine its temples became churches, but after being sacked by the Arabs in 748, and more completely pillaged by Tamerlane in 1401, it sank into hopeless decay. The work of destruction was completed by an earthquake in 1759.

Baal-zebub. See *Beelzebub*.

Baba, a cape near the north-west point of Asia Minor.

Babadagh (ba' -ba' -däg'), a town of Roumania, capital of the Dobrudsha, carrying on a considerable Black Sea trade; it was bombarded by the Russians in 1854. Pop. 10,000.

Bab'bage, Charles, English mathematician and inventor of the calculating-machine, born 1792, died 1871. He graduated at Cambridge in 1814, and occupied the Lucasian chair of mathematics at Cambridge for eleven years, but delivered no lectures. As early as 1812 he conceived the idea of calculating numerical tables by machinery, and in 1823 he received a grant from Government for the construction of such a machine. After a series of experiments lasting eight years, and an expenditure of £17,000 (£6000 of which was sunk by himself, the balance voted by Government), Babbage abandoned the undertaking in favour of a much more enlarged work, an analytical engine, worked with cards like the jacquard-loom; but the project was never completed. The incompleted machine is now in the South Kensington Museum. Among the many treatises he published on subjects connected with mathematics and mechanics few can be regarded as finished performances. Babbage was instrumental in founding the Astronomical and Statistical Societies (1820 and 1834).

Babbitt-metal, a soft metal resulting from alloying together certain proportions of copper, tin, and zinc or antimony, used with the view of as

far as possible obviating friction in the bearings of journals, cranks, axles, &c., invented by Isaac Babbit (1799-1862), a goldsmith of Taunton, Massachusetts.

Ba'bel. See *Babylon*.

Ba'bel, Tower of, according to the 11th chapter of *Genesis*, a structure in the Plain of Shinar, Mesopotamia, commenced by the descendants of Noah subsequent to the deluge, but which was not allowed to proceed to completion. It has commonly been identified with the great temple of Belus, or Bel, that was one of the chief edifices in Babylon, and the huge mound called Birs Nimrud is generally regarded as its site, though another mound, which to this day bears the name of Babil, has been assigned by some as its site. Babel means literally 'gate of God'. The meaning 'confusion' assigned to it in the Bible really belongs to a word of similar form. See *Babylon*.

Bab-el-Mandeb ('gate of tears', from being dangerous to small craft), a strait, 15 miles wide, between the Indian Ocean and the Red Sea, formed by projecting points of Arabia in Asia, and Abyssinia in Africa. The Island of Perim is here.

Ba'ber, first Grand Mogul, the founder of the Mogul dynasty in Hindustan, born in 1483, died 1530. He was a grandson of the great Tartar prince, Timur or Tamerlane, and was sovereign of Cabul. He invaded Hindustan, and in 1525 overthrew and killed Sultan Ibrahim, the last Hindu emperor of the Pathan or Afghan race. He made many improvements, social and political, in his empire, and left a valuable autobiography (English translation, 1826).

Babeuf (ba' -beuf), François Noel, a French political agitator, born in 1764; started a democratic journal in Paris, called *Le Tribun du Peuple*, par '*Gracchus*' Babeuf, and wrote with great severity against the Jacobins. After the fall of Robespierre, to which he powerfully contributed, he openly attacked the terrorists, and advocated the most democratic principles. He was accused of a conspiracy against the directorial Government, condemned to death, and guillotined in 1797. He was one of the pioneers of Socialism in France.

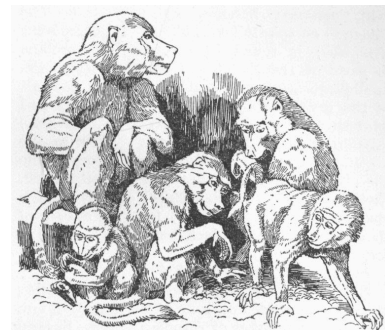
Bab'ington, Anthony, a Catholic gentleman of Derbyshire, born 1561. He associated with others of his own persuasion to assassinate Queen Elizabeth, and deliver Mary, Queen of Scots. The plot being discovered, the conspirators were executed in 1586.

Babiroussa. See *Babyroussa*.

Bab'ism, the doctrines of a Mohammedan sect whose head-quarters is Persia, founded by Seyd Ali Mohammed in 1844. He took the name of Bab-ed-din, 'the gate of the faith', and afterwards that of Nokteh, 'the point', as not merely the recipient of a new divine revelation, but the focus in which all preceding dispensations would converge. One of his most successful disciples was a highly-gifted woman, Gurred-ul-Ayn, 'consolation of the eyes', who perished with many others during a persecution in 1852. The Bab himself had been executed about two years before this, and was succeeded by a noble youth, Mirza Yahya. The sect holds that all individual existence is an emanation from the supreme deity, by whom it will be ultimately reabsorbed. The morality of the sect is pure and cheerful, and it shows great advancement in the treatment of woman. Moses, Christ, and Mohammed are acknowledged as prophets, though only mere precursors of the Bab. A schism divided the followers of Babism into two sects, Bahais and Ezelis. The former have carried on an active propaganda in America.

Ba'boo, or **Babu**, a Hindu title of respect equivalent to *sir* or *master*, usually given to wealthy and educated native gentlemen, especially when of the mercantile class.

Baboon', a common name applied to a division of old-world quadrumana (apes and monkeys), comprehending the genera *Cynocephalus* and *Papio*. They have elongated abrupt muzzles like a dog, strong tusks or canine teeth, usually short tails, cheek-pouches, small deep eyes with large eyebrows, and naked callosities on the buttocks. Their hind and fore feet are well proportioned, so that they run easily on all fours, but they do not maintain themselves in an upright posture with facility. They are generally of the size of a



Family of Common Baboons
(*Cynocephalus babouin*)

moderately large dog, but the largest, the mandrill, is, when erect, nearly of the height of a man. They are almost all African, ugly, sullen, fierce, lascivious, and gregarious, defending themselves by throwing stones, dirt, &c. They live on fruits and roots, eggs and insects. They include the chacma, drill, common baboon, and mandrill. The chacma or pig-tailed baboon (*Cynocephalus porcarius*) is found in considerable numbers in parts of the S. African colonies, where the inhabitants wage war against them on account of the ravages they commit in the fields and gardens. The common baboon (*C. babouin*) inhabits a large part of Africa farther to the north. It is of a brownish-yellow colour, while the chacma is greyish-black, or in parts black. The hamadryas (*C. hamadryas*) of Abyssinia is characterized by long hair, forming a sort of shoulder-cape. The black baboon (*C. niger*) is found in Celebes.

Babour (bä'bu'r). See *Barber*.

Bab'rius, a Greek poet who flourished during the second or third century of the Christian era, and wrote a number of Æsopian fables. Several versions of these made during the Middle Ages have come down to us as *Æsop's Fables*. In 1840 a manuscript containing 120 fables by Babrius, previously unknown, was discovered on Mount Athos.

Babuya'nes Islands, a group in the Pacific Ocean, between Luzon and Formosa, belonging to United States. Pop. 12,000.

Baby-farming. The law relating to the protection of children and young persons has been subjected to amendment and consolidation by the Children Act of 1908, which wholly repealed the Infant Life Protection Act of 1897 dealing with baby-farming and infant life protection.

The Act of 1908 (Part I) deals with baby-farming, and requires that notice shall be given to the Local Authority (in Scotland the Parish Council) by the person undertaking for reward the nursing and maintenance of any infant or infants under seven years of age apart from their parents, within forty-eight hours after reception, unless the period for which it is received be only forty-eight hours or less. The notice requires the name, sex, date and place of birth of the child to be stated, where it is to be kept, and from whom it is received. The Act also provides that such a person shall notify any change in his residence or the removal or death of the infant to the Local Authority

within forty-eight hours. In regard to all these matters the Act is retrospective, and necessitates persons who had undertaken nursing and maintenance of such infants before its coming into operation to comply with its provisions within a month after the commencement thereof. But it exempts any person who may have given the similar notice required by the Infant Life Protection Act of 1897, although it does not exempt any person whose duty it was to have given notice thereunder from any liability which such a person may have incurred thereunder.

A duty is imposed upon Local Authorities to inquire regarding persons willing to undertake the nursing and maintenance of infants, to appoint infant-protection visitors of either sex in so far as it provides for infant life protection and gives powers to such Local Authorities and visitors for fulfilling the requirements of the Act. It is an offence for the person undertaking the nursing to refuse to allow such visitors access to the infant or the premises in which it is kept; and, if need be, application may be made to the court for a warrant to enter the house in which an infant is farmed out, and where there is reason to believe that the Act is being contravened. It is an offence for any infant to be kept (1) by any person from whose care an infant has been removed under the Act or the Infant Life Protection Act, or (2) by any person convicted of any offence under Part II of the Children Act of 1908 or the Prevention of Cruelty to Children Act, 1904.

The Local Authority may fix the number of infants under the age of seven years which may be kept in any dwelling, and it is an offence to keep more. Provision is also made for the removal of an infant from overcrowded, dangerous, or insanitary premises, or from the custody of a person unfit to take charge of the child; and an application may be made by a visitor to a magistrate for a removal order enforceable by the visitor or a constable. Should the infant die, notice of the fact must be given by the person with whom it is farmed out to the Procurator-fiscal of the district, if in Scotland, and, if in England, to the Coroner of the district, within twenty-four hours. He shall hold an inquest unless a certificate by a medical practitioner specifying the cause of death shall be forwarded. Failure to give such notice is punishable under the Act.

A person nursing an infant for reward shall have no interest in the life of the child for the purposes of life assurance. It is not permissible for such a

person or insurance company to insure the life of such a child. To do so renders both the person and the insurance company liable to prosecution.

Any person knowingly or wilfully making any false statement in any notice required to be given under the Act commits an offence under the Act.

Imprisonment for a term not exceeding six months, or a fine not exceeding £25, may be imposed upon any person found guilty upon summary conviction of an offence against Part I of the Children Act, 1908; in addition the court may order the removal of the infant to a place of safety. All fines are payable to the Local Authority for the purposes of the Act.

Legal guardians of an infant, as well as institutions established for the protection and care of infants, are exempted from the provisions of Part I of the Children Act, 1908.

Bab'ylon, the capital of Babylonia, on the left of the Euphrates, one of the largest and most splendid cities of the ancient world, now a scene of ruins, and earth-mounds containing them. Babylon was a royal city sixteen hundred years before the Christian era; but the old city was almost entirely destroyed in 683 B.C. A new city was built by Nebuchadnezzar nearly a century later. This was in the form of a square, each side 15 miles long, with walls of such immense height and thickness as to constitute one of the wonders of the world. It contained splendid edifices, large gardens and pleasure-grounds, especially the 'hanging-gardens', a sort of lofty terraced structure supporting earth enough for trees to grow, and the celebrated Tower of Babel or Temple of Belus, rising by stages to the height of 625 feet. (See *Babel, Tower of*.) After the city was taken by Cyrus in 538 B.C., and Babylonia made a Persian province, it began to decline, and had suffered severely by the time of Alexander the Great. He intended to restore it, but was prevented by his death, which took place here in 323 B.C., from which time its decay was rapid. Interesting discoveries have been made on its site, more especially of numerous and valuable inscriptions in the cuneiform or arrow-head character. The modern town of Hillah is believed to represent the ancient city, and the plain here for miles round is studded with vast mounds of earth and brick and imposing ruins. The greatest mound is Birs Nimrud, about 6 miles from Hillah. It rises nearly 200 feet, is crowned by a ruined tower, and is commonly believed to be the remains of

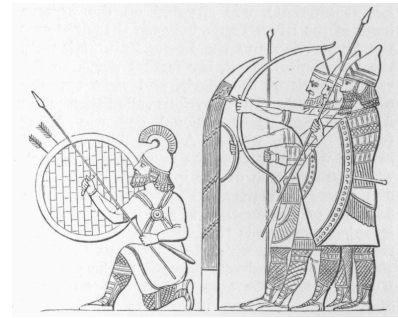
the ancient Temple of Belus. Another great ruin-mound, called Mujellibeh, has also been assigned as its site.

Babylonia and Assyria. These ancient seats of Mesopotamian civilization lay across the western Asian trade-routes, Babylonia being in the south and Assyria in the north. The area, in which they flourished is embraced by the Rivers Tigris and Euphrates, and extends from the head of the Persian Gulf to the frontiers of Armenia and Northern Syria, is bordered on the east by Persia and on the west by the Syro-Arabian desert. 'Mesopotamia' is a term borrowed by the ancient Greeks from the Semites, and was first applied to the north-western region. There was a Roman province of that name. The northern area of the Tigris-Euphrates valley is partly mountainous and partly steppe land, with wide stretches of elevated grazing-lands and fertile districts on the banks of rivers. Assyria, derived from A-usar ('the river-bank region'), had origin in north-eastern Mesopotamia. Its most ancient capital, Asshur (modern Kal'at Sherkat), was situated on the western bank of the River Tigris, between the tributaries the Upper Zab and the Lower Zab. Nineveh, the last capital of all, lay farther north on the eastern bank of the Tigris, and right opposite modern Mosul. The Assyrians called themselves Asshurai, and their national god was Ashur (earlier Ashir), written A-Shur, but confused in time with Ash-Shur, the name of the capital. The southern area of Mesopotamia, below Bagdad, where the Tigris and Euphrates come within 35 miles of one another, is a flat alluvial plain. Babylonia proper is that fish-shaped region between the rivers, which broadens out to about 100 miles and gradually narrows to the point at Kurnah where the rivers meet and form the Shatt-el-Arab. From Kurnah to the head of the Persian Gulf is the ancient area of Chaldea, which means 'Sea-land'. The inhabitants called themselves Kaldu (Sea-landers) and were known to the Hebrews as the Kasdim, and to the Greeks as the Chaldaioi. Babylonia, and the rest of the alluvial plain, is the 'gift' of the rivers which rise in flood each year when the snow melts on the Armenian mountains. Both the Tigris and the Euphrates bring down enormous quantities of sediment—indeed five times as much as does the Egyptian Nile—and they have withal more destructive tendencies. The accumulating silt tends to divert the flow of the rivers in the south, and a process of land-making which began at the close of the Ice Age is still in progress, thrusting back the head of the Persian Gulf. At one time the Tigris and Euphrates entered

the gulf by separate mouths, and Chaldea was then a narrow fringe of steppe land, plain, and marshes. The River Tigris is about 1146 miles long, and begins to rise early in March, reaching its height in May, and subsiding before the end of June. The Euphrates is a slower river, about 1780 miles in length. It begins to rise a fortnight later than the Tigris, is longer in flood, and does not reach its lowest level until September. As there is a drop of only 120 feet between Bagdad and the sea, a distance, as the crow flies, of about 300 miles, the ancient Babylonians did their utmost to conserve the water which came down in such abundance and was rapidly drained away, after doing much damage. Near Bagdad the Tigris is on a higher level than the Euphrates, and could be run into it through a canal; farther south the Euphrates, being on the higher level, could be run by canal into the Tigris. The ancient engineers cut these and other canals; indeed, they covered the whole valley with a network of them. To store the Euphrates water for the season of drought and great heat, they formed canals which carried the roaring flood into two depressions in the western desert between the modern towns of Kerbela and Rama'di. These depressions are utilized for the British irrigation scheme. The Babylonians did their utmost to control the Tigris by erecting earthen dams so as to hold up as much of the water as possible. Earthen dykes were also erected to raise its banks. The right bank protected the farms from disastrous flooding. These ancient irrigation works made Babylonia the greatest grain-yielding area in the ancient world. Its vast surplus of food stimulated trade and brought Babylonia immense wealth, while its strategic situation made it very powerful. Its cultural influence flowed along the trade routes, eastward across Persia and the Iranian plateau, northward through Assyria to Armenia (ancient Urartu) and beyond, and north-westward along the Euphrates banks into Syria and Palestine, and into Asia Minor, the land of the Hittites, and then along the highway to Europe, called later by the Persians 'the royal road'. The road to Egypt ran southward from North Syria through Palestine, skirting the maritime valley by 'the way of the Philistines'. The merchants of many nations met in the city of Babylon, the London of the ancient western Asian world, and in the trading centres of North Syria, including Carchemish.

The earliest agriculturists and traders of Babylonia were the Sumerians, a non-Semitic people, who built a number of cities in the irrigated valley and founded colonies in Assyria. No trace has been found of a Neolithic Age in

Babylonia. Its civilization, so far as our knowledge goes, began after copper was introduced. At an early period Semites filtered into the valley. They absorbed Sumerian civilization, and ultimately became politically predominant in the northern part of the valley, which was called Akkad (the biblical Accad). The southern part was then known as Sumer or Sumeria (the biblical 'Plain of Shinar'). Akkad embraced the city States of Agade, Babylon, Borsippa, Sippar, Kish, Opis, and Kutha. The Sumerian city States included Eridu, Ur (later 'of the Chaldees' and the birthplace of Abraham), Lagash, Shuruppak, Erech, Umma, and Adab. Inter-state wars resulted time and again in the formation of confederacies which were more of political than racial character, and the predominance of one city or another. Religious ideas were fused, and local pantheons appear to have reflected local politics. The chief seat of early Sumerian civilization was the city of Eridu, a name signifying 'on the seashore'. Its site is marked by modern Abu-Shahreïn on the eastern bank of the Euphrates, and south-east of Ur, modern Mukayyar. Eridu was originally a seaport on the Persian Gulf, from which it is now separated by about 125 miles of land formed by river silt. The chief god of Eridu was Ea, a name also rendered Ae and Aa, and apparently meaning 'waters' or 'house (region) of waters'. That the term 'waters' had a special significance for the inhabitants of a water-formed land is evident by the fact that the Semitic equivalent of Ea is Enki, which means 'lord of earth'. Other titles of the god were 'lord of Heaven and Earth', 'ruler of the land', 'the deep', 'god of the abyss', and 'king of the river'. He was withal the 'Creator', water having created Sumeria, and Nadimmud, 'lord of everything', as the Egyptian Osiris in his character of god of the Nile was 'Neb-er-Zer', also 'lord of everything'. Ea had a ship with a crew including his son Marduk (Merodach or Marad, the Ni-marad or Nimrod of legend) and In-ab, 'the pilot of Eridu'. The ship was hailed, in religious literature, as the bringer of fertility and joy. Evidently the beginning of Sumerian civilization had some connection with ancient seafarers. Of special interest in this connection are the references made by the Greek writer Berosus to Oannes, a sea-dweller who came daily from the Persian Gulf to instruct the early Babylonians how to irrigate the land, grow crops, erect houses, make



Assyrian Soldiers

laws, write, calculate and measure, and engage in trade. Texts have been discovered which tend to identify Oannes with the god Ea. As the early Sumerians bore a strong physical resemblance to the ancient Egyptians, and there were links between the early religious beliefs and customs of the two peoples, it is suspected that they represented two branches of the same ancient stock. Some think Sumerian civilization was founded by colonists from Egypt, while others believe that Egyptian civilization was stimulated by settlers from Sumeria. The question as to who first invented ships has an important bearing in this connection. When we turn to Sumero-Babylonian religious myths we find that some reflect local natural phenomena. In one the Euphrates is hailed as the River (god) "who didst create all things". Another, referring to the creation of man, says that Marduk (earlier the god Gilimma) "tied reeds on the face of the waters, he formed dust and poured it out beside the reeds ... he formed mankind". Another version is that man was created from the god's blood and bone, or, as Berosus has it, from the god's blood and earth. The oldest Sumerian creation myth states that all things came from primeval water. Then many deities were created by the pre-existing triad of gods, Anu (the sky-god), Enlil ('the elder Bel', the earth-god, and 'lord of grain'), and Ea, who was associated with the goddess Ninkharsagga. Ea had several other wives, including Ninshar or Ninkurra, Ninella, Nintu, and Damkina. These goddesses were ultimately all absorbed by Ishtar, the Semitic name of the love and mother goddess. In the later Babylonian 'Seven Tablets' version of the creation myth a conflict is introduced. The female chaos-dragon Tiamat is slain by Marduk, the chief god of Babylon, 'the later Bel', who cuts up her body and uses part to form the sky and part to keep the waters in their place. After a time a deluge took place. In the Sumerian myth a priest-king called Ziusudu (earlier named Tatug or Uttu) corresponds both to Adam and to Noah, and builds a large vessel, having been warned by Ea of the approaching disaster. The Gilgamesh Epic refers to the hero as Ut-na-pishtim (or Pir-na-pishtim). He and his wife afterwards become immortal, and they are visited on their island by Gilgamesh (the Babylonian Hercules), when, smitten by disease, he searches for the Plant of Life, which he finds and afterwards loses. Ishtar, the goddess who plays a part in this epic, is also the heroine of the poetic legend which tells of her descent to Hades in quest of Tammuz, whose story resembles somewhat that of Adonis and has Osirian characteristics. The Babylonian Persephone was Eresh-ki-gal with whom

Tammuz dwelt in her underworld for part of the year. Associated with her was Nergal, chief god of Kutha, whose planet was Mars. He conquered Hades. He and other city gods appear to have been developed from attributes vaguely shared by the earlier triad Anu, Bel-Enlil, and Ea. The moon-god of Ur was Sin (Sumerian Nannara), whose name clings to the desert of Sinai. Shamash was the Semitic name of the sun-god of Sippar, the Sumerian being Utu; Gishnu, the Light, and Ma-banda-anna, 'the boat of heaven' (like the Egyptian Ra), and Mitra were among his other non-Semitic names. Dagan, the god of the Euphrates, is regarded as a form of Ea, and as probably the same deity as Dagon of the Philistines. In later times the astrologers coloured with their doctrines the religious beliefs of Babylonia.

Among the early empire-builders of Babylonia were Lugal-zaggisi of Sumerian Erech, who claimed to have subdued all lands from the Lower Sea (Persian Gulf) to the Upper Sea (the Mediterranean). He flourished about 2800 B.C. Another was Sargon of Akkad, who lived about a century and a half later. He must not be confused with the much later biblical Sargon of Assyria. His son or grandson, Naram-Sin, recorded his great trading activities and victories over his enemies, including the Elamites, whose chief stronghold was Susa in south-western Persia. Elam was a rival power of considerable strength. About 2200 B.C. it had overrun a great part of Sumeria, and Akkad fell to the invading bands of Amorites. These Amorites founded the Hammurabi Dynasty, of which the king of that name was sixth in succession; Hammurabi extinguished the last sparks of Elamite power in Sumeria, which, with Akkad, was united to form the kingdom of Babylonia, named after Babylon, the capital. It was during this period that Abraham migrated from Ur to Palestine. Hammurabi (c. 2123-2081 B.C.) codified the ancient Sumerian laws and did much to develop trade. The chief god of his kingdom was Marduk, 'the later Bel', whose temple was called Esagilla. Babylon became the greatest trading centre in Western Asia and a rival of the Egyptian Memphis. Sumerian gradually ceased to be a spoken language, being supplanted by Semitic, but remained, like Latin, in mediæval Europe, the language of law, culture, and religion, while the Babylonian language became the language of trade and diplomacy, and was used in international correspondence by all the great Powers. The Hammurabi dynasty came to an end about 1926 B.C., when the Hittites

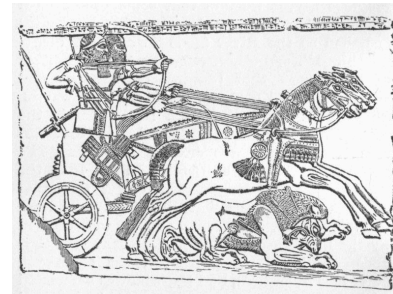
raided Babylonia and carried off the statues of Marduk and his consort. Kassites, assisted by Elamites, had been attacking Babylonia from the east, and the Sea-landers, who were Arabians mixed with Sumerians, established their independence. In the end the Kassites, who were probably Aryan by race, conquered Babylonia and established a new dynasty. Like the Trojans they were 'tamers of horses'. Their military successes are believed to have been due to the use of the horse, which was a rarity in Babylonia before their time, but became common as a beast of burden as soon as the Kassite dynasty was established. Kassite supremacy lasted for over 570 years, and during that period Babylonia was known as Karduniash. One of its early kings brought back from Khani (Mitanni) the statues of Marduk and his consort which had been carried off by the Hittites. Babylonian civilization was not changed by Kassite conquest, and its trade went on as of yore. The Kassites formed an aristocracy like the Normans in England. During the Kassite period Assyria was a growing power. It had origin as a Sumerian colony, but above the relics of the Sumerians at Asshur have been found those of a people whose kings had such non-Semitic and probably Aryan names as Ushpia, Kikia, and Adasi. The Amorites afterwards swept into Assyria, which they Semitized. During the Hammurabi dynasty it was subject to Babylonian overlordship. It grew independent during the latter period of the Kassite dynasty, and its kings formed compacts with the Kassites. Their western neighbours were by this time the Mitannians. An Aryan military aristocracy had formed a powerful State in north-western Mesopotamia, where they worshipped Indra, Varuna, and Mitra, gods which figure in the mythology of the Aryan invaders of India. Egypt, under Thothmes III, extended her Syrian empire to the borders of Mitannia and a friendship sprang up between the two Powers. Assyria was during the period subject to Mitanni. The Tell-el-Amarna letters reveal the fact that Egyptian monarchs had married princesses from Mitanni, and the famous Akhenaton had Mitannian blood in his veins.

The Hittites conquered Northern Syria and overran Mitannia when the Egyptian Asian empire went to pieces. Assyria then became powerful and independent again, the first great king of its new age being Ashur-uballit, who was strong enough to interfere with Babylonia's domestic politics. Shalmaneser I of Assyria (c. 1300 B.C.) conquered the whole of the Mitanni kingdom and extended the Assyrian empire westward across the Babylonian



Tablet recording
the Wars of
Sennacherib

caravan road to North Syria. He built a new capital at Kalah (Nimrud). His son, Tulkulti—Nineb I—(c. 1275 B.C.), conquered Babylonia and reigned over it for seven years. In the end he was murdered by political conspirators, with whom his son was associated. Civil war ensued, and Assyria's history



Ashur-natsir-pal in Chariot,
hunting Lions about 884 B.C.
From a relief in the Palace of
Nimrod

is found to be obscure for a century afterwards. In Babylon the Kassite dynasty, under which trade flourished, came to an end as a result of an Elamite invasion about 1185 B.C. A new Babylonian dynasty then arose, and the third king, Nebuchadnezzar I, revived the old empire and waged war against Assyria. But the future lay with Assyria, which was organized as a military State. It did not have Babylonia's natural resources, and could only exist as a great Power by imposing tribute on weaker States. A standing army was the basis of its strength. The national god was Ashur, who was symbolized by a winged disk, an adaptation of the Egyptian winged disk of Horus. It was carried to battle with the king, so that wherever the king was, there was the national god of war. The heavily-bearded Assyrians were a fiercer and more war-like people than the mild shaven Babylonians, who were ever influenced by Sumerian modes of thought. The Assyrian temperament is reflected in its art, which at its best is characterized by vigorous realism and brilliant ferocity. Much of it is harsh, exaggerated, and pompous. A sharp contrast is presented by the calmer and more idealistic art of the Sumero-Babylonians. The records of Assyrian monarchs deal mainly with extensive conquests. They did their utmost to set up a reign of terror in Western Asia, and when subject States revolted the chief men in them were flayed alive or impaled on stakes. The wholesale destruction of trading city States and the massacres of thousands of innocents were fit subjects for an Assyrian conqueror to boast of in his inscriptions. Hittites, North Syrians, Babylonians, and Urtians (Armenians), were time and again plundered and subjected. A great conqueror of the Middle Period was Tiglath-Pileser I (c. 1100 B.C.), and

other monarchs of the same name repeated his conquests and atrocities during the last century of Assyria's existence. The military glory of such a monarch as Ashur-natsir-pal III (885-860 B.C.), whose armies swept through western Asia like cyclones, has a lurid background of savage cruelty and oppression. He fought for no greater cause than plunder and the maintenance of Assyria's ability to wring tribute from the oppressed. His son, Shalmaneser III, extended the empire but died under a cloud of internal revolt. Subsequent rulers dominated Babylonia, and one, Adad-nirari IV (810-782 B.C.), married the Babylonian princess, Sammurammât, the Semiramis of tradition. Tiglath-Pileser IV (745-727 B.C.) was overlord of Babylon, and, indeed, Babylonia was more or less under Assyrian sway for the next century. Sargon II ('Sargon the later') did his utmost to break the national spirit in subject States by transporting whole communities from one to another. He removed a portion of the 'ten lost tribes' to the Median hills, and dispatched Babylonians from Kutha to Samaria, where they made Nergal their chief god. His campaigns extended to Phœnicia, Cilicia, and Armenia. Sennacherib, his son, defeated Egypt and her allies in Palestine and besieged Hezekiah in Jerusalem. A revolution in Babylonia broke out, and he swept into it and devastated Babylon, so that Nineveh might become the chief city in Mesopotamia. But Babylon had to be rebuilt by his son Esarhaddon, so that its dislocated trade might be restored. Esarhaddon, recognizing that the great commercial rival of Babylonia was Egypt, and that Egypt instigated the Palestinian and Syrian revolts against Assyria, invaded the Nile valley and captured and sacked Memphis. Ashur-bani-pal (668-626 B.C.), the next Assyrian emperor, punished the Egyptian rebels by sweeping southwards and capturing and sacking Thebes, which is referred to as No (Nu-Amon) in the Bible (*Nahum*, iii, 8-10): "Populous No ... Yet was she carried away ... her young children also were dashed in pieces at the top of all the streets: and they (the Assyrians) cast lots for her honourable men, and all her great men were bound in chains". Egypt became for a time an Assyrian province. Elam was similarly dealt with, the great capital Susa being treated like Thebes. On the north and east Assyrian power was established with characteristic Assyrian ferocity and thoroughness. The result was that the predatory nomadic tribes of Scythians and Medes were able to sweep through the devastated territories and strike at the very heart of the empire. Ashur-bani-pal was the last great monarch of Assyria. In private life he was cultured and scholarly, and his library, which

contained copies of many ancient records and literary works, is one of the sources of Mesopotamian history. Although the Sardanapulus of Greek tradition, he was not the actual monarch who perished when Nineveh was captured and sacked about 606 B.C. by the fiery hillmen from the east. The Assyrian military organization collapsed and Assyria ceased to be. The factors that led to the downfall of Assyria can only be guessed at. It may be that its many wars of conquest had left it in a state of exhaustion. It is also possible that the new generation of the aristocratic class that grew up in wealth and luxury was disorganized by intrigues and corruption. After Ashur-bani-pal died, no great leader arose, and the vast unwieldy empire suffered from internal decay. Babylonia was the first to foster organized revolt. It was overrun by the Chaldeans from the south, who, two years after Ashur-bani-pal's death, set on the throne at Babylon their leader, King Nabopolassar. With him began the Neo-Babylonian dynasty which lasted for 86 years. He had an ambitious and capable son named Nebuchadnezzar, who set out, as soon as Nineveh fell, to seize the western portion of the Assyrian Empire—the Medes, with whom the Chaldeans had an agreement, being content to retain north-eastern Mesopotamia. The Egyptians, having overcome the Assyrian garrisons in the Nile valley, were by this time moving northward through Palestine with the idea of re-establishing their ancient empire in Asia. Memphis and Babylon were rival trading centres, and the struggle that ensued was one which was to decide which should have preeminence on the western Asian trade-routes, and especially the 'clearing-houses' of North Syria. In 604 B.C. Nebuchadnezzar met and defeated Pharaoh Necho's Egyptian army at Carchemish. The Egyptians retreated in confusion and were pursued to their frontier. Nabopolassar's death in Babylon caused Nebuchadnezzar to return home and ascend the throne. Eight years later, however, he was back in Palestine again. Judah had been plotting with Egypt, its ally, and the Babylonian monarch besieged and captured Jerusalem. Thousands of Jews were carried off as captives to Babylonia. In 587 B.C. Egypt involved Judah



Israelites before Shalmaneser. From the Black Obelisk of Shalmaneser II.

in a second revolt. Nebuchadnezzar II again captured Jerusalem. This time he destroyed the city and carried away the greater part of the remnant of the Jewish people to serve his artisans and agriculturists "by Babel's streams". Babylon was triumphant as a political and trading centre, and, during Nebuchadnezzar's long reign of 42 years, prospered greatly. Its products and its culture were carried far and wide, and Chaldean astrologers as well as merchants became known in Egypt and Greece. Babylonia was the birthplace of the science of astronomy, which developed from astrology. Nebuchadnezzar was a great builder. It is the ruins of his Babylon that have been excavated in our own day. Although Greek descriptions of the city have been found to be somewhat exaggerated, it was yet made very strong and impressive. The royal palace was rebuilt, and the Hanging Garden on its terraced platform became one of the seven wonders of the world. Trade flourished in the capital, and a harbour was built at the head of the Persian Gulf to promote sea traffic. In 539 B.C. Cyrus, the Persian conqueror, having defeated the Babylonian army, entered Babylon in state and was welcomed by the priests, Nabonidus, the last king of the Neo-Babylonian empire, having made them his enemies. The independence of Babylonia then came to an end, but the importance of its capital as a trading centre did not suffer sudden decline. When it was captured by Alexander the Great in 331 B.C. it had been much damaged in consequence of revolts against Persia. The Macedonian conqueror resolved to make it his capital, but his early death, and subsequent political developments, hastened its decay. The Greeks had removed the rocks that blocked the Tigris, which then could be navigated by vessels from the Persian Gulf. Seleucus built the new capital of Seleucia on the right bank of the Tigris, and the traders followed the officials to the new Babylon. In time Ctesiphon arose on the left bank, and after the fall of the Parthian empire it was the principal city of the province. The Moslem conquest of Mesopotamia in the seventh century of our era brought about the rise of Bagdad, a few miles to the north of Ctesiphon. It will be noted that all these cities are strategically situated in the neck of land, between the rivers, which is crossed by the caravan roads leading east and west. During the reign of the famous Harun al Rashid (786-809) Bagdad was at the height of its splendour. A portion of this ruler's great palace still remains, and the tomb of Zobeide, his queen, is a prominent feature on the outskirts of the city. Bagdad declined as a trading centre after Vasco da Gama discovered the sea-route to India, and diverted the merchandise previously

carried to Europe across the overland routes across Persia and through Mesopotamia and Asia Minor. Bagdad afterwards depended chiefly on the traffic caused by pilgrim caravans to and from Mecca. The disastrous neglect of Babylonia's irrigation system occurred under Turkish rule. Towards the close of the nineteenth century Germany obtained concessions from the Turks and planned the Bagdad railway, so as to reopen the ancient overland trade-route for Indo-Persian trade. British interests in the Persian Gulf were seriously threatened by this scheme. When war broke out in 1914, a British force landed in ancient Chaldea and pushed northward. Bagdad, the modern Babylon, was captured in 1917, and Mosul (Nineveh) was occupied in 1918. The Peace Treaty between the Allied Powers and Turkey placed Mesopotamia under the protection of Great Britain, which thus secured control of the Bagdad railway. An extensive irrigation scheme in the southern alluvial valley promises to restore ancient Babylonia to its importance as one of the great grain-yielding areas of the world.—

BIBLIOGRAPHY: R. W. Rogers, *History of Babylonia and Assyria*; H. Winckler, *History of Babylonia and Assyria*; L. W. King and H. R. H. Hall, *Egypt and Western Asia in the Light of recent Discoveries*; P. S. P. Handcock, *The Latest Light on Bible Lands*; L. W. King, *History of Babylon*; D. A. Mackenzie, *Myths of Babylonia and Assyria*; T. G. Pinches, *The Religion of Babylon and Assyria*; *The Old Testament in the Light of the Records of Assyria and Babylonia*; A. H. Sayce, *Religions of Ancient Egypt and Babylonia*; M. Jastrow, *Aspects of Religious Belief and Practice in Babylonia and Assyria*; R. Koldeweg, *The Excavations of Babylon*.

Babyroussa (bab-i-ru's'a; a Malay word signifying stag-hog), a species of wild hog (*Sus* or *Porcus Babyrussa*), a native of the Indian Archipelago. From the outside of the upper jaw spring two teeth 12 inches long, curving upwards and backwards like horns, and almost touching the forehead. The tusks of the lower jaw also appear externally, though they are not so long as those of the upper jaw. Along the back are some weak bristles, and on the rest of the body only a sort of wool. These animals live in herds, feed on herbage, are sometimes tamed, and their flesh is well flavoured.

Bac'carat, a gambling card game of French origin, played by any number of players, or rather betters, and a banker. The latter deals two cards to each player and two to himself, and covers the stakes of each with an equal sum. The cards are then examined, and according to the scores made the players

take their own stake and the banker's, or the latter takes all or a certain number of the stakes.

Baccarat (ba´k-a´-rä), a town of France, about 15 miles south-east of Lunéville, department Meurthe-et-Moselle, on the River Meurthe, with a handsome new church, and one of the largest works for plate-glass and crystal in France. Pop. 7277.

Bacchana´lia, or **Dionysia**, feasts in honour of Bacchus or Dionysus, characterized by licentiousness and revelry, and celebrated in ancient Athens. In the processions were bands of Bacchantes of both sexes, who, inspired by real or feigned intoxication, wandered about rioting and dancing. They were clothed in fawn-skins, crowned with ivy, and bore in their hands *thyrsi*, that is spears entwined with ivy, or having a pine-cone stuck on the point. These feasts passed from the Greeks to the Romans, who celebrated them with still greater dissoluteness till the Senate abolished them, 187 B.C.

Bacchante (bak-an´te), a person taking part in revels in honour of Bacchus. See *Bacchanalia*.

Bacchiglione (ba´k-kil´yō-nā), a river of Northern Italy, rises in the Alps, passes through the towns of Vicenza and Padua, and enters the Adriatic near Chioggia, after a course of about 90 miles.

Bacchus (bak´us; in Greek, generally *Dionýsus*), the god of wine, son of Zeus (Jupiter) and Sēmēlē. He first taught the cultivation of the vine and the preparation of wine. To spread the knowledge of his invention he travelled over various countries and received in every quarter divine honours. Drawn by lions (some say panthers, tigers, or lynxes), he began his march, which resembled a triumphal procession. Those who opposed him were severely punished, but on those who received him hospitably he bestowed rewards. His love was shared by several; but Ariadne, whom he found deserted upon Naxos, alone was elevated to the dignity of a wife, and became a sharer of his immortality. In art he is represented with the round, soft, and graceful form of a maiden rather than with that of a young man. His long waving hair is knitted behind in a knot, and wreathed with sprigs of ivy and vine leaves. He is usually naked; sometimes he has a loose mantle hung negligently round his shoulders; sometimes a fawn-skin hangs across his

breast. He is often accompanied by Silenus, Bacchantes, Satyrs, &c. See *Bacchanalia*.—BIBLIOGRAPHY: Farnell, *Cults of the Greek States*; J. E. Harrison, *Prolegomena to the Study of Greek Religion*; Sir J. G. Frazer, *The Golden Bough*.

Bacchylides (bak-kil'i-dēz), born in the Island of Cos, about the middle of the fifth century B.C., the last of the great lyric poets of Greece, a nephew of Simonides and a contemporary of Pindar. The extant remains of his odes, hymns, pæans, &c., were augmented in 1897 by papyrus discoveries.

Bacciocchi (ba't-chok'ē), Maria Anne Eliza Bonaparte, sister of Napoleon, born at Ajaccio, 1777, died near Trieste, 1820; a great patroness of literature and art. She married Captain Bacciocchi, who in 1805 was created Prince of Lucca and Piombino. She virtually ruled these principalities herself, and as Grand-Duchess of Tuscany she enacted the part of a queen. She fell with the Empire.

Baccio Delia Porta (ba'ch'ō). See *Bartolommeo*.

Bach (bah), Johann Sebastian, one of the greatest of German musicians, born in 1685 at Eisenach, died in 1750 at Leipzig. Being the son of a musician he was early trained in the art, and soon distinguished himself. In 1703 he was engaged as a player at the Court of Weimar, and subsequently he was musical director to the Duke of Anhalt-Köthen, and afterwards held an appointment at Leipzig. He paid a visit to Potsdam on the invitation of Frederick the Great. As a player on the harpsichord and organ he had no equal among his contemporaries; but it was not till a century after his death that his greatness as a composer was fully recognized. His compositions, many of which are pieces of sacred music, show great originality. They include pieces, vocal and instrumental, for the organ, piano, stringed and keyed instruments; church cantatas, oratorios, masses, passion music, &c. More than fifty musical performers have proceeded from this family. Sebastian himself had eleven sons, all distinguished as musicians. The most renowned were the following:—Wilhelm Friedemann, born in 1710 at Weimar, died at Berlin in 1784. He was one of the most scientific harmonists and most skilful organists.—Karl Philipp Emmanuel, born in 1714 at Weimar, died in 1788 at Hamburg. He composed mostly for the piano, and published melodies for Gellert's hymns; he also wrote *The True*

Manner of Playing the Harpsichord.—Johann Christoph Friedrich, born at Weimar, 1732, died in 1795, a great organist, is known also by the music he published.—Johann Christian, born in 1735 at Leipzig, died in London, 1782, was a favourite composer and conductor with the English public.—BIBLIOGRAPHY: P. Spitta, *Johann Sebastian Bach*; A. Pirro, *L'Esthétique de Jean Sebastian Bach*.

Bacharach (ba´h´a´-ra´h), a small place of 2000 inhabitants on the Rhine, 12 miles s. of Coblenz. The vicinity produces excellent wine, which was once highly esteemed. The view from the ruins of the castle is one of the sublimest on the Rhine.

Bach´elor, a term applied anciently to a person in the first or probationary stage of knighthood who had not yet raised his standard in the field. It also denotes a person who has taken the first degree in the liberal arts and sciences, or in divinity, law, or medicine, at a college or university; or a man of any age who has not been married.—A *knight bachelor* is one who has been raised to the dignity of a knight without being made a member of any of the orders of chivalry such as the Garter, the Thistle, or the Bath.

Bachelor's Buttons, the double-flowering buttercup (*Ranunculus acris*), with white or yellow blossoms, common in gardens.

Bachian (ba´ch´a´n), one of the Molucca Islands, immediately s. of the equator, s.w. of Gilolo; area, 800 sq. miles. It is ruled by a native sultan under the Dutch.

Bachmut (ba´h-möt'), a town in the Ukraine, government of Ekaterinoslav, with a trade in cattle, tallow, &c., and coal and rock-salt mines. Pop. 20,000.

Bacilla´ria, a genus of microscopic algæ, belonging to the class Diatomaceæ, the siliceous remains of which abound in cretaceous, tertiary, and more recent geological deposits.

Bacil´lus. See *Bacteria*.

Back, Admiral Sir George, eminent English Arctic discoverer, born 1796, died 1878. He accompanied Franklin and Richardson in their northern expeditions, and in 1833-4 headed an expedition to the Arctic Ocean

through the Hudson Bay Company's territory, on which occasion he wintered at the Great Slave Lake, and discovered the Back or Great Fish River.

Backergunge. See *Bakarganj*.

Backgam'mon, a game played by two persons upon a table or board made for the purpose, with pieces or men, dice-boxes, and dice. The table is in two parts, on which are twenty-four black and white spaces called points. Each player has fifteen men of different colours for the purpose of distinction. The movements of the men are made in accordance with the numbers turned up by the dice.

Backhaus, Wilhelm, one of the greatest living pianists, born at Leipzig in 1884. At the age of twenty-one he was appointed a professor at the Royal College of Music, Manchester. This position, however, he soon gave up, and since 1905 has devoted his time to concert tours.

Backhuysen (ba ' k'hoi-zn), Ludolf, a painter of the Dutch school, particularly celebrated for sea pieces, born in 1631, died 1709. His most famous picture is a sea piece which the burgomasters of Amsterdam commissioned him to paint as a present to Louis XVI. It is still at Paris.

Backwarda'tion, a stock exchange term signifying the rate paid by a speculative seller of stock for the privilege of carrying over or continuing a bargain from one fortnightly account to another, instead of closing it on the appointed day.

Bacninh, a town of Tongking, on the Red River, fortified and containing a French garrison, being in an important strategic position. Pop. 7000.

Ba'con, Anthony, elder brother of the celebrated Lord Chancellor, was born in 1558 and died in 1601. He was a skilful politician, and much devoted to learned pursuits. He became personally acquainted with most of the foreign literati of the day, and gained the friendship of Henri IV of France. Francis Bacon dedicated to him the first edition of the *Essays*.

Bacon, Francis, Baron Verulam, Viscount St. Albans, and Lord High Chancellor of England, was born at London in 1561, died at Highgate in 1626. His father, Nicholas Bacon, was Keeper of the Great Seal under

Queen Elizabeth. He was educated at Trinity College, Cambridge, and in 1575 was admitted to Gray's Inn. From 1576 to 1579 he was at Paris with Sir Amyas Paulet, the English ambassador. The death of his father called him back to England, and being left in straitened circumstances he zealously pursued the study of law, and was admitted a barrister in 1582. In 1584 he became member of Parliament for Melcombe Regis, and soon after drew up a *Letter of Advice* to Queen Elizabeth, an able political memoir. In 1586 he was member for Taunton, in 1589 for Liverpool. A year or two after he gained the Earl of Essex as a friend and patron. Bacon's talents and his connection with the Lord-Treasurer Burleigh, who had married his mother's sister, and his son, Sir Robert Cecil, First Secretary of State, seemed to promise him the highest promotion; but he had displeased the queen, and when he applied for the attorney-generalship, and next for the solicitor-generalship (1595), he was unsuccessful. Essex endeavoured to indemnify him by the donation of an estate in land. Bacon, however, forgot his obligations to his benefactor, and not only abandoned him as soon as he had fallen into disgrace, but without being obliged took part against him on his trial, in 1601, and was active in obtaining his conviction. He had been chosen member for the county of Middlesex in 1593, and for Southampton in 1597, and had long been a Queen's Counsel. The reign of James I was more favourable to his interest. He was assiduous in courting the king's favour, and James, who was ambitious of being considered a patron of letters, conferred upon him in 1603 the honour of knighthood. In 1604 he was appointed King's Counsel, with a pension of £60; in 1606 he married; in 1607 he became Solicitor-General, and six years after Attorney-General. Between James and his Parliament he was anxious to produce harmony, but his efforts were without avail, and his obsequiousness and servility gained him enmity and discredit. In 1617 he was made Lord-Keeper of the Seal; in 1618 Lord High Chancellor of England and Baron Verulam. In this year he lent his influence to bring a verdict of guilty against Raleigh. In 1621 he was made Viscount St. Albans. Soon after this his reputation received a fatal blow. A new Parliament was formed in 1621, and the Lord Chancellor was accused before the House of bribery, corruption, and other malpractices. It is difficult to ascertain the full extent of his guilt; but he seems to have been unable to justify himself, and handed in a 'confession and humble submission', throwing himself on the mercy of the Peers. He was condemned to pay a fine of £40,000, to be committed to the Tower during

the pleasure of the king, declared incompetent to hold any office of State, and banished from Court for ever. The sentence, however, was never carried out. The fine was remitted almost as soon as imposed, and he was imprisoned for only a few days. He survived his fall a few years, during this time occupying himself with his literary and scientific works, and vainly hoping for political employment. In 1597 he published his celebrated *Essays*, which immediately became very popular, were successively enlarged and extended, and translated into Latin, French, and Italian. The treatise on the *Advancement of Learning* appeared in 1605; *The Wisdom of the Ancients* in 1609 (in Latin); his great philosophical work, the *Novum Organum* (in Latin), in 1620; and the *De Augmentis Scientiarum*, a much enlarged edition (in Latin) of the *Advancement*, in 1623. His *New Atlantis* was written about 1614-7; *Life of Henry VII* about 1621. Various minor productions also proceeded from his pen. Numerous editions of his works have been published, by far the best being that of Messrs. Spedding, Ellis, and Heath, 1857-9 (reprinted, 1879-90). Bacon was great as a moralist, a historian, a writer on politics, and a rhetorician; but it is as the father of the inductive method in science, as the powerful exponent of the principle that facts must be observed and collected before theorizing, that he occupies the grand position he holds among the world's great ones. His moral character, however, was not on a level with his intellectual, self-aggrandizement being the main aim of his life. We need do no more than allude to the preposterous attempt that has been made to prove that Bacon was the real author of the plays attributed to Shakespeare, an attempt that only ignorance of Bacon and Shakespeare could uphold and tolerate.—BIBLIOGRAPHY: J. Spedding, *Life and Letters of Francis Bacon*; R. W. Church, *Bacon* (in English Men of Letters Series); Sir Sidney Lee, *Great Englishmen of the Sixteenth Century*; J. M. Robertson, *Short History of Free-thought*.

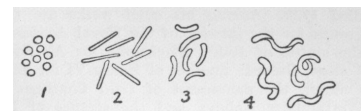
Bacon, John, English sculptor, born 1740, died 1799. Among his chief works are two groups for the interior of the Royal Academy; the statue of Judge Blackstone for All Souls College, Oxford; another of Henry VI for Eton College; the monument of Lord Chatham in Westminster Abbey; and the statues of Dr. Johnson and Mr. Howard in St. Paul's Cathedral.

Bacon, Sir Nicholas, father of Francis Bacon, Lord-Keeper of the Great Seal, born 1510, died 1579. Henry VIII gave him several lucrative offices, which he retained under Edward VI. He lived in retirement during the reign

of Mary, but Queen Elizabeth appointed him Lord-Keeper for life. He was the intimate friend of Lord Burleigh, a sister of whose wife he married, and by her became the father of the great Chancellor.

Bacon, Roger, an English monk, and one of the most profound and original thinkers of his day, was born about 1214, near Ilchester, Somersetshire, died at Oxford in 1294. He first entered the University of Oxford, and went afterwards to that of Paris, where he is said to have distinguished himself and received the degree of Doctor of Theology. About 1250 he returned to England, entered the order of Franciscans and fixed his abode at Oxford; but having incurred the suspicion of his ecclesiastical superiors, he was sent to Paris and kept in confinement for ten years, without writing-materials, books, or instruments. The cause seems to have been simple enough. He had been a diligent student of the chemical, physical, and mathematical sciences, and had made discoveries, and deduced results, which appeared so extraordinary to the ignorant that they were believed to be works of magic. This opinion was countenanced by the jealousy and hatred of the monks of his fraternity. In subsequent times he was popularly classed among those who had been in league with Satan. Having been set at liberty, he enjoyed a brief space of quiet while Clement IV was Pope; but in 1278 he was again thrown into prison, where he remained for at least ten years. Of the close of his life little is known. His most important work is his *Opus Majus*, where he discusses the relation of philosophy to religion, and then treats of language, metaphysics, optics, and experimental science. He was undoubtedly the earliest philosophical experimentalist in Britain; he made signal advances in optics; was an excellent chemist; and in all probability discovered gunpowder. He was intimately acquainted with geography and astronomy, as appears by his discovery of the errors of the calendar, and their causes, and by his proposals for correcting them, in which he approached very nearly to truth.—Cf. E. Charles, *Roger Bacon, sa vie, ses ouvrages, ses doctrines*.

Bacte'ria (Gr. *baktērion*, a rod), a general term applied to the lowest visible forms of life; other names given are *germs* or *microbes*. They are divided into three classes: (1) *cocci*—these are minute spheres about one twenty-five thousandth of an inch in diameter, arranged in chains or



Bacteria—Forms of cells

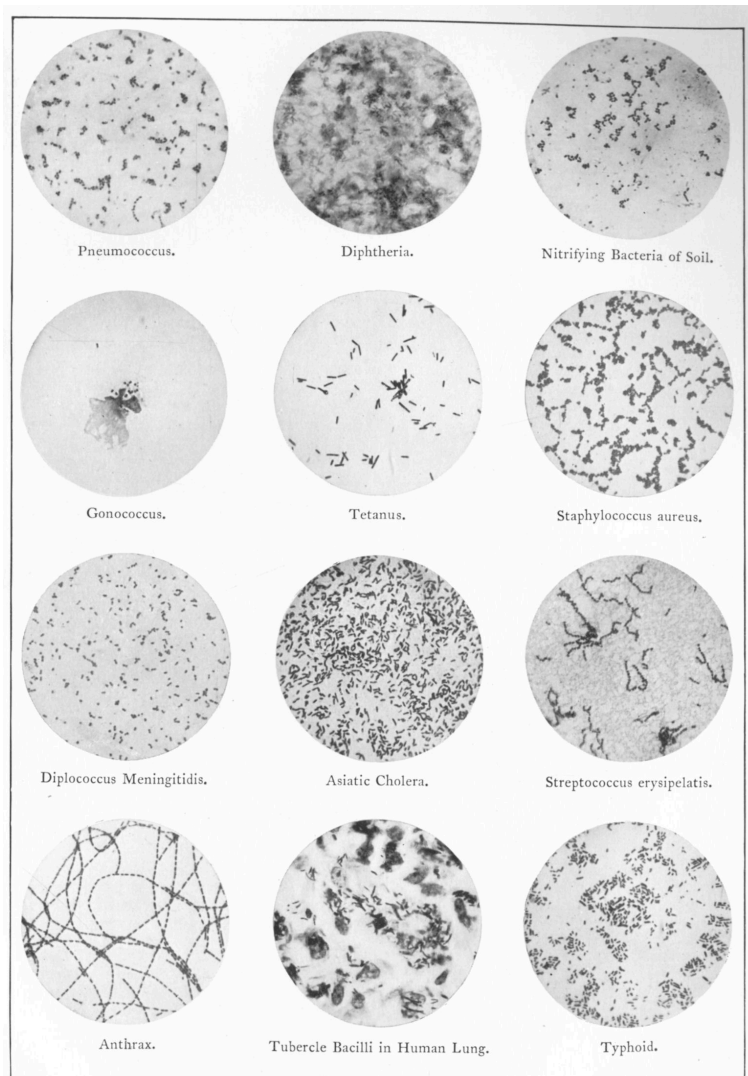
clusters; (2) *bacilli*—these are straight rods about the breadth of a coccus and two to four times as long; (3) *spirilla*—here there are two forms; one is like a bacillus, but curved (spirillum proper or vibrio); the other is a very fine spiral thread (spirochaete) often only half the thickness of a vibrio and two to six times as long. Bacteria consist of protoplasm surrounded by a membrane, and many show active movements; they all multiply asexually by splitting, and so rapidly may the process occur that 17,000,000 organisms or more may be formed from one bacterium in twenty-four hours. The chief part played by bacteria in nature is to cause putrefaction. They are found on all organic matter, and are therefore omnipresent. When grown artificially (see *Bacteriology*) they usually require organic matter for their food, and this they split up into simpler chemical compounds, some of which are very poisonous to animals (toxins); they can also split up sugars to produce various alcohols, acids, and gases. When certain of the bacilli are starved, the living matter gathers itself up into a spore surrounded by a very resisting membrane, and it can survive for a long time in this condition. All ordinary bacteria, including most of those which produce disease, can be killed by a few minutes' boiling, but the sporing varieties (e.g. many putrefactive forms) withstand much longer boiling—up to one and a half hours; they can also be killed by chemical means (see *Disinfection*). Many bacteria are useful to man, and a few give rise to disease (see *Germ Theory*).

1. Coccus. 2. Bacillus. 3. Spirillum (vibrio). 4. Spirillum (spirochaete).

Bacteriology, the branch of science which deals with the structure and properties of bacteria. Though the existence of bacteria had been known from the time when Leeuwenhock arranged lenses so as to form a microscope, it was through the work of Pasteur that modern knowledge regarding the group originated. Pasteur's discovery that yeasts are the cause of the formation of alcohol from sugar was followed by his showing that bacteria are responsible for other processes of a similar character, e.g. the formation of vinegar from alcohol, the occurrence of the rancid change in butter, &c. When he went on to prove that a bacterium was the cause of a disease of silkworms, a stimulus was given to the germ theory of infection, and bacteriology was thus early brought intimately into contact with medical science. Another stimulus to bacteriological inquiry came from the controversy on spontaneous generation, and Pasteur was one of those who

helped to establish the principle that the new development of living things, as represented by bacteria, out of non-living matter does not take place. This led to the inevitable deduction that all putrefactive processes are due to bacteria. On this Lister founded his antiseptic treatment of wounds, the practical effects of which focused attention on the important part played by bacteria in human affairs. The modern science of bacteriology really dates from 1877, when Koch proved that *Bacillus anthracis* is the cause of anthrax in cattle. The outstanding feature of Koch's research was that in it, for the first time,

methods were employed whereby all extraneous organisms were excluded from the inquiry, and the way was thus opened up for the study of the properties and effects of one species of bacterium acting by itself. The chief aspects of his work were as follows: (1) the determination of the temperatures at which bacteria are killed; the knowledge of this enables investigators to render free from living organisms their apparatus and the food materials (usually called *media*) necessary to support bacterial life (this is called *sterilization*); (2) the discovery that the insertion of a plug of cotton-wool in the neck of a vessel (e.g. a flask or test-tube) before its sterilization constitutes a sufficient filter, which prevents bacteria in the



BACTERIA

All the above are magnified 560 times.

outside air gaining access to its interior or to any food medium placed in it for sterilization; (3) the elaboration of a series of media capable of acting as food for all the commonly occurring bacteria; (4) the introduction of a simple method of separating from one another bacterial species which in nature are often growing side by side. This is effected by making media containing a jelly (e.g. gelatine) which melts at a relatively low temperature. The bacterial mixture is well diluted in the melted medium, and this is poured out in a thin layer on a sterile glass plate and allowed to set. Each individual bacterial cell can thus multiply apart from neighbouring cells, and, it may be, in a few hours the resulting growth (a 'colony') is visible to the naked eye. A minute part of the colony is picked off with a heated needle and placed on a mass of sterile medium. In this way a 'pure culture' of each species present in the original mixture can be obtained. Koch's work was chiefly concerned with disease bacteria, and between 1880 and 1895 many such organisms were isolated in pure culture and their properties investigated (see *Germ Theory of Disease*).

While in the evolution of knowledge the progress of bacteriology has been most associated with that of medicine and surgery, the science has fundamental relations with many other human interests, and the same methods are used in investigating such relationships. The adaptation of these methods on a commercial scale underlies all industrial processes concerned in the preserving of milk and in the canning and bottling of meat and fruit (apart from the addition of chemical preservatives). Since bacteriology came into existence, many of the most primitive domestic procedures have been improved and placed on a scientific basis. Thus, butter-making involves a previous souring of the milk; this is due to bacterial action, and standard mixtures of milk-souring bacteria are now regularly supplied by the laboratory as 'starters' of the process. It is recognized that the disorders of milk, e.g. 'ropiness', 'blue milk', &c., are due to certain 'wild' bacteria gaining access to the dairy. The ripening of such cheeses as Gorgonzola and Stilton is due to bacteria and allied organisms, and here again light has been shed on the 'diseases' of cheese by bacteriological investigation. The pickling of foods by salt, or vinegar, or nitre, is due to the action of these substances as antiseptics, and a similar principle, aided by boiling, underlies the making of jams. Again, certain kinds of bacteria which only flourish at high temperatures are responsible

for the heating of hay and corn stacks, and for the production of ensilage. Bacteriology is also throwing important light on factors concerned in the fertility of soils. For example, it has been shown that the little nodules on the roots of plants of the pea tribe are caused by bacteria; these, in growing, take up nitrogen from the air, and form compounds which promote the growth of the plant on which they are parasites and remain to enrich the soil when the plant dies. In short, there is no industry, which depends on natural putrefaction being prevented or modified so as to stop at a particular stage, which bacteriology cannot guide towards securing the most perfect results.—BIBLIOGRAPHY: Fischer, *The Structure and Functions of Bacteria*; Abbot, *Principles of Bacteriology*; Muir and Ritchie, *Manual of Bacteriology*; E. R. Stitt, *Practical Bacteriology*; P. H. Hiss and H. Zinsser, *A Textbook of Bacteriology*.

Bactria´na, or **Bactria**, a country of ancient Asia, south of the Oxus and reaching to the west of the Hindu Kush. It is often regarded as the original home of the Indo-European races. A Græco-Bactrian kingdom flourished about the third century B.C., but its history is obscure.

Baculi´tes, a fossil genus allied to the ammonites, characteristic of cretaceous strata, having a straight tapering shell. See *Cephalopoda*.

Ba´cup, a municipal borough of England, in Lancashire, 18 miles N. of Manchester. The chief manufacturing establishments are connected with cotton-spinning and power-loom weaving; there are also ironworks, Turkey-red dyeing-works, and in the neighbourhood numerous coal-pits and immense stone-quarries. Pop. (1921), 21,256.

Badagry, seaport in the British province of Southern Nigeria, West Africa, 50 miles E.N.E. of Whydah. Pop. about 10,000.

Badajoz (ba´-da´-hōth´; ancient, **Pax Augusta**), the fortified capital of the Spanish province of Badajoz, on the left bank of the Guadiana, which is crossed by a stone bridge of twenty-eight arches. It is a bishop's see, and has an interesting cathedral. During the Peninsular War Badajoz was besieged by Marshal Soult, and taken in March, 1811. It was twice attacked by the English, on 5th and 29th May, 1811, and was besieged by Wellington on 16th March, and taken 6th April, 1812. Pop. 37,600.—The province of Badajoz has an area of 8451 sq. miles. Pop. 644,220.

Badakshan', a territory of Central Asia, tributary to the Ameer of Afghanistan. It has the Oxus on the north, and the Hindu Kush on the south, and has lofty mountains and fertile valleys; the chief town is Faizabad. The inhabitants profess Mahommedanism. Pop. 20,957.

Badalona (bä-da'-lō'na), a Mediterranean seaport of Spain, 5 miles from Barcelona. Pop. 19,240.

Baden (bä'dèn), formerly a Grand-Duchy and one of the more important States of the German Empire, situated in the s.w. of Germany, to the west of Württemberg. It is divided into four districts, Constance, Freiburg, Karlsruhe, and Mannheim. It has an area of 5819 sq. miles, and a pop. of 2,142,833. It is mountainous, being traversed to a considerable extent by the lofty plateau of the Schwarzwald or Black Forest, which attains its highest point in the Feldberg (4904 feet). The nucleus of this plateau consists of gneiss and granite. In the north it sinks down towards the Odenwald, which is, however, of different geological structure, being composed for the most part of red sandstone. The whole of Baden, except a small portion in the s.e., in which the Danube takes its rise, belongs to the basin of the Rhine, which bounds it on the south and west. Numerous tributaries of the Rhine intersect it, the chief being the Neckar. Lakes are numerous, and include a considerable part of the Lake of Constance. The climate varies much. The hilly parts, especially in the east, are cold and have a long winter, while the valley of the Rhine enjoys the finest climate of Germany. The principal minerals worked are coal, salt, iron, zinc, and nickel. The number of mineral springs is remarkably great, and of these not a few are of great celebrity. The vegetation is peculiarly rich, and there are magnificent forests. The cereals comprise wheat, oats, barley, and rye. Potatoes, hemp, tobacco, wine, and sugar-beet are largely produced. Several of the wines, both white and red, rank in the first class. Baden has long been famous for its fruits also. Of the total area 55.4 per cent is under cultivation, 39.4 per cent under forest, and 5.2 per cent uncultivated (houses, roads, water, &c.). The farms are mostly quite small. The manufactures are important. Among them are textiles, tobacco and cigars, chemicals, machinery, pottery ware, jewellery (especially at Pforzheim), wooden clocks, confined chiefly to the districts of the Black Forest, musical boxes and other musical toys. The capital is Karlsruhe, about 5 miles from the Rhine; the other chief towns are Mannheim, Freiburg-im-Breisgau (with a Roman Catholic university),

Baden, and Heidelberg. Baden has warm mineral springs, which were known and used in the time of the Romans. Heidelberg has a university (Protestant), founded in 1386, the oldest in Germany. The railways have a length of 1200 miles, and are all State property. In the time of the Roman Empire southern Baden belonged to the Roman province of Rhætia. Under the old German Empire it was a margraviate, which in 1533 was divided into Baden-Baden and Baden-Durlach, but reunited in 1771. The title of grand-duke was conferred by Napoleon in 1806, and in the same year Baden was extended to its present limits. Until 1918 the executive power was vested in the grand-duke, the legislative in a house of legislature, consisting of an upper and a lower chamber. The former consisted partly of hereditary members; the latter of elected representatives of the people. In the break-up of the German Empire in Nov., 1918, Baden was among the first States to feel the shock of revolution. The Grand-Duke Frederick I abdicated, and the Provisional Government, under the Socialist Herr Jeiss, proclaimed Baden a Republic. Two-thirds of the population are Roman Catholics, the rest Protestants.

Baden (or **Baden-Baden**, to distinguish it from other towns of the same name; German *Bad*, a bath), a town and watering-place, in the former Grand-Duchy of Baden, 18 miles s.s.w. of Karlsruhe, built in the form of an amphitheatre on a spur of the Black Forest, overhanging a valley, through which runs a little stream Oosbach. Baden has been celebrated from the remotest antiquity for its thermal baths; and it used also to be celebrated for its gaming-saloons. It has many good buildings, and a castle, the summer residence of the former grand-dukes. Pop. 22,066.

Baden, a town of Austria, 15 miles s.w. of Vienna. It has numerous hot sulphurous springs, used both for bathing and drinking, and much frequented. Pop. 19,000.

Baden, a small town of Switzerland, canton Aargau, celebrated for its hot sulphurous baths, which attract many visitors. Pop. 8318.

Baden-Powell. See *Powell*.

Badge (baj), a distinctive device, emblem, mark, honorary decoration, or special cognizance, used originally to identify a knight or distinguish his followers, now worn as a sign of office or licensed employment, as a token

of membership in some society, or generally as a mark showing the relation of the wearer to any person, occupation, or order. See *Cockade*, *Heraldry*, *Totems*.

Badger (baj'ér), a plantigrade, carnivorous mammal, allied both to the bears and to the weasels, of a clumsy make, with short thick legs, and long claws on the fore-feet. The common badger (*Meles vulgāris*) is as large as a middling-sized dog, but much lower on the legs, with a flatter and broader body, very thick tough hide, and long coarse hair. It inhabits the north of Europe and Asia, burrows, is indolent and sleepy, feeds by night on vegetables, small quadrupeds, &c. Its flesh may be eaten, and its hair is used for artists' brushes in painting. The American badger belongs to a separate genus. *Badger baiting*, or *drawing the badger*, is a barbarous sport, long a favourite amusement in England, but declared illegal in the middle of last century, and yet to some extent practised, generally as an attraction to public-houses of the lowest sort. A badger is put in a barrel, and one or more dogs are put in to drag him out. When this is effected he is returned to his barrel, to be similarly assailed by a fresh set. The badger usually makes a most determined and savage resistance.



Common Badger (*Meles vulgāris*)

Badger Dog, a long-bodied, short-legged dog, with rather large pendulous ears, usually short haired, black, and with yellow extremities; often called by its German name *Dachshund*.

Bad'minton, a game closely resembling lawn-tennis, but played with battledore and shuttle-cock instead of ball and racket: named after a seat of the Duke of Beaufort, in Gloucestershire.

Badrinath (-ät'), a peak of the main Himalayan range, in Garhwāl District, in the United Provinces, 23,210 feet above the sea. On one of its shoulders, at an elevation of 10,400 feet, stands a celebrated temple of Vishnu, which some years attracts as many as 50,000 pilgrims.

Baedeker (bā'de-kér), Karl, a German publisher, born 1801, died 1859; originator of a celebrated series of guide-books for travellers.

Baelz's Disease, an affection of the mouth characterized by swelling of the mucous glands of the mouth and lips, leading to the formation of small abscesses and ulcers.

Baena (ba´-ā´na´), a town of Spain, in Andalusia, province of Cordova, and 24 miles S.S.E. of the city of that name. Pop. 14,730.

Baeza (ba´-ā´tha´; anciently, **Beatia**), a town of Spain, in Andalusia, 22 miles E.N.E. of Jaen, with 15,843 inhabitants. The principal edifices are the cathedral, the university (now suppressed), and the old monastery of St. Philip de Neri.

Baffa (ancient, **Paphos**), a seaport on the S.W. coast of Cyprus. Pop. 1000. It occupies the site of New Paphos, which, under the Romans, was full of beautiful temples and other public buildings. Old Paphos stood a little to the south-east.

Baffin, William, an English navigator, born 1584, famous for his discoveries in the Arctic regions; in 1616 he ascertained the limits of Baffin Bay. He was killed at the siege of Ormuz, in the East Indies, 1622.

Baffin Bay, on the N.E. of North America, between Greenland and Baffin Land and other islands that lie on the N. of the continent; discovered by Baffin in 1616.

Bafulabe (ba´-fu´-lä-bā), a trading-place in French West Africa, on the Upper Senegal, and on the railway that runs from Kayes on the Senegal to Bammako on the Niger.

Bagamo´yo, a seaport of former German East Africa, a short distance north-west of Dar-el-Salaam. There is no proper harbour, only a shallow and rather exposed roadstead where ships have to anchor some distance from the shore, yet it is the chief trading-centre of the colony, and as the terminus of the caravan route to Tanganyika is a place of some importance. Pop. about 25,000. See *European War*.

Bagasse´, the sugar-cane in its dry crushed state as delivered from the mill, and after the main portion of its juice has been expressed; used as fuel in the sugar factory, and called also *cane-trash*.

Bagatelle', a game played on a long flat board covered with cloth like a billiard-table, with spherical balls and a cue or mace. At the end of the board are nine cups or sockets of just sufficient size to receive the balls. These sockets are arranged in the form of a regular octagon, with the ninth in the middle, and are numbered consecutively from one upwards. Nine balls are used, generally one black, four white, and four red, the distinction between white and red being made only for the sake of variety. In the ordinary game, at starting, the black ball is placed on a point in the longitudinal middle line of the board, a few inches in front of the nearest of the sockets, and the player places one of his eight balls on a corresponding point at the other end of the board, and tries to strike the black ball into one of the sockets with his own. After this his object is to place as many of his balls as possible in the sockets. Each ball so placed counts as many as the socket is numbered for, and the black ball always counts double. He who first makes the number of points agreed on wins.

Bagdad', or **Baghdad**, a city of a vilayet of same name (54,540 sq. miles; pop. 900,000) in what was anciently Mesopotamia. The greater part of it lies on the eastern bank of the Tigris, which is crossed by a bridge of boats; old Bagdad, the residence of the caliphs (now in ruins), was on the western bank of the river. The city occupies a large area. The houses are mostly built of brick, the streets unpaved and very narrow, and there are few buildings of any note. The palace of the governor is spacious. Of the mosques only a few attract notice, many are in ruins; the bazaars are large and well stocked; that of Dawd Pasha still ranks as one of the most splendid in the world. Manufactures: leather, silks, cottons, woollens, carpets, &c. Steamers ply on the river between Bagdad and Bassorah, and the town exports wheat, dates, galls, gums, mohair, carpets, &c., to Europe, the chief import being cotton goods. The estimated population is over 200,000; of these about 86,000 are Mahommedans, 52,000 Jews, and 7000 Christians. The city has been frequently visited by the plague. Bagdad was founded in 762 by the Caliph Almansur, and raised to a high degree of splendour in the ninth century by Harun Al Rashid. It is the scene of a number of the tales of the *Arabian Nights*. It was long the chief city of the Mahommedan world, and at one time is said to have had two millions of inhabitants. In the thirteenth century it was stormed by Hulaku, grandson of Genghis-Khan, who caused the reigning caliph to be slain, and destroyed the caliphate. It was again laid

waste by Timur in 1401. For a considerable period it was the object of contention between the Persians and the Turks, until it was besieged and captured by Sultan Murad IV in 1638. Since that date it remained a part of the Turkish Empire. It was captured by the British on 11th March, 1917.

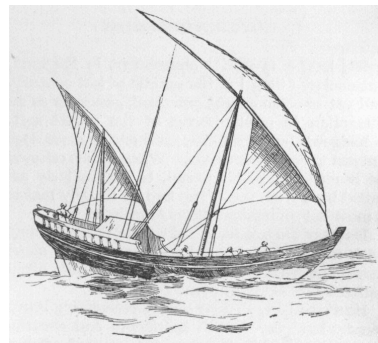
Bagdad Railway, a railway which, under arrangement with the Turkish Government, was intended to run from Konia, which is the terminus of the Anatolian railway system, to Bagdad, Basra (Bassorah), and the Persian Gulf. It was built chiefly by means of funds provided by German and French financiers, in association with the Anatolian Railway Company, a German company. The line has been constructed continuously as far as Kara Bunar in Cilicia. Farther on, the following sections have been completed: Dorak to Bagtché, on the Ordana side of the Taurus Mountains; Radjun, via Muslimié, to Jerablus, on the Euphrates, with a branch from Muslimié to Aleppo; Jerablus to El-Abiad, and Bagdad to Samara. A branch line has also been completed from Alexandretta to Toprak Kalé. Of the distance from Konia to Bagdad (1509 miles), 1117 miles (with gaps) are already working. In 1919 the railway system passed to the Allies, and in 1920 it was announced that the line between Basra and Bagdad had been completed.

Bagehot (bag'ot), Walter, English economist and journalist, born at Langport, Somerset, 1826, died at the same place 1877. He studied at Bristol, and at University College, London, and graduated as B.A. and M.A. at the London University (1848). He was for some time associated with his father in the banking business at Langport, and for a number of years he acted as London agent for the bank. He was one of the editors of the *National Review* (1855-64), and from 1860 till his death he was editor and part proprietor of the *Economist*. His chief works are: *Physics and Politics*; *The English Constitution*; *Lombard Street*; and *Studies, Literary, Biographic, and Economic*. He was a high authority on economics, banking, and finance, and was often consulted by public men.

Bag'gala, a two-masted Arab boat, generally 200-250 tons burden, used for trading in the Indian Ocean, Red Sea, &c.

Baggesen (bag'e-sen), Jens, a Danish poet, who also wrote much in German, born 1764, at Korsör, died at Hamburg, 1826. He led a rather

wandering and unsettled life, spending some time in England as well as in France and Germany. He tried lyric, epic, and dramatic poetry, but was most successful as a humorist and satirist. One of his best poems is his famous song *There was a Time when I was very Little*.



Baggala

Bagging is the name of the textile material which is used in the making of bags. Baltic hemp was the first material used for the purpose, but now all kinds of fibres are used, though the majority of bags are made of jute fibre. The style of the bagging used depends, naturally, upon the requirements of the bag; for the export of raw cotton from the United States of America we find the coarsest of all cotton-bagging used. When intended to carry grain, spices, sugar, &c., the bag must be made of closely-woven material. In many cases bags to carry sugar are lined with paper.

Baghelkand, a tract of country in Central India, occupied by a collection of native States (Rewah being the chief), under the Governor-General's agent for Central India; area, 11,323 sq. miles. Pop. 1,772,574.

Bagheria (ba'-gā'rē-a'), a town of Sicily, province of Palermo, 9 miles by railway from the city of Palermo. Pop. 21,212.

Bagimont's Roll, a rent-roll of Scotland, made up in 1275 by *Baiamund* or *Boiamond* de Vicci, vulgarly called *Bagimont*, who was sent from Rome by the Pope, in the reign of Alexander III, to collect the tithe of all the church livings in Scotland for an expedition to the Holy Land. It remained the statutory valuation, according to which the benefices were taxed, till the Reformation. A copy of it as it existed in the reign of James V is in the Advocates' Library, Edinburgh.

Bagirmi (ba'-gir'mē), or **Baghermi**, a Mohammedan Negro State in Central Africa, situated to the south-east of Lake Tchad. It is mostly a plain; has an area of about 20,000 sq. miles, and about 100,000 inhabitants. The people are industrious, and fairly civilized. The country is in the French sphere of influence, but the people are still under a native ruler.

Bagnara (ba´-nyä´ra´), a seaport near the s.w. extremity of Italy. Pop. 11,000.

Bagnères de Bigorre (ba´n-yār dé bē-gorr), a watering-place, France, department Hautes-Pyrénées, on the left bank of the Adour. It owes its chief celebrity to its sulphurous and saline baths. The springs were known to and used by the Romans. The site is one of the most romantic in the Pyrénées, and the town is well built and well provided with accommodation for visitors. Pop. 8455.

Bagnères de Luchon (ba´n-yār dé lu-shōn), a town, France, department Haute-Garonne, in a picturesque valley surrounded by wooded hills, one of the principal watering-places of the Pyrénées, having sulphurous thermal waters, beneficial in rheumatic complaints. There is a large bathing establishment, and a large and splendid casino building, comprising a theatre, concert and ball rooms, &c. Resident pop. 4000.

Bagni (bän´yē), Italian for 'baths', a name in Italy for various places which possess natural baths or thermal springs, with some designation added to mark the particular locality. *Bagni di Lucca*, in the province of Lucca, and about 13 miles north-east of the city of Lucca, is one of the most frequented of these bathing-places, its waters being recommended against gout, rheumatism, affections of the skin, &c.

Bagpipe, a musical wind-instrument of very great antiquity, having been used among the ancient Greeks, and being a favourite instrument over Europe generally in the fifteenth century. It still continues in use among the country people of Poland, Italy, the south of France, and in Scotland and Ireland. Though now often regarded as the national instrument of Scotland, especially Celtic Scotland, it is only Scottish by adoption, being introduced into that country from England. It consists of a leathern bag, which receives the air from the mouth, or from bellows; and of pipes, into which the air is pressed from the bag by the performer's elbow. In the common or Highland form, one pipe (called the *chanter*) plays the melody; of the three others (called *drones*) two are in unison with the lowest A of the chanter, and the third and longest an octave lower, the sound being produced by means of reeds. The chanter has eight holes, which the performer stops and opens at pleasure, but the scale is imperfect and the tone harsh. The Highland

bagpipe is a powerful instrument, and calls for great exertion of the lungs in order that the air may be supplied in sufficient quantity. There are several other species of bagpipes, as the soft and melodious Irish bagpipe, supplied with wind by a bellows, and having several keyed drones and a keyed chanter; the old English bagpipe (now no longer used); the Italian bagpipe, a very rude instrument, &c. The Irish bagpipe is, musically speaking, the most perfect of all.—BIBLIOGRAPHY: Sir J. G. Dalyell, *Musical Memoirs of Scotland*; E. de Bricqueville, *Les Musettes*.

Bagration (ba´g-rä´tyōn), Peter, Prince, distinguished Russian general, descended from a noble Georgian family. He was born in 1765, entered the Russian army in 1782, and henceforth was constantly engaged in active service, distinguishing himself in many actions and gradually rising in military rank. He fought in Poland, in Italy, and against the Turks, was engaged in the battles of Austerlitz, Eylau, and Friedland, and at last was mortally wounded at the battle of Borodino, Sept., 1812.

Bagshot-sand, in geology, the collective name for a series of sands of the Eocene age, occupying extensive tracts round Bagshot, in Surrey, and in the New Forest, Hampshire, the whole reposing on the London-clay; generally devoid of fossils.

Bagster, Samuel, publisher, founder of the firm of Bagster & Sons, celebrated for their bibles, was born in 1772, and died in 1851. He began business as a London bookseller in 1794, and soon turned his attention to the publication of bibles, bringing out a Hebrew bible, the *Septuagint* (Greek) version, and the English version, with 60,000 parallel references, followed by his great polyglot bible, which in its final form showed eight languages at the opening of the volume. Separate versions in different languages were also brought out, with various other aids to the study of Scripture; a polyglot *Book of Common Prayer*, in eight languages; &c.

Bahā´dur Shah, the last of the Grand Moguls of India, a descendant of Tamerlane. In 1857, during the Indian Mutiny, the Mahommedans, who wished to restore the Empire of the Moguls, placed him, then a very old man, at the head of the movement in Delhi; but the city was soon retaken by the British, and the Emperor was banished to Rangoon, where he died in 1862.

Baha'ma Islands, or **Lucayos**, a group of islands in the West Indies, forming a colony belonging to Britain, lying N.E. of Cuba and S.E. of the coast of Florida, the Gulf Stream passing between them and the mainland. They extend a distance of upwards of 600 miles, and are said to be twenty-nine in number, besides keys and rocks innumerable. The principal islands are Grand Bahama, Great and Little Abaco, Andros Islands, New Providence, Eleuthera, San Salvador, Great Exuma, Watling Island, Long Island, Crooked Island, Acklin Island, Mariguana Island, Grand Inagua. Of the whole group about twenty are inhabited, the most populous being New Providence, which contains the capital, Nassau; the largest being Andros, 100 miles long, 20 to 40 broad. They are low and flat, and have in many parts extensive forests. Total area, 4400 sq. miles. The soil is a thin but rich vegetable mould, and an important product is pineapples, which form a large export (both canned and green). Other fruits are also grown, with sisal hemp (a valuable export), cotton, sugar, maize, yams, ground nuts, coconuts, &c. Sponges are obtained in large quantity, and are at present the chief export. Total exports about £382,140 (1919). The Bahama Islands are a very favourite winter resort for persons suffering from lung trouble. San Salvador, or Cat Island, is generally believed to be the same as Guanahani, the land first touched on by Columbus (12th Oct., 1492) on his first great voyage. The first British settlement was made on New Providence towards the close of the seventeenth century. A number of loyal Americans settled in the islands after the War of Independence. Pop. 55,944, including 14,000 whites.

Baha'walpur, a town of India, capital of State of same name in the Punjab, 2 miles from the Sutlej; surrounded by a mud wall and containing the extensive palace of the Nawab. Pop. 18,700. The State has an area of 15,000 sq. miles, of which 10,000 is desert, the only cultivated lands lying along the Indus and Sutlej. Pop. 780,641.

Bahia (ba'-ē'a'; formerly **St. Salvador**), a town of Brazil, on the Bay of All Saints, State of Bahia. It consists of a lower town, which is little more than an irregular, narrow, and dirty street, stretching about 4 miles along the shore; and an upper town, larger and much better built, with various outlying suburbs. The harbour is one of the best in South America; and the export trade, chiefly in sugar, cotton, coffee, tobacco, hides, piassava, and tapioca, is very extensive. Pop. 348,000. The State, area 164,600 sq. miles,

pop. 3,013,007, has much fertile land, both along the coast and in the interior.

Bahia Blanca, a seaport of the Argentine Republic, on a bay of same name opening into the Atlantic, about 350 miles south-west of Buenos Ayres; it is a rising place connected by railway with the capital, and a terminus of other railways from the interior. It carries on a considerable trade directly with Europe, exporting mutton, wheat, &c. Pop. 75,000.

Bahr (bär), an Arabic word signifying sea or large river; as in Bahr-el-Huleh, the Lake Merom in Palestine; Bahr-el-Abiad, the White Nile, Bahr-el-Azrek, the Blue Nile, which together unite at Khartoum.

Bahraich (bä-rāch'), a flourishing town of India, in Oudh, Faizabad division. It carries on a good local trade, and has a shrine that attracts many Hindu and Mohammedan pilgrims. It is the seat of an American Methodist mission. Pop. 27,304.

Bahrein (bä-rīn) **Islands**, a group of islands in the Persian Gulf, in an indentation on the Arabian coast, since 1867 under British protection. The principal island, usually called Bahrein, is about 27 miles in length and 10 in breadth. It is in general very flat and low, and the soil is not very fertile except in a few places; but irrigation is employed and excellent dates are grown. The principal town is Menamah or Manama; pop. 30,000. The smaller island of Moharrek, separated by a shallow strait 2 miles wide, contains the town of Moharrek, the present seat of government; pop. 25,000. The Bahrein Islands are chiefly noted for their pearl-fisheries, which were known to the ancients, and which employ in the season from 2000 to 3000 boats manned by from 8 to 20 men each. Total pop. estimated at 110,000.

Bahr-el-Ghazal, a large river of Central Africa, a western tributary of the White Nile (or Bahr-el-Abiad). It flows through a very swampy region, and is liable to inundations. The head of steam navigation is Meshra-er-Rek. The river gives its name to a province of the Anglo-Egyptian Sudan, yielding ivory, rubber, timber, &c., and suited for cotton growing.

Baiadeer. See *Bayadere*.

Baiaë (bī'ē), an ancient Roman watering-place on the coast of Campania, 10 miles west of Naples. Many of the wealthy Romans had country houses at Baiaë, which was a favourite resort of Horace. Ruins of temples, baths, and villas still attract the attention of archæologists.

Baikal (bī'ka'l), a large freshwater lake in Eastern Siberia, 360 miles long, and about 50 in extreme breadth, interspersed with islands; in the line of the great Siberian Railway. It is surrounded by rugged and lofty mountains; contains seals, and many fish, particularly salmon, sturgeon, and pike. Its greatest depth is over 4000 feet. It receives the waters of the Upper Angara, Selenga, Barguzin, &c., and discharges its waters by the Lower Angara. It is frozen over in winter.

Baikie, William Balfour, born in the Orkney Islands 1824, died at Sierra Leone 1863. He joined the British navy, and was made surgeon and naturalist of the Niger Expedition, 1854. He took the command on the death of the senior officer, and explored the Niger for 250 miles. Another expedition, which started in 1857, passed two years in exploring, when the vessel was wrecked, and all the members, with the exception of Baikie, returned to England. With none but native assistants he formed a settlement at the confluence of the Benué and the Quorra, in which he was ruler, teacher, and physician, and within a few years he opened the Niger to navigation, made roads, and established a market.

Bail, the person or persons who procure the release of a prisoner from custody by becoming surety for his appearance in court at the proper time; also, the security given for the release of a prisoner from custody. In the United States bail is a matter of right in all cases where a sentence of death cannot be pronounced, and even in such a case it may be allowed by one of the judges.

Baildon, an urban district in the W. Riding of Yorkshire, about 5 miles north of Bradford, with worsted manufactures, chemical works, and quarries. Pop. (1921), 6527.

Bailen (bī-len'), a town of Southern Spain, province Jaen, with lead-mines. Pop. 8334.

Bailey (bā'li), the name given to the courts of a castle formed by the spaces between the circuits of walls or defences which surrounded the keep.

Bailey, or **Baily**, Nathaniel, an English lexicographer, school teacher at Stepney, and author of several educational works. His *Dictionary*, published in 1721, passed through a great many editions.

Bailey, Philip James, English poet, born near Nottingham, 1816, called to the bar 1840, died 1902. He published *Festus*, his best work, in 1839; *The Mystic*, 1855; *The Age*, 1858; and *The Universal Hymn*, 1867.

Bailie, or **Baillie**, a municipal officer or magistrate in Scotland, corresponding to an *alderman* in England. The criminal jurisdiction of the provost and bailies of royal burghs extends to breaches of the peace, drunkenness, adulteration of articles of diet, thefts not of an aggravated character, and other offences of a less serious nature.

Bailiff, a civil officer or functionary, subordinate to someone else. There are several kinds of bailiffs, whose offices widely differ, but all agree in this, that the keeping or protection of something belongs to them. In England the sheriff is the monarch's bailiff, and his county is a bailiwick. The name is also applied to the chief magistrates of some towns, to keepers of royal castles, as of Dover, to persons having the conservation of the peace in hundreds and in some special jurisdictions, as Westminster, and to the returning officers in the same. But the officials commonly designated by this name are the *bailiffs* of sheriffs, or sheriffs' officers, who execute processes, &c.

Bailleul (ba' -yeul), an ancient French town, department of Nord, near the Belgian frontier, about 19 miles west of Lille. Much fighting took place in its vicinity during the European War, but it was not destroyed until the campaign of 1918. Pop. 12,828.—A village of the same name in department Orne gave its name to the Baliol family.

Baillie, Joanna, a Scottish authoress, born at Bothwell, Lanarkshire, in 1762, died at Hampstead, 1851. She removed in early life to London, where her brother, Matthew Baillie, was settled as a physician. Here in 1798 she published her first work, entitled *A Series of Plays*, in which she attempted to delineate the stronger passions by making each passion the subject of a

tragedy and a comedy. The series was followed up by a second volume in 1802, and a third in 1812. A second series appeared in 1836, and a complete edition of her whole dramatic works in 1850. She also published a volume of miscellaneous poetry, including songs, in 1841. Her only plays performed on the stage were a tragedy entitled the *Family Legend*, brought out at Edinburgh under the patronage of Sir Walter Scott; and *De Montfort*, brought out by John Kemble, who acted in it with Mrs. Siddons.

Baillie, Matthew, M.D., physician and anatomist, brother of the preceding, was born 1761 at Shotts, Lanarkshire, died at Cirencester, Gloucestershire, in 1823. In 1773 he was sent to the University of Glasgow. He afterwards studied anatomy under his maternal uncles, John and William Hunter, and entered Oxford, where he graduated as M.D. In 1783 he succeeded his uncle as lecturer on anatomy in London, where he acquired a high reputation as a teacher and demonstrator, having also a large practice. In 1810 he was appointed physician to George III. His work on *The Morbid Anatomy of Some of the Most Important Parts of the Human Body* gave him a European reputation.

Baillie, Robert, an eminent Scottish Presbyterian clergyman, was born at Glasgow in 1599, died 1662. Though educated and ordained as an Episcopalian, he resisted the attempt of Archbishop Laud to introduce his *Book of Common Prayer* into Scotland, and joined the Presbyterian party. In 1638 he represented the presbytery of Irvine in the General Assembly at Glasgow, which dissolved Episcopacy in Scotland. In 1640 he was selected to go to London, with other commissioners, to draw up the accusation against Archbishop Laud. Of this, and almost all the other proceedings of his public life, he has left a minute account in his letters and journals, which form a most valuable collection for the history of his time. In 1642 he was appointed professor of divinity at Glasgow. He was a member of the Westminster Assembly of Divines, and attended its sittings from 1643-6. After the Restoration, though made principal of his college through Court patronage, he did not hesitate to express his dissatisfaction with the reintroduction of Episcopacy.

Baillie, Robert, of Jerviswood, in Lanarkshire, a Scottish patriot of the reign of Charles II. He brought himself into notice by opposing the tyrannical measures of Archbishop Sharpe against the Nonconformists, for which he

was fined 6000 merks (or £500) and imprisoned for four months. In 1683 he went to London in furtherance of a scheme of emigration to South Carolina taken up by a number of Scottish gentlemen, as being the only way of escaping the tyranny of the Government. He became associated with Monmouth, Sydney, Russell, and the rest of that party, and was charged with complicity in the Rye-house plot. After a long imprisonment, during which vain attempts were made to obtain evidence against him, he was brought before the Court of Justiciary (23rd Dec., 1684), was found guilty, and condemned to be executed that afternoon.

Baillieston, a town of Scotland, in Lanarkshire, a few miles east of Glasgow, with extensive collieries, in which many of the inhabitants are employed. Pop. 5500.

Bailly (ba' -yē), Jean Sylvain, French astronomer and statesman, born at Paris 1736. After some youthful essays in verse, he was induced by Lacaille to devote himself to astronomy, and on the death of the latter in 1753, being admitted to the Academy of Sciences, he published a reduction of Lacaille's observations on the zodiacal stars. In 1764 he competed ably but unsuccessfully for the Academy prize offered for an essay upon Jupiter's satellites, Lagrange being his opponent; and in 1771 he published a treatise on the light reflected by these satellites. In the meantime he had won distinction as a man of letters by his *Éloges* on Pierre Corneille, Leibnitz, Molière, and others; and the same qualities of style shown by these were maintained in his *History of Astronomy* (1775-87), his most extensive work. In 1784 the French Academy elected him a member. The revolution drew him into public life. Paris chose him, 12th May, 1789, first deputy of the *tiers-état*, and in the Assembly itself he was made first president, a post occupied by him on 20th June, 1789, in the session of the Tennis Court, when the deputies swore never to separate till they had given France a new constitution. As Mayor of Paris his moderation and impartial enforcement of the law failed to commend themselves to the people, and his forcible suppression of mob violence, 17th July, 1791, aroused a storm which led to his resignation and retreat to Nantes. In 1793 he attempted to join Laplace at Melun, but was recognized and sent to Paris, where he was condemned by the revolutionary tribunal, and executed on 12th Nov.

Bailment, in law, is the delivery of a chattel or thing to a person in trust, either for the use of the bailer or person delivering, or for that of the bailee or person to whom it is delivered. A bailment always supposes the subject to be delivered only for a limited time, at the expiration of which it must be re-delivered to the bailer, the responsibility of the bailee being dependent, in some degree, upon the contract on which the bailment is made. Pledging and letting for hire are species of bailment. *Agistment*, or the taking in of cattle or live stock to feed at a rate of so much per head, is a contract of bailment.

Baily, Edward Hodges, an English sculptor, born at Bristol 1788, died at London 1867. He became a pupil of Flaxman in 1807, gained the Academy Gold Medal in 1811, and was elected R.A. in 1821. Principal works: *Eve at the Fountain*, *Eve Listening to the Voice*, *Maternal Affection*, *Girl Preparing for the Bath*, *The Graces*, &c. The bas-reliefs on the south side of the Marble Arch, Hyde Park, the statue of Nelson on the Trafalgar Square monument, and other public works, were by him.

Baily, Francis, astronomer, born in Berkshire 1774; settled in London as a stockbroker in 1802. While thus actively engaged he published *Tables for the Purchasing and Renewing of Leases*, *The Doctrine of Interest and Annuities*, *The Doctrine of Life Annuities and Assurances*, and an epitome of universal history. On retiring from business with an ample fortune in 1825 he turned his attention to astronomy, became one of the founders of the Astronomical Society, contributed to its *Transactions*, and in 1835 published a life of Flamsteed. He died in 1844.

Baily's Beads, a phenomenon attending eclipses of the sun, the unobscured edge of which appears discontinuous and broken immediately before and after the moment of complete obscuration. It is classed as an effect of irradiation, and is caused by the sun shining through the depressions between the lunar mountains.

Bain, Alexander, Scottish philosopher and educationalist, was born at Aberdeen in 1818. He was educated at Marischal College (then a separate university), Aberdeen; was for some years a deputy professor in the university; subsequently held official posts in London; and in 1860 was appointed professor of logic and English in Aberdeen University, a post

which he held till his resignation in 1881. His most important works are: *The Senses and the Intellect* (1855); *The Emotions and the Will* (1859), together forming a complete exposition of the human mind; *Mental and Moral Science* (1868); *Logic, Deductive and Inductive* (1870); *Mind and Body* (1873); *Education as a Science* (1879); *James Mill, a Biography* (1881); *John Stuart Mill, a Criticism with Personal Recollections* (1882); besides an *English Grammar*; *English Composition and Rhetoric*; an *Autobiography*, &c. Bain was at once grammarian, rhetorician, educationalist, and logician, but his fame rests on his contributions to psychology. He died in 1903.

Bainsizza Plateau. See *European War*.

Bairam (bī'ram), the Easter of the Mohammedans, which follows immediately after the Ramadan or Lent (a month of fasting), and lasts three days. This feast during the course of thirty-three years makes a complete circuit of all the months and seasons, as the Turks reckon by lunar years. Sixty days after this first great Bairam begins the lesser Bairam. They are the only two feasts prescribed by the Mohammedan religion.

Baird, Sir David, a distinguished British soldier, was born in Aberdeenshire in 1757, and entered the army 1772. Having been promoted to a lieutenancy in 1778 he sailed for India, distinguished himself as a captain in the war against Hyder Ali, was wounded and taken prisoner, and confined in the fortress of Seringapatam for nearly four years. He and his fellow-prisoners were treated with great barbarity, and many of them died or were put to death, but at last (in 1784) all that survived were set at liberty. In 1787 he became major, and in 1791 joined the army under Cornwallis as lieutenant-colonel, and was appointed to the command of a brigade in the war against Tippoo. After much hard service he was made a colonel in 1795, went in 1797 to the Cape of Good Hope as brigadier-general, and in 1798, on his appointment as major-general, returned to India. In 1799 he commanded the storming party at the assault of Seringapatam, and, in requital, was presented with the State sword of Tippoo Saib. Being appointed in 1800 to command an expedition to Egypt, he landed at Kosseir in June, 1801, crossed the desert, and, embarking on the Nile, descended to Cairo, and thence to Alexandria, which he reached a few days before it surrendered to General Hutchinson. Next year he returned to India, but being soon after

superseded by Sir Arthur Wellesley (Wellington), he sailed for Britain, where he was knighted and made K.B. With the rank of lieutenant-general he commanded an expedition in 1805 to the Cape of Good Hope, and in 1806, after defeating the Dutch, he received the surrender of the colony. He commanded a division at the siege of Copenhagen, and after a short period of service in Ireland sailed with 10,000 men for Corunna, where he formed a junction with Sir John Moore. He commanded the first division of Moore's army, and in the battle of Corunna lost his left arm. By the death of Sir John Moore, Sir David succeeded to the chief command, receiving for the fourth time the thanks of Parliament, and a baronetcy. In 1814 he was made a general. He died in 1829.

Baird, Spencer Fullerton, American naturalist, born 1823, died 1887. From 1850 to 1878 he was assistant secretary, and then became secretary, of the Smithsonian Institution, Washington, and was also chief Government commissioner of fish and fisheries. He wrote much on natural history, his chief works being *The Birds of North America* (in conjunction with John Cassin), *The Mammals of North America*, *Review of American Birds in the Smithsonian Institution*, and (with Messrs. Brewer and Ridgeway) *History of North American Birds*.

Baireuth (bī roit), a well-built and pleasantly-situated town of Bavaria, on the Red Main, 41 miles north-east of Nürnberg. The principal edifices, besides churches, are the old and the new palace, the opera-house, the gymnasium, and the national theatre, constructed after the design of the composer Wagner, and opened in 1876 with a grand performance of his tetralogy of the *Nibelungen Ring*. Industries: cotton-spinning, sugar-refining, musical instruments, sewing-machines, leather, brewing, &c. There is a monument to Jean Paul F. Richter, who died here. Pop. 34,547.

Baius, or **De Bay**, Michael, Catholic theologian, was born 1513, in Hainaut, educated at Louvain, made professor of theology there in 1552, and chosen a member of the Council of Trent in 1563. Leaving the scholastic method, he founded systematic theology directly upon the Bible and the Christian fathers, of whom he particularly followed St. Augustine. His doctrines of original sin and of salvation by grace led to his persecution as a heretic by the old Scotists, and the Jesuits, who succeeded in obtaining a Papal bull in 1567, condemning the doctrines imputed to him. Baius, however, remained

in the possession of his dignities, was appointed in 1575 Chancellor of Louvain University; and the King of Spain even conferred upon him the office of Inquisitor-General in the Netherlands. He died in 1589. His Augustinian views descended to the Jansenists, while his doctrine of pure undivided love to God formed the staple of Quietism. His system is called *Baianism*.

Baize, a sort of coarse woollen fabric with a rough nap, now generally used for linings, and mostly green or red in colour.

Baja (bä'ya'), a market town of Hungary, district of Bacs, on the Danube, with a trade in grain and wine, and a large annual hog fair. Pop. 20,361.

Bajaderes. See *Bayaderes*.

Bajazet (ba'-ya'-zet'), or **Bayezid, I**, Turkish emperor, who, in 1389, having strangled his brother Jacob, succeeded his father Murad or Amurath, who fell in the battle of Kossovo while fighting against the Serbians. From the rapidity of his conquests he received the name of *Ilderim*, the Lightning. In three years he overcame Bulgaria, part of Serbia, Macedonia, Thessaly, and the States of Asia Minor, and besieged Constantinople for ten years, defeating Sigismund and the allied Hungarians, Poles, and French in 1395. The attack of Timur (Tamerlane) on Natolia, in 1400, saved the Greek Empire, Bajazet being defeated and taken prisoner by him near Ancyra, Galatia, 1402. The story of his being carried about in a cage by Timur is improbable; but Bajazet died, in 1409, in Timur's camp, in Caramania. His successor was Soliman I.

Bajazet II succeeded his father, Mohammed II, Sultan of the Turks, in 1481. He increased the Turkish Empire by conquests on the N.W. and in the E., took Lepanto, Modon, and Durazzo in a war against the Venetians, and ravaged the coasts of the Christian States on the Mediterranean, to revenge the expulsion of the Moors from Spain. Having abdicated in favour of his younger son Selim, he died on his way to a residence near Adrianople in 1513. He did much for the improvement of his empire and the promotion of the sciences.

Bajimont's Roll. See *Bagimonts Roll*.

Bajocco, or **Baiocco** (ba'-yok'o), was a copper coin in the Papal States, the hundredth part of a scudo, or rather more than a halfpenny. The name was also given in Sicily to the Neapolitan *grano*, the hundredth part of the ducato (3s. 4d.).

Bajus. See *Baius*.

Bajza (boi'za), Anton, Hungarian lyric poet, historian, and critic, born 1804, died 1858. As contributor to and editor of various periodicals he played an important part in the development of modern Hungarian literature and drama. A volume of his poems, of high merit, was published in 1835. He also translated a collection of foreign dramas, and edited a series of historical works.

Bakalaha'ri, a Bechuana tribe inhabiting the Kalahari Desert, S. Africa.

Bak'arganj, a maritime district and town in Eastern Bengal; chief rivers: Ganges, Brahmaputra, and Meghna. Area, 3649 sq. miles. Pop. 2,292,000. The town now lies in ruins. Pop. 7000.

Bakau (bä'kou), a town of Roumania, on the Bistritza. Pop. 16,187.

Bakchisarai (ba'k-chi-sa'-rī'), or **Bagtcheserai** (Bakh-chi-Sarai) (bäg-che-se-rī; Turk., 'Garden Palace'), an ancient town of Russia, in the Crimea, picturesquely situated at the bottom of a narrow valley, hemmed in by precipices. It contains the palace of the ancient Crimean khans, restored by the Russian Government. Pop. 15,000.

Baker, Sir Richard, an English historian, born in Kent in 1568, educated at Oxford, knighted in 1603 by James I, and in 1620 appointed High Sheriff of Oxfordshire, where he had estates. Having given security for a debt incurred by his wife's family, he was thrown into Fleet Prison, where, after remaining some years, he died in 1645. During his imprisonment he wrote some devotional books and his *Chronicle of the Kings of England*, first published in 1641, and afterwards continued by Edward Phillips, the nephew of Milton, and others—a work of great popularity in its day, though of no permanent value.

Baker, Sir Samuel White, a distinguished English traveller, born in 1821. He resided some years in Ceylon; in 1861 began his African travels, which

lasted several years, in the Upper Nile regions, and resulted, among other discoveries, in that of Albert Nyanza Lake in 1864, and of the exit of the White Nile from it. In Africa he encountered Speke and Grant after their discovery of the Victoria Nyanza. On his return home he was received with great honour and was knighted. In 1869 he returned to Africa as head of an expedition sent by the Khedive of Egypt to annex and open up to trade a large part of the newly-explored country, being raised to the dignity of pasha. He returned in 1873, having finished his work, and was succeeded by the celebrated Gordon. His writings include: *The Rifle and the Hound in Ceylon*; *Eight Years' Wanderings in Ceylon*; *The Albert Nyanza*; *The Nile Tributaries of Abyssinia*; *Ismailia: a Narrative of the Expedition to Central Africa*; *Cyprus as I saw it in 1879*; also, *Cast up by the Sea*, a story published in 1868. He died 30th Dec., 1893.

Baker, Thomas, antiquary, born 1656, educated at Cambridge. As a non-juror he lost his living at Long-Newton in 1690, and was compelled to resign his fellowship on the accession of George I, but continued to reside at St. John's College till his death in 1740. His *Reflections on Learning* (1709-10) went through seven editions. He left in MS. forty-two folio volumes of an *Athenæ Cantabrigienses*, from which a *History of St. John's College* was edited by Professor Mayor in 1869.

Bakewell, an ancient market town, England, county of Derby, between Buxton and Matlock, possessing a fine Gothic church, a chalybeate spring, a cotton-mill erected by Arkwright, and a large marble-cutting industry. Pop. 3062.

Bakewell, Robert, an English agriculturist, celebrated for his improvements in the breeding of sheep, cattle, and horses, was born in Leicestershire in 1725, and died in 1795. He commenced experiments in breeding sheep, about 1755, upon his father's farm at Dishley, and for fifty years devoted himself to the acquisition and diffusion of information upon the subject. He was the originator of the new Leicestershire breed of sheep, which has since been so well known, and also of a breed of cattle very famous in their day. The demand for his rams increased so considerably, that whereas he let them for the season at £16 per head in 1755, by 1789 they were hired at 6000 guineas. Various improvements in farm management were also introduced by him.

Bakhmut. See *Bachmut*.

Bakhuisen. See *Backhuysen*.

Baking, a term meaning 'to cook by dry heat', and primarily applied to the baking of bread. A common application of the term is to a mode of cooking food in a close oven, baking in this case being opposed to roasting or broiling, in which an open fire is used. The oven should not be too close, but ought to be properly ventilated. Baking is also applied to the hardening of earthenware or porcelain by fire. See *Bread*.

Baking Powder, a mixture of bicarbonate of soda and tartaric acid, usually with some flour added. The water of the dough causes the liberation of carbonic acid, which makes the bread 'rise'.

Bakony (ba'-kon'yè) **Wald**, a thickly wooded mountain range dividing the Hungarian plains, famous for the herds of swine fed on its mast.

Baksheesh', an Eastern term for a present or gratuity. A demand for baksheesh meets travellers in the East everywhere from Turkey and Egypt to Hindustan.

Baku (bä-kö'), a Russian port on the western shore of the Caspian, occupying part of the Peninsula of Apsheron. The naphtha or petroleum springs of Baku have long been known; and the Field of Fire, so called from emitting inflammable gases, has long been a place of pilgrimage with the Guebres or Fire-worshippers. From the development of the petroleum industry, Baku has greatly increased, and is now a large and flourishing town. About 1500 oil-wells are in operation, producing immense quantities of petroleum (6,448,000 tons in 1917), much of which is led direct in pipes from the wells to the refineries in Baku, and it is intended to lay a pipe for its conveyance all the way to the Black Sea at Batoum, which is already connected with Baku by railway. Some of the wells have had such an outflow of oil as to be unmanageable, and the Baku petroleum now competes successfully with any other in the markets of the world. Baku, formerly the station of the Caspian fleet, is strongly fortified, and has a large shipping trade. The district of Baku lies within the limits of the new Republic of Azerbaijan. Pop. 237,000; pop. of the province, 1,119,600.

Baku'nin, Michael, Russian anarchist, the founder of Nihilism, born 1814 of a rich and noble family, entered the army, but threw up his commission after two years' service, and studied philosophy at Moscow, with his friends Herzen, Turgenev, Granovski (historian), and Byelinski (critic). Having adopted Hegel's system as the basis of a new revolution, he went in 1841 to Berlin, and thence to Dresden, Geneva, and Paris, as the propagandist of anarchism. Wherever he went he caused disturbance, and after undergoing imprisonment in various States, was handed over to Russia in 1851 by Austria, imprisoned for five years, and finally sent to Siberia. Escaping thence through Japan, he joined Herzen in London on the staff of the *Kolokol*. His extreme views, however, ruined the paper and led to a quarrel with Marx and the Internationale; and having fallen into disrepute with his own party in Russia, he died suddenly and almost alone at Berne in 1878. He demanded the entire abolition of the State as a State, the absolute equalization of individuals, and the extirpation of hereditary rights and of religion, his conception of the next stage of social progress being purely negative and annihilatory.

Bala, a lake 4 miles long, and an urban district of N. Wales, in Merionethshire. Pop. 1408.

Balaam (bā'lam), a prophet, invited by Balak, King of Moab, to curse the Israelites, but compelled by miracle to bless them instead (*Num.* xxii-xxiv). In another account he is represented as aiding in the perversion of the Israelites to the worship of Baal, and as being, therefore, slain in the Midianitish war (*Num.* xxxi; *Josh.* xiii). He is the subject of many rabbinical fables, the Targumists and Talmudists regarding him, as most of the fathers did, in the light of an impious and godless man.

Bala Beds, or **Bala Series**, a deposit of Upper Ordovician Age, named from the Bala district, North Wales, consisting of slates, grits, sandstones, and limestones, there being two limestones separated by sandy and slaty rocks about 1400 feet thick. They contain trilobites of many species, and other marine fossils. The lower Bala limestone (25 feet thick) may be traced over a large area in North Wales. See *Caradoc Series*, *Ordovician System*.

Balachong', an Oriental condiment, composed of small fishes, or shrimps, pounded up with salt and spices and then dried.

Balæ´na, the genus which includes the Greenland or right whale, type of the family Balænidæ, or whale-bone whales.

Balæ´niceps ('whale-head'), a genus of wading birds belonging to the Sudan, intermediate between the herons and storks, and characterized by an enormous bill, broad and swollen, giving the only known species (*B. rex*), also called shoe-bird, a peculiar appearance. It feeds on fishes, water-snakes, carrion, &c., and makes its nest in reeds or grass adjoining water. The bill is yellow, blotched with dark-brown, the general colour of the plumage dusky-grey, the head, neck, and breast slaty, the legs blackish.

Balænop´tera. See *Rorqual*.

Balagarh (bä-lä-gar'), town of Hindustan, in the Punjab. Pop. 11,233.

Balaghat´, a district of India, in the Central Provinces, Nagpur division; area, 3146 sq. miles; capital, Burha, a small place. The surface is diversified, comprising extensive uplands as well as lowlands; forests are extensive, and a comparatively small part of the area is under cultivation, rice being the chief crop. Iron exists in considerable quantities, and it is worked in a small way; gold is also worked. Pop. 330,000, almost entirely rural.

Balaguer (ba´-la´-gār'), Victor, Spanish author and politician, important writer in the Catalanian dialect, born 1824, died 1901. Born at Barcelona, he studied law there, and, becoming learned in Catalanian history, was appointed archivist and soon after professor of history. In 1869 he entered the Cortes as a Liberal; in 1872 was Minister of Public Works; from 1886 to 1888 Colonial Minister. He wrote much both in prose and verse, his prose comprising historical works, novels, &c., his verse lyrics, ballads, tragedies, odes, &c. His tragedies were partly on subjects taken from Catalanian history, partly on subjects connected with Greek and Roman history or literature. Among his lyrical and other poems are: *El Trovador de Montserrat*; *Primavera de Ultimo Trovador Catalan*; *Poesias Completas* (1874); *Obras Poeticas* (1880). *Don Juan de Serrallonga* is the most popular of his novels. Among other works of his are: *Historia Politica y Literaria de los Trovadores*; *Historia de Cataluña*; *Los Pirineos*; *Cristobal Colon*; *Estudios Historicos y Politicos*; *Historias y Tradiciones*;

Instituciones y Reyes de Aragon. As a poet he was imitative, reminding us of Quintana, Zorilla, and Byron.

Balaklava (ba´-la´-klä'va), a small seaport in the Crimea, 8 miles S.S.E. Sevastopol. In the Crimean war it was captured by the British, who used the harbour as a base of supplies. The famous battle of Balaklava was fought some distance to the north of the seaport, on 25th Oct., 1854. The Russians brought a force of 25,000 men against the allies, a much larger force than they had to encounter, and the chief incidents in the battle were as follows: The Russians captured a series of positions occupied by the Turks in front of the British position; a Russian cavalry charge was repulsed by the 93rd Highlanders; a great mass of Russian cavalry was defeated by a charge of British heavy cavalry; the British light cavalry brigade charged and took a Russian battery, and put to flight the cavalry behind it, but were compelled by overpowering force to retreat with heavy loss, their retreat being covered by a brilliant charge of the French. The celebrated charge of the light brigade, or of the 'six hundred', though a brilliant feat of arms, was made under an erroneous interpretation of orders; that of the heavy brigade was equally glorious, and contributed far more to the final repulse of the enemy.

Balako'vo, a river-port of South-Eastern Russia, on the left bank of the Volga, government of Samara, with a very important trade in grain. Pop. 16,000.

Balalai'ka, a musical instrument of very ancient Slavonic origin, common among the Russians and Tartars. It is a narrow, shallow guitar with only two strings.

Bal'ance, an instrument for determining the relative weights of bodies. Balances are of various forms; in that most commonly used a horizontal beam rests so as to turn easily upon a fulcrum in the middle. From the extremities of the beam, called the centres of suspension, hang the scales; and a slender metal pointer between them, and in front of the fulcrum, indicates when the beam is level. The characteristics of a good balance are: (1) that the beam should rest in a horizontal position when the scales are either empty or loaded with equal weights; (2) that a very small addition of weight put into either scale should cause the beam to deviate from the level, which property is denominated the sensibility of the balance; (3) that when

the beam is deflected from the horizontal position by inequality of the weights on the scales, it should have a tendency speedily to restore itself and come to rest in the level, which property is called the stability of the balance. To secure these qualities the arms of the beam should be exactly similar, equal in weight and length, and as long as possible; the centres of gravity and suspension should be in one straight line, and the fulcrum immediately above the centre of gravity; and the fulcrum and the centres of suspension should cause as little friction as possible. The fulcrum ought to be a knife-edge; and if the balance requires to be very delicate, the centres of suspension ought to be knife-edges also. If the balance has no tendency to one position more than another, when the scales are either loaded, empty, or off altogether, it is proof that the centre of gravity and the fulcrum coincide, and the remedy is to lower the centre of gravity. If the beam is disturbed by a small addition of weight to either scale, and exhibits no tendency to resume the horizontal position, we may infer that the centre of gravity is above the fulcrum. If it requires a considerable excess of weight to deflect the beam from the level, we may infer either that there is too much friction at the fulcrum, or that the centre of gravity is too low. If two weights are found to be in equipoise, one being in each scale, and if, when that which is in one scale is put into the other, there is no longer equilibrium, then we may infer that the arms of the beam are of unequal lengths. For purposes of accuracy, balances have occasionally means of raising or depressing the centre of gravity, of regulating the length of the arms, &c., and the whole apparatus is usually enclosed in a glass case, to prevent the heat from expanding the arms unequally, or currents of air from disturbing the equilibrium. A refinement in weighing is obtained by the use of the milligram rider, a short bent wire which can be moved along a scale engraved on the beam.

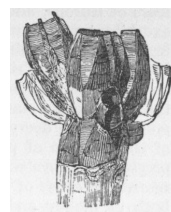
Of the other forms of balance, the Roman balance, or steel-yard, consists of a lever moving freely upon a suspended fulcrum, the shorter arm of the lever having a scale or pan attached to it, and the longer arm, along which slides a weight, being graduated to indicate quantities. It is commonly used for weighing loaded carts, for luggage at railway stations, &c. A variety of this, the Danish balance, has the weight fixed at the end of the lever, the fulcrum being movable along the graduated index. The spring-balance registers the weight of an article by the extent to which it draws out or

compresses a spiral spring. It is of service where a high degree of exactness is not required, as in its domestic use, and it also finds application in the dynamometer for measuring horse-power of machinery. An extremely ingenious balance, used in the Mint and the Bank of England, for weighing 'blanks' and sovereigns, distributes them automatically into three compartments according as they are light, heavy, or the exact weight. The Roberval balance is a form in common use for weighing letters and parcels.—BIBLIOGRAPHY: E. A. Brauer, *The Construction of the Balance*; E. Nicholson, *Men and Measures: a history of weights and measures, ancient and modern*.

Balance of Power, a political principle which first came to be recognized in modern Europe in the sixteenth century, though it appears to have been also acted on by the Greeks in ancient times, in preserving the relations between their different States. The object in maintaining the balance of power is to secure the general independence of nations as a whole, by preventing the aggressive attempts of individual States to extend their territory and sway at the expense of weaker countries. The first European monarch whose ambitious designs induced a combination of other States to counteract them was the Emperor Charles V, similar coalitions being formed in the end of the seventeenth century, when the ambition of Louis XIV excited the fears of Europe, and a century later against the exorbitant power and aggressive schemes of the first Napoleon. Since that time we have the instance of the Crimean War, entered into to check the ambition of Russia. Of late years there has been a marked tendency among British politicians to decry and impugn the principle of the balance of power, as calculated only to propagate a system of mutual hostility, and retard the cause of progress, by the expenditure both of money and life thus occasioned. An equilibrium between the various Powers is, of course, essential to the very existence of international law. The war of 1914-8 has proved to the world that in the absence of any central authority neither treaties nor signatures could prevent a State sufficiently powerful from ignoring the law and acting solely according to its interests and ambitious designs. See *Society of Nations*.—BIBLIOGRAPHY: Hume, *Essay on the Balance of Power*; Von Gentz, *Fragments on the Balance of Power*; Professor L. Oppenheim, *International Law*, vol. i; Vattel, *Le droit des gens*.

Balance of Trade, the difference between the stated money values of the exports and imports of a country. The balance is erroneously said to be 'in favour' of a country when the value of the exports is in excess of that of the imports, and 'against it' when the imports are in excess of the exports. The phrases date from the days of the mercantile system, the characteristic doctrine of which alleged the desirability of regulating commerce with a view to amassing treasure by exporting produce largely, importing little merchandise in return, and receiving the balance in bullion. In certain conceivable political and industrial conditions this may have had beneficial results; but its importance was greatly overestimated, and the state of this balance came to be regarded as an invariable criterion of the industrial condition of a country. The false analogy of the successful merchant who gains more than he spends became the basis of popular reasoning, the products of a country being mistakenly identified with its exports, its consumption with its importation. It is now generally recognized that if bullion be exported from a country it is because it is at the time the cheapest commodity available for export; and further, that there are certain natural limits to its undue exportation, in that the increased scarcity of money is attended with a fall in the money-value of other commodities, which thus in turn become preferable objects of exportation, while bullion flows back. The excess of the value of imports over that of exports, which is regarded by some as an adverse and alarming symptom in British trade, is in large part readily accounted for on the ground of shipping receipts, insurance returns, interest on capital employed in foreign trade, merchants' profits, and the income derived from foreign investments.

Bal´anus ('acorn-shells'), a genus of sessile cirripeds, family Balanidæ, of which colonies are to be found on rocks at low water, on timbers, crustaceans, shells of mollusca, &c. They differ from the barnacles in having a symmetrical shell, and being destitute of a flexible stalk. The shell consists of six plates, with an operculum of four valves. They pass through a larval state, in which they are not fixed, moving by means of swimming feet which disappear in the final state. All the Balanidæ are hermaphrodite. A South American species (*Balānus psittācus*) is eaten on the coast of Chile, the *Balānus tintinnabŭlum* by the Chinese. The old Roman epicures esteemed the larger species.



Group of
Balānus
tintinnabŭlum

Balapur´, town of India, in Akola district, Berar, with strong fort and fine pavilion of black stone. Pop. 10,500.

Bal´as, a name used to distinguish the rose-coloured species of ruby from the ruby proper.

Balator´, a seaport town, Hindustan, presidency of Bengal, province of Orissa, head-quarters of a district and subdivision bearing the same name. It carries on a considerable traffic with Calcutta. Pop. 21,362.

Bala´ta, a gum yielded by *Mimŭsops Balata*, a tree growing abundantly in British, French, and Dutch Guiana, Honduras and Brazil, obtained in a milky state by 'tapping' the tree, and hardening to a substance like leather. Used for similar purposes to india-rubber, and in the United States chewed as a masticatory.

Bal´aton, or **Plattensee**, a lake of Hungary, 55 miles s.w. of Pesth; length, 50 miles; breadth, 3 to 10 miles; area, about 390 sq. miles. Of its thirty-two feeders the Szala is the largest, and the lake communicates with the Danube by the Rivers Sio and Sarviz. It abounds with a species of perch.

Balbec. See *Baalbek*.

Balbi, Adrien, geographer and statistician, born at Venice in 1782. In 1808 his first work on geography, *Prospetto Politico-Geografico*, procured his appointment as professor of geography in the College of San Michele at

Murano, and he became in 1811 professor of natural philosophy in the Lyceum at Fermo. In 1820 he proceeded to Portugal, and collected there materials for his *Essai Statistique sur le Royaume de Portugal et d'Algarve* and *Variétés Politiques et Statistiques de la Monarchie Portugaise*, both published in 1822 at Paris, where he resided till 1832. He then settled in Padua, where he died in 1848. Balbi's admirable *Abrégé de Géographie* was written at Paris, and translated into the principal European languages.

Balbi, Gaspāro, a Venetian dealer in precious stones, born about the middle of the sixteenth century, who travelled first to Aleppo and thence down the Euphrates and Tigris to the Malabar coast, sailing finally for Pegu, where he remained for two years. His *Viaggio all' Indie Orientale*, published on his return to Venice in 1590, contains the earliest account of India beyond the Ganges.

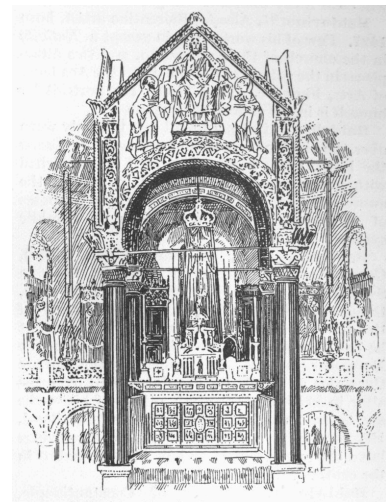
Balbo, Cesāre, Italian author and statesman, born in 1789 at Turin. After holding one or two posts under the patronage of Napoleon, he devoted himself to history, publishing a history of Italy prior to the period of Charlemagne, a compendium of Italian history, &c. His *Speranze d'Italia* (1843), a statement of the political condition of Italy, and of the practicable ideals to be kept in view, gave him a wide reputation. He died in 1853.

Balbo'a, Vasco Nuñez de, one of the early Spanish adventurers in the New World, born 1475. Having dissipated his fortune, he went to America, and was at Darien with the expedition of Francisco de Enciso in 1510. An insurrection placed him at the head of the colony, but rumours of a western ocean and of the wealth of Peru led him to cross the isthmus. On 25th Sept., 1513, he saw for the first time the Pacific, and after annexing it to Spain, and acquiring information about Peru, returned to Darien. Here he found himself supplanted by a new governor, Pedrarias Davila, with much consequent grievance on the one side, and much jealousy on the other. Balboa submitted, however, and in the following year was appointed Viceroy of the South Sea. Davila was apparently reconciled to him, and gave him his daughter in marriage, but shortly after, in 1517, had him beheaded on a charge of intent to rebel. Pizarro, who afterwards completed the discovery of Peru, served under Balboa.

Balbriggan, a seaport and favourite watering-place, Ireland, county of Dublin; celebrated for its hosiery. Pop. 2273.

Bal'cony, in architecture, is a gallery projecting from the outer wall of a building, supported by columns or brackets, and surrounded by a balustrade. Balconies were not used in Greek and Roman buildings, and in the East the roof of the house has for centuries served similar purposes on a larger scale. Balconies properly so styled came into fashion in Italy in the Middle Ages, and were introduced into Britain in the sixteenth century.

Bal'dachin (-kin; It. *baldachino*), a canopy or tent-like covering of any material, either suspended from the roof, fastened to the wall, or supported on pillars over altars, thrones, pulpits, beds, portals, &c. Portable baldachins of rich materials were formerly used to shield the heads of dignitaries in processions, and are still so used in the processions of the Roman Catholic Church, and in the East. The enormous bronze baldachin designed by Bernini for Pope Urban VIII, and placed over the tomb of the apostles in St. Peter's at Rome, is one of the most famous, though surpassed in beauty by many in other European cathedrals and churches.



Baldachin: Church of St.
Ambrogio, Milan

Balder, or **Baldur**, a Scandinavian divinity, represented as the son of Odin and Frigga, beautiful, wise, amiable, and beloved by all the gods. His mother took an oath from every creature, and even from every inanimate object, that they would not harm Balder, but omitted the mistletoe. Balder was therefore deemed invulnerable, and the other gods in sport flung stones and shot arrows at him without harming him. But the evil god Loki fashioned an arrow from the mistletoe and got Balder's blind brother Hōder to shoot it, himself guiding his aim. Balder fell dead, pierced to the heart, to the deep grief of all the gods. He is believed to be a personification of the brightness and beneficence of the sun.

Baldi, Bernardino, mathematician, theologian, geographer, historian, poet, &c., born at Urbino 1533; studied at Padua; became abbot of Guastalla. He

knew upwards of twelve languages, and is said to have written over a hundred works, most of which remain in MS. His works include a poem on *Navigation*, various translations and commentaries, *Lives of Celebrated Mathematicians*, &c. He died in 1617.

Baldness, loss of the hair, complete or partial, usually the latter, and due to various causes. Most commonly it results as one of the changes belonging to old age, due to wasting of the skin, hair-sacs, &c. It may occur as a result of some acute disease, or at an unusually early age, without any such cause. In both the latter cases it is due to defective nourishment of the hair, owing to lessened circulation of the blood in the scalp. The best treatment for preventing loss of hair seems to consist in such measures as bathing the head with cold water and drying it by vigorous rubbing with a rough towel and brushing it well with a hard brush. Various stimulating lotions are also recommended, especially those containing cantharides. But probably in most cases senile baldness is unpreventable. When extreme scurfiness of the scalp accompanies loss of the hair, an ointment that will clear away the scurf will prove beneficial.

Baldovinet`ti, Alessio, Florentine artist, born 1427. Few of his works remain except a *Nativity* in the church of the Annunziato, and two altarpieces in the gallery of the Uffizi and the Academy of Arts, Florence. Died 1499. His portrait by himself is in the gallery at Bergamo.

Baldric (ba`ld`rik), a broad belt formerly worn over the right or left shoulder diagonally across the body, often highly decorated and enriched with gems, and used not only to sustain the sword, dagger, or horn, but also for purposes of ornament, and as a military or heraldic symbol.

Bal`dung, Hans, or **Hans Grün** (grün), German painter and wood engraver, born in Swabia 1470, died in Strasburg 1552. His work, though inferior to Dürer's, possessed many of the same characteristics, and on this account he has been sometimes considered a pupil of the Nuremberg master. His principal paintings are the series of panels (of the date 1516) over the altar in Freiburg Cathedral; others of his works are to be found at Berlin, Colmar, and Basel. His numerous and often fantastic engravings have the monogram H and B, with a small G in the centre of the H.

Baldwin I, Emperor of Constantinople, founder of the short-lived dynasty of Latin sovereigns of the Eastern Empire, was born in 1172, and was hereditary Count of Flanders and Hainault. His courage and conduct in the fourth crusade led to his unanimous election as Emperor of the East after the capture of Constantinople by the French and Venetians in 1204. In the absence of Baldwin's brother with a large part of the army, the Greeks rose in revolt under the instigation of Joannices, King of Bulgaria. Baldwin marched on Adrianople, but was taken prisoner and died in captivity, 1205. Baldwin was succeeded by his brother Henry.—**Baldwin II**, fifth and last Latin Emperor of Constantinople, was born 1217. During his minority John de Brienne was regent, but on his assuming the power himself the Empire fell to pieces. In 1261 Constantinople was taken by the forces of Michael Palæologus, and Baldwin retired to Italy, dying in 1270.

Baldwin I, King of Jerusalem, reigned 1100-18, having assumed the title which his elder brother, Godfrey de Bouillon, had refused. He subdued Cæsarea, Ashdod, Tripolis, and Acre.—**Baldwin II**, his cousin and successor, reigned from 1118-31. During his reign the reduction of Tyre and institution of the order of Templars took place.—**Baldwin III**, King of Jerusalem from 1143 to 1162, was son and successor of Foulques of Anjou, and the embodiment of the best aspects of chivalry. After defeating Nouredin in 1152, and again in 1157, he was enabled to devote himself to the hopeless task of improving the kingdom and establishing the Christian chivalry in the East. His death in 1162 was almost immediately followed by the total collapse of the kingdom.

Baldwin, James Mark, American philosopher, born in Columbia, 12th Jan., 1861. He was educated at Princeton College, Leipzig and Berlin Universities; was instructor of German and French at Princeton College, 1885-7; professor of philosophy in Lake Forest University, 1887-9; professor of psychology at Princeton University, 1893-1903. From 1903-9 he was professor of philosophy and psychology at the Johns Hopkins University, Baltimore. Professor Baldwin is best known by his experimental psychology and his theories of genetic logic. He distinguishes between genetic logic, as theory of thought, and genetic philosophy, as theory of reality. Genetic sciences, according to Baldwin, are psychology, sociology, ethics, &c., and his point of view he terms 'Æstthonomical Idealism'. Among his numerous works are: *Handbook of Psychology* (2 vols., 1891-2); *Mental*

Development in the Child and the Race (3rd edition, 1906); *Social and Ethical Interpretations in Mental Development* (1904); *Thought and Things, or Genetic Logic* (1906-8). He also edited the *Dictionary of Philosophy and Psychology* (1901-9) and *The Psychological Review* (1894-1909).

Bâle (bäl). See *Basel*.

Bale, John, an English ecclesiastic, born in Suffolk in 1495, died in 1563. Although educated a Roman Catholic, he became a Protestant, and the intolerance of the Catholic party drove him to the Netherlands. On the accession of Edward VI he returned to England, was presented to the living of Bishop's Stoke, Southampton, and soon after nominated Bishop of Ossory, in Ireland. Here, on his preaching the reformed religion, the popular fury against him reached such a pitch that in one tumult five of his domestics were murdered in his presence. On the accession of Mary he lay some time concealed in Dublin, and after many hardships found refuge in Switzerland. At her death he was appointed by Elizabeth a prebend of Canterbury, where he died. His fame as an author rests upon his *Scriptorum Illustrium Majoris Britanniae Catalogus*, or *An Account of the Lives of Eminent Writers of Britain*, commencing with Japhet the son of Noah, and ending with the year 1557. It is compiled from various writers, chiefly from the antiquary Leland. He was also the author of nineteen miracle plays, printed in 1558. One of his plays, *Kynge Johan*, is a link between the old morality plays and the historical drama.

Baleař'ic Crane (*Baleař'ica pavonĩna*), a handsome species of crested crane inhabiting North-West Africa.

Baleař'ic Islands, an archipelago of four large and eleven small islands, south-east of Spain, including Majorca, Minorca, Iviza, and Formentera. The popular derivation of the ancient name Baleares (Gr. *ballein*, to throw) has reference to the repute of the inhabitants for their skill in slinging, in which they distinguished themselves both in the army of Hannibal and under the Romans, by whom the islands were annexed in 123 B.C. After being taken by the Vandals, under Genseric, and in the eighth century by the Moors, they were taken by James I, King of Aragon, 1220-34, and constituted a kingdom, which in 1343 was united to Spain. The islands now

form a Spanish province, with an area of 1935 sq. miles, and 330,167 inhabitants. See separate articles.

Baleen´. See *Whale-bone*.

Bale-fire, any great fire kindled in the open air, especially the fire of a funeral pile, or a beacon-fire.

Balen (bä´len), Hendrik van, painter, born at Antwerp 1560, died 1632. His works, chiefly classical, religious, and allegorical—some of them executed in partnership with Breughel—are to be found in most of the European galleries. He was the first master of Van Dyck and Snyders. Three of his sons also followed the art.

Balfe (balf), Michael William, composer, was born in Dublin, 15th May, 1808. In his seventh year he performed in public on the violin, and at sixteen took the part of the Wicked Huntsman in *Der Freischütz* at Drury Lane. In 1825 he went to Italy, wrote the music for a ballet, *La Pérouse*, for the Scala at Milan, and in the following year sang at the Théâtre-Italien, Paris, with moderate success. He returned to Italy, and at Palermo was given his first opera, *I Rivali* (1829). For five years he continued singing and composing operas for the Italian stage. In 1835 he came to England, and composed a number of operas, amongst others *The Bohemian Girl* (1843), *The Rose of Castile* (1857), *Satanella* (1858), and *The Talisman* (first performed in 1874). He died 20th Oct., 1870. His operas are melodious, and many of the airs are excellent.

Balfour (bal-för´), Sir Andrew, Bart., a Scottish botanist and physician, born in Fifeshire in 1630. After completing his studies at St. Andrews and London, and travelling on the Continent, he settled at Edinburgh, where he planned, with Sir Robert Sibbald, the Royal College of Physicians, and was elected its first president. He also laid the foundation of a hospital in Edinburgh, which expanded into the Royal Infirmary. He died in 1694. His familiar letters were published in 1700.

Bal´four, Right Hon. Arthur James, First Earl of (created 1922), son of J. M. Balfour of Whittingehame, Haddingtonshire, and of Lady Blanche Cecil, sister of the third Marquess of Salisbury, was born 25th July, 1848, and educated at Eton and Trinity College, Cambridge. He represented

Hertford from 1874-5, East Manchester from 1885 to 1906, and has been representative for the City of London since 1906. For a time he was private secretary to his uncle, Lord Salisbury, under whom he was President of the Local Government Board, and afterwards Secretary for Scotland (1886-7), with a seat in the cabinet. He showed much firmness and ability as Chief Secretary for Ireland during Lord Salisbury's administration from 1887-91. He succeeded Rt. Hon. W. H. Smith as leader of the House of Commons and First Lord of the Treasury in 1891-2, and held the position again from 1895 till 1900, and from that year till 1902. On the retirement of Lord Salisbury in 1902 he became Prime Minister, a position which he retained till Dec., 1905. Under him as Premier was passed the new English Education Act (which owed much to his personal influence and exertions), that for London, and the new Licensing Act, and he advocated a change of our fiscal policy, at least as far as having recourse to retaliation, denying that he in any way favoured protection. He became First Lord of the Admiralty in the Coalition Government of 1915, and Foreign Secretary in 1916. He was one of the British delegates to the Peace Conference in 1919. He is given to studies bearing on philosophy and religion, and has published a *Defence of Philosophic Doubt* (1879), *Essays and Addresses* (1893), *The Foundations of Belief* (1895), *Economic Notes on Insular Free-Trade* (1903), *Criticism and Beauty* (1909), *Theism and Humanism* (1915), &c. He was President of the British Association in 1904, and Gifford Lecturer, Glasgow University, 1913-4. In 1919 he was elected to succeed Lord Rayleigh, O.M., as Chancellor of Cambridge University.

Balfour, Francis Maitland, writer on embryology, brother of the foregoing, born in 1851, early distinguished himself in his special study, and in 1874, in conjunction with Dr. M. Foster, published *The Elements of Embryology*; but the promise of his chief work, *Comparative Embryology* (1880-1) was unfulfilled, as in 1882 he was killed by a fall on Mont Blanc.

Balfour, Sir James, Scottish lawyer and politician, born about 1522, took part in the conspiracy against Cardinal Beaton, and was condemned with Knox to the galleys; but after his release found it to his interest to change his opinions, and was appointed, through the favour of Queen Mary, Lord of Session and member of the Privy Council. In 1567 he was appointed Governor of Edinburgh Castle, but had no scruple in surrendering it to Murray, who made him President of the Court of Session. He was charged

with a share in the murder of Darnley, and helped to bring Regent Morton to his death. He died in 1583. *The Practicks of Scots Law*, attributed to him, was long a textbook.

Balfour, John Hutton, a distinguished botanist, born 1808, died 1884. He graduated at Edinburgh University in arts and in medicine; from 1841 to 1845 was professor of botany in Glasgow University; and in the latter year removed to Edinburgh to occupy a similar post, resigning his chair in 1879. He wrote valuable botanical textbooks, including *Elements*, *Outlines*, *Manual*, and *Class-book*, besides various other works.

Balfroosh´, or **Barfurush´**, a town, Persia, province of Mazanderan, about twelve miles from the Caspian, a great emporium of the trade between Persia and Russia. Pop. estimated at 50,000.

Ba´li, an island of the Indian Archipelago east of Java, belonging to Holland; greatest length, 85 miles, greatest breadth, 55 miles; area, about 2160 sq. miles. It consists chiefly of a series of volcanic mountains, of which the loftiest, Agoong (11,326 feet), became active in 1843 after a long period of quiescence. Principal products: rice, cocoa, coffee, indigo, cotton, &c. The people are akin to those of Java and are mostly Brahmans in religion. It is divided into eight provinces under native rajahs, and since 1882 has formed one colony with Lombok, the united pop. being 1,363,000, of whom about 900,000 belong to Bali.

Bal´iol, or **Balliol**, John de, of Barnard Castle, Northumberland, father of King John Baliol, a great English (or Norman) baron in the reign of Henry III, to whose cause he strongly attached himself in his struggles with the barons. In 1263 he laid the foundation of Balliol College, Oxford, which was completed by his widow Devorguila or Devorgilla. She was daughter and co-heiress of Allan of Galloway, a great baron of Scotland, by Margaret, eldest daughter of David, Earl of Huntingdon, brother of William the Lion. It was on the strength of this genealogy that his son John Baliol became temporary King of Scotland. He died 1269.

Bal´iol, or **Balliol**, John, King of Scotland, born about 1249, died 1315. On the death of Margaret, the Maid of Norway and grandchild of Alexander III, Baliol claimed the vacant throne by virtue of his descent from David, Earl of Huntingdon, brother to William the Lion, King of Scotland (see above

article). Robert Bruce (grandfather of the king) opposed Baliol; but Edward I's decision was in favour of Baliol, who did homage to him for the kingdom, 20th Nov., 1292. Irritated by Edward's harsh exercise of authority, Baliol concluded a treaty with France, then at war with England; but after the defeat at Dunbar he surrendered his crown into the hands of the English monarch. He was sent with his son to the Tower, but, by the intercession of the Pope in 1297, obtained liberty to retire to his Norman estates, where he died.—His son, Edward, in 1332 landed in Fife with an armed force, and, having defeated a large army under the Regent Mar (who was killed), got himself crowned king, but was driven out in three months.

Balis'tidæ. See *Trigger-fishes*.

Balize (ba-lēz'). See *Belize*.

Bal'kan (ancient, **Hæmus**), a rugged chain of mountains extending from Cape Emineh, on the Black Sea, in Eastern Roumelia, westwards to the borders of Serbia, though the name is sometimes used to include the whole mountain system from the Black Sea to the Adriatic, the region south of Austria and Russia, or south of the Danube and Save, forming the Balkan Peninsula. The range, which is over 200 miles in length, forms the watershed between the streams flowing northward into the Danube and those flowing southward to the Ægean, the chief of the latter being the Maritza. The average height is not more than 5000 feet, but the highest point, Tchat-al-dagh, is 8340 feet. In the Russo-Turkish war of 1877-8 the Russian troops managed to cross it without great difficulty, though they had to encounter a stubborn resistance at the Shipka Pass, where a Turkish army of 32,000 men ultimately surrendered to them.

Balkan League, The. After the annexation of Bosnia and Herzegovina by Austria, in 1908, the Balkan League of Bulgaria, Serbia, Montenegro, and Greece was formed, apparently at the instigation of the Powers of the Triple Entente, primarily to constitute a bulwark against Austrian aggression towards the southeast. The League, however, directed its activities against Turkey, taking advantage of Turkish embarrassments in Libya and of the success of an insurrection in Albania to attempt in 1912 the overthrow of the Turkish regime in Europe. The declaration of hostilities by the League in Oct., 1912, began the First Balkan War (q.v.). The League was

maintained during the two campaigns of the war, but jealousy of Bulgarian pre-eminence led to its disruption in June, 1913, when Serbia, Montenegro, and Greece went to war with their former ally.

Balkan States. See *Bulgaria, Greece, Roumania, Serbia, Turkey, and Yugo-Slavs.*

Balkan War, The. The First Balkan War (Oct., 1912-May, 1913) was the effort of the Balkan League (q.v.) to dismember Turkey in Europe. Each member of the League was allotted a definite strategic object, and each gained a considerable measure of success. Bulgaria defeated the Turks at Kirk Kilisseh and Lule Burgas, and drove them behind the lines of Tchataldja. The Greeks captured Salonica; the Serbians forced their way to the sea at Durazzo. The Montenegrins invested Scutari. An armistice in December was followed by indecisive negotiations, in which the Great Powers took part. Renewed hostilities in the spring of 1913 (during which the Turks lost Adrianople) came to an end with the Treaty of London in May. Before its promises could be carried out, Serbia, Greece, and Roumania joined to crush Bulgaria in the Second Balkan War, alleging that her claims to conquered territory in Macedonia were excessive. The Treaty of Bucharest (Aug., 1913) finally concluded the war. The net results were: (1) to confine Turkey to Constantinople and the vilayet of Adrianople; (2) to extend Bulgarian territory to the Ægean and to increase greatly both Serbia and Montenegro; (3) to grant Salonica to Greece; (4) to create an autonomous Albania.—BIBLIOGRAPHY: E. Ashmead-Bartlett, *With the Turks in Thrace*; *The Balkan War Drama*, by a Special Correspondent; D. J. Cassvetti, *Hellas and the Balkan Wars*; Sir J. R. L. Rankin, *The Inner History of the Balkan War*.

Balkash´, or **Balkhash** (ba´l-ha´sh´), a salt lake in Russian Central Asia, surrounded by steppes and plains; length about 330 miles, area 8500 sq. miles, depth nowhere more than 80 feet; formerly of much greater area and gradually growing smaller; receives the Ili and other smaller streams.

Balkh (bälk or bälh), a city in the north of Afghanistan, in Afghan Turkestan, at one time the emporium of the trade between India, China, and Western Asia. It was long the centre of Zoroastrianism, and was also an important Buddhist centre. In 1220 it was sacked by Genghis Khan, and

again by Timur in the fourteenth century. The remains of the ancient city extend for miles. The town is now merely a village, but a new town has been built an hour's journey north of the old, the residence of the Afghan governor, with a pop. of about 20,000. The district, which formed a portion of ancient Bactria, lies between the Oxus and the Hindu-Kush, with Badakshan to the east and the desert to the west. In the vicinity of the Oxus, where there are facilities for irrigation, the soil is rich and productive, and there are many populous villages.

Bal'kis, the Arabian name of the Queen of Sheba who visited Solomon. She is the central figure of innumerable Eastern legends and tales.

Ball, Game of. Ball-playing was practised by the ancients, and old and young amused themselves with it. The Phæacian damsels are represented in the *Odyssey* as playing it to the sound of music; and Horace represents Mæcenas as amusing himself thus during a journey. In the Greek gymnasia, the Roman baths, and in many Roman villas, a *sphæristerium* (a place appropriated for playing ball) was to be found; the games played being similar to those of the present day. In the Middle Ages the sport continued to be very popular both as an indoor and outdoor exercise, and was a favourite Court pastime until about the end of the eighteenth century. In England football and tennis are mentioned at an early date, and a favourite game prior to the English revolution was one in which a *mall* or mallet was used, hence the name *pall-mall* (It. *palla*, Lat. *pila*, a ball) for the game and the place where it was played. The most popular modern forms are cricket, base-ball, football, golf, lawn-tennis, fives, and polo.

Ball, John, an itinerant preacher of the fourteenth century, excommunicated about 1367 for promulgating "errors, schisms, and scandals against the Pope, archbishops, bishops, and clergy". He was one of the most active promoters of the popular insurgent spirit which found vent under Wat Tyler in 1381, and the couplet

"When Adam delved and Eve span
Who was then the gentleman?"

is attributed to him. He was hanged, drawn, and quartered on 15th July, 1381.

Ball, Sir Robert Stawell, astronomer, born at Dublin 1840, educated at Chester and Trinity College, Dublin. In 1865 he was appointed Lord Rosse's astronomer at Parsonstown, and subsequently held various official posts, including those of Andrews professor of astronomy in the University of Dublin and Astronomer Royal for Ireland (1874), Lowndean professor of astronomy and geometry in the University of Cambridge and director of the observatory (1892). He became F.R.S. in 1873, and was knighted in 1886. Besides many memoirs and articles, he published *The Story of the Heavens*, *Starland*, *In Starry Realms*, *Time and Tide*, *The Story of the Sun*, *Great Astronomers*, *The Earth's Beginning*, *Popular Guide to the Heavens* (numerous plates with accompanying text), *Natural Sources of Power*, &c. He died in 1913.

Bal'lad, a term loosely applied to various poetic forms of the song type, but in its most definite sense a poem in which a short narrative is subjected to simple lyrical treatment. It was, as indicated by its name, which is related to the Low Latin *ballare* and O. French *baller*, to dance, originally a song accompanied by a dance. The ballad, like the nursery tales and the *Märchen*, is probably one of the earliest forms of rhythmic poetic expression, constituting a species of epic in miniature, out of which by fusion and remoulding larger epics were sometimes shaped. Their present form is, of course, relatively recent. As in the folk-tales, so in the ballads of different nations, the resemblances are sufficiently numerous and close to point to the conclusion that they have often had their first origin in the same primitive folk-lore or popular tales. But in any case, excepting a few modern literary ballads of a subtler kind, they have been the popular expression of the broad human emotions clustering about some strongly-outlined incidents of war, love, crime, superstition, or death. It is probable that in the Homeric poems fragments of older ballads are embedded; but the earliest ballads, properly so called, of which we have record were the *ballistea* or dancing-songs of the Romans, of the kind sung in honour of the deeds of Aurelian in the Sarmatic war by a chorus of dancing boys. In their less specialized sense of lyric narratives, their early popularity among the Teutonic race is evidenced by the testimony of Tacitus, of the Gothic historian Jornandes, and the Lombard historian Paulus Diaconus; and many appear to have been written down by order of Charlemagne and used as a means of education. Of the ballads of this period, however, only a general

conception can be formed from their traces in conglomerates like the *Nibelungenlied*; the more artificial productions of the Minnesingers and Meistersingers overlying the more popular ballad until the fifteenth century, when it sprang once more into vigorous life. A third German ballad period was initiated by Bürger, under the inspiration of the revived interest in the subject shown in Great Britain and the publication of the Percy *Reliques*; and the movement was sustained by Herder, Schiller, Goethe, Heine, Uhland, and others. The earlier German work is, however, of inferior value to that of Scandinavia, where, though comparatively few manuscripts have survived, and those not more than three or four centuries old, a more perfect oral tradition has rendered it possible to trace the original stock of the twelfth century.

Of the English and Scottish ballads anterior to the thirteenth century there are few traces beyond the indication that they were abundant, if indeed anything can be definitely asserted of them earlier than the fourteenth century. Among the oldest may be placed *The Little Gest of Robin Hood*, *Hugh of Lincoln*, *Sir Patrick Spens*, and the *Battle of Otterbourn*. In the fifteenth century specimens multiply rapidly: ballad-making became in the reign of Henry VIII a fashionable amusement, the king himself setting the example; and though in the reign of Elizabeth ballads came into literary disrepute and ballad singers were brought under the law, yet there was no apparent check upon the rate of their production. Except perhaps in the north of England and south of Scotland, there was, however, a marked and increasing tendency to vulgarization as distinct from the preservation of popular qualities. The value of the better ballads was lost sight of in the flood of dull, rhythmless, and frequently scurrilous verse. The modern revival in Britain dates from the publication of Allan Ramsay's *Evergreen* and *Tea-table Miscellany* (1724-7) and of the selection *Reliques* made by Bishop Percy from his seventeenth-century MSS. (1765), a revival not more important for its historic interest than for the influence which it has exercised upon all subsequent poetry.

The threefold wave discernible in German, if not in British, ballad history, is equally to be traced in Spain, which alone among the Latinized countries of Europe has songs of equal age and merit with the British historic ballads. The principal difference between them is, that for the most part the Spanish romance is in trochaic, the British ballad in iambic metre. The ballads of the

Cid date from about the end of the twelfth and beginning of the thirteenth century; and then followed an interval of more elaborate production, a revival of ballad interest in the sixteenth century, a new declension, and finally a modern and still-persisting enthusiasm.

The French poetry of this kind never reached any high degree of perfection, the romance, farce, and lyric flourishing at the expense of the ballad proper. Of Italy much the same may be said, though Sicily has supplied a great store of ballads; and nearly all the Portuguese poetry of this kind is to be traced to a Spanish origin. The Russians have lyrico-epic poems, of which some, in old Russian, are excellent, and the Serbians are still in the ballad-producing stage of civilization. Modern Greece has also its store of ballads, published in several collections. In Greece, Russia, and elsewhere the old habit of improvising song as an accompaniment to dance still exists.—
BIBLIOGRAPHY: Professor Child, *English and Scottish Popular Ballads*; Professor Gummere, *The Beginnings of Poetry*; Sir W. Scott, *Minstrelsy of the Scottish Border* (edited by T. F. Henderson); T. F. Henderson, *The Ballad in Literature*.

Ballade (bal-a´d´), the earlier and modern French spelling of *ballad*, but now limited in its use to a distinct verse-form introduced into English literature of late years from the French, and chiefly used by writers of *vers-de-société*. It consists of three stanzas of eight lines each, with an *envoy* or closing stanza of four lines. The rhymes, which are not more than three, follow each other in the stanzas thus: a, b, a, b; b, c, b, c, and in the envoy, b, c, b, c; and the same line serves as a refrain to each of the stanzas and to the envoy. There are other varieties, but this may be regarded as the strictest, according to the precedent of Villon and Marot.

Bal´lentyne, James, the printer of Sir Walter Scott's works, born at Kelso 1772, died at Edinburgh 1833. Successively a solicitor and a printer in his native town, at Scott's suggestion he removed to Edinburgh, where the high perfection to which he had brought the art of printing, and his connection with Scott, secured him a large trade. The printing firm of James Ballantyne & Co. included Scott, James Ballantyne, and his brother John (who died in 1821). For many years he conducted the *Edinburgh Weekly Journal*. His firm was involved in the bankruptcy of Constable & Co., by which Scott's fortunes were wrecked, but Ballantyne was continued by the creditors'

trustee in the literary management of the printing-house. He survived Scott only about four months.

Ballantyne, Robert Michael, writer of books for boys, born at Edinburgh 1825, died in Rome 1894, was a nephew of James Ballantyne, the printer of Sir Walter Scott's works. He was for some years in North America in the service of the Hudson's Bay Company, and his experiences there supplied him with materials for some of his earlier books, especially *Hudson's Bay, or Life in the Wilds of North America* (1848); *The Young Fur Traders*; and *Ungava, a Tale of Eskimo Land*. For many years he continued to produce popular and instructive boys' books, dealing with scenes and subjects of the most varied kind; and he also published a volume entitled *Personal Reminiscences of Book-Making* (1893).

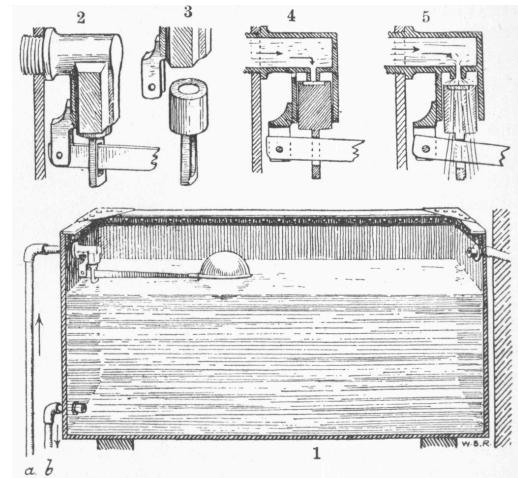
Ballarat', or **Ballaarat**, an Australian town in Victoria, chief centre of the gold-mining industry of the colony, and next in importance to Melbourne, from which it is distant W.N.W. about 60 miles direct. It consists of two distinct municipalities, Ballarat West and Ballarat East, separated by the Yarrowee Creek, and has many handsome buildings, and all the institutions of a progressive and flourishing city, including hospital, mechanics' institute and library, free public library, Anglican and Roman Catholic cathedrals, &c. Gold was first discovered in 1851, and the extraordinary richness of the field soon attracted hosts of miners. The surface diggings having been exhausted, the precious metal is now got from greater depths, and there are mines as deep as some coal-pits, the gold being obtained by crushing the auriferous quartz. There are also foundries, woollen mills, flour-mills, breweries and distilleries, &c. Pop. 42,252.

Bal'last, a term applied (1) to heavy matter, as stone, sand, iron, or water placed in the bottom of a ship or other vessel to sink it in the water to such a depth as to enable it to carry sufficient sail without oversetting. (2) The sand placed in bags in the car of a balloon to steady it and to enable the aeronaut to lighten the balloon by throwing part of it out. (3) The material used to fill up the space between the rails on a railway in order to make it firm and solid.

Ball'ater, a village and favourite summer resort in Aberdeenshire, on the Dee, at the terminus of the Deeside Railway, about 43 miles from Aberdeen,

a centre from which Balmoral, Lochnagar, Braemar, &c., are easily reached. The Pananich chalybeate springs are adjacent. Pop. (1921), 1542.

Ball-cock, a kind of self-acting stopcock opened and shut by means of a hollow sphere or ball of metal attached to the end of a lever connected with the cock. Such cocks are often employed to regulate the supply of water to cisterns. The ball floats on the water in the cistern by its buoyancy, and rises and sinks as the water rises and sinks, shutting off the water in the one case and letting it on in the other.

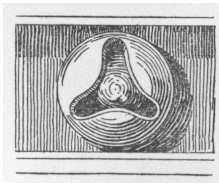


Ball-cock

1. Cistern with ball-cock attached: *a*, rising main; *b*, supply to house. 2. Outside view of valve (cistern full). 3. Piston with rubber disk. 4. Section of valve (cistern full). 5. Section of valve with piston pulled down.

Bal'lentyne, or **Bellenden**, John, a Scottish poet; was a native of Lothian, and appears to have been born towards the close of the fifteenth century. He was in the service of James V from the king's earliest years, and at his request he translated Boece's *Latin History*, which had been published at Paris in 1526, the translation being printed in 1536. As a reward he was made Archdeacon of Moray and a canon of Ross. He was a bitter opponent of the Reformation, and is said to have died at Rome in 1550.

Ballet (bal'ā), a species of dance, usually forming an interlude in theatrical performances, but principally confined to opera. Its object is to represent, by mimic movements and dances, actions, characters, sentiments, passions, and feelings, in which several dancers perform together. The ballet is an invention of modern times, though pantomimic dances were not unknown to the ancients. The dances frequently introduced into operas seldom deserve the name ballet, as they usually do not represent any action, but are destined only to give the dancers an opportunity of showing their skill. The modern ballet was developed and perfected in France, and introduced into England in the eighteenth century. From an artistic point of view, the modern ballet is often a very low-class entertainment.



Ball-flower

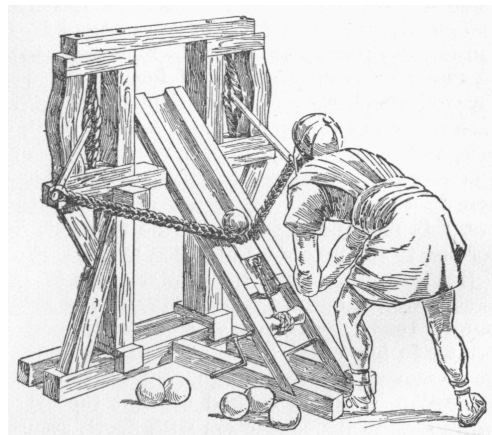
Ball-flower, an architectural ornament resembling a ball placed in a circular flower, the three petals of which form a cup round it; usually inserted in a hollow moulding, and generally characteristic of the Decorated Gothic style of the fourteenth century.

Ballia, a town of India, in the United Provinces, on the Ganges, the administrative head-quarters of a district of the same name. Pop. 16,680.

Ballina', a town and river-port, Ireland, County Mayo, on both banks of the Moy, about 5 miles above its mouth in Killala Bay, with a considerable local and also a little coasting and foreign trade. Pop. 4662.

Ballinasloe' (-slō), a town, Ireland, in Galway and Roscommon counties, 15 miles south-west of Athlone, on both sides of the Suck, noted for its cattle fair, from 5th till 9th Oct., the most important in Ireland. The town contains remains of a castle of Elizabethan date. Pop. 5170.

Ballis'ta, a machine used in military operations by the ancients for hurling heavy missiles, thus serving in some degree the purpose of the modern cannon. The motive power appears to have been obtained by the torsion of ropes, fibres, catgut, or hair. They are said to have sometimes had an effective range of a quarter of a mile, and to have thrown stones weighing as much as 300 lb. The ballistæ differed from the *catapultæ*, in that the latter were used for throwing darts.



Ballista as used by the Romans

Ballis'tic Pendulum, an apparatus for ascertaining the velocity of military projectiles, and consequently the force of fired gunpowder. A piece of ordnance is fired against bags of sand supported in a strong case or frame suspended so as to swing like a pendulum. The arc through which it vibrates is shown by an index, and the amount of vibration forms a measure of the force or velocity of the ball.

Ballistics, the science of the motion of projectiles. *Interior ballistics* deals with the motion of the shot while inside the gun, and with the results of the explosion of the charge. *Exterior ballistics* comprises a study of the trajectory or path of the projectile in the air, and the causes which affect the shape of this path, such as the density of the air, shape and weight of the shot, rifling of the barrel, angle of fire and speed of the projectile. If the air-resistance be neglected, it can be shown that the trajectory is a *parabola*. This result, given by Galileo, is of little use in gunnery except when taken along with a knowledge of air-resistance. Robins, in 1742, first measured the muzzle-speed of the shot by firing into a ballistic pendulum and noting the swing of the pendulum. Wheatstone, in 1840, suggested measuring the speed by causing the shot to cut successive wire screens, each of which formed part of an electric circuit. The Boulangé chronograph is a modern development of this idea. Rev. F. Bashforth's experiments, carried out between 1865 and 1880, form the basis of our knowledge of air-resistance. He found that the latter was proportional to the square of the diameter of the shot, and that, although the resistance is expressible in terms of a power of the speed, this power changes with the speed. The air-resistance is a maximum at about the speed at which sound travels, viz. 1100 feet per second. Artillerists employ ballistic tables based on the results obtained by Bashforth and others, by means of which, given the kind of projectile, its initial velocity, and angle of fire, they can calculate the range and time of flight of the projectile. See *Gunnery*.—BIBLIOGRAPHY: F. Bashforth, *The Motion of Projectiles*; J. M. Ingalls, *Interior Ballistics*; P. Charbonnier, *Balistique Extérieure*.

Balloon-fish (*Tetraodon lineatus*), ord. Plectognathi, a curious tropical fish that can inflate itself so as to resemble a balloon.

Balloons, the most elementary form of lighter-than-air air-craft, unprovided with any means of propulsion or steering. Deriving its power of sustentation from the gas contained in the gas-bag, the free balloon is spherical in form. It is usually filled with coal-gas, which has a lifting-force of 35 lb. per 1000 cu. feet. Varying in size from 5000 to 20,000 cu. foot capacity, the gas-bag is covered with a 'net', from which is hung the 'basket' for the accommodation of the aeronaut, or pilot, and passengers. The spherical gas-bag has at the bottom an elongation, known as the 'neck', which is open to the air in order that the gas may escape as it expands during the rise into the

rarefied atmosphere of the upper air. A 'valve' is fitted at the top, which may be opened by a cord from the basket when it is desired to allow gas to escape in order to descend. Bags containing sand are carried in order that ballast may be dropped when it is desired to rise. A free balloon is completely at the mercy of the wind, and simply drifts where the wind takes it. Balloons are usually constructed of cotton fabric, though occasionally of gold-beater's skin, in which case they are commonly filled with hydrogen in place of coal-gas. A 'trail-rope' is usually carried, with which the practice of 'trailing' may be effected over open country or the sea. This consists in allowing the end of the trail-rope to drag on the ground. As the balloon rises it raises more of the rope off the ground, the additional weight bringing it down again. *Per contra*, as it falls the weight of the trail-rope is taken by the ground and so lightens the balloon. This tends to maintain the balloon at a uniform height. The greatest height ever attained by a balloon is 37,000 feet, reached by Glaisher and Coxwell in Sept., 1862. Both aeronauts nearly perished from the effects of the cold and rarefaction of the atmosphere. In 1895 Parseval and Siegsfield, two officers of the Prussian army, invented the 'Drachen-ballon' or 'Kite-balloon' (Fr. *Saucisse*). This consists of a captive balloon comprising an elongated gas-bag fitted with an automatic air-chamber, or 'ballonet', for maintaining the pressure, and automatic stabilizers to prevent yawing off the wind, rolling from side to side, and pitching horizontally. This was improved upon by Captain Caquot of the French army, who in 1916 produced an improved type, offering many advantages, which was used in large numbers by the British, French, and American armies and navies during the Great European War of 1914-8. See also *Aeronautics, Air-ships*.—BIBLIOGRAPHY: *British Military Manual of Ballooning*; Rev. J. M. Bacon, *The Dominion of the Air*; H. Coxwell, *My Life and Balloon Experiences*; J. Glaisher, *Travels in the Air*; G. Tissandier, *Histoire de mes Ascensions*.

Bal'lot, Voting by, signifies literally voting by means of little balls (called by the French *ballottes*), usually of different colours, which are put into a box in such a manner as to enable the voter, if he chooses, to conceal for whom or for what he gives his suffrage. The method is adopted by most clubs in the election of their members—a white ball indicating assent, a black ball dissent. Hence, when an applicant is rejected, he is said to be *blackballed*. The term voting by ballot is also applied in a general way to

any method of secret voting, as, for instance, when a person gives his vote by means of a ticket bearing the name of the candidate whom he wishes to support. In this sense vote by ballot is the mode adopted in electing the members of legislative assemblies in most countries, as well as the members of various other bodies. In ancient Greece and Rome the ballot was in common use. In Britain it had long been advocated in the election of members of Parliament and of municipal corporations, but it was only introduced by an Act passed in 1872.

Ballyme'na, a town of Ireland, County Antrim, 22 miles from Belfast, with a considerable trade in linens and linen yarns, the manufacture of which is carried on to a great extent. Pop. 11,380.

Ballymo'ney, a town of Ireland, County Antrim, 38 miles N.W. of Belfast; linen, chemicals, tanning, and brewing. Pop. 3100.

Ballyshan'on, a small seaport of Ireland, County Donegal, on the Erne, about 1 mile from the S.E. shore of Donegal Bay. Pop. 2471.

Balm (bäm; *Melissa officinālis*), a plant, belonging to the Labiatae; formerly in great repute for its medicinal virtues. A native of the south of Europe, it is cultivated in English gardens. It is a herbaceous perennial, with an erect branching stem about 2 feet high. The leaves arise with the flower-stems from a thick joint at the extremity of the stalk. The flowers are whitish; they are produced in a round terminal umbel, and appear in June. The stems and leaves are slightly stimulating and tonic. They contain an essential oil of a yellowish colour and with a fragrant smell, called oil of balm.

Balm of Gilead, the exudation of a tree, *Balsamodendron gileadense*, nat. ord. Amyridaceæ, a native of Arabia Felix, and also obtained from the closely-allied species *Balsamodendron Opobalsānum*. The leaves of the former tree yield when bruised a strong aromatic scent; and the balm of Gilead of the shops, or balsam of Mecca or of Syria, is obtained from it by making an incision in its trunk. It has a yellowish or greenish colour, a warm, bitterish, aromatic taste, and an acidulous fragrant smell. It is valued as an odoriferous unguent and cosmetic by the Turks. It is frequently adulterated for market.—The *balm of Gilead fir*, which produces a turpentine called Canada balsam, is the *Abies balsamifera*, a North

American species, whose range is from Virginia to Canada. The name is derived from *Jeremiah*, viii, 22.

Balmain', a populous western suburb of Sydney, N.S.W., with extensive docks and various industrial works. Pop. 33,000.

Balmer'ino, Arthur Elphinstone, Lord, a Scottish Jacobite, born 1688, executed 1746. He took part in the Jacobite rebellion of 1715, and fought at Sheriffmuir. Having joined the young Pretender in 1745, he was taken prisoner at Culloden, tried at Westminster, found guilty, and beheaded. His title was from Balmerino in Fife.

Balmor'al Castle, a private residence of the British sovereign, beautifully situated on the s. bank of the Dee, in Aberdeenshire, 45 miles w. by s. of Aberdeen. It stands in the midst of fine and varied mountain scenery, is built of granite in the Scottish baronial style, mainly in two connected blocks, and has an imposing appearance. The estate, purchased by the Prince Consort in 1852 and bequeathed to Queen Victoria, extends to about 40,000 acres, mostly deer forest.

Balnaves', Henry, of Halhill, a Scottish reformer, was born at Kirkcaldy, educated at St. Andrews, and became a lord of session and a member of the Scottish Parliament in 1538. He was one of the commissioners appointed in 1543 to treat of the proposed marriage between Edward VI and Mary. In 1547 he was one of the prisoners taken in the castle of St. Andrews and exiled to France. Recalled in 1554, he busily engaged in the establishment of the reformed faith, assisted in revising the *Book of Discipline*, and accompanied Murray to England in connection with Darnley's murder. He died in 1579.

Balrampur. See *Bulrampur*.

Balsa, a kind of raft or float used on the coasts and rivers of Peru and other parts of South America for fishing, for landing goods and passengers through a heavy surf, and for other purposes where buoyancy is chiefly wanted. It is formed generally of two inflated seal-skins, connected by a sort of platform on which the fisherman, passengers, or goods are placed.

Bal'sam, the common name of succulent plants of the genus *Impatiens*, family Balsaminaceæ, having beautiful irregular flowers, cultivated in

gardens and green-houses. *Impatiens balsamīna*, native of the East Indies, is a common cultivated species. *I. noli-me-tangere* grows wild in Britain, but is not native. See *Impatiens*.

Balsam, an aromatic, resinous substance, flowing spontaneously or by incision from certain plants. A great variety of substances pass under this name. But in chemistry the term is confined to such vegetable juices as consist of resins mixed with volatile oils, and yield the volatile oil on distillation. The resins are produced from the oils by oxidation. A balsam is thus intermediate between a volatile oil and a resin. It is soluble in alcohol and ether, and capable of yielding benzoic acid. The balsams are either liquid or more or less solid; as, for example, the balm of Gilead, and the balsams of copaiba, Peru, and Tolu. Benzoin, dragon's-blood, and storax are not true balsams, though sometimes called so. The balsams are used in perfumery, medicine, and the arts. See *Copaiba*, &c.—*Balsam of Gilead* or *of Mecca*, balm of Gilead (q.v.).—*Canada balsam*. See *Canada balsam*.

Balsam Fir. See *Balm of Gilead*.

Balsa'mo, Joseph. See *Cagliostro*, Count.

Balsamoden'dron, a genus of trees or bushes, ord. Amyridaceæ, species of which yield such balsamic or resinous substances as balm of Gilead, bdellium, myrrh, &c.

Balta, a town in the Ukraine, government of Podolia, on the Kodema, an affluent of the Bug, 115 miles N.N.W. of Odessa. Pop. 18,450.

Baltic, Battle of the. See *Copenhagen*; *Parker*, Sir Hyde.

Baltic Provinces, a term commonly given to the former Russian governments of Courland, Livonia, and Esthonia (q.v.).

Baltic Sea, an inland sea or large gulf connected with the North Sea, washing the coasts of Denmark, Germany, Russia, and Sweden; nearly 900 miles long, extending to 200 broad; superficial extent, together with the Gulfs of Bothnia and Finland, 171,743 sq. miles. Its greatest depth is 126 fathoms; mean, 44 fathoms. A chain of islands separates the southern part from the northern, or Gulf of Bothnia. The water of the Baltic is colder and clearer than that of the ocean: it contains a smaller proportion of salt, and

the ice obstructs the navigation three or four months in the year. Among the rivers that enter it are the Neva, Dwina, Oder, Vistula, and Niemen. Islands: Samsoe, Moen, Bornholm, Langeland, Laaland, which belong to Denmark (besides Zealand and Funen); Gottland and Oeland, belonging to Sweden; Rügen, belonging to Prussia; Dagoe and Oesel, belonging to Russia; and the Åland Islands. The Sound, the Great and the Little Belt lead from the Kattegat into the Baltic. The Baltic and North Sea were long connected by the Eider and a canal from it to the neighbourhood of Kiel, but this has been superseded by the great ship canal, starting from the Elbe near its mouth and ending in Kiel Bay.

Bal'timore, a city and port in Maryland, United States, finely situated on the N. side of the Patapsco, 14 miles above Chesapeake Bay. Baltimore takes its name from Lord Baltimore, the founder of Maryland; it was first laid out as a town in 1729; and was erected into a city in 1797. It is well built, chiefly of brick, and is known as the 'monumental city', from the public monuments which adorn it, the principal being the Washington monument. Among its buildings are the city hall, built in Renaissance style, of white marble, with a tower and dome rising 240 feet; the Peabody Institute, containing a library, art gallery, &c.; the Maryland Institute; the custom-house; the post office; the United States court-house and jail, the Johns Hopkins hospital, the Roman Catholic cathedral, &c. The chief educational institution, now one of the most important in the States, is the Johns Hopkins University, endowed with 3,500,000 dollars by its founder (whose name it bears). There is a Roman Catholic archbishop with the rank of primate, and a Protestant episcopal bishop. Industries: shipbuilding; manufactures of iron, wool, cotton, pottery, &c.; sugar-refining, distilling, tanning, the making of agricultural implements, canning oysters and fruits, &c. As a flour market Baltimore is an important centre; and it does an immense trade in exporting tobacco and other products. The harbour is very extensive. Pop. (1920), 733,826.

Baltimore, George Calvert, Lord, born in Yorkshire about 1580, died in London 1632. He was for some time Secretary of State to James I, but this post he resigned in 1624 in consequence of having become a Roman Catholic. Notwithstanding this he retained the confidence of the king, who in 1625 raised him to the Irish peerage, his title being from Baltimore, a fishing-village of Cork. He had previously obtained a grant of land in

Newfoundland, but as this colony was much exposed to the attacks of the French he left it, and obtained another patent for Maryland. He died before the charter was completed, and it was granted to his son Cecil, who deputed the governorship to his brother Leonard (1606-47).

Baltimore Bird, an American bird, the *Icterus Baltimorii*, family Icteridæ, nearly allied to the Sturnidæ, or starlings. It is a migratory bird, and is known also by the names of 'golden robin', 'hang-bird', and 'fire-bird'. It is about 7 inches long; the head and upper parts are black, the under parts of a brilliant orange hue. It builds a pouch-like nest, very skilfully constructed of threads deftly interwoven, suspended from a forked branch and shaded by overhanging leaves. It feeds on insects, caterpillars, beetles, &c. Its song is a clear, mellow whistle.

Baluchistan (ba-lö'chi-stän), a country in Asia, the coast of which is continuous with the north-western seaboard of India, bounded on the north by Afghanistan, on the west by Persia, on the south by the Arabian Sea, and on the east by Sindh. It has an area of about 135,000 sq. miles, and a population estimated at 834,703. The general surface of the country is rugged and mountainous, with some extensive intervals of barren sandy deserts, and there is a general deficiency of water. Various kinds of grain are cultivated, and there is abundance of excellent fruits. The main body of the inhabitants are divided into two great branches, the Baluchis and Brahuïs, differing in language, figure, and manners. The Baluchi language resembles the modern Persian, the Brahui presents points of agreement with the Dravidian languages of India. The Baluchis in general have tall figures, long visages, and prominent features; the Brahuïs have shorter, thicker bodies, with round faces and rather flat features. Both races are Mahommedans, hospitable and brave. There are also many Afghans and others in the country. The Khan of Kelat, the nominal ruler of the whole land, in 1877 concluded a treaty with Britain, in virtue of which he became a feudatory of India. Since then Quetta, in the north-east, occupying an important position, and certain portions of territory, have been absolutely annexed, and in 1887 a separate administration was constituted, and British authority established, under the Governor-General's agent. The area of British Baluchistan, including tribal areas, is 54,228 sq. miles; pop. 414,412. The country nominally under the Khan of Kelat has an area of 72,280 sq. miles; pop. 359,086. Under British administration roads and

railways have been made, with telegraphs and a postal service. Wool, fruits, &c., are exported, cottons and other goods imported, and there is a fair transit trade.—BIBLIOGRAPHY: Floyer, *Unexplored Baluchistan*; T. Thornton, *Life of Sandeman*.

Baluster, a small column or pilaster, of various forms and dimensions, often adorned with mouldings, used for balustrades.

Balustrade, a range of balusters, together with the cornice or coping which they support, used as a parapet for bridges or the roofs of buildings, or as a mere ornament; also serving as a fence or enclosure for balconies, terraces, staircases, &c.

Baluze (ba' -lüz), Étienne, French historian and miscellaneous writer, born 1630, died 1718. For more than thirty years he was librarian to M. de Colbert, and was appointed professor of canon law in the Royal College, but displeasing Louis XIV with his *Histoire Générale de la Maison d'Auvergne*, he was thrown into prison and his property confiscated. He recovered his liberty in 1713, but did not regain his position. He left some 1500 MSS. in the national library of Paris, besides forty-five printed works, including *Regum Francorum Capitularia* (2 vols.) and *Miscellanea* (7 vols.).

Balzac (ba' l-za' k), Honoré de, a celebrated French novelist, was born at Tours in 1799, died 1850. Before completing his twenty-fourth year he had published a number of novels under various assumed names, but the success attending all was very indifferent; and it was not till 1829, by the publication of *Le Dernier Chouan*, a tale of La Vendée, and the first novel to which Balzac appended his name, that the attention of the public was diverted to the extraordinary genius of the author. A still greater popularity attended his *Physiologie de Mariage*, a work full of piquant and caustic observations on human nature. He wrote a large number of novels, all marked by a singular knowledge of human nature and distinct delineation of character, but apt to be marred by exaggeration. Among his best-known works are: *Scènes de la Vie de Province*, *Scènes de la Vie Parisienne*, *Le Père Goriot*, *Eugénie Grandet*, and *Le Médecin de Campagne*. The publication of this last, in 1835, led to a correspondence between Balzac and the Countess Eveline de Hanska, a Polish lady whom, after about

fifteen years, he visited and married. A collected edition of his works, under the title *La Comédie Humaine*, was published in 45 vols., Paris, 1856-9, and another in 1869. In 1899 appeared the first volume of Balzac's *Œuvres Posthumes*, and in 1906 the second.—BIBLIOGRAPHY: F. Wedmore, *Balzac*; M. F. Sandars, *Balzac*; W. H. Helm, *Aspects of Balzac*; Taine, *Essay on Balzac*.

Balzac (ba´l-za´k), Jean Louis Guez de, French writer, born 1594, died 1654. His writings, which had a great reputation in their day owing to the elegance of his style, are now neglected. The most esteemed are his *Familiar Letters*, *Le Prince*, *Le Socrate Chrétien*, and *Aristippe*.

Bamako, capital of the French colony of Upper Senegal and Niger since 1904. It is an important trade-centre and the terminus of the Niger-Senegal railway. Pop. 8000.

Bamba, a district of the Congo, w. coast of Africa, lying to the south of the River Ambriz. It is thickly populated, and is rich in gold, silver, copper, salt, &c.

Bambar´ra, a territory of Western Africa, on the Upper Niger, first visited by Mungo Park, now in the French portion of the Sudan. The country is generally very fertile, producing wheat, rice, maize, yams, &c. The inhabitants are of negro or mixed race, and some are Mohammedans. Excellent cotton cloth is made. The chief town is Segou. Pop. estimated at 2,000,000. See *Senegal*.

Bam´berg, a town of Germany, Bavaria, charmingly situated on several hills, on the navigable River Regnitz, some 3 miles from its mouth in the Main. It is the seat of a Catholic archbishop; the cathedral, founded in 1004 (the present structure dates from 1111), is one of the finest churches in Germany. The royal library contains 100,000 volumes and valuable MSS. Bamberg carries on a large trade; its industries are cotton-spinning, tobacco-manufacture, brewing, &c. Pop. 48,063.

Bambino (ba´m-bē´nō; It., an infant), the figure of our Saviour represented as an infant in swaddling-clothes. The *Santissimo Bambino* in the church of Ara Cœli at Rome, a richly-decorated figure carved in wood, is believed to be able to perform miracles and heal the sick. According to popular belief,

the painting was miraculously done by St. Luke or by an angel. Bambinos are set up for the adoration of the faithful in many places in Catholic countries.

Bambocciades (bam-boch-ādz'), pictures, generally grotesque, of common, rustic, or low life, such as those of Peter Van Laar, a Dutch painter of the seventeenth century, who on account of his deformity was called *Bamboccio* (cripple). Teniers is the great master of this style.

Bamboo, the common name of the arborescent grasses belonging to the genus *Bambusa*. There are many species, belonging to the warmer parts of Asia, Africa, and America, and growing from a few feet to as much as 100 feet, requiring much moisture to thrive properly. A peculiarity of the bamboo family is that they flower on reaching a certain age, generally about 130 years. The best-known species is *B. arundinacĕa*, common in tropical and sub-tropical regions. From the creeping underground rhizome, which is long, thick, and jointed, spring several round jointed stalks, which send out from their joints several shoots, the stalks also being armed at their joints with one or two sharp rigid spines. The oval leaves, 8 or 9 inches long, are placed on short footstalks. The flowers grow in large panicles from the joints of the stalk. Some stems grow to 8 or 10 inches in diameter, and are so hard and durable as to be used for building purposes. The smaller stalks are used for walking-sticks, flutes, &c.; and indeed the plant is used for innumerable purposes in the East Indies, China, and other Eastern countries. Cottages are almost wholly made of it; also, bridges, boxes, water-pipes, ladders, fences, bows and arrows, spears, baskets, mats, paper, masts for boats, &c. The young shoots may be either pickled or salted and eaten with rice, or candied and preserved in sugar; the seeds of some species are also eaten. The substance called *tabasheer* is a siliceous deposit that gathers at the



Bamboo (*Bambusa arundinacĕa*)

internodes of the stems. The bamboo is imported into Europe for various purposes, and is grown in British gardens.

Bambook´, a country in Western Africa between the Falémé and Senegal Rivers, about 140 miles in length, by 80 to 100 in breadth. It is on the whole hilly and somewhat rugged. The valleys and plains are remarkably fertile. The natives are Mandingoes, mostly professed Mahommedans ruled by independent chieftains, most of whom acknowledge the supremacy of France. Gold and ivory are exchanged for European goods.

Bambook-butter, shea-butter (q.v.).

Bam´borough Castle, an ancient English castle on the coast of Northumberland, formerly with connected estate the property of the Forsters, and forfeited to the Crown in 1715, both being purchased by Lord Crewe, Bishop of Durham, and bequeathed by him for charitable purposes.

Bambu´sa. See *Bamboo*.

Bam´ian, a valley and pass of Afghanistan, the latter at an elevation of 8496 feet, the only known pass over the Hindu Kush for artillery and heavy transport. The valley is one of the chief centres of Buddhist worship, and contains two remarkable colossal statues and other ancient monuments.

Bamo. See *Bhamo*.

Bampton Lectures, a course of lectures established in 1751 by John Bampton, Canon of Salisbury, who bequeathed certain property to the University of Oxford for the endowment of eight divinity lectures to be annually delivered. The subjects prescribed are mainly connected with the evidences of Christianity, and the lecturer must have taken the degree of M.A. at Oxford or Cambridge. The first course of lectures was delivered in 1780, and they have been delivered every year since, with the exceptions of 1834, 1835, and 1841. Among the more remarkable lectures were those by Dr. White in 1784, by Dr. Mant in 1812, by Reginald Heber in 1815, Whately in 1822, Milman in 1827, Dr. Hampden in 1832, H.L. Mansel in 1858, Canon Liddon in 1866, Canon Gore in 1891, J. R. Illingworth in 1894, and W. R. Inge in 1899. A similar course of lectures, the Hulsean, is annually delivered at Cambridge.

Ban, in political law, is equivalent to excommunication in ecclesiastical. In Teutonic history the *ban* was an edict of interdiction or proscription: thus, to put a prince under the ban of the empire was to divest him of his dignities, and to interdict all intercourse and all offices of humanity with the offender. Sometimes whole cities have been put under the *ban*, that is, deprived of their rights and privileges.

Ban, anciently, a title given to the military chiefs who guarded the eastern marches of Hungary; till 1919 the title of the Governor of Croatia and Slavonia. A province over which a ban is placed is called *banat*, and is equivalent to the German term *mark*. During the European War, Serbia occupied the banat to the south of Hungary, and her claims to that region were disputed both by Hungary and Roumania. The territory is bounded almost entirely by the Danube, Theiss, and Maros Rivers, except on the east, where it is shut in by the Transylvanian Carpathians. The banat was ultimately divided between Yugo-Slavia and Roumania. See *Temesvar*.

Bana'na, a plant of the genus *Musa*, nat. ord. Musaceæ, being *M. sapientum*, while the plantain is *M. paradisiāca*. It is originally indigenous to the East Indies, and an herbaceous plant with an underground stem. The apparent stem, which is sometimes as high as 30 feet, is formed of the closely-compacted sheaths of the leaves. The leaves are 6 to 10 feet long and 1 or more broad, with a strong midrib, from which the veins are given off at right angles; they are used for thatch, basket-making, &c., besides



Banana (*Musa sapientum*)

yielding a useful fibre. The spikes of the flowers grow nearly 4 feet long, in bunches, covered with purple-coloured bracts. The fruit is 4 to 10 or 12 inches long, and 1 inch or more in diameter; it grows in large bunches, weighing often from 40 to 80 lb. The pulp is soft and of a luscious taste; when ripe it is eaten raw or cooked. The banana is cultivated in tropical and sub-tropical countries, and is the food of millions. It is estimated that in Jamaica alone 40,000 acres are devoted to its culture. Excellent biscuits

may be made from banana meal, but it is not suitable for bread. Vast quantities are imported by Britain and the United States. Manila hemp is the product of a species of banana.

Bana´na, an African port, in Belgian Congo, at the mouth of the Congo.

Banana-bird, a pretty insessorial bird (*Icterus leucopteryx*), a native of the West Indies and the warmer parts of America. It is a lively bird, easily domesticated, tawny and black in colour, with white bars upon the wings.

Banat. See *Ban*.

Ban´bridge, a town of Ireland, County Down, 22 miles s.w. of Belfast, on the Bann. The manufacture of linen is carried on to a great extent in town and neighbourhood. Pop. 5101.

Banbury (ban'be-ri), a town of England, in Oxfordshire, long celebrated for its cheese, its cakes, and its ale; a parliamentary borough till 1885, and now giving name to a parliamentary division of the county. Pop. (municipal borough) (1921), 13,347.

Banca, an island belonging to the Dutch East Indies, between Sumatra and Borneo; area, 4446 sq. miles; pop. 113,658, of which a considerable proportion are Chinese. It is celebrated for its excellent tin, of which the annual yield is above 6000 tons; but it produces nothing else of any importance.

Banco, in commerce, a term employed to designate the money in which the banks of some countries keep or kept their accounts, in contradistinction to the current money of the place, which might vary in value or consist of light and foreign coins. The term was applied to the Hamburg bank accounts before the adoption (in 1873) of the new German coinage. The mark *banco* had a value of 1s. 5½*d.*; but there was no corresponding coin. See *Bank*.

Ban´croft, George, American historian, born near Worcester, Mass., 1800, died 1891. He was educated at Harvard and in Germany, where he made the acquaintance of many literary men of note. In 1823 he published a translation of Heeren's *Politics of Ancient Greece*, and a small volume of poems, and was also meditating and collecting materials for a history of the United States. Between 1834 and 1840 three volumes of his history were

published. In 1845 he was appointed Secretary of the Navy, and effected many reforms and improvements in that department. He was American ambassador to Britain from 1846 to 1849, when the University of Oxford conferred on him the honorary degree of D.C.L. He took the opportunity while in Europe to perfect his collections on American history. He returned to New York in 1849, and began to prepare for the press the fourth and fifth volumes of his history, which appeared in 1852. The sixth appeared in 1854, the seventh in 1858, the eighth soon after, but the ninth did not appear till 1866. From 1867 to 1874 he was Minister Plenipotentiary at the Court of Berlin. The tenth and last volume of his great work appeared in 1874. An additional section appeared first as a separate work in 1882: *History of the Formation of the Constitution of the United States*, and the whole came out in 6 vols. in 1884-5. He has also published many essays in the *North American Review* and other periodicals, a selection from which was published in 1855 under the title of *Miscellanies*.

Ban'croft, Richard, born in Lancashire 1544, died 1610, studied at Cambridge, entered the Church, and rose rapidly during the reign of Elizabeth till he obtained the see of London in 1597. James I made him Archbishop of Canterbury on the death of Whitgift. He suppressed the Puritans mercilessly, and they in return never ceased to abuse him.

Ban'croft, Sir Squire, English actor, born in London in 1841. After playing in Dublin and elsewhere he made his first appearance in London, at the Prince of Wales's Theatre, in 1865. He married Miss Marie Effie Wilton in 1867, and was knighted in 1897. In collaboration with his wife he has written *The Bancrofts: Recollections of Sixty Years* (1909).

Banda, a town and district of India, in the United Provinces. The town stands on a plain on the right bank of the Ken River, 95 miles s.w. from Allahabad, and is a considerable cotton mart. Pop. 22,000.—Area of district 2965 sq. miles. Pop. 660,000.

Banda Islands, a group belonging to Holland, Indian Archipelago, south of Ceram, Great Banda, the largest, being 12 miles long by 2 broad. They are beautiful islands, of volcanic origin, yielding quantities of nutmeg. Goenong Api, or Fire Mountain, is a cone-shaped volcano which rises 2320 feet above the sea. Pop. 10,000.

Bandajan´, a pass over a range of the Himálaya, Kashmir State, 14,854 feet above sea-level.

Bandan´na, a variety of silk handkerchief having a uniformly-dyed ground, usually of bright red or blue, ornamented with white or yellow circular, lozenge-shaped, or other simple figures produced by discharging the groundcolour.

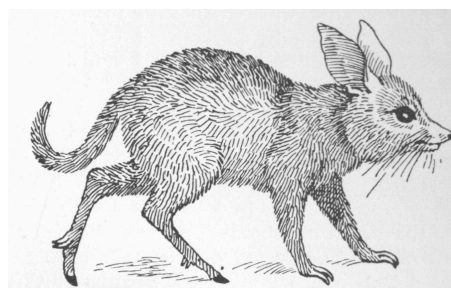
Banda Oriental. See *Uruguay*.

Bandel´lo, Matteo, an Italian writer of *novelle* or tales, born about 1480, died about 1562. He was, in his youth, a Dominican monk, and having been banished from Italy as a partisan of the French, Henry II of France gave him in 1550 the bishopric of Agen. He left the administration of his diocese to the Bishop of Grasse, and employed himself, at the advanced age of seventy, in the completion of his *novelle*. He also wrote poetry, but his fame rests on his *novelle*, which are in the style of Boccaccio's *Decameron* and the Queen of Navarre's *Heptameron*, and have been made use of by Shakespeare, Massinger, and Beaumont and Fletcher.

Bande Noire. See *French Revolution*.

Band-fish, the popular name of fishes of the genus *Cepōla*, from their long, flat, thin bodies. *C. rubescens*, a very fragile creature, is sometimes cast up on British shores. Also called *Snake-fish*, *Ribbon-fish*.

Ban´dicoot (*Perameles*), a genus of small Australian marsupials. The most common species (*P. nasuta*), the long-nosed bandicoot, measures about 1½ feet from the tip of the snout to the root of the tail. It is not unlike an overgrown rat, and is most destructive to farm-produce.



Pig-footed Bandicoot (*Chæropus castanotis*)

Ban´dicoot Rat, the *Mus gigantēus*, the largest known species of rat, attaining the weight of 2 or 3 lb., and the length, including the tail, of 24 to 30 inches. It is a native of India, and is very abundant in Ceylon. Its flesh is said to be delicate and to resemble young pork, and is a favourite article of diet with the coolies. It is destructive upon rice-fields and gardens.

Bandinelli, Baccio, Italian sculptor, born at Florence 1493, died there 1560. He was jealous of and strove to rival Michael Angelo. Among his works are a *Hercules and Cacus*, *The Dead Body of Christ held up by an Angel*, *Adam and Eve*, &c.

Ban´dit, It. *bandito*, originally an *exile*, banished man, or outlaw, and hence, as persons outlawed frequently adopted the profession of brigand or highwayman, the word came to be synonymous with brigand, and is now applied to members of the organized gangs which infest some districts of Italy, Sicily, Spain, Greece, and Turkey.

Band of Hope. See *Temperance*.

Ban´dolier, a large leathern belt or baldrick, to which were attached a bag for balls and a number of pipes or cases of wood or metal covered with leather, each containing a charge of gunpowder. It was worn by ancient musketeers, and hung from the left shoulder under the right arm with the ball bag at the lower extremity, and the pipes suspended on either side. The name is now given to a similar belt, worn by mounted troops, for carrying cartridges. Another form of the word is *bandoleer*.

Ban´doline, a gummy perfumed substance used to impart gloss and stiffness to the hair.

Ban´don, a town, Ireland, County Cork, on both sides of the Bandon. Pop. 3122.

Bands, a small article of clerical dress, made of linen, going round the neck and hanging down in front for a short distance in two pieces with square ends, supposed to be a relic of the amice.

Baneberry, or **Herb Christopher** (*Actæa spicāta*), a European plant, ord. Ranunculaceæ, local in England, with a spike of white flowers and black, poisonous berries. Two American species are considered remedies for rattlesnake bite.

Banér (ba´-nār´), Johan Gustafsson, a Swedish general in the Thirty Years' War, born 1596, died 1641. He made his first campaigns in Poland and Russia, and accompanied Gustavus Adolphus, who held him in high esteem, to Germany. After the death of Gustavus in 1632 he had the chief command

of the Swedish army, and in 1634 invaded Bohemia, defeated the Saxons at Wittstock, 24th Sept., 1636, and took Torgau. He ravaged Saxony again in 1639, gained another victory at Chemnitz, and in 1640 defeated Piccolomini. In Jan., 1641, he very nearly took Ratisbon by surprise.

Banff (bamf), county town of Banffshire, Scotland, a seaport on the Moray Firth at the mouth of the Deveron. It is well built, carries on some boat-building, and has ropeworks, a brewery, &c., fishing and shipping. Near the town are the county lunatic asylum, and Duff House, a seat of the Duke of Fife, now public property; on the east side of the Deveron is the town of Macduff, where an extensive fishing trade is carried on. Banff is one of the Elgin burghs, which together return a member to Parliament. Pop. 3517.—The county has an area of 403,053 acres. In the south it is mountainous; but the northern part is comparatively low and fertile; principal rivers, the Spey and Deveron; principal mountains, Cairngorm (4095 feet) and Ben Macdhui (4296 feet), on its southern boundary. Little wheat is raised, the principal crops being barley, oats, turnips, and potatoes. Fishing is an important industry; as is also the distilling of whisky. Serpentine abounds in several places, especially at Portsoy, where it is known as 'Portsoy marble', and Scotch topazes or cairngorm stones are found on the mountains in the south. Banffshire returns one member to Parliament. Pop. (1921), 57,293.

Banff, a town of N. W. Canada, at the foot of the Rocky Mountains, on the Canadian Pacific Railway, with National Park, hot springs, coal-mines, &c.

Bang. See *Bhang*.

Bangalore, a town of Hindustan, capital of Mysore, and giving its name to a considerable district in the east of Mysore State. The town stands on a healthy plateau 3000 feet above sea-level, has a total area of nearly 14 sq. miles, and is one of the pleasantest British stations in India. In the old town stands the fort, reconstructed by Hyder Ali in 1761, and taken by Lord Cornwallis in 1791. Under English administration the town has greatly prospered in recent times. There are manufactures of silks, cotton cloth, carpets, gold and silver lace, &c. Pop. 189,485.—The Bangalore district has an area of 3070 sq. miles, of which more than half represent cultivable land. Pop. about 760,000.

Bangkok´, or **Bankok**, the capital of the kingdom of Siam, extending for several miles on both sides of the Menam, which falls into the Gulf of Siam about 15 miles below. The inner city occupies an island surrounded with walls and bastions, and contains the palace of the king and other important buildings. The dwellings of the common people are of wood or bamboo, often raised on piles; a large portion of the population, however, dwell in boats or wooden houses erected on bamboo rafts moored in the river, and forming a floating town. Temples are numerous and lavishly decorated. Houses in the European style have been erected in modern times, and the telegraph, telephone, gas, fire-engines, and omnibuses introduced. A university with eight faculties has recently been established. The trade, both inland and foreign, is very extensive, the exports consisting chiefly of rice, sugar, silk, cotton, tobacco, pepper, sesame, ivory, aromatic wood, cabinet woods, tin, hides, &c.; and the imports consisting chiefly of British cotton, woollen, and other goods. Pop. 628,675, of whom about a half are Chinese. See *Siam*.

Ban´gor, a city of North Wales, in Carnarvonshire, picturesquely situated near the northern entrance of the Menai Strait. It appears to have possessed a cathedral in the sixth century, which was destroyed by the Normans in 1071. The present cathedral—the third—only dates from the reign of Henry VII. It has a college of the University of Wales (founded in 1903), and a Normal College. Bangor has risen into importance as a popular resort; its principal trade is in the export of slates from the neighbouring quarries. Pop. (1921), 11,032.

Ban´gor, a seaport town, Ireland, County Down, on the south side of Belfast Lough. Principal trade: cotton, linen, and embroideries. Pop. 7776.

Ban´gor, a port of the United States, in Maine, on the w. side of Penobscot River, a flourishing and pleasantly-situated town, and one of the largest lumber depots in the world. The river is navigable to the town for vessels of the largest size. Pop. (1920), 25,948.

Bango´rian Controversy, a controversy stirred up by a sermon preached before George I in 1717 by Dr. Hoadly, Bishop of Bangor, from the text "My kingdom is not of this world" (*John*, xviii, 36), in which the bishop contended in the most pronounced manner for the spiritual nature of Christ's

kingdom. The controversy was carried on with great heat for many years, and resulted in an enormous collection of pamphlets.

Bangs´ring. See *Banxring*.

Bangweo´lo, or **Bangweulu, Lake**, in South Africa, the southernmost of the great lake reservoirs of the Congo, heard of in 1798, was reached by Livingstone in 1868, an oval-shaped shallow sheet of water, said to be 150 miles in length along its greater axis, from east to west, and about 75 miles in width, but its exact limits are uncertain. It was first circumnavigated by Poulett Weatherley in 1896. See *Congo*.

Ban´ian, or **Ban´yan**, an Indian trader or merchant, one engaged in commerce generally, but more particularly one of the great traders of Western India, as in the seaports of Bombay, Kurrachee, &c., who carry on a large trade by means of caravans with the interior of Asia, and with Africa by vessels. They form a class of the Vaisya caste, wear a peculiar dress, and are strict in the observance of fasts and in abstaining from the use of flesh. Hence, *Banian days*, days in which sailors in the navy had no flesh meat served out to them. Banian days are now abolished, but the term is still applied to days of poor fare.

Banian Tree. See *Banyan*.

Ba´nim, John, an Irish novelist, dramatist, and poet, born in 1798, died 1842. His chief early work was a poem, *The Celt's Paradise* (1821). Having settled in London, he made various contributions to magazines and to the stage; but his fame rests on his novels, particularly the *O'Hara Tales*, in which Irish life is admirably portrayed. In these, as in some of his other publications, his brother, Michael Banim (born 1796, died 1874), had an important share, if not an equal claim to praise. The two brothers have been justly called 'the first national novelists of Ireland'.

Banishment. See *Exile*.

Ban´jarmassin, a district and town in the south-east of Borneo, under the government of the Dutch. The town is situated on an arm of the Banjar, about 14 miles above its mouth, in a marshy locality, the houses being built on piles, and many of them on rafts. Exports: pepper, benzoin, bezoar,

ratans, dragon's-blood, birds'-nests, &c.; imports: rice, salt, sugar, opium, &c. Pop. 52,000, mostly Dyaks.

Ban'jo (a negro corruption of *bandore*; It. *pandora*, from Lat. *pandura*, a three-stringed instrument), the favourite musical instrument of the negroes of the Southern States of America. It is six-stringed, has a body like a tambourine and a neck like a guitar, and is played by stopping the strings with the fingers of the left hand and twitching or striking them with the fingers of the right. The upper or octave string, however, is never stopped. The banjo was introduced into England in 1846.

Banjoemas (ba'n'yö-ma's), a town in Java, near the centre of the island, well built and of commercial importance; it is 22 miles from the coast, and is the residence of a Dutch governor. Pop. 6500.

Banking and Banks. Banks are establishments for the deposit, custody, and repayment on demand of money, and obtain the bulk of their profits from the investment of sums thus derived and not in immediate demand. The term is a derivative of the *banco* or bench of the early Italian money-dealers, being analogous in its origin to the terms *trapezītai* (*trapeza*, a bench or table) applied to the ancient Greek money-changers, and *mensarii* (*mensa*, a table) applied to the public bankers of Rome.

In respect of constitution there is a broad division of banks into public and private; public banks including such establishments as are under any special state or municipal control or patronage, or whose capital is in the form of stock or shares which are bought and sold in the open market; private banks embracing those which are carried on by one or more individuals without special authority or charter and under the laws regulating ordinary trading companies. In respect of function three kinds of banks may be discriminated: (1) banks of deposit merely, receiving and returning money at the convenience of depositors; (2) banks of discount or loan, borrowing money on deposit and lending it in the discount of promissory notes, bills of exchange, and negotiable securities; (3) banks of circulation or issue, which give currency to promissory notes of their own, payable to bearer and serving as a medium of exchange within the sphere of their banking operations. The more highly-organized banks discharge all three functions, but all modern banks unite the two first. For the successful working of a

banking establishment certain resources other than the deposits are of course necessary, and the capital paid up by shareholders on their shares and forming the substantial portion of their claim to public credit is held upon a different footing from the sums received from depositors. It is usually considered that for sound banking this capital should not be traded with for the purpose of making gain in the same way as the moneys deposited in the bank; and it is for the most part invested in Government or other securities subject to little fluctuation in value and readily convertible into money. But in any case prudence demands that a *reserve* be kept sufficient to meet all probable requirements of customers in event of commercial crises or minor panics. The reserve of the banking department of the Bank of England is always in coin, or in notes against which an equivalent value of coin and bullion is lying in the issue department. In other English banks the reserve is usually kept partly in gold and partly in Government stocks and Bank of England notes; but it sometimes lies as a deposit in the Bank of England. The working capital proper of a bank is constituted by moneys on deposit, for which the bank may or may not pay interest; the advantages of security, of ease in the transmission of payments, &c., being regarded in the cases of banks little affected by competition as a sufficient return to the depositor. Thus the Bank of England pays no interest on deposits, while the contrary practice has prevailed in Scotland since 1729, where interest is paid on deposits although not on current accounts.

Of the methods of making profit upon the money of depositors, one of the most common is to advance it in the discounting of bills of exchange not having long periods (seldom more than three months with the Bank of England) to run; the banker receiving the amounts of the bills from the acceptors when the bills arrive at maturity. Loans or advances are also often made by bankers upon exchequer bills or other Government securities, on railway debentures or the stock of public companies of various kinds, as well as upon goods lying in public warehouses, the dock-warrant or certificate of ownership being transferred to the banker in security. In the case of a well-established credit they may be advanced upon notes of hand without other security. Money is less commonly advanced by bankers upon mortgages on land, in which the money loaned is almost invariably locked up for a number of years. To banks of issue a further source of profit is open in their note circulation, inasmuch as the bank is enabled to lend these

notes, or promises to pay, as if they were so much money and to receive interest on the loan accordingly, as well as to make a profitable use of the money or property that may be received in exchange for its notes, so long as the latter remain in circulation. It is obvious, however, that this interest on its loaned notes may not run over a very extended period, in that the person to whom they are issued may at once return them to the bank to lie there as a deposit and so may actually draw interest on them from the bank of issue; or he may present them to be exchanged for coin, or by putting them at once into circulation may ensure a certain number speedily finding their way back through other hands or other banks to the establishment from which he received them. A considerable number of the notes issued will, however, be retained in circulation at the convenience of the public as a medium of exchange; and on this circulating portion a clear profit accrues. This rapid return of notes through other banks, &c., in exchange for portions of the reserve of the issuing bank, is one of the restraints upon an issue of notes in excess of the ability of the bank to meet them. In the United Kingdom a more obvious restraint upon an unlimited note issue, originating partly in a desire for greater security, partly in the belief that the note augmentation of the currency might lead to harmful economic results in its influence upon prices, is to be found in the Bank Acts of 1844 and 1845, which impose upon banks of issue the necessity of keeping an equivalent in gold for all notes issued beyond a certain fixed amount. The wisdom of these legal restrictions, which are not uniform throughout the kingdom, and the desirability of the acquisition and control by the State of the whole business of issue, are still matters of debate.

In specific relation to his customer the banker occupies the position of debtor to creditor, holding money which the customer may demand at any time in whole or in part by means of a cheque payable at sight on presentation during banking hours. For the refusal to cash a cheque from the erroneous supposition that he has no funds of his customer's in his hands, or for misleading statements respecting the position in which the bank stands, the banker is legally responsible. Moreover, the law regards him as bound to know his customer's signature, and the loss falls upon him in event of his cashing a forged cheque. In their relations to the community, the chief services rendered by banks are the following: By receiving deposits of money they are the means by which the surplus capital of one part of a

country is transferred to another where it may be advantageously employed in stimulating industry; they enable vast and numerous money transactions to be carried on without the intervention of coin or notes at all, thus obviating trouble, risk, and expense. The mechanism by which the last of these benefits is secured is to be found in perfection in the London Clearing House.

The modern tendency of banks is towards amalgamation. The large English banking institutions have absorbed many of the smaller banks. They have also made working arrangements with Scotch and Irish banks and with similar institutions abroad.

The result of these amalgamations is to give control of immense financial resources to a smaller number of banks. In 1918 the Government appointed a Committee to inquire into the effect of such amalgamations and absorptions. The terms of reference were: "To consider and report to what extent, if at all, amalgamations between banks may affect prejudicially the interests of the industrial and mercantile community, and whether it is desirable that legislation should be introduced to prohibit such amalgamations or to provide safeguards under which they might continue to be permitted"; and the Committee reported: "That legislation be passed requiring that the prior approval of the Government must be obtained before any amalgamations are announced or carried into effect".

In order, however, that such legislation may not have the effect of producing secret amalgamations, the Committee decided that "all proposals for agreements which would alter the status of a bank as regards its separate entity and control, or for purchase by one bank of the shares of another bank, be also submitted for the prior approval of the Government before they are carried out".

Another development is the registration of the British Overseas Bank, Limited, with a nominal capital of £5,000,000. The principal object of this institution is to carry on general banking business in all parts of the world. Four banks—two English, one Scotch, and one having connections with South America—are financially interested in the Overseas Bank.

Banking operations on a considerable scale appear to have been conducted by the ancients, and recent excavations have proved that in the seventh

century B.C. banking was practised at Bagdad by a firm of Egibi & Sons. Modern banking, however, must be regarded as having had an independent origin in the reviving civilization of the Middle Ages. In the twelfth century almost the whole trade of Europe was in the hands of the Italian cities, and it was in these that the need of bankers was first felt. The earliest public bank, that of Venice, established in 1171 and existing down to the dissolution of the Republic in 1797, was for some time a bank of deposit only, the Government being responsible for the deposits, and the whole capital being in effect a public loan. In the early periods of the operations of this bank deposits could not be withdrawn, but the depositor had a credit at the bank to the amount deposited, this credit being transferable to another person in place of money payment. Subsequently deposits were allowed to be withdrawn, the original system proving inconvenient outside the Venetian boundaries. It is, however, less from the Bank of Venice than from the Florentine bankers of the thirteenth and fourteenth centuries that modern banking specially dates, the magnitude of their operations being indicated by the fact that between 1430 and 1433, 76 bankers of Florence issued on loan nearly 5,000,000 gold florins. The Bank of St. George at Genoa also furnished a striking chapter in financial history. The important Bank of Amsterdam, taken by Adam Smith as a type of the older banks, was established in 1509, and owed its origin to the fluctuation and uncertainty induced by the clipped and worn currency. The object of the institution (established under guarantee of the city) was to give a certain and unquestionable value to a bill on Amsterdam; and for this purpose the various coins were received in deposit at the bank at their real value in standard coin, less a small charge for recoinage and expense of management. For the amount deposited a credit was opened on the books of the bank, by the transfer of which payments could be made, this so-called *bank money* being of uniform value as representing money at the mint standard. It bore, therefore, an *agio* or premium above the worn-coin currency, and it was legally compulsory to make all payments of 600 guilders and upwards in bank money. The deposits were supposed to remain in the coffers of the bank, but they were secretly traded with in the eighteenth century till the collapse of the bank in 1790. Banks of a similar character were established at Nuremberg and other towns, the most important being the Bank of Hamburg, founded in 1619. In England there was no corresponding institution, the London merchants being in the habit

of lodging their money at the Mint in the Tower, until Charles I appropriated the whole of it (£200,000) in 1640. Thenceforth they lodged it with the goldsmiths, who began to do banking business in a small way, encouraging deposits by allowing interest for their use, lending money for short periods, discounting bills, &c. The bank-note was first invented and issued in 1690 by the Bank of Stockholm, founded by Palmstruck in 1688, and one of the most successful of banking establishments. About the same time the banks of England and Scotland began to take shape, opening up a new era in the financing of commerce and industry.

Bank of England.—The Bank of England, the most important banking establishment in the world, was projected by William Paterson, who was afterwards the promoter of the disastrous Darien scheme. It was the first public bank in the United Kingdom, and was chartered in 1694 by an Act which, among other things, secured certain recompenses to such persons as should advance the sum of £1,500,000 towards carrying on the war against France. Subscribers to the loan became, under the Act, stockholders, to the amount of their respective subscriptions, in the capital stock of a corporation, denominated the *Governor and Company of the Bank of England*. The company thus formed advanced to the Government £1,200,000 at an interest of 8 per cent—the Government making an additional bonus or allowance to the bank of £4000 annually for the management of this loan (which, in fact, constituted the capital of the bank), and for settling the interest and making transfers, &c., among the various stockholders. This bank, like that of Venice, was thus originally an engine of the Government, and not a mere commercial establishment. Its capital has been added to from time to time, the original capital of £1,200,000 having increased to £14,553,000 in 1816, since which no further augmentation has taken place. There exists besides, however, a variable 'rest' of over £3,000,000. The charter of the bank was originally granted for eleven years certain, or till a year's notice after 1st Aug., 1705. It was subsequently renewed for various periods in 1697, 1708, 1713, 1742, 1764, 1781, 1800, 1833, and 1844, certain conditions which the bank had to fulfil being specified at each renewal. On this last occasion it was continued till twelve months' notice from 1855. At the same time the issue department of the bank was established as distinct from the general banking department, the sole business entrusted to the former being the issue of notes. By this

arrangement the bank was authorized to issue notes to the value of £14,000,000 upon securities specially set apart, the most important of the securities being the sum of £11,015,100 due to the bank by the Government, together with so much of the coin and bullion then held by the bank as was not required by the banking department. The bank has since been permitted to increase its issue on securities to £18,450,000, but for every note that the issue department may issue beyond the total sum of £18,450,000 an equivalent amount of coin or bullion must be paid into the coffers of the bank. For its special privileges the bank has to pay to the Government an annual sum of £180,000, and the profit derived from the issue of notes beyond £14,000,000. The Bank of England notes are practically equivalent to gold. They are (like all English bank-notes) of the value of £5 and upwards, and are legal tender throughout England. Notes once issued by the bank and returned to it are not reissued but are destroyed—a system adopted in order to facilitate the keeping of an account of the numbers of the notes in circulation, and so prevent forgery. On 7th Aug., 1914, Government notes for £1, and ten shillings, were issued convertible into gold at the Bank of England.

In compliance also with the Bank Charter Act of 1844 the bank is compelled to publish a weekly account, of which the following is a specimen:—

<i>Dr.</i>		ISSUE DEPARTMENT. Week ending 11th August, 1920	
		<i>Cr.</i>	
Notes issued	£139,980,460	Government debt	£11,015,100
		Other securities	7,434,900
		Gold coin and bullion	121,530,460
		Silver bullion	—
	£139,980,460		£139,980,460

<i>Dr.</i>		BANKING DEPARTMENT	
		<i>Cr.</i>	
Capital	£14,553,000	Government securities	£68,250,449

Rest	3,458,795	Other securities	73,805,565
Public deposits	15,999,059	Notes	14,452,665
Other deposits	124,018,988	Gold and silver coin	1,537,092
Seven-day and other bills	15,929		
	£158,045,771		£158,045,771

The total of the notes given out by the issue department is called the 'issue circulation', the portion of it in the hands of the public being the 'active circulation', and that still in the banking department being the 'note reserve'. This 'note reserve' represents really the amount of bullion in the issue department available for the use of the banking department. Of the other items in the account it may be noted that the proprietors' 'rest' is a varying surplus increased always by accumulated profits up to 5th April and 10th Oct., when the bank dividends are paid to the shareholders; and that the public deposits, which include sums lodged on account of the customs, inland revenue, &c., increase through revenue receipts until the dividend terms in January, April, July, and October. The other or private deposits comprise those of bankers, merchants, and other persons. An increase in these private deposits indicates an increase of monetary ease, while a decrease informs us that bankers, merchants, and traders have calls upon them for money. A better indication of the demand for money is furnished, however, by the advances on commercial securities, and it is by this and the condition of the reserve that the bank rate of discount is regulated. When the reserve is high and the advances moderate the discount rate is low, and it is raised according as the reserve falls and advances are more in request, especially during an adverse foreign exchange and drain of gold. Gold is thus restrained from going abroad, and its influx into the country is encouraged. In addition to the profit which the bank may make by ordinary banking business, by the Bank Act of 1892 and the Revenue Act of 1906, it receives an allowance for the management of the national debt, at the rate of £325 per million on 500 millions, and £100 per million on all debt above that sum. The annual sum is not to be less than £160,000. It also derives a profit from foreign gold coin and bullion brought to it, for which it pays £3, 17s. 9d., or 1½d. per ounce less than the real value.

The management of the bank is in the hands of a governor, deputy-governor, and twenty-four directors, elected by stock-holders who have held £500 of stock for six months previous to the election. A director is required to hold £2000, a deputy-governor £3000, and a governor £4000 of the stock. The court or board of directors meets every Thursday, when the weekly account is presented. The Bank of England has branches in Liverpool, Manchester, Newcastle, Birmingham, Leeds, Hull, Bristol, Plymouth, and Portsmouth.

The other English banks consist of numerous joint-stock and private banks in London and the provinces, many of the provincial establishments of both kinds having the right to issue notes. Private banks in London with not more than six partners have never been prevented from issuing notes, but they could not profitably compete with the Bank of England. The maximum issues of the provincial banks are limited to a certain amount, against which they are not compelled to hold gold in reserve, and they have no power to issue against specie in excess of the fixed circulation. Their actual issues are considerably below this amount. No union can take place between a joint-stock bank and a private bank, or between two joint-stock banks of issue, without one of them losing its issue. Their total authorized circulation is about £2,958,900, but the actual circulation of these banks is now only about £1,200,000, being distributed among about eighteen private and about twenty joint-stock banks. The notes of these banks are payable in Bank of England paper. The greater number of joint-stock banks are of limited liability, though their liability in respect of their notes is unlimited. Some of them have a large number of branches. All the joint-stock banks allow interest on money deposited with them. The total paid-up capital and reserves of the English joint-stock banks is about £100,000,000.

Scotland.—In Scotland there are no private banks, the only banks in that portion of the United Kingdom being eight joint-stock banks of issue, and their branches (the Mercantile and Savings not being banks of issue). By the Act of 1845 new banks of issue were prohibited, a monopoly being given to such establishments (nineteen in number) as existed in the year previous to 1st May, 1845. At the same time the issue of each was limited to the amount of its average circulation during that year, together with the specie held at the head office. Any bank issuing notes in excess of this limit is supposed to hold an equivalent amount of gold. The aggregate authorized circulation is

now £2,676,350; the average actual circulation is fully £7,000,000. The Bank of Scotland, established by Act of Parliament in 1695, had for its original capital only £100,000, increased to £200,000 in 1744; but it now has a capital of £1,325,000 paid up. It remained the only bank in Scotland till the Royal Bank of Scotland was established in 1727, with an original capital of £151,000, which has grown to £2,000,000 paid up. The British Linen Company was incorporated in 1746, for the purpose of promoting the linen manufacture, but soon became a general banking company; capital, £1,250,000 paid up. These three banks claim to be by their charters banks of limited liability. All the other Scottish banks have been established within the nineteenth century. They are all incorporated by royal charter or Act of Parliament, which enables them to sue and be sued as a corporation, and latterly they have all become banks of limited liability, except that their liability is not to be limited in respect to their note issue. The total paid-up capital of the Scotch banks is £9,300,000. A large number of one-pound notes circulate in Scotland, thus tending to keep the requirements for gold low. From allowing a moderate rate of interest on deposit accounts (although not on current accounts) it is not uncommon for depositors in Scottish banks to lodge their money permanently as an investment; and the habit of keeping an account with a banker is much more general in Scotland than in England, branch offices of the banks being very numerous. Several of the Scotch banks have branch offices in London, but of course they cannot issue their own notes from these offices. The Scotch banks have enjoyed a high reputation for stability, and though public confidence was somewhat shaken by the failure of the Western Bank in 1857, and even more rudely by that of the City of Glasgow Bank in 1878, their shares are generally looked upon as a safe and remunerative investment. Their total deposits amount to fully £107,000,000.

Ireland.—The banks in Ireland consist of one public or national bank, the Bank of Ireland, and of sundry joint-stock and private banks. The authorized note circulation is arranged on the same footing as that of the Scotch banks. If any bank discontinues its issue and issues notes of the Bank of Ireland, the circulation of the latter may be to an equal amount increased. The authorized circulation is £6,354,494; the actual circulation is sometimes a little above, sometimes a little below. The Bank of Ireland, which was established by charter in 1783 with similar privileges to those

granted to the Bank of England, has lent the greater portion of its capital to Government. Its capital is £2,769,230 (or £3,000,000 Irish); it has also a rest or reserve of over £1,000,000. The bank allows interest on deposits when lodged for a stated period.

Banque de France.—The Bank of France is second in importance only to the Bank of England. It was established in the beginning of the nineteenth century, at first with a capital of 45,000,000 francs, and with the exclusive privilege in Paris of issuing notes payable to bearer, a privilege which was extended in 1848 to cover the whole of France. It has numerous branches in the larger towns, a number of these having been acquired in 1848, when certain joint-stock banks of issue were by Government decree incorporated with the Bank of France, the capital of which was then increased to 91,250,000 francs (£3,650,000), in 91,250 shares of 1000 francs each. In 1857 the capital was doubled, and besides this it has a large surplus capital or rest. Like the Bank of England, it is a bank of deposit, discount, and circulation, and is a large creditor of the State. The Government appoints the governor and the two deputy-governors, who are all required to be stockholders. There is also a body of fifteen directors and three censors, nominated by the shareholders. The value of its note circulation in March, 1922, was 36,225,852,000 francs.

British Colonies.—With regard to the banks in British colonies little need be said. All the more important are joint-stock concerns, and they are carried on subject to Acts passed by the respective colonial legislatures. Some of them have their head-quarters in London, and have been established by English capital. In Canada the banks are not allowed to issue notes of lower denominations than five dollars, notes for one and two dollars and higher amounts being issued by the Dominion Government; and the banking laws are such that there is no possibility of holders of bank-notes being losers by them. The total paid-up capital of the Canadian banks is about £20,000,000; their total deposits about £135,000,000.

United States.—The more important of the banks of the United States are what are called national banks, established in accordance with an Act passed in 1863. Associations of this kind at starting must invest at least a third of their paid-up capital in Government bonds, which pay them an interest of 4 per cent more or less. They then obtain from the Government

bureau, established for the purpose, 90 per cent of paper-money sheets, which they sign and pay out, this constituting their note circulation. These banks pay no interest to depositors. Besides the notes of these banks a large portion of the currency of the United States consists of Government notes issued from the national treasury. There are also banks chartered by the different States and private banks.

Savings-banks are banks established for the receiving of small sums, so as to be taken advantage of by the poorer classes, and they are carried on entirely for the benefit of the depositors. They are of comparatively recent origin, one of the earliest having been an institution in which small sums were received, and interest allowed on them, established by Mrs. Priscilla Wakefield, at Tottenham, near London, in 1803. The first savings-bank in Scotland was formed in 1810 by the Rev. Henry Duncan, of Ruthwell, Dumfriesshire. In 1814 the Edinburgh savings-bank was established on the same principles, and the system soon spread over the kingdom. The first Act relating to savings-banks was passed in 1817. By it all deposits in savings-banks, as soon as they reached £50, were placed in the hands of the National Debt Commissioners, who allowed interest on them. In 1824 it was enacted that the deposits for the first year should not exceed £50, nor those in subsequent years £30, the total deposits being limited to £150, and interest ceasing to be paid when accrued interest made the total £200. By the Act of 1893 the limit of annual deposit was raised to £50, and the interest is now rather less than 3 per cent. An Act of 1833 had provided for the purchase of Government annuities by depositors either for life or for a term of years; and an annuity of any amount up to £100 may now be obtained. Depositors in savings-banks (by an Act of 1880) can also have their money invested for them in Government stock, the banks collecting and paying the dividends; and when accrued interest raises an account above £200 the excess is now so invested for the depositor. The total amount invested by one person in Government stock is not to exceed £500, nor £200 in one year. These banks are managed by local trustees having no personal interest in the business, and by certain paid officers. A new class of savings-banks, namely, *Post Office Savings-banks*, was established in Britain in connection with the money-order department of the Post Office in 1861. Any sum not less than a shilling is received, provided that the total amount banked does not exceed £50 in one year, or more than £200 in all;

the excess of accrued interest above this being invested in Government stock. Interest is paid on every complete pound at the rate of 2½ per cent. For the deposits the Government is responsible, and they may be drawn from any Post Office Savings-bank in the kingdom. These savings-banks have become very numerous, and much of the funds formerly in the trustees' savings-banks has been transferred to them. The total amount deposited in the old banks is now about £52,000,000, in the new about £157,600,000. The regulations regarding the purchase of Government stock and annuities correspond with those given above. Savings-banks are now well known in all civilized countries, and the good they have done is incalculable. In the United States there is an enormous amount of money deposited in them. Post Office Savings-banks have been proposed to be established in the States, but have not yet been so. In Canada, Australia, and other British colonies they are established, as well as savings-banks of several other kinds. School savings-banks are the most recent institutions of this kind, and have had a marked effect for good.

Bank Holidays, in England and Ireland, Good Friday, Easter Monday, Whit Monday, first Monday in August, Christmas and following day, or 27th (if next day is a Sunday); in Scotland, New-Year's Day, first Monday of May and August and Christmas.

Bankrupt (from It. *banca rotta* 'bank broken' or 'bench broken'. Dr. Johnson said that the word originated from the Italian custom of breaking the bench of an insolvent money-changer; but *rotta* also means 'interrupted' or 'stopped', and is here used more in the sense of our colloquial word *broken*, and means 'insolvent'), a person whom the law does or may take cognizance of as unable to pay his debts. Properly it is of narrower signification than *insolvent*, an insolvent person simply being unable to pay all his debts. In England up till 1861 the term *bankrupt* was limited to an insolvent trader, and such traders were on a different footing from other insolvent persons, the latter not getting the same legal relief from their debts. In all civilized communities laws have been passed regarding bankruptcy. At present bankruptcy in England is regulated by the Bankruptcy Acts of 1883 and 1890, which have as one chief feature the intervention of the Board of Trade in the proceedings, with the object of obtaining full official supervision and control. A bankruptcy petition may be presented either by a creditor or a debtor. A creditor's petition must be

founded on a debt of not less than fifty pounds, due to one or more creditors, and on an 'act of bankruptcy' committed by the debtor within three months before the presentation of the petition. A debtor commits an act of bankruptcy if he makes a conveyance of his property to a trustee for the benefit of his creditors; if he makes a fraudulent transfer of any part of his property; if, to defeat or delay his creditors, he conceal himself either at home or abroad; if execution issued against him has been enforced by seizure and sale of his goods under process in an action in any court; if he files in court a declaration of inability to pay his debts, or presents a bankruptcy petition against himself; if a creditor has obtained a final judgment against him for any amount and he fail to pay the judgment debt without satisfactory reason; or if the debtor gives notice to any of his creditors that he has suspended, or is about to suspend, payment of his debts. In London jurisdiction in bankruptcy now rests with the High Court of Justice, while the county courts continue to have jurisdiction in bankruptcy outside the London district. When the court is satisfied as to the petition, a 'receiving order' is issued to protect the debtor's estate by constituting the official appointed by the Board of Trade receiver of the debtor's property, and to stay the remedies of all creditors until the meeting of creditors. The debtor must make out a full statement of his affairs, accounting as best he can for his insolvency. The official receiver summons the meeting of creditors, a summary of the debtor's affairs being sent to each creditor with the notice of the meeting, which is also advertised in the *London Gazette*. The creditors must send to the official receiver, one day before the meeting, sworn proofs of their claims to enable them to vote. At the meeting the creditors (unless the debtor's proposal for a composition or scheme be entertained) pass a resolution adjudging the debtor bankrupt, and appoint a trustee of the bankrupt's property, with a committee of inspection selected from their own body to superintend the administration of the bankrupt's property by the trustee, who divides the available realized assets amongst all creditors who have sent sworn proofs of claims. Rates, assessments, and taxes, and all wages or salary of a clerk, servant, labourer, or workman during four months before the date of the receiving order not exceeding £50 are paid in priority to all other debts. The trustee is required to give satisfactory security to the Board of Trade, by which his accounts are audited not less than twice in each year. All moneys received by the trustee under the bankruptcy must be paid forthwith to an account kept at

the Bank of England by the Board of Trade, called the 'Bankruptcy Estates Account'. The debtor is bound to be publicly examined upon oath in court, and any creditor who has tendered a proof, or his representative, may take part in the examination. Until the debtor has passed his public examination he cannot apply for an order of discharge, and upon proof of misdemeanour the court refuses or suspends the discharge. A discharged bankrupt is disqualified for five years from acting as member of Parliament, justice of peace, alderman, overseer of the poor, county councillor, or as a member of any school, highway, or burial board. An undischarged bankrupt obtaining credit to the extent of £20 or upwards from any person, without informing such person of his status, is guilty of a misdemeanour. By the Act of 1883 the creditors may at the first meeting resolve to entertain a proposal for a composition or scheme of arrangement of the debtor's affairs, but the composition or scheme shall not be binding on the creditors, unless confirmed at a second meeting by a majority in number representing three-fourths in value of all the creditors who have proved. The composition or scheme has then to be formally brought before the court for approval, which may be refused. A composition or scheme may be sanctioned by the court after the debtor's adjudication as a bankrupt, and in this case the bankruptcy is annulled. Though imprisonment for debt has been abolished, fraudulent bankrupts may be punished, and the conduct of prosecutions for offences arising out of any bankruptcy proceeding falls to the public prosecutor. The estates of persons dying insolvent may be administered according to the law of bankruptcy. The Act of 1913 embodies suggestions of the Bankruptcy Law Amendment Committee of 1906. Thus, the Summary Jurisdiction Act can now be applied to offenders under the Debtors' Act. Whereas previously conviction for such offences as failure to keep proper accounts and concealment of debt could only be by trial before a jury, they are punishable after summary trial before a stipendiary magistrate or justice of the peace, the onus of proving innocence falling on the debtor. A married woman, too, may be made bankrupt even if not trading separately from her husband.

According to Scots law bankruptcy is notorious insolvency, that is, a public acknowledgment of inability to discharge obligations. By a judicial proceeding, called *sequestration*, authorized by the Court of Session or sheriff court, on the petition of the debtor himself with the concurrence of one creditor swearing to a debt of £50, two whose debts together amount to

£70, or of any number of creditors whose debts together amount to £100; or on the petition of a creditor or creditors to the foregoing extent without the concurrence of the debtor, if he has clearly shown himself to be insolvent (or a *notour bankrupt*), the whole estates and effects of the debtor, real and personal, are legally taken for behoof of the creditors. The debtor's estate is then made over to a trustee chosen by the creditors, the trustee being charged to bring the whole estate into the form of money, with certain precautions, and to receive, investigate, and reject or admit the claims of the creditors, subject to review of the Court of Session or sheriff court by summary petition. The debtor, and all who can give information as to the estate, must submit to public examination on oath before the sheriff of the county, and the debtor may thereafter, or by petition after six, twelve, or eighteen months from sequestration, be discharged of all debts by the court with consent of the creditors or a number of them, or at the expiry of two years without consent. These proceedings may be partly superseded by 'composition' if such be assented to by a majority in number and nine-tenths in value of creditors, or by a majority in number and four-fifths in value of the creditors, according to the period at which such arrangement may be proposed. They may also be terminated by a deed of arrangement entered into between the bankrupt and a majority in number and four-fifths in value of his creditors, approved of by the court. Before a discharge is given there must be a report from the trustee as to the conduct of the bankrupt, whether he has complied with the provisions of the Act, whether his bankruptcy is culpable or not, &c. Before the abolition of imprisonment for ordinary civil debts by Act passed in 1880, an insolvent debtor often took advantage of a form of process by which, on making a complete *cessio bonorum*, or surrender to his creditors of all his property, he might obtain protection from imprisonment. Though no person can now be imprisoned for ordinary debts, a creditor of a *notour bankrupt* may present a petition to the sheriff, praying him to decern that the debtor assign over all his goods for behoof of his creditors and that a trustee be appointed; and this proceeding is still designated a process of *cessio bonorum*. The Act of 1880 also provides for the better punishment of fraudulent debtors in Scotland. If the debtor's liabilities exceed £200, the sheriff may award *sequestration* of the debtor's estate instead of *cessio bonorum*. There is no appeal from the sheriff's decision with respect to discharge.

In Ireland there is a special code of bankruptcy contained in special Acts, differing to some extent from the regulations prevailing both in England and Scotland. Thus in Ireland it is not an offence for an undischarged bankrupt to obtain credit to any amount without disclosing the fact of his bankruptcy. All bankruptcy business comes before the court at Dublin, Belfast, or Cork. Imprisonment for debt was abolished in 1872.

Official returns in Great Britain show a tendency towards a smaller number of failures, but a larger amount of money is involved.

In the different British colonies the laws regulating bankruptcy naturally differ, and the same is the case with the individual States of the American Union, though Congress has the power of legislating for the whole country in regard to this, and has oftener than once done so.—BIBLIOGRAPHY: H. Goudy, *Treatise of the Law of Bankruptcy in Scotland*; Sir R. H. J. Palgrave, *Dictionary of Political Economy*; E. T. Baldwin, *The Law of Bankruptcy and Bills of Sale*; Sir R. L. V. Williams, *The Law and Practice of Bankruptcy*; R. Ringwood, *The Principles of Bankruptcy*.

Banks, Sir Joseph, Baronet, a distinguished naturalist, born in London 1743. After studying at Harrow and Eton he went to Oxford in 1760, and formed there amongst his fellow-undergraduates a voluntary class in botany, &c. He was elected Fellow of the Royal Society in 1766, and soon after went to Newfoundland and Hudson's Bay to collect plants. In 1768, with Dr. Solander, a Swedish gentleman, pupil of Linnæus, and then assistant librarian at the British Museum, he accompanied Cook's expedition as naturalist. In 1772 he visited Iceland along with Dr. Solander, and during this voyage the Hebrides were examined, and the columnar formation of the rocks of Staffa first made known to geologists. In 1777 Banks was chosen President of the Royal Society, and in 1781 was made a baronet, and in 1795 received the Order of the Bath. He wrote only essays, papers for learned societies, and short treatises. He died in 1820, and bequeathed his collections to the British Museum.

Banks, Thomas, an English sculptor, born in 1735, died in 1805. He studied sculpture in the Royal Academy, where he obtained the gold medal for a bas-relief of the *Rape of Proserpine*. In 1772 he went to Italy, where he executed several excellent pieces, particularly *Cupid tormenting a Butterfly*,

which was afterwards purchased by the Empress Catherine. On leaving Italy he spent two unsatisfactory years in Russia, and then returned to England, where he was soon after made an Academician. One of his best-known works is *Shakespeare attended by Painting and Poetry*. Among his other works was a colossal statue—*Achilles mourning the loss of Briseis*—in the hall of the British Institution, and the monument of Sir Eyre Coote in Westminster Abbey. His bust of Warren Hastings is in the National Gallery.

Bank'sia (named after Sir Joseph Banks), a genus of Australian shrubs and trees, order Proteaceæ, with leathery leaves generally dark-green on the upper surface and pale below, often cultivated in conservatories for their peculiar foliage and flowers. They are named "honeysuckles" by the colonists, from the honey the flowers contain.

Banks'ring. See *Banxring*.

Bankura', a town of Bengal, on the Dhalkisor River, healthy and with a considerable trade Pop. 21,000.

Bann, Upper and Lower, two rivers in the N. of Ireland, the former rising in the mountains of Mourne, County Down, and, after flowing 38 miles in a northerly direction, falling into Lough Neagh, the latter being the outlet of Lough Neagh, and falling into the Atlantic Ocean 4 miles below Coleraine, after a course of nearly 40 miles.

Ban'natyne Club, a literary society instituted in Edinburgh (1823) by Sir Walter Scott (its first president), David Laing (secretary till its dissolution in 1861), Archibald Constable, and Thomas Thomson. It started with thirty-one members, subsequently extended to 100, having as its object the printing of rare works on Scottish history, literature, geography, &c. It derived its name from George Bannatyne (1545-1609), the collector of the famous MS. of early Scottish poetry.

Ban'neret, formerly, in England, a knight made on the field of battle as a reward for bravery, with the ceremony of cutting off the point of his pennon and making it a banner. The first banneret in England was made by King Edward I, and the last (John Smith) by Charles I after the battle of Edgehill in 1642.

Ban'nock, a cake made of oatmeal, barley-meal, or peasemeal baked on an iron plate or griddle over the fire. From a supposed resemblance the turbot is sometimes called in Scotland the *Bannock-fluke*.

Bannockburn, a village of Scotland, in Stirlingshire, 2 miles S.E. Stirling, famous for the decisive battle in which King Robert Bruce of Scotland defeated Edward II of England, on the 24th June, 1314. It has manufactures of woollens, such as tartans, carpets, &c.; pop. 4103.

Banns of Marriage, public notice of the intended celebration of a marriage given either by proclamation, viva voce, by a clergyman, session-clerk, or precentor in some religious assembly, or by posting up written notice in some public place. Dissent of parents or guardians renders null and void the publication of the banns of minors. In France the banns must be published on two distinct Sundays, and the marriage cannot take place until three days after the second publication. In America the practice is confined to the Roman Catholics, although it is still recognized in the statutes of some of the States.

Bannu, a district in the north-west of Hindustan, traversed by the Indus; area, 3847 sq. miles; pop. 390,000, nearly half being Afghans.

Banquette (bang-ket') in fortification, the elevation of earth behind a parapet, on which the garrison or defenders may stand. The height of the parapet above the banquette is usually about 4 feet 6 inches; the breadth of the banquette from 2½ or 3 feet to 4 or 6 feet, according to the number of ranks to occupy it. It is frequently made double, that is, a second is made still lower.

Banshee', or **Benshi'**, a fairy woman believed in Ireland and some parts of Scotland to attach herself to a particular house, and to appear or make her presence known by wailing before the death of one of the family.

Ban'tam, a residency occupying the whole of the w. end of the Island of Java. It formed an independent kingdom, governed by its own sultan, till 1683, and the Dutch exercised suzerainty with brief intermission until its formal incorporation by them at the beginning of last century. It produces rice, coffee, sugar, cinnamon, &c. Serang is its capital. The town Bantam

was the first Dutch settlement in Java (1595), and for some time their principal mart, though now not so prosperous.

Ban'tam Fowl, a small but spirited breed of domestic fowl, first brought from the East Indies, supposed to derive its name from Bantam in Java. Most of the sub-varieties have feathered legs; but these are not to be preferred. In point of colour the black and nankeen varieties are the best. A well-bred bantam does not weigh more than a pound.

Banteng' (*Bos Banteng* or *Sondaicus*), a wild species of ox, native of Java and Borneo, having a black body, slender white legs, short sleek hair, sharp muzzle, and the back humped behind the neck.

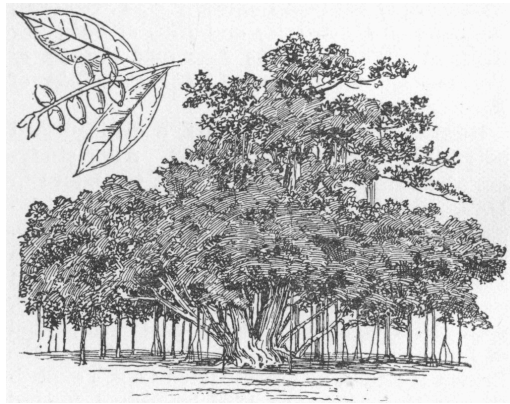
Banting System, a course of diet for reducing superfluous fat, adopted and recommended in 1863 by W. Banting of London. The dietary recommended was the use of butcher-meat principally, and abstinence from beer, farinaceous food, and vegetables. See *Corpulence*.

Ban'try, a seaport near the head of Bantry Bay, County Cork, Ireland. Pop. 3159.—The *bay*, one of three large inlets at the s.w. extremity of Ireland, affords an unsurpassed anchorage, and is about 25 miles long by 4 to 6 broad, and from 10 to 40 fathoms deep, with no dangerous rocks or shoals.

Bantu (bän-tö'), the ethnological name of a group of African races below about 6° N. latitude, and including the Kaffirs, Zulus, Bechuanas, the tribes of the Loango, Congo, &c., but not the Hottentots.—BIBLIOGRAPHY: Deniker, *Races of Man*; J. Roscoe, *The Northern Bantu*.

Banu. See *Bannu*.

Banx'ring (genus *Tapaia*), a quadruped belonging to the Insectivora, inhabiting the Indian Archipelago, bearing some resemblance externally to a squirrel, but having a long pointed snout.



Banyan (*Ficus benghalensis*)

Ban'yan (*Ficus benghalensis*), a tree of India, of the fig genus. The most peculiar feature of this tree is its method of throwing out from the horizontal branches supports which take root as soon as they reach the ground, enlarge into trunks, and, extending branches in their turn, soon cover a prodigious extent of ground. A celebrated banyan tree has been known to shelter 7000 men beneath its shade. The wood is soft and porous, and from its white glutinous juice bird-lime is sometimes prepared. Both juice and bark are regarded by the Hindus as valuable medicines, and the tree itself is sacred. The banyan tree is described by Southey in his poem *The Curse of Kehama*.

Ba'obab (*Adansonia digitāta*) or **Monkey-bread Tree**, a tree belonging to the nat. ord. (or sub-ord.) Bombaceæ, and the best-known species of its genus, which was named after the naturalist Adanson. Though not tall, it is one of the bulkiest of trees, its trunk sometimes measuring 30 feet in diameter, and as the profusion of leaves and drooping boughs sometimes almost hides the stem, the whole forms a hemispherical mass of verdure 140 to 150 feet in diameter and 60 to 70 feet high. It is a native of the African savannahs, and is cultivated in many of the warmer parts of the world. The roots are of extraordinary length, a tree 77 feet in girth having a tap-root



Baobab or Monkey-bread Tree (*Adansonia digitata*)

110 feet in length. The leaves are deep green, divided into five unequal parts lanceolate in shape, and radiating from a common centre. The flowers resemble the white poppy, having snowy petals and violet-coloured stamens; and the fruit, which is large and of an oblong shape, is said to taste like gingerbread, with a pleasant acid flavour. The wood is pale-coloured, light, and contains a large store of water, which enables the tree to withstand prolonged drought. The tree is liable to be attacked by a fungus which, growing in the woody part, renders it soft and pithlike. By the negroes of the west coast these trunks are hollowed into chambers, and dead bodies are suspended in them. There they become perfectly dry and well preserved, without further preparation or embalming. The baobab is emollient and mucilaginous; the pulverized leaves constitute *lalo*, which the natives mix with their daily food.

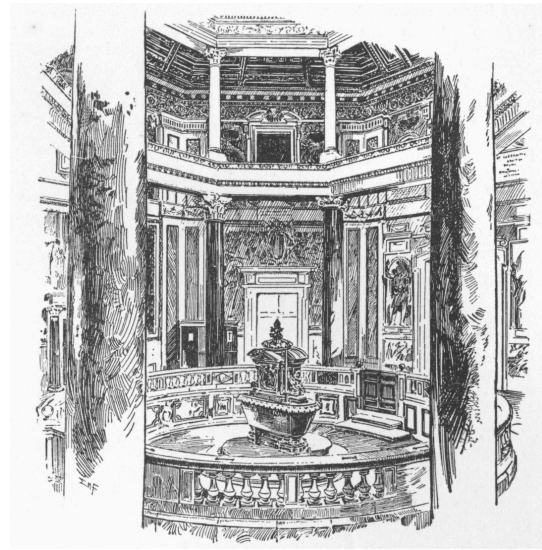
Bapaume, a French town in the department of Pas-de-Calais. It has been the scene of several battles. In 1793 the French were compelled by the allied troops to abandon the place, and in the Franco-German war one of the most closely-contested battles was fought there on 3rd Jan., 1871. Bapaume, occupied by the Germans in the European War, was recaptured by British troops on 29th Aug., 1918, the great battle of Bapaume having begun on 21st Aug. Pop. (1911) 2917.

Baph'omet, the imaginary idol or symbol which the Templars were accused of employing in their mysterious rites, and of which little is known. The word is supposed to be a corruption of Mahomet, to whose faith the Templars were accused of having a leaning. See *Gnosticism*.

Baptism (from the Gr. *baptizō*, from *baptō*, to immerse or dip), a rite which is generally thought to have been usual with the Jews even before Christ, being administered to proselytes. All that can be said, however, with some certainty is that the Jewish baptism was not of later origin than the Christian baptism. Anyhow, from this baptism that of St. John the Baptist differed, because he baptized Jews also as a symbol of the necessity of perfect purification from sin. Christ himself never baptized, but directed his disciples to administer this rite to converts (*Mat.* xxviii, 19); and baptism, therefore, became a religious ceremony among Christians, taking rank as a sacrament with all sects which acknowledge sacraments. It should, however, be borne in mind that among many peoples a rite is found which

has considerable likeness to Christian baptism. In the primitive Church the person to be baptized was dipped in a river or in a vessel, with the words which Christ had ordered, generally adopting a new name further to express the change. Sprinkling, or, as it was termed, *clinic* baptism, was used only in the case of the sick who could not leave their beds. The Greek Church and Eastern schismatics retained the custom of immersion; but the Western Church adopted or allowed the mode of baptism by pouring or sprinkling, since continued by most Protestants. This practice can be traced back certainly to the third century, before which its existence is disputed. Since the Reformation there have been various Protestant sects called Baptists, holding that baptism should be administered only by immersion, and to those who can make a personal profession of faith. The Montanists in Africa baptized even the dead, and in Roman Catholic countries the practice of baptizing church bells—a custom of tenth-century origin—continues to this day. Being an initiatory rite, baptism is only administered once to the same person. The Roman and Greek Catholics consecrate the water of baptism, but Protestants do not. The act of baptism is accompanied only with the formula that the person is baptized in the name of the Father, Son, and Holy Ghost; but, among most Christians, it is preceded by a confession of faith made by the person to be baptized if an adult, and by his parents or sponsors if he be a child. The Roman Catholic form of baptism is far more elaborate than the Protestant. This Church teaches that all persons not baptized are damned, and that even unbaptized infants are not admitted into Heaven; but for those whose chief fault was that they had not been baptized, even St. Augustine himself believed in a species of mitigated damnation. Protestants hold that though the neglect of the sacrament is a sin, yet the saving new birth may be found without the performance of the rite which symbolizes it. Naming the person baptized forms no essential part of the ceremony, but has become almost universal, probably from the ancient custom of renaming the catechumen.—BIBLIOGRAPHY: E. B. Tylor, *Primitive Culture* (vol. ii); S. Baring-Gould, *Origin and Development of Religious Belief*; J. E. Hanaver, *Baptism, Jewish and Christian*; Harnack, *History of Dogma*; article *Baptism* in *Encyclopedia of Ethics and Religion*; Corblet, *Histoire du Sacrement du Baptême*; P. Mauro, *Baptism: its place and importance in Christianity*.

Bap'tistry, a building or a portion of a building in which is administered the rite of baptism. In the early Christian Church the baptistry was distinct from the basilica or church, but was situated near its west end, and was generally circular or octagonal in form, and dome-roofed. About the end of the sixth century the baptistry began to be absorbed into the church, the font being placed within and not far from the western door. Some detached baptisteries still remain in use, as those of the Lateran, Rome (the earliest in existence), at Pisa, Parma, Ravenna, Aquileia, Siena, Florence, &c., that of Florence being 108 feet in diameter externally, and richly decorated. Baptisteries were dedicated to St. John the Baptist.



Baptistry of S. Giovanni in Laterano,
Rome

Bap'tists, a Protestant sect, distinguished by their opinions respecting the mode and subjects of baptism. With regard to the mode, they maintain the necessity of immersion, and with regard to the subjects, they consider that baptism ought not to be administered to children at all, nor to adults in general, but to those only who profess repentance and faith. They are sometimes called *Anti-pædobaptists*, to express their variance from those who defend infant baptism, and who are called *Pædobaptists*. Apart from the special sect of that name, Baptists are to be found equally among Calvinists and Arminians, Trinitarians and Unitarians. The Baptists as a whole adopt the Independent or Congregational form of church government, and their ecclesiastical assemblies are held for the purpose of mutual stimulus and intercourse, and not for the general government of the body, or for interference with individual churches. The Particular Baptists of England (so called from believing that Christ died only for the elect), the Baptists of Scotland and Ireland, the Associated Baptists of America, and some of the Seventh-day Baptists, are Calvinistic. The other classes, such as the General Baptists (who believe that Christ died for all), are Arminian, or at least not Calvinistic. Most Baptists profess to be Trinitarians. The Free-

will Baptists, the Christian Society, and most of the General Baptists of England admit of open communion; the other bodies decline communion with any Christians but Baptists. The Associated or Calvinistic Baptists long ranked in the United States as the most numerous denomination of Christians. The Seventh-day Baptists, or Sabbatarians, observe the seventh day of the week. The Free-will Baptists profess the doctrine of free salvation. The Anabaptists of the Reformation period are not to be confounded with the Baptists, by whom their principles were expressly disclaimed. The first regular Baptist Church appears to have been formed in the reign of Elizabeth, but we may date their first public acknowledgment as distinct from the Anabaptists from their petition to Parliament in 1620. The year 1633 provides the earliest record of the formation of a Particular Baptist Church in London. In 1689 a Baptist General Assembly, held in London, formulated a confession of thirty-two articles and a catechism. The Baptist Union, formed in 1832, comprehends the greater number of members of the sect in Great Britain and Ireland. In July, 1905, a world-congress of Baptists was held in London, and the Baptist World Alliance was constituted. The total number of members of Baptist Churches in the United Kingdom was 405,104 in 1922, and 408,029 in 1920. There are nine colleges for training ministers, of which the chief are: Bristol Baptist College; Regent's Park; Rawdon, Bradford; and the Metropolitan Pastors' College. The Regular Baptists in the United States numbered 7,504,447 members in 1922, and nearly 6,000,000 in 1920, in addition to which there are Anti-mission Baptists, Free-will Baptists, and Seventh-day Baptists. In Canada there are in all about 128,730 Baptists.—BIBLIOGRAPHY: H. C. Vedder, *A Short History of the Baptists*; A. H. Newman, *History of the Baptist Churches in the United States*; *The Baptist Handbook* (annually); *The American Baptist Year-book* (annually); W. E. MacIntyre, *Baptist Churches, 1500-1914*.

Bar, in law, the railing that encloses the place which counsel occupy in courts of justice; hence the phrase, *at the bar of the court*, that is, in open court. Hence also persons duly admitted as pleaders or advocates before the courts of England are denominated *barristers* (see *Barrister*), and the whole body of such barristers or advocates are called the *bar*. The enclosed place or dock in which persons accused of crimes stand in court is also called the bar. Near the door of both Houses of Parliament there is also a bar, beyond

which none but the members and clerks are admitted, and at which counsel, witnesses, offenders against privilege, &c., are heard.

Bar, in music, is a line drawn through the stave to mark the rhythm of small portions; the notes composing these are also called a bar.

Bar, in heraldry, an ordinary resembling the fesse, stretching like it horizontally across the shield but narrower.

Bar, Confederation of. See *Poland*.

Bar'aba, the name of a great steppe in the West Siberian governments of Tomsk, Akmolinsk, and Tobolsk.

Barabin'zians, a rude, uncultivated tribe of Tartars, living on the banks of the River Irtysh, and subsisting chiefly on the produce of their herds and on fish supplied by the lakes of the Baraba steppe.

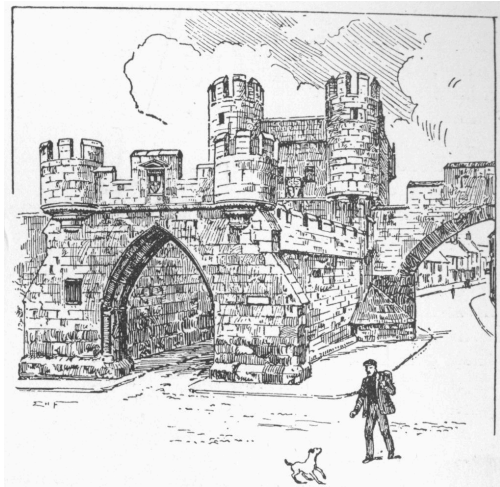
Baraguey-d'Hilliers (ba' -ra' -gā-dēl-yā), Louis, a distinguished French general under the first Empire, born in Paris, 1764. After serving under Custine and other generals he joined the army of Italy, and took Bergamo and Venice, of which he became Governor. He took part in the expedition to Egypt, served in the campaigns in Germany and Spain, and commanded a division of the "Great Army" in the Russian campaign of 1812. He was entrusted with the direction of the vanguard in the retreat, but was compelled to capitulate. Napoleon ordered him to return to France as under arrest, but, overcome with grief and fatigue, he died at Berlin on the way, Dec., 1812.

Baranovitchi, a town in Russia, government of Minsk. A battle was fought here between the Russians and the Germans in July, 1916. See *European War*.

Barb, a horse of the Barbary breed, introduced by the Moors into Spain, and of great speed, endurance, and docility.

Bar'bacan, or **Barbican**, generally an advanced work defending the entrance to a castle or fortified town, as before the gate or drawbridge, and often of formidable size and strength.

Barba'does, or **Barbados**, the most eastern of the West India Islands, first mentioned in 1518, and occupied by the British in 1625. Length, 21 miles; breadth, 13; area, 106,470 acres or 166 sq. miles, about 74,000 being under cultivation. It is divided into eleven Church of England parishes; capital, Bridgetown. It is one of the most densely-populated areas in the world, the population in 1917 being estimated at 186,656, or nearly 1100 to the square mile. The climate is very hot, though moderated by the constant trade-winds; and the island is subject to dreadful hurricanes. The surface is broken, now without forests, and with few streams; the highest point is 1145 feet above the sea-level. There are few indigenous mammals or birds. The black lowland soil gives great returns of sugar in favourable seasons. The chief exports, besides sugar, are molasses and rum; imports: rice, salt meat, corn, butter, flour, &c. Both exports and imports were over £2,000,000 in 1918. Barbadoes has a considerable transit trade, being in some measure the central mart for all the Windward Islands. It is the see of a bishop and the head-quarters of the British forces in the West Indies. There is a railway across the island, also tramways, telephones, &c. The island forms a distinct government under a Governor, an Executive and a Legislative Council of 9 members, and a House of Assembly of 24 members. Liberal provision is made for education both by old foundations and by annual vote.



Barbacan. Walmgate Bar, York

Barbadoes Leg, a form of elephantiasis.

Bar'bara, St., according to the legend, belonged to Nicomedia, in Asia Minor, and was beheaded by her father for having turned Christian, he being immediately thereafter struck dead by lightning. The latter part of this legend caused her, probably, to be invoked in storms, and to be considered the patron saint of artillerymen.

Barbarelli. See *Giorgione*.

Barbarian (Gk. *barbaros*), a name given by the Greeks, and afterwards by the Romans, to everyone who spoke an unintelligible language; and hence coming to connote the idea of *rude, illiterate, uncivilized*. This word, therefore, did not always convey the idea of something odious or savage; thus a Greek character in Plautus calls Nævius a barbarous poet, because he had not written in Greek; and Cicero terms illiterate persons without taste 'barbarians'.

Barbarossa ('red-beard'), the name of two famous Turkish corsairs of the sixteenth century, who ravaged the shores of the Mediterranean, and established themselves in Algiers. The elder of the brothers, Aruch or Horuk, was killed in 1518; the younger and more notorious, Hayraddin, who captured Tunis, died in 1546. See *Frederick Barbarossa*.

Bar'bary, a general name for the most northerly portion of Africa, extending about 2600 miles from Egypt to the Atlantic, with a breadth varying from about 140 to 550 miles, comprising Morocco, Algeria, Tunisia, and Libya (Tripolitania, Barqa, and Fezzan). The principal races are: the Berbers, the original inhabitants, from whom the country takes its name; the Arabs, who conquered an extensive portion of it during the times of the caliphs; the Bedouins, Jews, Turks, and the French colonists of Algeria, &c. The country, which was prosperous under the Carthaginians, was, next to Egypt, the richest of the Roman provinces, and the Italian States enriched themselves by their intercourse with it. In the fifteenth century, however, it became infested with adventurers who made the name of Barbary corsair a terror to commerce, a condition of things finally removed by the French occupation of Algeria.

Barbary Ape (*Inūus ecaudātus*), a species of ape, or tailless monkey, with greenish-brown hair, of the size of a large cat, remarkable for docility, also called the *magot*. It is common in Barbary and other parts of Africa, and some used to live formerly on Gibraltar Rock, being the only European monkey, though probably not indigenous. It has been the 'showman's ape' from time immemorial.

Bar'bastel, or **Barbastelle**, a bat with hairy lips (*Barbastellus commūnis*), a native of England.

Barbas'tro, a city, Spain, Arragon, province of Huesca, with an interesting cathedral, and some trade and manufactures. Pop. 7202.

Bar'bould, Anna Letitia, English poet and general writer, was born in Leicestershire, 1743, daughter of a Presbyterian minister named Aikin. She published a small volume of miscellaneous poems in 1772, and in 1773, in conjunction with her brother, Dr. John Aikin, a collection of pieces in prose. In 1774 she married the Rev. Rochemont Barbould. Her *Early Lessons and Hymns for Children*, and various essays and poems, won considerable popularity. She edited a collection of English novels, with critical and biographical notices; a selection from the British essayists of the reign of Anne, and another from Richardson's correspondence. Her last long poem, *Eighteen Hundred and Eleven*, appeared in 1812. She died at Stoke-Newington, 1825.

Bar'becue, a word of West Indian origin, meaning a hog, or other large animal, roasted whole. In the United States the word means a large gathering of people in the open air, for a social or political feast, where whole animals are roasted and eaten.

Barbed Wire, wire-rope for fencing or other purposes, having fixed in it short bent pieces of wire with sharp projecting points, which serve to keep animals from pressing against it. There is an Act of Parliament (1893) to prevent the use of barbed-wire fences that form a nuisance on a public road or path; and a person employing barbed wire for fencing may render himself liable for damages caused by it to another person who is legitimately using the adjoining ground. Barbed wire plays an important part in the construction of obstacles in modern warfare. During the European War, all systems of trenches were protected by thick belts of wire. This wire was laid out in various different ways, two of the best being known as the Concertina Fence and the Double-apron Fence.

Barbel (*Barbus*), a genus of freshwater fishes of the carp family, distinguished by the four fleshy filaments growing from the lips, two at the nose and one at each corner of the mouth, forming the kind of beard to which the genus owes its name. Of the several species the European *Barbus vulgāris*, common in most rivers, has an average length of from 12 to 18 inches, and in form and habits strongly resembles the pike. Its body is

elongated and rounded, olive-coloured above and bluish on the sides, and covered with small scales. The upper jaw, which is much longer than the lower, forms a snout, with which it bores into the mud for worms, insects, aquatic plants, &c. It is common in the Thames, where it gives good sport to the angler; but its flesh is very coarse, and at the time of spawning the roe is dangerous to eat.

Barber, one whose occupation is to shave or trim the beard, or to cut and dress hair. The practice of surgery was formerly a part of the craft, and by an Act of Henry VIII the Company of Barbers was incorporated with the Company of Surgeons—the company being then known as the Barber-surgeons—with the limitation, however, that the surgeons were not to shave or practise 'barbery', and the barbers were to perform no higher surgical operation than blood-letting and tooth-drawing. This continued till the time of George II. The signs of the old profession—the pole which the patient grasped, its spiral decoration in imitation of the bandage, and the basin to catch the blood—are still sometimes retained. The barbers' shops, always notorious for gossip, were in some measure the news centres of classic and mediæval times. The ancient hall in London, formerly used by the Barber-surgeons, is still standing in Markwell Street, Cripplegate.

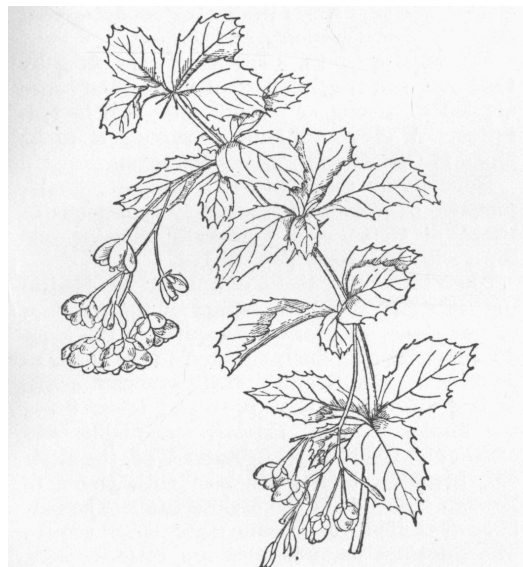
Barberini (ba`r-be-rē`nē), a celebrated Florentine family, which, since the pontificate of Maffeo Barberini (Urban VIII, 1623-44), has occupied a distinguished place among the nobility of Rome. During his reign Urban was chiefly intent on the aggrandizement of his three nephews, of whom two were appointed cardinals, and the third Prince of Palestrina.

Bar`berry, a genus of shrubs, nat. ord. Berberidaceæ. The common barberry (*Berbēris vulgāris*) has serrate, pointed leaves (the primary ones reduced to tripartite spines), hanging clusters of yellow flowers, and small, orange-red berries, which are sometimes made into preserves. The inner bark and roots yield a fine yellow dye (berberine). The shrub is a native of temperate Asia, but is now generally diffused through Europe and N. America. In Britain it has been almost universally banished from hedgerows, owing to its connection with the black rust of wheat. Numerous other species inhabit Asia and America; those of the section *Mahonia* have pinnate leaves. *B. aristata* is an erect bush, with furrowed red-brown

branches, and pendulous clusters of yellow flowers; it thrives best in the south. See *Irritability* and *Rust*.

Bar'berton, the chief mining-centre of De Kaap gold-fields, Transvaal, about 80 miles from Lydenburg, and 150 to 160 from Delagoa Bay. British prisoners were confined here in the Boer War (1899-1902). Pop. about 2000 (whites).

Bar'berts (Bucconidæ), a family of climbing birds with a thick conical beak, having tufts of bristles at its base. Their wings are short and their flight somewhat heavy. They have been divided into three sub-genera: The *barbicans* (Pogonias), inhabiting India and Africa, and feeding chiefly on fruit; the *barbets* proper (Bucco), found in Africa and America, and nearly related to the woodpeckers; and the *puff-birds* (Tamatia), inhabiting America, and feeding on insects.



Barberry (*Berberis aristata*)

Barbette (ba' r-bet'), an elevation of earth behind the breastwork of a fortification, from which the artillery may be fired over the parapet instead of through an embrasure. In ships of war the name is given to a strong breastwork of armour over which heavy guns are fired. See *Fortification*.

Barbeyrac (ba' r-bā-ra'k), Jean, an able writer on jurisprudence and natural law, translator of Grotius and Cumberland, and translator and annotator of Pufendorf. Born 1674; professor of law at Lausanne and Groningen; died 1744.

Barbican. See *Barbacan*.

Barbié du Bocage (ba' rb-yā dü bō-kāzh), Jean Denis, a distinguished geographer, born in Paris in 1760, who laid the foundation of his fame in 1788 by his *Atlas* to Barthélemy's *Voyage du Jeune Anacharsis*. His maps and plans to the works of Thucydides, Xenophon, &c., exhibit much erudition, and materially advanced the science of ancient geography. He

also prepared many modern maps, and published various excellent dissertations. He held many honourable posts, and died in 1825.

Barbier (ba`rb-yā), Antoine Alexandre, bibliographer (1765-1825). He was appointed keeper of the library of the Conseil d'État in 1798; Napoleon made him his librarian in 1807; and he was afterwards librarian to Louis XVIII. His *Catalogue de la Bibliothèque du Conseil d'État* (1801-3), and *Dictionnaire des Ouvrages Anonymes et Pseudonymes* (1806-9), are both valuable works.

Barbieri (ba`r-bē-ā`rē), Giovanni Francesco (otherwise known as *Guercino* (the squinter) *da Cento*), an eminent and prolific historical painter, born near Bologna 1590, died in 1666. His style showed the influence of Caravaggio and of the Caracci, his best work being of the latter school. Chief work, a *St. Petronilla* in the Capitol at Rome; but most of the large galleries have pictures by him.

Barbieri, Paolo Antonio, a celebrated still-life and animal painter, was a brother of Guercino; born 1596, died 1640.

Barbitone. See *Veronal*.

Barbizon (ba`r-bē-zōn), a French village about 25 miles south by east of Paris, on the borders of the Forest of Fontainebleau, and 6 miles north-west of the town of that name. It has given its name to a French school of painters, including Corot, Millet, Rousseau, Jules Dupré, and Daubigny.

Barbour, John, an ancient Scottish poet, contemporary with Chaucer, born about 1316. By 1357 he was Archdeacon of Aberdeen, and in the following year was appointed a commissioner to treat for the ransom of David II. He appears as Auditor of the Exchequer more than once; he travelled through England on several occasions, and was pensioned by Robert II. His chief poem, *The Bruce* (Brus), written about 1375, was first published in 1571, and the chief manuscripts of the poem are those in the Advocates' Library, Edinburgh, dated 1489, and in St. John's College, Cambridge. Of another long poem, setting forth the Trojan origin of the Scottish kings, no MS. remains, unless a portion of two Troy books in the Cambridge and Bodleian libraries may be ascribed to Barbour. He has also been credited with having compiled a *Book of Legends of Saints*, existing in a single MS. at

Cambridge, and published in 1881-2 by Horstman in his *Legenden Sammlung*. He died in 1395. He was the father of Scottish poetry and history, and his *Bruce* is linguistically of high value. Though wanting in the higher qualities of poetry, it is truthful and natural, and often exhibits a high moral dignity.—BIBLIOGRAPHY: Andrew Lang, *History of Scotland*; Irving, *History of Scottish Poetry*.

Barbuda (ba´r-bö´da), one of the West Indies, annexed by Britain in 1628; about 15 miles long and 8 miles wide; lying north of Antigua; pop. 580. It is flat, fertile, and healthy. Corn, cotton, pepper, and tobacco are the principal produce, but the island is only partially cleared for cultivation. There is no harbour, but a well-sheltered roadstead on the west side. It is a dependency of Antigua.

Barbusse, Henri, a French author born at Asnières in 1874. He served in the French army during the European War, and his descriptions of the realities of modern warfare have made him famous. His works include *Le Feu* and *Clarté*.

Barby (bär´bē), a German town on the Elbe, in the government of Magdeburg, with an old castle. Pop. 5202.

Bar´ca, a division of N. Africa, between the Gulf of Sidra and Egypt, annexed by Italy in 1911-2; capital, Benghazi. It formed a portion of the ancient Cyrenaica, and from the time of the Ptolemies was known as Pentapolis from its five Greek cities. It became part of Tripoli, and later a separate province under Turkish dominion. On 18th Oct., 1912, it was formally recognized as a dependency of Italy, and forms with Tripoli the Italian colony of Libya. The country consists of a rocky plateau. A large portion of it is desert, but some parts, especially near the coast, are fertile, and yield abundant crops and excellent pasture, the chief being wheat, barley, dates, figs, and olives. Flowering shrubs, roses, honeysuckles, &c., occur in great variety. There are hardly any permanent streams, but the eastern portion is tolerably well watered by rains and springs. The exports are grain and cattle, with ostrich feathers and ivory from the interior. Next to Benghazi (pop. 35,000), the seaport of Derna is the chief town (pop. 8000). The population probably does not exceed 300,000. See *Cyrenaica* and *Tripoli*.

Barcarolle (-rol'), a species of song sung by the *barcaruoli*, or gondoliers, of Venice, and hence applied to a song or melody composed in imitation. A very well-known example is to be found in Offenbach's *Tales of Hoffmann*.

Barcellona (ba' r-chel-ō'na'), seaport, Sicily, province of Messina, immediately contiguous to Pozzo di Gotto, and practically forming one town with it. Joint pop. 26,172.

Barcelona (ba' r-thel-ō'na'), next to Madrid the chief city of Spain, capital of the province of Barcelona, and formerly capital of the kingdom of Catalonia; finely situated on the northern portion of the Spanish Mediterranean coast. It is divided into the upper and lower town; the former modern, regular, stone-built, and often of an English architectural type, the latter old, irregular, brick-built, and with traces of Eastern influence in the architecture. The harbour is suited to accommodate large vessels. The principal manufactures are cottons, silks, woollens, machinery, paper, glass, chemicals, stoneware, soap; exports manufactured goods, wine and brandy, fruit, oil, &c.; imports coals, textile fabrics, machinery, cotton, fish, hides, silks, timber, &c. The city contains a university, several public libraries, a museum, a large arsenal, cannon-foundry, &c. Barcelona was, until the twelfth century, governed by its own counts, but was afterwards united with Arragon. In 1640, with the rest of Catalonia, it placed itself under the French Crown; in 1652 it submitted again to the Spanish Government; in 1697 it was taken by the French, but was restored to Spain at the Peace of Ryswick. It has had several severe visitations of cholera and yellow fever, and has been the scene of many serious and sanguinary revolts, as in 1836, 1840, and 1909. Pop. 618,766.—The province has an area of 2968 sq. miles; pop. 1,191,386 (1918). It is generally mountainous, but well cultivated, and among the most thickly peopled in Spain.

Barcelona, a town of Venezuela, near the mouth of the Neveri, which is navigable for vessels of small size, but larger vessels anchor off the mouth of the river. Pop. 13,000.

Barcelona Nuts, hazel-nuts exported from the Barcelona district of Spain.

Bar'clay, Alexander, a poet of the sixteenth century, most probably a native of Scotland, born about 1475; for some years a priest and chaplain of St. Mary Ottery, in Devonshire; afterwards a Benedictine monk of Ely;

subsequently a Franciscan, and also the holder of one or two livings; died 1552. His principal work was a satire, entitled *The Shyp of Follys of this Worlde*, part translation and part imitation of Brandt's *Narrenschiff* (Ship of Fools), and printed by Pynson in 1509. He also wrote *The Myrroure of Good Maners*, and some *Egloges* (Eclogues), both printed by Pynson, as well as translations, &c.

Barclay, John, poet and satirist, son of a Scottish father, born at Pont-à-Mousson (Lorraine) in 1582, and probably educated in the Jesuits' College there. Having settled in England, he published a Latin politico-satirical romance, entitled *Euphormionis Satyricon*, having as its object the exposure of the Jesuits. In 1616 he left England for Rome, received a pension from Pope Paul V, and died in 1621. His chief work is a singular romance in Latin, entitled *Argenis* (Paris, 1621), thought by some to be an allegory bearing on the political state of Europe at the period. Several seventeenth-century romances were indebted to this work, among others Fénelon's *Télémaque* and Calderon's *Argenis y Poliarco*. It has been translated into several modern languages. His shorter poems appeared in the *Delitiæ Poetarum Scotorum*,

Barclay, Robert, the celebrated apologist of the Quakers, born in 1648 at Gordonstown, Moray, and educated at Paris, where he became a Roman Catholic. He was recalled home by his father, whose example he followed in becoming a Quaker. His first treatise in support of his adopted principles, published at Aberdeen in the year 1670, under the title of *Truth Cleared of Calumnies*, together with his subsequent writings, did much to rectify public sentiment in regard to the Quakers. His chief work, in Latin, *An Apology for the True Christian Divinity*, was soon reprinted at Amsterdam, and quickly translated into German, Dutch, French, and Spanish, and, by the author himself, into English. His fame was now widely diffused; and, in his travels with William Penn and George Fox through England, Holland, and Germany, to spread the opinions of the Quakers, he was received everywhere with the highest respect. The last of his productions, *On the Possibility and Necessity of an Inward and Immediate Revelation*, was not published in England until 1686; from which time Barclay lived quietly with his family. He died, after a short illness, at his own house of Ury, Kincardineshire, in 1690. He was a friend of and had influence with James II.

Barclay, Sir Thomas, British barrister, born in 1853. Educated at the Universities of London, Paris, Bonn, and Jena, he was sent to Paris in 1876 as correspondent of *The Times*. He has been greatly instrumental in bringing about a better understanding between England and France. He was knighted in 1904, and entered Parliament in 1910. His works include: *Problems of International Practice and Diplomacy*; *The Turco-Italian War and its Problems*; *Law and Usage of War*.

Barclay de Tolly, Michael, Prince, a distinguished general and field-marshal of Russia, born in 1755. His family, of Scottish origin, had been established in Livonia since 1689. He entered the army at an early age, served in various campaigns against the Turks, Swedes, and Poles, and in 1811 was named Minister of War. On the invasion of Napoleon he was transferred to the chief command of the army, and adopted a plan of retreat; his forces did not greatly exceed 100,000 men, but the Court became impatient, and after the capture of Smolensk by the French he was superseded by Kutusov. Sinking all personal feeling, he asked leave to serve under his successor, commanded the right wing at the battle of the Moskwa, maintained his position, and covered the retreat of the rest of the army. After the battle of Bautzen, in 1813, he was reappointed to the chief command, which he had soon after to resign to Prince Schwarzenberg. He forced the surrender of General Vandamme after the battle of Dresden, took part in the decisive battle of Leipzig, and was made a field-marshal in Paris. In 1815 he received from the emperor the title of prince, and from Louis XVIII the badge of the Order of Military Merit. He died in 1818.

Bar-cochba (ba´r-koh´ba´), or **Bar-cochebas**, Simon, a Jewish impostor, who pretended to be the Messiah, raised a revolt, and made himself master of Jerusalem about A.D. 132, and of about fifty fortified places. Hadrian sent to Britain for Julius Severus, one of his ablest generals, who gradually regained the different forts and then took and destroyed Jerusalem. Bar-cochba retired to a mountain fortress, and perished in the assault of it by the Romans three years after, about 135.

Bar´coo. See *Cooper's Creek*.

Bard, one of an order among the ancient Celtic tribes, whose occupation was to compose and sing verses in honour of the heroic achievements of

princes and brave men, generally to the accompaniment of the harp. Their verses also frequently embodied religious or ethical precepts, genealogies, laws, &c. Their existence and function was known to the Romans two centuries B.C.; but of the Gallic bards only the tradition of their popularity survives. The first Welsh bards of whom anything is extant are Taliesin, Aneurin, and Llywarch, of the sixth century. A considerable lacuna then occurs in their history until the order was reconstituted in the tenth century by King Howel Dha, and again in the eleventh by Gryffith ap Conan. Edward I is said to have hanged all the Welsh bards as promoters of sedition. Some modern attempts have been made in Wales to revive bardism, and the Cambrian Society was formed in 1818, for this purpose and for the preservation of the remains of the ancient literature. The revived Eisteddfodau, or bardic festivals, have been so far exceedingly popular. In Ireland there were three classes of bards: those who sang of war, religion, &c., those who chanted the laws, and those who gave genealogies and family histories in verse. They were famous harpists. In the Highlands of Scotland there are considerable remains of compositions supposed to be those of their old bards still preserved.—BIBLIOGRAPHY: E. Jones, *Relics of the Welsh Bards*; Walker, *Memoirs of the Irish Bards*; Dottin, *Manuel de l'Antiquité Celtique*; John Rhys and D. Brynmor-Jones, *The Welsh People*; D. D. Evans, *Ancient Bards of Britain*.

Bardesa'nes, a Syrian Gnostic, poet, astrologer, and philosopher, born 154 or 164 at Edessa, died 222. His system started with the statement that from the union of God with matter sprang Christ and a female Holy Ghost, from whom in turn sprang various existences. He propagated his doctrines in Syrian hymns (*Hymn on the Soul*), the first in the language. His son, Harmonius, was also an able hymn-writer. The Bardesanists maintained themselves till the twelfth century, the later adherents having practically adopted Manichæism.

Bardwan', or **Burdwan'**, a division of Bengal, upon the Hugli, comprising the six districts of Bardwan, Hugli, Howrah, Midnapur, Bankura, and Birbhum. Area, 13,850 sq. miles; pop. 8,245,000.—The district *Bardwan* has an area of 2697 sq. miles, and a pop. of 1,532,500. Apart from its products, rice, grain, hemp, cotton, indigo, &c., it has a noted coal-field of about 500 sq. miles in area, with an annual output of about half a million

tons.—The town of *Bardwan* has a fine palace of the Maharajah and a pop. of 35,022.

Barebone, or **Barbone**, Praise-God, the name of a leather-seller in Fleet Street, London, who obtained a kind of lead in the convention which Cromwell substituted for the Long Parliament, and which was thence nicknamed the Barebone Parliament. After its dissolution he disappeared till 1660, when he presented a petition to Parliament against the restoration of the monarchy. In 1661 he was committed to the Tower for some time, but his subsequent history is unknown.

Barefooted Friars. See *Monasticism*.

Barége (ba-rāzh´), a light, open tissue of silk and worsted, or cotton and worsted, for women's dresses, originally manufactured near Baréges. It is now made chiefly at Bagnères-de-Bigorre.

Baréges (ba´-rāzh), a watering-place, south of France, department Hautes-Pyrénées, about 4000 feet above the sea, celebrated for its thermal springs, which are frequented for rheumatism, scrofula, &c. The place is hardly inhabited except in the bathing season, June till September.

Baregine (ba-rāzh´in; from *Baréges*), a gelatinous product of certain algæ growing in sulphuric mineral springs, and imparting to them the colour and odour of flesh-broth.

Bareilly (ba-rā´li), a town of Hindustan, in the United Provinces, capital of a district of same name, on a pleasant and elevated site. It has a fort and cantonments, a Government college, and manufactures sword-cutlery, gold and silver lace, perfumery, furniture and upholstery. On the outbreak of the Indian Mutiny the native garrison took possession of the place, but it was retaken by Lord Clyde in May, 1858. Pop. 129,462.—The district has an area of 1595 sq. miles; pop. 1,089,550.

Bar´ents, William, a Dutch navigator of the end of the sixteenth century, who, on an expedition intended to reach China by the North-East Passage, discovered Novaya Zemlya. He wintered there in 1596-7, and died before reaching home.

Baret'ti, Joseph, an Italian writer, born at Turin, 1719. In 1748 he came to England, and in 1753 published in English a *Defence of the Poetry of Italy against the Censures of M. Voltaire*. In 1760 he brought out a useful *Italian and English Dictionary*. After an absence of six years, during part of which he edited the *Frusta Letteraria* (Literary Scourge) at Venice, he returned to England, and in 1768 published an *Account of the Manners and Customs of Italy*. Not long after, in defending himself in a street brawl, he stabbed his assailant and was tried for murder at the Old Bailey but acquitted, Johnson, Burke, Goldsmith, Garrick, Reynolds, and Beauclerk giving testimony to his good character. An *English and Spanish Dictionary*, and various other works, followed before his death in 1789.

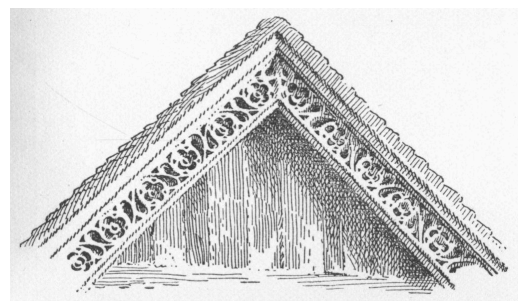
Barfleur (ba`r-fleur), at one time the best port on the coast of Normandy, and the reputed port from which William the Conqueror sailed. It was destroyed in the year 1346 by Edward III. Present pop. 1304.

Barfrush´, or **Barfurush´**, Same as *Balfroosh*.

Bargain and Sale, a legal term denoting the contract by which lands, tenements, &c., are transferred from one person to another. See *Conveyancing*.

Barge, a term similar in origin to *barque*, but generally used of a flat-bottomed boat of some kind, whether used for loading and unloading vessels, or as a canal-boat, or as an ornamental boat of state or pleasure.

Barge-board (perhaps a corruption of *verge-board*), in architecture, a board generally pendent from the eaves of gables, so as to conceal the rafters, keep out rain, &c. They are sometimes elaborately ornamented. The portion of the roof projecting from the wall at the gable-end, and beneath which the barge-board runs, is termed the *barge-course*.



Barge-board or gable-end of Manor House, Bramley, Hants.

Barham (bar'am), Rev. Richard Harris, a humorous writer, born in 1788 at Canterbury; educated at St. Paul's School, London, and at Brasenose, Oxford; appointed in succession curate of Ashford, curate of Westwell,

rector of Snargate in Romney Marsh, and one of the minor canons of St. Paul's Cathedral. He published two novels, *Baldwin* and *My Cousin Nicholas*, wrote nearly a third of the articles in Gorton's *Biographical Dictionary*, and contributed to *Blackwood's Magazine*. In 1824 he was appointed priest in ordinary of the Chapel Royal, and afterwards rector of St. Mary Magdalene and St. Gregory-by-St.-Paul, London. In 1837, on the starting of *Bentley's Miscellany*, he laid the main foundation of his literary fame by the publication in that periodical of the *Ingoldsby Legends*. He died in 1845.

Barhebræ'us. See *Abulfaragius*.

Bari (bä'rē; ancient, **Barium**), a seaport, S. Italy, on a small promontory of the Adriatic, capital of the province Terra di Bari. It was a place of importance as early as the third century B.C., and has been thrice destroyed and rebuilt. The present town, though poorly built for the most part, has a large Norman castle, a fine cathedral and priory, &c. It manufactures cotton and linen goods, hats, soap, glass, and liqueurs; has a trade in wine, grain, almonds, oil, &c., and is now an important seaport. In 1915 the town suffered considerably from floods. Pop. 109,218.—The province has an area of 2048 sq. miles, and is fertile in fruit, wine, oil, &c.; pop. 935,982.

Bari, a negro people of Africa, dwelling on both sides of the White Nile, between lat. 3° 30' and 6° N. They practise agriculture, cattle-rearing, smithwork, &c. Their country was conquered by Baker for Egypt.

Baril'la, is the commercial name for impure sodium carbonate imported from Spain and the Levant. At one time all sodium carbonate required for industrial purposes was prepared from barilla obtained by burning various marine plants (*Salsola*, *Salicornia*, and others). Only 25 per cent of the weight of the ash contained soda, and consequently its import was expensive and considerably added to the price of the manufacture of glass and soap. During the Napoleonic wars the price of barilla rose to such a height that Napoleon offered a reward for the discovery of a process for the manufacture of sodium carbonate. A chemist, Leblanc, in 1791 invented a new process by which sodium carbonate could be obtained from common salt, and this entirely superseded barilla. Plants are still grown in Southern France and Spain to obtain soda and other substances from the ash; but

almost all the sodium carbonate of the present day is manufactured chemically.

Baring, the name of a famous English family of bankers and financiers, statesmen and diplomats, from which has arisen the financial house of Baring Brothers & Co., and of which members now hold four peerages, namely, two earldoms—Northbrook and Cromer—and two baronies—Ashburton and Revelstoke. The founder of the house was Francis Baring (1740-1810), whose father was a cloth manufacturer near Exeter, and grandfather pastor of the Lutheran Church at Bremen, the family being thus of German origin. Francis Baring settled in London, attained a high position in the mercantile and financial world, was long a member of Parliament and a director of the East India Company, and was made a baronet in 1793. Of his sons, one, Alexander, was created Baron Ashburton (in 1835), and rendered important political services in connection with the boundary treaty concluded between Britain and the United States, and known by his title (see *Ashburton, Ashburton Treaty*). His son, the second Lord Ashburton (1799-1864), held a position of some note in politics, and his first wife, a daughter of the sixth Earl of Sandwich, was fond of the society of both politicians and literary men, among those who were on friendly terms with her being Carlyle and Thackeray. Sir Francis Baring's eldest son, Thomas, was father of Francis Thornhill Baring, first Lord Northbrook (1796-1866), whose eldest son, Thomas, was created Earl of Northbrook in 1876, and held successively a number of high positions, among them that of Viceroy of India (1872-6) and First Lord of the Admiralty (1880-5). (See *Northbrook*.) Another son of Sir Francis, Henry Baring, was father of Edward Baring, the first Baron Revelstoke (born in 1828, died in 1897), who was long one of the most conspicuous members of the financial world of London, and was raised to the peerage as baron in 1885. He had full management of the great house of Baring Brothers, which under him was long most prosperous, but in 1890 fell into temporary difficulties. Henry Baring was father also of Evelyn Baring, Earl Cromer (q.v.).

Baring-Gould (bā-ring-gōld'), Sabine, English clergyman and author, born at Exeter, 1834. Educated at Cambridge, he held several livings in the English Church, and in 1881 became rector of Lew Trenchard, Devon. After publishing several books on theological and miscellaneous subjects, he distinguished himself as a novelist. Among his works are: *Iceland, its*

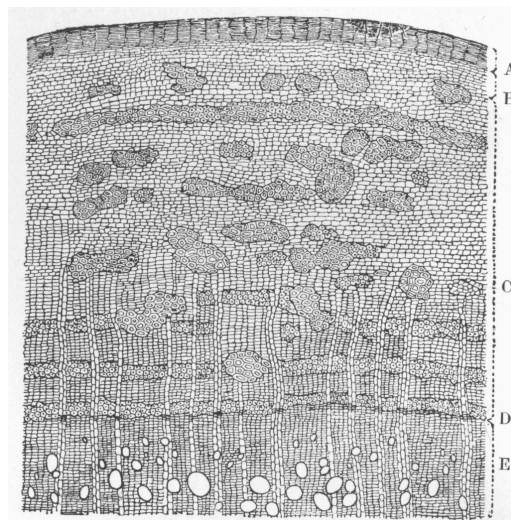
Scenes and Sagas; Curious Myths of the Middle Ages; The Origin and Development of Religious Belief; Lives of the Saints (in 15 vols.); *Village Sermons; The Vicar of Morwenstowe* (an account of the Rev. R. S. Hawker); *The Mystery of Suffering; History of the Church in Germany; The Tragedy of the Cæsars; Cliff Castles and Cave Dwellings; The Church Revival; &c.*; besides the novels *Mehalah, John Herring, Richard Cable, The Gaverocks, Court Royal, The Pennycomequicks, &c.*; and short stories or novelettes.

Baringo, a lake in Africa, N.E. of the Victoria Nyanza, about 20 miles long.

Bar'itone, or **Bar'ytone**, a male voice, the compass of which partakes of those of the common bass and the tenor, but does not extend so far downwards as the one, nor to an equal height with the other. Its best tones are from the lower A of the bass clef to the lower F in the treble.

Ba'rium (from the Gr. *barys*, heavy), the metal of which *baryta* is the oxide; specific gravity 4; symbol Ba. It is only found in compounds, such as the common sulphate (or *barytes*) and carbonate, and was isolated by Davy in 1808. It is a yellow malleable metal, which readily oxidizes, decomposes water, and fuses at a low temperature. Its nitrate and chlorate are used in pyrotechny. Barium salts, when brought into a non-luminous flame, burn with a yellowish-green colour. See *Baryta*.

Bark, properly the *periderm*, or outer covering, of woody stems and roots, composed of dead tissues cut off from the living parts of the organ by one or more layers of impervious cork. It serves to prevent loss of water, and also affords protection against animals and fungi. The outermost bark is continually sloughed in the form of scales (Scots Pine) or sheets (Common Birch), being regenerated from within by the activity of the cork-cambium (*phellogen*). Bark is usually rich in waste-products, many of which are of economic value, such as tannin, gums, &c. Bottle-cork is the



Magnified Section of Oak-bark.
A, Epidermis. B, Cells containing chlorophyll or colouring matter. C, Liber or Bast. D, Cambium. E, Sapwood.

bark of the cork-oak. Bark for tanning is obtained from oak, hemlock-spruce, species of acacia, &c. Peruvian and Angostura barks, cinnamon, &c., are other useful barks. As commonly used, bark denotes all the tissues external to the *cambium*.

Bark. See *Barque*.

Bark, Peruvian. See *Cinchona*.

Barker, Granville, playwright, was born in London in 1877. His principal plays are: *The Voysey Inheritance* (1905), *Waste* (1907), *The Madras House* (1910). He has published a book (in which William Archer collaborated) called *A National Theatre*, and another entitled *The Red Cross in France*. His revivals of Shakespeare have introduced some new ideas of mounting and stage-effects.

Barker's Mill, also called Scottish turbine, a hydraulic machine on the principle of what is known as the hydraulic tourniquet. This consists of an upright vessel free to rotate about a vertical axis, and having at its lower end two discharging pipes projecting horizontally on either side and bent in opposite directions at the ends, through which the water is discharged horizontally, the direction of discharge being mainly at right angles to a line joining the discharging orifice to the axis. The backward pressures at the bends of the tubes, arising from the two issuing jets of water, cause the apparatus to revolve in an opposite direction to the issuing fluid. The machine was invented by a Dr. Barker towards the end of the seventeenth century.

Barking, a town, England, county of Essex, on the Roding, 7 miles N.E. of London, with some important manufacturing works. Near it is the outfall of the sewage of a large part of London. Pop. (1921), 35,543.

Barkston Ash, a parliamentary division of the West Riding of Yorkshire.

Bark-stove, or **Bark-bed**, a sort of hothouse for forcing or for growing plants that require a great heat combined with moisture, both of which are supplied by the fermentation that sets up in a bed of spent tanner's bark contained in a brick pit under glass.

Bar'laam and Jos'aphat, a famous mediæval spiritual romance, which is in its main details a Christianized version of the Hindu legends of Buddha. The story first appeared in Greek in the works of Joannes Damascenus in the eighth century. It is, however, more probably the work of a monk of the Sabbas monastery, near Jerusalem, and written at the beginning of the seventh century. The compilers of the *Gesta Romanorum*, Boccaccio, Gower, and Shakespeare have all drawn materials from it.—BIBLIOGRAPHY: Max Müller, *Chips from a German Workshop*; J. Jacobs, *Barlaam and Josaphat*, (There is a translation by Rev. G. R. Woodward and H. Mattingly in the Loeb Library).

Bar-le-duc (ba ' r-l-du ' k), a town of North-East France, capital of department Meuse, with manufactures of cotton and woollen stuffs, leather, confectionery, &c. Pop. 17,068.

Barlet'ta, a seaport in South Italy, province of Bari, on the Adriatic, with a fine Gothic cathedral; it has a considerable export trade in grain, wine, almonds, &c. Pop. 44,233.

Bar'ley, the name of several cereal plants of the genus *Hordeum*, ord. Gramineæ: (grasses), yielding a grain used as food and also for making malt, from which are prepared beer, porter, and whisky. Barley has been known and cultivated from remote antiquity, and beer was made from it among the Egyptians. It is mentioned in *Exodus*, in connection with the ten plagues; and it is referred to in ancient Chinese records. Excellent barley is produced in Britain. The species principally cultivated are *Hordëum distichum*, two-rowed barley; *H. vulgāre*, four-rowed barley; and *H. hexastichum*, six-rowed, of which the small variety is the sacred barley of the ancients. The varieties of the four- and six-rowed species are generally coarser than those of the two-rowed, and adapted for a poorer soil and more exposed situation. Some of these are called *bere* or *bigg*. In Britain barley occupies more than 22½ per cent of the land under corn, but in N. America the extent of it as a crop is comparatively small, being in Canada, however, relatively greater than in the States, and the Canadian barley is of very high quality. Barley is better adapted for cold climates than any other grain, and some of the coarser varieties are cultivated where no other cereal can be grown. Russia devotes a surface of over 20,000,000 acres to barley, i.e. a little under 11 per cent of the cereal area of the country. Germany has

4,000,000 and France 1,800,000 acres of barley. Some species of the genus, three of which are natives of Britain, are mere grasses. *Pot* or *Scotch barley* is the grain deprived of the husk in a mill. *Pearl barley* is the grain polished and rounded and deprived of husk and pellicle. *Patent barley* is the farina obtained by grinding pearl barley. *Barley-water*, a decoction of pearl barley, is used in medicine as possessing emollient, diluent, and expectorant qualities.

Barley-sugar. See *Confectionery*; *Sugar*.

Bar'low, Joel, an American poet and diplomatist, born 1755. After an active and changeful life as chaplain, lawyer, editor, land-agent, lecturer, and consul, he went to Paris and acquired a fortune. On his return to America he was appointed Minister Plenipotentiary to France (1811), but died near Cracow in 1812 on his way to meet Napoleon. His principal poem, *The Columbiad (The Vision of Columbus)*, dealing with American history from the time of Columbus, was published in 1807.

Barm. See *Yeast*.

Bar'mecides (-sīdz), a distinguished Persian family, whose virtues and splendour form a favourite subject with Mahommedan poets and historians. Two eminent members of this family were Khaled-ben-Barmek, tutor of Haroun al Rashid, and his son Yahya, grand vizier of Haroun. The expression *Barmecide Feast*, meaning a visionary banquet or make-believe entertainment, originates from the Barber's story of his Sixth Brother in the *Arabian Nights' Entertainments*.

Bar'men, a German city on the Wupper, in the Prussian Rhine Province, government of Düsseldorf, and forming a continuation of the town of Elberfeld, in the valley of Barmen. It has extensive ribbon and other textile manufactures; also dyeworks, manufactures of chemicals, metal wares, buttons, yarns, iron, machines, pianos, organs, soap, &c. In 1919 Barmen was the scene of Spartacist disorders. Pop. 169,214.

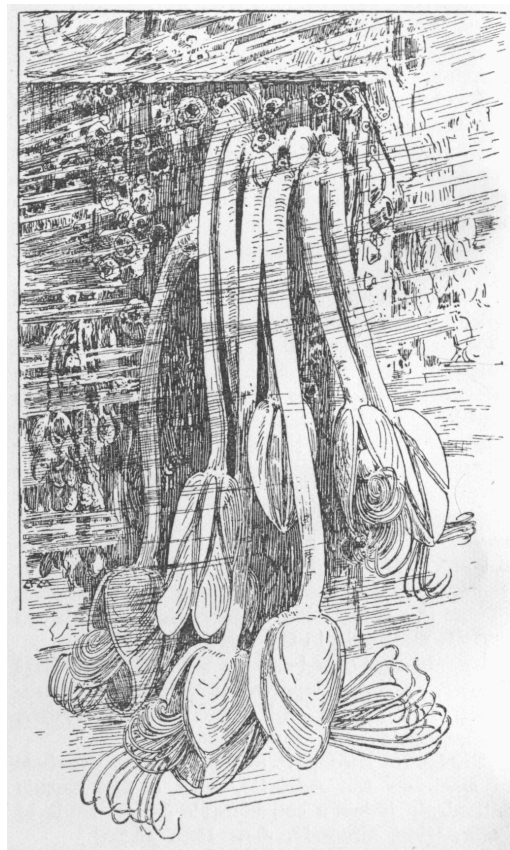
Barmouth, Welsh name **Abermaw**, a small seaport and watering-place of Wales, in Merionethshire, at the entrance of Barmouth Bay, a fine estuary. It is picturesquely situated, and has become a favourite resort of tourists and others. Pop. (1921), 3559.

Bar'nabas, the surname, according to *Acts*, iv, 36-7, given by the apostles to Joseph, a fellow-labourer of St. Paul, and, like him, ranked as an apostle. He is said to have founded at Antioch the first Christian community, to have been first Bishop of Milan, and to have suffered martyrdom at Cyprus. His festival is held on the 11th June.

Barnabas, Saint, Epistle of, an epistle in twenty-one chapters unanimously ascribed to Barnabas by early Christian writers, but without any support of internal evidence. It was probably written about A.D. 130 (or, according to others, between A.D. 70 and 79) by one who was not a Jew, and under the influence of Alexandrian Judaistic thought. See *Codex (Sinaiticus)*.

Barnabites, an order of monks founded in Milan in 1530 and named after the Milan church of St. Barnabas which was allotted them to preach in. On their expulsion from France in 1905, the majority of the Barnabites sought refuge in England. A few monasteries of the order still exist in Italy, Belgium, Austria, and Spain.

Bar'nacle, the name of a family (*Lepadidæ*) of marine crustaceous animals, ord. *Cirripedia*. They are enveloped by a mantle and shell, composed of five principal valves and several smaller pieces, joined together by a membrane attached to their circumference; and they are furnished with a long, flexible, fleshy stalk or peduncle, provided with muscles, by which they attach themselves to ships' bottoms, submerged timber, &c. They feed on small marine animals, brought within their reach by the water and secured by their tentacula. Some of the larger species are edible. According to an old fable, these animals produced barnacle geese.



Barnacle Goose (*Anser Bernicla* or *leucopsis*), a summer visitant of the

Barnacles hanging from timber

northern seas, in size rather smaller than the common wild goose, and having the forehead and cheeks white, the upper body and neck black. A fable asserts that the crustaceans called barnacles changed into geese, and various theories have been framed to account for its origin. Max Müller supposes the geese were originally called *Hiberniculæ* or Irish geese, and that barnacle is a corruption of this; but the resemblance of a barnacle to a goose hanging by the head may account for it. The Brent Goose is also sometimes called the Barnacle Goose, but the two should not be confused.

Barnard-Castle, a town, England, County Durham, giving name to a parliamentary division of the county. There are a large thread-mill and carpet manufactories; the Bowes Museum and Art Gallery, endowed by private munificence, and costing over £80,000; and the Northern Counties School, richly endowed. The castle was originally built about 1178 by Bernard Baliol, grandfather of John Baliol. Pop. 4737.

Barnar'do, Thomas John, philanthropist, born in Ireland in 1845, qualified as a medical practitioner, studying in London, Edinburgh, and Paris, established his first home for neglected London children in 1867, and for the rest of his life was engaged in the same class of work. Up to his death in 1905 he had nearly 60,000 destitute waifs cared for and trained in the institutions with which his name is connected, many of the children having been sent to Canada or other colonies. The Barnardo homes and institutions have become very numerous, being established in various English counties as well as in London, a few being in Canada. A naval training-school for boys is included among them. It was a rule of the founder that no destitute child should be refused admission. The homes have since 1899 been under a council or body incorporated under the title of the National Incorporated Waifs Association, but continue to be managed on the same lines as before. Dr. Barnardo published *Something Attempted, Something Done; The Rescue of Waifs*; &c.—Cf. J. H. Batt, *Dr. Barnardo, the Foster-father of Nobody's Children*.

Barnaul', a town of Siberia, government of Tomsk, on the Barnaulski, near its influx into the Obi. The town is of wood but well built, with museum,

observatory, &c. It is an important mining centre for lead, copper, and silver, has a copper-mint, kilns, and factories. Pop. 61,330.

Barnave (ba´r-näv), Antoine-Pierre-Joseph-Marie, a distinguished French revolutionist, born 1761, died 1793. He successfully maintained against Mirabeau the right of the National Assembly, as against that of the king, to declare for peace or war, but afterwards asserted the inviolability of the king's person. He was impeached, condemned, and guillotined.

Barnes, an urban district of England, in Surrey, on the right bank of the Thames, a short distance above London, connected with Middlesex by Hammersmith Suspension Bridge. There is here a common of 120 acres, one of the preserved open spaces of the metropolitan area. Pop. (1921), 34,281.

Barnes (bärnz), Albert, theologian, born in the State of New York, 1798. In 1825 he was ordained pastor of the Presbyterian Church of Morristown, New Jersey, and from 1830 till his death in 1870 had charge of the first Presbyterian Church in Philadelphia. He is chiefly known by his *Notes on the New Testament* and *Notes on the Old Testament*.

Barnes, Rt. Hon. George Nicoll, British Labour leader, born at Dundee, in Scotland, in 1859. After working as an engineer he became assistant secretary of the Amalgamated Society of Engineers in 1892, and was general secretary from 1896 to 1906. He entered Parliament in 1906, became Minister for Pensions in 1916, member of War Cabinet in 1917, and was Minister without portfolio from 1919 to Jan., 1920. He was a British representative at the Peace Conference in Paris.

Barnes, William, English dialect poet and philologist, born in Dorsetshire in 1800, died 1886. Of humble birth, he first entered a solicitor's office, then taught at a school in Dorchester, and having taken orders became rector of Winterbourne Came in his native county and died there. He acquired a knowledge of many languages, and published works on Anglo-Saxon and English, as *An Anglo-Saxon Delectus*, *A Philological Grammar* (grounded upon English), *Grammar and Glossary of the Dorset Dialect*, &c., but is best known by his *Poems of Rural Life in the Dorset Dialect* and *Poems of Rural Life in Common English*.

Bar'net, a town of England, in Herts, 11 miles from London, where was fought in 1471 a battle between the Yorkists and Lancastrians, resulting in the defeat of the latter and the death of Warwick, Edward IV being thus established king. Pop. (1921), 11,772.

Barnett, John, musical composer, son of Bernhard Beer, a German, born at Bedford 1802, died 1890. He composed music as a lad, and his songs and ballads soon had great vogue. In 1834 his opera *The Mountain Sylph* was produced with success, and three years after was followed by the less successful *Fair Rosamond*. For many years before his death he was little heard of.

Barnett, John Francis, nephew of the former, born in 1837, died in 1916, was also a distinguished musician and teacher, author of cantatas: *The Ancient Mariner*, *Paradise and the Peri*, *The Raising of Lazarus*, *The Building of the Ship*, and of various other works.

Barneveldt (bär'ne-velt), Johan van Olden, grand pensionary of Holland during the struggle with Philip II of Spain, born in 1547. After the assassination of William of Orange, and the conquest of the south provinces by the Spaniards under Parma, he headed the embassy to secure English aid. Finding, however, that the Earl of Leicester proved a worse than useless ally, he secured the elevation of the young Maurice of Nassau to the post of Stadtholder, at the same time by his own wise administration doing much to restore the prosperity of the State. After serving as ambassador to France and England, he succeeded in 1607 in obtaining from Spain a recognition of the independence of the States, and two years later in concluding with her the twelve years' truce. Maurice, ambitious of absolute rule and jealous of the influence of Barneveldt, was interested in the continuance of the war, and lost no opportunity of hostile action against the great statesman. In this he was aided by the strongly-marked theologic division in the State between the Gomarites (the Calvinistic and popular party) and the Arminians, of whom Barneveldt was a supporter. Maurice, who had thrown in his lot with the Gomarites, encouraged the idea that the Arminians were the friends of Spain, and procured the assembly of a synod at Dort (1618), which violently condemned them. Barneveldt and his friends Grotius and Hoogerbeets were arrested, and subjected to a so-called trial; and Barneveldt, to whom the country owed its political existence and the

Commons their retention of legislative power, was beheaded on 13th May, 1619. His sons, four years later, attempted to avenge his death; one was beheaded, the other escaped to Spain.

Barnoldswick, an urban district of England, West Riding of Yorkshire, 9 miles south-west of Skipton, with cotton manufactures and other industries. Pop. (1921), 11,951.

Barns'ley, a municipal borough of England, West Riding of Yorkshire. Its staple industries are the manufacture of linens, iron, and steel, and there are numerous collieries in the neighbourhood. Created a parliamentary borough in 1918, Barnsley returns one member to Parliament. Pop. (1921), 53,670.

Barn'stapse, a county and municipal borough in England, county of Devon, giving its name to a parliamentary division of the county on the right bank of the Taw, where it receives the Yeo; manufactures of lace, paper, pottery, furniture, toys and turnery, and leather. Pop. 14,409.

Baroach. See *Broach*.

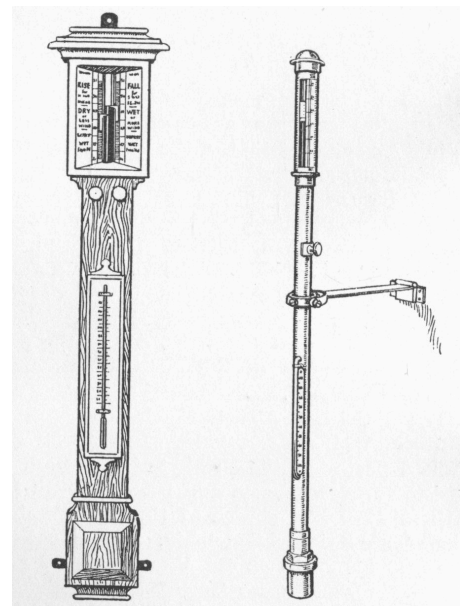
Baro'da, a non-tributary State, but subordinate to the Indian Government, situated in the north of the Bombay presidency. It consists of a number of detached territories in the province of Guzerat, and is generally level, fertile, and well cultivated, producing luxuriant crops of grain, cotton, tobacco, opium, sugar-cane, and oil-seeds. There is a famous breed of large white oxen used as draught cattle. Area, 8182 sq. miles; pop. (1911) 2,032,798. The ruler is called the *Gaekwár*. The dissensions of the Baroda family have more than once called for British intervention, and in 1875 the ruling Gaekwár was tried and deposed in connection with the charge of attempting to poison the British Resident. Since 1908 there is a Legislative Council of 17 members.—*Baroda*, the capital, is the third city in the Bombay presidency. It consists of the city proper within the walls and the suburbs without, and is largely composed of poor and crowded houses, but has also some fine buildings, and is noted for its Hindu temples kept up by the State. Pop. 99,345.—Cf. E. St. C. Weeden, *A Year with the Gaekwár of Baroda*.

Barograph, a kind of aneroid barometer, which, by means of special mechanism and appliances, is made to furnish automatically a continuous

record of the successive changes in atmospheric pressure. The paper that receives the record is made to move by clockwork while in contact with the index pencil, which rises and falls according to alterations of atmospheric pressure.

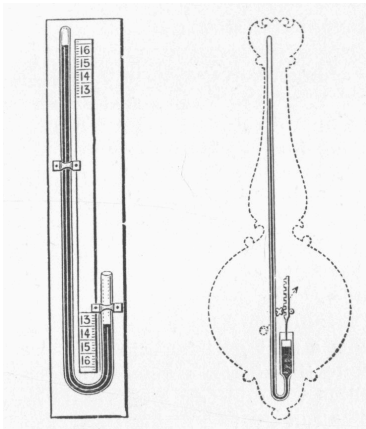
Barom'eter, an instrument for measuring the weight or pressure of the atmosphere and thus determining changes in the weather, the height of mountains, and other phenomena. It had its origin about the middle of the seventeenth century in an experiment of Torricelli, an Italian, who found that if a glass tube about 3 feet in length, open at one end only, and filled with mercury, were placed vertically with the open end in a cup of the same fluid metal, a portion of the mercury descended into the cup, leaving a column only about 30 inches in height in the tube. He inferred, therefore, that the atmospheric pressure on the surface of the mercury in the cup forced it up the tube to the height of 30 inches, and that this was so because the weight of a column of air from the cup to the top of the atmosphere was only equal to that of a column of mercury of the same base and 30 inches high. Pascal confirmed the conclusion in 1645; six years afterwards it was found by Perrier that the height of the mercury in the Torricellian tube varied with the weather; and in 1655 Boyle proposed to use the instrument to measure the height of mountains.

The common or *cistern* barometer, which is a modification of the Torricellian tube, consists of a glass tube 33 inches in length and about one-third of an inch in diameter, hermetically sealed at the top, and having the lower end resting in a small vessel containing mercury, or bent upwards and terminating in a glass bulb partly occupied by the mercury and open to the atmosphere. The tube is first filled with purified mercury, and then inverted, and there is affixed to it a scale to mark the height of the mercurial column, which comparatively seldom rises above 31 inches or sinks below 28 inches. In general the rising of the mercury presages fair weather, and its



Common Upright Barometer; Marine Barometer

falling the contrary, a great and sudden fall being the usual presage of a storm. The weather-points on the ordinary barometric scale are as follows: At 28 inches, stormy weather; $28\frac{1}{2}$, much rain or snow; 29, rain or snow; $29\frac{1}{2}$, changeable; 30, fair or frost; $30\frac{1}{2}$, settled fair or frost; 31, very dry weather or hard frost. Certain attendant signs, however, have also to be noted: thus, when fair or foul weather follows almost immediately upon the rise or fall of the mercury, the change is usually of short duration; while if the change of weather be delayed for some days after the variation in the mercury, it is usually of long continuance. The direction of the wind has also to be taken into account.



Siphon Barometer; Wheel Barometer

The *siphon* barometer consists of a bent tube, generally of uniform bore, having two unequal legs, the longer closed, the shorter open. A sufficient quantity of mercury having been introduced to fill the longer leg, the instrument is set upright, and the mercury takes such a position that the difference of the levels in the two legs represents the pressure of the atmosphere. In the best siphon barometers there are two scales, one for each leg, the divisions on one being reckoned upwards, and on the other downwards from an intermediate zero point, so that the sum of the two readings is the difference of levels of the

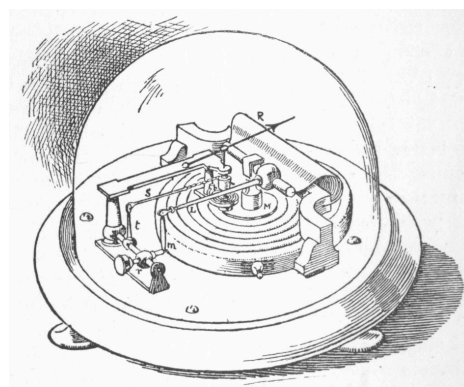
mercury in the two branches.

The *wheel* barometer is the one that is most commonly used for domestic purposes. It is far from being accurate, but it is often preferred for ordinary use on account of the greater range of its scale, by which small differences in the height of the column of mercury are more easily observed. It usually consists of a siphon barometer having a float resting on the surface of the mercury in the open branch, a thread attached to the float passing over a pulley, and having a weight as a counterpoise to the float at its extremity. As the mercury rises and falls the thread and weight turn the pulley, which again moves the index of the dial.

The *mountain* barometer is a portable mercurial barometer with a tripod support and a long scale for measuring the altitude of mountains. To prevent

breakage, through the oscillations of such a heavy liquid as mercury, it is usually carried inverted, or it is furnished with a movable basin and a screw, by means of which the mercury may be forced up to the top of the tube. For delicate operations, such as the measurement of altitudes, the scale of the barometer is furnished with a nonius or vernier, which greatly increases the minuteness and accuracy of the scale. For the rough estimate of altitudes the following rule is sufficient: As the sum of the heights of the mercury at the bottom and top of the mountain is to their difference, so is 52,000 to the height to be measured, in feet. (See also *Heights, Measurement of*.) In exact barometric observations two corrections require to be made, one for the depression of the mercury in the tube by capillary attraction, the other for temperature, which increases or diminishes the bulk of the mercury. In regard to the measurement of heights, the general rule is to subtract the ten-thousandth part of the observed altitude for every degree of Fahrenheit above 32°.

In the *aneroid* barometer, as its name implies (Gr. *a*, not, *nēros*, liquid), no fluid is employed, the action being dependent upon the susceptibility to atmospheric pressure shown by a flat circular metallic chamber from which the air has been partially exhausted, and which has a flexible top and bottom of corrugated metal plate. By an ingenious arrangement of springs and levers the depression or elevation of the surface of the box is



Aneroid Barometer

registered by an index on the dial, by which means it is also greatly magnified, being given in inches to correspond with the mercurial barometer. Aneroids are, however, generally less reliable than mercurial barometers, with which they should be frequently compared. The illustration shows an aneroid without its case. At the centre of the partially exhausted metal chamber is a small pillar [M], connected with a steel spring [R]. The rise and fall of the top of the chamber, due to changing atmospheric pressure, is transmitted by means of the levers [L], and [m] to a metallic axis [r], and this axis carries a lever [t], whose end is attached to a chain [s], which turns a drum on whose axis the index needle is fixed.—

BIBLIOGRAPHY: C. Abbe, *Meteorological Apparatus*; Marvin, *Barometers and the Measurement of Atmospheric Pressure*.

Bar'on, originally, in the feudal system, the vassal or immediate tenant of any superior; but the term was afterwards restricted to the king's barons, and again to the greater of these only, who attended the Great Council, or who, at a later date, were summoned by writ to Parliament. It was the second rank of nobility, until dukes and marquesses were introduced and placed above the earls, and viscounts also set above the barons, who therefore now hold the lowest rank in the British peerage. The present barons are of three classes: (1) barons by writ, whose ancestors have long sat in the Upper House; (2) by patent; (3) by tenure, i.e. holding the title as annexed to land. The coronet is a plain gold circle with six balls or 'pearls' on its edge, the cap being of crimson velvet.

Baron and feme, a term used for husband and wife in English law.

Bar'onet, a hereditary dignity in Great Britain and Ireland, next in rank to the peerage, originally instituted by James I in 1611, nominally to promote the colonization and defence of Ulster, each baronet, on his creation, being then obliged to pay into the treasury a sum of £1095, exclusive of fees. Baronets of Ireland were instituted in 1620, and of Scotland in 1625, the latter being called baronets of Nova Scotia, because their creation was originally intended to further the colonization of Nova Scotia. But the baronets of Scotland and of England have been baronets of Great Britain if created since 1707; since 1801 all creations have been known as baronetcies of the United Kingdom. A baronet has the title of 'Sir' prefixed to his Christian and surname, and his wife is 'Lady' so-and-so. Baronets rank before all knights except Knights of the Garter, the Thistle, and St. Patrick. They have as their badge a 'bloody hand' (the arms of Ulster), that is, a left hand, erect and open, cut off at the wrist, and red in colour.—BIBLIOGRAPHY: Pixley, *History of the Baronetage*; G. E. Cokayne, *Complete Baronetage*; *The Baronetage under 27 Sovereigns, 1309-1910*.

Baro'nus, or **Baronio**, Cæsar, Italian ecclesiastical historian, born 1538; educated at Naples; in 1557 went to Rome; was one of the first pupils of St. Philip of Neri, and member of the oratory founded by him; afterwards cardinal and librarian of the Vatican Library. He owed these dignities to the

services which he rendered the Church by his *Ecclesiastical Annals*, comprising valuable documents from the papal archives, on which he laboured from the year 1580 until his death, 30th June, 1607. They were continued, though with less power, by other writers, of whom Raynaldus takes the first rank; others are Laderchi and Theiner.

Baron of beef, two sirloins not cut asunder.

Barons' War, the war carried on for several years by Simon de Montfort and other barons of Henry III against the king, beginning in 1263.

Barony, a manor or landed estate under a baron, who formerly had certain rights of jurisdiction in his barony and could hold special courts. In Ireland baronies are still the chief subdivisions of the counties.

Barosma. See *Bucku*.

Barot'seland, a former kingdom of South Africa, now a part of Rhodesia.

Barouche (ba-rösh'), a four-wheeled carriage with a falling top and two inside seats in which four persons can sit, two facing two.

Barque (bärk), a three-masted vessel of which the fore-mast and main-mast are square-rigged, but the mizzen-mast has fore-and-aft sails only.

Barquisimeto (ba `r-kē-sē-mā'tō), a city in the north of the Republic of Venezuela, capital of the State of Lara. Pop. about 32,000.

Bar'ra, or **Bar**, a small kingdom in Africa, near the mouth of the Gambia. The Mandingoes, who form a considerable part of the inhabitants, are Mahommedans and the most civilized people on the Gambia. Pop. 200,000. The coast here belongs to Britain. The chief town is Barrinding, where the so-called king resides. See *Gambia*.

Barra, an island of the Outer Hebrides, west coast of Scotland, belonging to Inverness-shire; 8 miles long and from 2 to 5 miles broad, of irregular outline, with rocky coasts, surface hilly but furnishing excellent pasture. On the west coast the Atlantic, beating with all its force, has hollowed out vast caves and fissures. Large herds of cattle and flocks of sheep are reared on the island. The coasts of this and adjacent islands abound with fish, and

fishing is an important industry. The inhabitants, about 2500, are Roman Catholics, and speak Gaelic.

Barra, a town about 3 miles east of Naples. Pop. 12,080.

Barracan´, or **Barragan**, strictly, a thick strong stuff made in Persia and Armenia of camel's hair, but the name has been applied, by Byron and others, to various wool, flax, and cotton fabrics.

Bar´rack (Sp. *barraca*), originally a small cabin or hut for troops, but now applied to the permanent buildings in which troops are lodged. Despite the obvious evils of the quartering system, the introduction of barracks by George III met with considerable opposition in the British Parliament as dangerous to liberty, by estranging the soldier from the citizen, and fitting him to become a tool of despotism.

Barrackpur (-pör'), a town and military cantonment, Hindustan, on the left bank of the Hughli, 10 miles N.N.E. of Calcutta. The suburban residence of the Viceroy is in Barrackpur Park. Pop. 31,907.

Barracoon´, a negro barrack or slave depot, formerly plentiful on the west coast of Africa, in Cuba, Brazil, &c.

Barracu´da, a name for certain large and ferocious fishes of the genus *Sphyræna*, and allied to the mullets, inhabiting southern seas, and caught in abundance for food.

Barrafran´ca, a town of Sicily, province Caltanissetta. Pop. 11,170.

Barramun´da. See *Ceratodus*.

Barranquilla (ba´r-ra´n-kēl'ya´), a port of S. America, in Colombia, on a branch of the River Magdalena, near its entrance into the Caribbean Sea, connected by rail with the seaport Puerto Colombia. Pop. 64,543.

Barras (ba´-rä), Paul François Jean Nicholas, Comte de, member of the French National Convention and of the Executive Directory, born in Provence 1755, died 1829. After serving in the army in India and Africa, he joined the revolutionary party and was a deputy in the *tiers état*. He took part in the attack upon the Bastille and upon the Tuileries, and voted for the death of Louis XVI. In the subsequent events he displeased Robespierre,

and on this account joined the members of the committee, who foresaw danger awaiting them, and being entrusted with the chief command of the forces of his party he made himself master of Robespierre. On 4th Feb., 1795, he was elected president of the Convention, and on 5th Oct., when the troops of the sections which favoured the royal cause approached, Barras for a second time received the chief command of the forces of the Convention. On this occasion he employed General Bonaparte, for whom he procured the chief command of the army of the interior, and afterwards the command of the army in Italy. From the events of the 18th Fructidor (4th Sept., 1797) he governed absolutely until the 13th June, 1799, when Siéyès entered the Directory, and in alliance with Bonaparte procured his downfall in the revolution of the 18th Brumaire (9th Nov., 1799). He afterwards resided at Brussels, Marseilles, Rome, and Montpellier under surveillance. His *Memoirs* (in four volumes) were published in French and in English in 1895-6.

Bar´ratry, in commerce, any fraud committed by the master or mariners of a ship, whereby the owners, freighters, or insurers are injured; as by deviation from the proper course of the voyage, by the captain, for his own private purposes; fraudulent negligence; embezzlement of any part of the cargo, &c.

Barratry, Common, in law, the stirring up of lawsuits and quarrels between other persons, the party guilty of this offence being indictable as a *common barrator* or *barretor*. The commencing of suits in the name of a fictitious plaintiff is common barratry. In old Scots law it denotes the taking of bribes by a judge.

Barrel, a well-known variety of wooden vessel; but the term is also used as a definite measure and weight. A barrel of beer is 36 gallons, of flour 196 lb., of beef or pork 200 lb. The Italian *barile* varies from 7 to 31 English gallons; the French *barrique* of Bordeaux contains 50 English gallons = 228 French litres.

Barrel-organ, a musical instrument usually carried by street musicians, in which a barrel, studded with pegs or staples, when turned round, opens a series of valves to admit air to a set of pipes, or acts upon wire strings like those of the piano, thus producing a fixed series of tunes.

Barren Grounds, a large tract in the North-West Territories of Canada, extending northwards from Churchill River to the Arctic Ocean between Great Bear and Great Slave Lake and Hudson's Bay. It largely consists of swamps, lakes, and bare rock.

Barrhead', a town, Scotland, Renfrewshire, on the Levern, 7 miles s.w. of Glasgow; chief industries: printing of cottons, the spinning of cotton yarn, dyeing, bleaching, iron and brass founding, and sanitary ware. Pop. 11,387.

Barrie, a town of Canada, province Ontario, 55 miles N.W. Toronto, picturesquely situated on Lake Simcoe, a favourite summer resort. Pop. 6420.

Barrie, Sir James Matthew, a novelist and playwright, born in 1860 at Kirriemuir, Forfarshire; studied at Edinburgh University, graduating as M.A. in 1882. After working on a Nottingham journal he was a journalist in London. His first book, *Better Dead* (1887), a satire on London life, was followed by the highly successful *Auld Licht Idylls* (1888), with its sequel *A Window in Thrums* (that is, Kirriemuir) (1889). Among his novels and tales are *When a Man's Single*, *My Lady Nicotine*, *The Little Minister*, *Sentimental Tommy*, *Tommy and Grizel*, *The Little White Bird*. Successful plays are *Peter and Wendy*, *The Professor's Love Story*, *The Little Minister* (based on the novel), *The Admirable Crichton*, *Quality Street*, *Little Mary*, *Peter Pan*, *What Every Woman Knows*, *Dear Brutus*, *A Kiss for Cinderella*, *Mary Rose*. His plays on the whole have been even more successful than his other works. He was created a baronet in 1913, and elected Rector of the University of St. Andrews in 1919.—Cf. J. A. Hammerton, *J. M. Barrie and his Books*.

Barrier Reef, a coral reef, or rather series of reefs, extending for 1260 miles off the N. E. coast of Australia, at a distance from land ranging from 10 to 100 miles. In sailing along this coast, steamers generally take the route inside the reef, where there is a channel about 12 fathoms deep throughout, protected by the reefs themselves; the outer channel is safer for sailing vessels.

Barrier Treaty, the treaty (1718) by which, when the Spanish Netherlands were ceded to Austria, the Dutch secured the right to garrison several border

fortresses of the country at the expense of Austria, to serve as a barrier against France. It was declared void in 1781 by Joseph II.

Bar'rington, Daines, son of Viscount Barrington, lawyer, antiquarian, and naturalist, born 1727, died 1800. He wrote many papers for the Royal Society and the Society of Antiquaries; published some separate works, and was a correspondent of White of Selborne, who addressed to him a number of the letters in *The Natural History of Selborne*.

Bar'rister, in England or Ireland, an advocate or pleader, who has been admitted by one of the Inns of Court, viz. the Inner Temple, Middle Temple, Lincoln's Inn, or Gray's Inn, to plead at the bar. It is they who speak before all the higher courts, being instructed in regard to the case they have in hand by means of the *brief* which they receive from the solicitor who may happen to engage their services, and which has a certain fee endorsed upon it as the sum to be paid for the barrister's services in the case. Before a student can be admitted to the bar he must have been a member of one of those societies, and have kept twelve terms there. The examinations, which had dwindled into mere forms, have been revived and made more stringent. Barristers are sometimes called *utter* or *outer barristers*, to distinguish them from the king's counsel, who sit within the bar in the courts and are distinguished by a silk gown. Barristers are also spoken of as *counsel*, as in the phrase *opinion of counsel*, that is, a written opinion on a case obtained from a barrister before whom the facts have been laid. All judges are selected from the barristers. A barrister cannot maintain an action for his fees, which are considered purely honorary. A *revising barrister* is a barrister appointed to revise the list of persons in any locality who have a vote for a member of Parliament. The term corresponding to barrister is in Scotland *advocate*, in the United States *counsellor-at-law*; but the position of the latter is not quite the same.—Cf. J. R. V. Marchant, *Barrister-at-law*.

Bar'ros, João de, Portuguese historian, born 1496. He was attached to the Court of King Emmanuel, who, after the publication in 1520 of Barros' romance, *The Emperor Clarimond*, urged him to undertake a history of the Portuguese in India, which appeared thirty-two years later. King John III appointed Barros Governor of the Portuguese settlements in Guinea, and General Agent for these colonies, further presenting him in 1530 with the province of Maranhão, in Brazil, for the purpose of colonization. For his

losses by the last enterprise the king indemnified him, and he died in retirement in 1570. Besides his standard work, the *Decadas*, a history of the Portuguese in India (a complete edition of which appeared at Lisbon between 1778 and 1788), he wrote a moral dialogue on compromise, and the first *Portuguese Grammar*.

Barro'sa, a village, Spain, near the s. w. coast of Andalusia, near which General Graham, when abandoned by the Spaniards, defeated a superior French force in 1811.

Bar'row, a river in the south-east of Ireland, province Leinster, rising on the borders of the King's and Queen's Counties, and after a southerly course joining the Suir in forming Waterford harbour. It is next in importance to the Shannon, and is navigable for vessels of 200 tons for 25 miles above the sea. Its principal tributary is the Nore.

Bar'row, Isaac, an eminent English mathematician and divine, born in London in 1630; studied at the Charterhouse and at Trinity College, Cambridge, of which he became a fellow in 1649. After a course of medical studies he turned to divinity, mathematics, and astronomy, and took his M. A. degree in 1652, and, failing to obtain the Cambridge Greek professorship, went abroad. In 1659 he was ordained; in 1660 elected Greek professor at Cambridge; in 1662 professor of geometry in Gresham College; and in 1663 Lucasian professor of mathematics at Cambridge, a post which he resigned to Newton in 1669. In 1670 he was created D. D., in 1672 master of Trinity College, and in 1675 vice-chancellor of Cambridge University. He died in 1677. His principal mathematical works (written in Latin) were: *Euclidis Elementa* (1655); *Euclidis Data* (1657); *Mathematicæ Lectiones* (1664-6); *Lectiones Opticæ* (1669); *Lectiones Geometricæ* (1670); *Archimedis Opera*; *Apollonii Conicorum lib. iv*; *Theodosii Spherica*, (1675). All his English works, which are theological, were left in MS., and published by Dr. Tillotson in 1685, the best edition being that prepared by the Rev. A. Napier in 1859. As a mathematician Barrow was deemed inferior only to Newton. The Latin edition of his mathematical works was prepared by Whewell in 1860.

Barrow, Sir John, Bart., geographer and man of letters, born in 1764 in Lancashire. At the age of sixteen he went in a whaler to Greenland; and

later on was teacher of mathematics in a school at Greenwich. In 1792 he was sent with Lord Macartney, in his embassy to China, to take charge of philosophical instruments for presentation to the Chinese emperor. His account of this journey was of great value, and not less so was the account of his travels in South Africa, whither he went in 1797 as secretary to Macartney. In 1804 he was appointed second secretary to the Admiralty, a post occupied by him for forty years. In 1835 he was made a baronet; and he died in 1848, three years after his retirement. Besides the accounts of his own travels, he published lives of Earl Macartney, Lord Anson, and Lord Howe; *Voyages of Discovery and Research within the Arctic Regions*; an *Autobiography* written at the age of eighty-three, &c.

Bar'row-in-Fur'ness, a seaport, county and parliamentary borough of Lancashire, in the district of Furness, opposite the Island of Walney, a town that has increased from a fishing-hamlet with 100 inhabitants in 1848 to a town of 74,254 inhabitants in 1921. Its prosperity is due to the mines of red hematite iron-ore which abound in the district, and to the railway rendering its excellent natural harbour available. It has several large docks, besides graving-docks, a floating dock capable of receiving vessels of 3000 tons, a large timber pond, &c. There is an extensive trade in timber, cattle, grain, and flour; and iron-ore and pig-iron are largely shipped. It has numerous blast-furnaces, and one of the largest Bessemer-steel works in the world. Besides ironworks, a large business is done in shipbuilding, the making of railway wagons and rolling-stock, ropes, sails, bricks, &c. A town hall, erected at a cost of £60,000, was opened in 1887. Barrow-in-Furness returns one member to Parliament.

Bar'rows, mounds of earth or stones raised to mark the resting-place of the dead, and distinguished, according to their shape, as *long*, *bowl*, *bell*, *cone*, and *broad* barrows. The practice of barrow-burial is of unknown antiquity and almost universal, barrows being found all over Europe, in Northern Africa, Asia Minor, Afghanistan, Western India, and in America. In the earliest barrows the enclosed bodies were simply laid upon the ground, with stone or bone implements and weapons beside them. In barrows of later date the remains are generally enclosed in a stone cist. Frequently cremation preceded the erection of the barrow, the ashes being enclosed in an urn or cist. A detailed description of an ancient barrow-burial is given in the Anglo-Saxon poem *Beowulf*.—BIBLIOGRAPHY: Canon W. Greenwell,

British Barrows; J. Anderson, *Scotland in Pagan Times*; W. C. Borlase, *The Dolmens of Ireland*.

Barrow Strait, the connecting channel between Lancaster Sound and Baffin's Bay on the east and the Polar Ocean on the west. Named after Sir John Barrow.

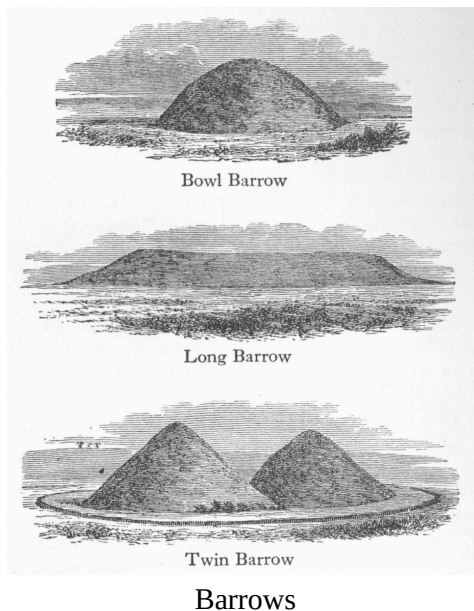
Barry, a seaport of S. Wales, Glamorganshire, 7 miles south-west of Cardiff, with large docks, exporting much coal. Pop. (1921), 38,927.

Bar'ry, Sir Charles, an English architect, born in London, 1795. After executing numerous important buildings, such as the Reform Clubhouse, London, St. Edward's School, Birmingham, &c., he was appointed architect of the new Houses of Parliament at Westminster, with the execution of which he was occupied for more than twenty-four years. He was elected a Royal Academician in 1844, was knighted in 1852, and died suddenly in 1860.

Barry, Comtesse Du. See *Du Barry*.

Barry, Edward Middleton, R.A. (1830-80), son of Sir Charles Barry, was also a distinguished architect, and designed many important buildings, though he was disappointed in regard to his plans for the Albert Memorial, National Gallery, and New Law Courts.

Barry, James, a painter and writer on art, born at Cork, 1741; studied abroad with the aid of Burke; was elected Royal Academician on his return; and worked seven years on the paintings for the hall of the Society for the Encouragement of the Arts. His first picture which attracted attention was *St. Patrick baptizing the King of Cashel*. In 1773 he published his *Inquiry into the Real and Imaginary Obstructions to the Increase of the Arts in England*, and in 1782 was elected professor of painting to the Academy. He was expelled in 1797 on the ground of his authorship of the *Letter to the Society of Dilettanti*. His chief painting was his *Victors at Olympia*. He died in 1806.



Barry Cornwall. See *Procter, Bryan Waller*.

Bar'sabbas, surnamed Justus, son of Alpheus, brother of James the Less and of Judas, and one of the candidates for the apostolical office left vacant by Judas Iscariot. According to tradition, he was afterwards Bishop of Eleutheropolis, near Jerusalem, and suffered martyrdom. Another Barsabbas, surnamed Judas, supposed to be the brother of the above, is mentioned in the *Acts* as a companion of St. Paul and Barnabas at Antioch. He is supposed to have died in Jerusalem at a very advanced age.

Bar-shot, a double-headed shot consisting of two pieces connected by a bar.

Barsi, a town of India, in Sholapur district, Bombay, 43 miles north of Sholapur and 128 miles east of Poona, with a trade in cotton, oil, &c. Pop. 21,000.

Bar-sur-Aube (ba' r-su' r-ōb), an ancient town, France, department Aube, where, in 1814, a hotly-contested action was fought between Napoleon and the Allies. Pop. 4533.

Bart, Barth, or Baert (bärt), Jean, a famous French sailor, born at Dunkirk, 1650, the son of a poor fisherman. He became captain of a privateer, and after some brilliant exploits was appointed captain in the royal navy. In recognition of his further services he was made commodore, subsequently receiving letters of nobility. Brusque, if not vulgar, in manner, and ridiculed by the Court for his indifference to ceremony, he made the navy of the nation everywhere respected, and furnished some of the most striking chapters in the romance of naval warfare. After the Peace of Ryswick he lived quietly at Dunkirk, and died there while equipping a fleet to take part in the War of the Spanish Succession, 1702.

Bartas (ba' r-tä), Guillaume de Salluste du, a French poet, termed 'the divine' by contemporary English writers, born 1544. Principal work, *La Semaine ou la Création du Monde*, a poem on the creation, translated into English by Joshua Sylvester. It is said to have had a considerable influence on Milton's *Paradise Lost*. He died of wounds received at Ivry, 1590.

Bartfeld (ba' rt'felt), an old town in Czecho-Slovakia, formerly Hungary, county of Saros, on the Tepl, with mineral springs in the neighbourhood.

Pop. 6160, mostly Slovaks.

Barth (bärt), Heinrich, African traveller, born at Hamburg, 1821, died in 1865. He graduated at the University of Berlin as Ph.D. in 1844, and set out in 1845 to explore all the countries bordering on the Mediterranean. The first volume of his *Wanderungen durch die Küstenländer des Mittelmeeres* was published in 1849, in which year he was invited by the British Government to join Dr. Overweg in accompanying Richardson's expedition to Central Africa. The expedition set out from Tripoli in Feb., 1850, and, in spite of the death both of Richardson and Overweg, Barth did not return to Tripoli till the autumn of 1855. His explorations, which extended over an area of about 2,000,000 sq. miles, determined the course of the Niger and the true nature of the Sahara. The English account of it was entitled *Travels and Discoveries in North and Central Africa* (5 vols., 1857-8). An important work on the African languages was left unfinished.

Barth, Jean. See *Bart*.

Barthélemy (ba´r-tāl-mē), Jean Jacques, French author, born 1716. He was educated under the Jesuits, for holy orders, but declined all offers of clerical promotion above the rank of Abbé. He gained considerable repute as a worker in philology and archæology; and after his appointment as Director of the Royal Cabinet of Medals, in 1753, spent some time travelling in Italy collecting medals and antiquities. His best-known work, not inaptly characterized by himself as an unwieldy compilation, was his *Voyage du Jeune Anacharsis en Grèce* (1787). It was very popular and was translated into various tongues. Though taking no part in the revolution, he was arrested on a charge of being an aristocrat in 1793, but was set at liberty, and subsequently offered the post of librarian of the National Library. He died in 1795.

Barthélmy-Saint-Hilaire (ba´r-tāl-mē-san-tē-lār), Jules, French scholar and statesman, born 1805, died in 1895. He was professor of Greek and Latin philosophy in the Collège de France, but resigned the chair after the *coup d'état* of 1852 and refused to take the oath. He was reappointed in 1862; and in 1869 was returned to the Corps Législatif. After the revolution he was a member of the National Assembly; and was elected Senator for

life in 1876. He published a translation of Aristotle, and works on Buddhism, Mohammed and Mohammedanism, the Vedas, &c.

Barthez (ba´r-tā), Paul Joseph, an eminent French physician, born at Montpellier 1734, died 1806. At Montpellier he founded a medical school, which acquired a reputation throughout all Europe. Having settled in Paris, he was appointed by the king consulting physician, and by the Duke of Orleans his first physician. The Revolution deprived him of the greatest part of his fortune, and drove him from Paris, but Napoleon brought him forth again, and loaded him in his advanced age with dignities. Among his numerous writings may be mentioned *Nouvelle Mécanique des Mouvements de l'Homme et des Animaux*; *Traitement des Maladies Goutteuses*; *Consultation de Médecine*, &c. His *Traité du Beau* was published posthumously (1807).

Bartholdi (ba´r-tol´dē), Auguste, French sculptor, born 1834, died 1904; best known for his colossal statue of Liberty, now overlooking the harbour of New York.

Bartholin (bär´to-lin), Kaspar, Swedish writer, born 1585, died 1630. He studied medicine, philosophy, and theology; was made doctor of medicine at Basel in 1610, rector of the University of Copenhagen 1618, and professor of theology 1624. His *Institutiones Anatomicæ* was for long a standard textbook in the universities.—His son, Thomas, born at Copenhagen 1616, died 1680, was equally celebrated as a philologist, naturalist, and physician. He was professor of anatomy at Copenhagen, 1648; physician to the king, Christian V, in 1670; and Councillor of State, 1675.—His sons, Kaspar (born 1654, died 1704) and Thomas (born 1659, died 1690) were also highly distinguished—the former as an anatomist, the latter as an archæologist.

Bartholomew, Massacre of St., the brutal slaughter of the French Protestants, which began on 24th Aug., 1572, by secret orders from Charles IX, at the instigation of his mother, Catherine de' Medici, and in which, according to Sully, 70,000 Huguenots, including women and children, were murdered throughout the country. During the minority of Charles and the regency of his mother, a long war raged in France between the Catholics and Huguenots, the leaders of the latter being the Prince of Condé and

Admiral Coligny. In 1570 overtures were made by the Court to the Huguenots, which resulted in a treaty of peace. This treaty blinded the chiefs of the Huguenots, particularly Admiral Coligny, who was tired of the civil war. The king appeared to have entirely disengaged himself from the influence of the Guises and his mother; he invited Coligny to his Court, and honoured him as a father. The most artful means were employed to increase this delusion. The sister of the king was married to the Prince de Béarn (18th Aug., 1572) in order to allure the most distinguished Huguenots to Paris. On 22nd Aug. a shot from a window wounded the admiral. The king hastened to visit him, and swore to punish the author of the villainy; but on the same day he was induced by his mother to believe that the admiral had designs on his life. "God's death!" he exclaimed; "kill the admiral; and not only him, but all the Huguenots; let none remain to disturb us." The following night Catherine held the council, which fixed the execution for the night of St. Bartholomew, 24th Aug., 1572. After the assassination of Coligny, a bell from the tower of the royal palace at midnight gave to the assembled companies of burghers the signal for the general massacre of the Huguenots. The Prince of Condé and the King of Navarre saved their lives by going to mass and pretending to embrace the Catholic religion. By the king's orders the massacre was extended throughout the whole kingdom; and the horrible slaughter continued for thirty days in almost all the provinces.—BIBLIOGRAPHY: Henry White, *Massacre of St. Bartholomew and History of the Religious Wars*; Lavissee, *Histoire de France* (vol. vi).

Barthol'omew, St., the apostle, is probably the same person as *Nathanael*, mentioned in the *Gospel of St. John* as an upright Israelite and one of the first disciples of Jesus. He is said to have taught Christianity in the south of Arabia, into which, according to Eusebius, he carried the *Gospel of St. Matthew* in the Hebrew language, and to have suffered martyrdom. The ancient Church had an apocryphal gospel bearing his name, of which nothing has been preserved. A festival is held in his memory on 24th Aug.

Bartholomew, St., or **St. Barthélemy**, an island, one of the West Indies, in the Leeward group, belonging to France, about 24 miles in circumference. It produces some tobacco, sugar, cotton, indigo, &c. Pop. 3000. The only town is Gustavia. The island, occupied by France in 1648, was ceded to Sweden in 1784, but was again acquired by France in 1877 at the cost of 275,000 francs.

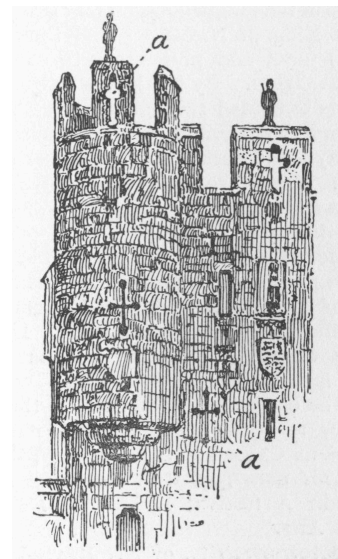
Bartholomew Fair, a celebrated fair, established in the reign of Henry I (1133), formerly held in West Smithfield, London, on St. Bartholomew's Day (24th Aug., O. S.), but abolished since 1855.

Bartholomew's Hospital, St., one of the great hospitals of London, formerly the priory of St. Bartholomew, and made a hospital by Henry VIII in 1547. On an average over 6000 patients are annually admitted to the hospital, while about 150,000 out-patients are treated at its out-door dispensaries. A medical school is attached to it, attended by more than four hundred students.

Barthou, Louis, French statesman, born at Orloron-Sainte-Marie in 1862. He practised as an advocate until he entered the Chamber of Deputies in 1889. He was a member of several Cabinets, and was appointed Premier in March, 1913, but resigned in December of the same year. He was Minister of Foreign Affairs in Painlevé's Cabinet during the European War. In April, 1922, he headed the French delegation to the Genoa Conference. His works include *Mirabeau*; *Lamartine*, *Orateur*; *Les Amours d'un Poète*, &c.

Bar'tizan, a small overhanging turret pierced with one or more apertures for archers, projecting generally from the angles on the top of a tower, or from the parapet, or elsewhere, as in a mediæval castle. The word, probably a corruption of *bratticing*, was apparently first used by Sir Walter Scott.

Bart'lett, William Henry, an English artist, born 1809, died, on a voyage from Malta to Marseilles, 1854. He travelled extensively abroad, and the illustrated works descriptive of the countries visited by him (Switzerland, the Bosphorus and the Danube, Syria and Palestine, Egypt, Canada, United States, &c.) obtained great success with the public, the engravings being from sketches by his own pencil.



Bartizan, Micklegate Bar,
York.
a, a, Balistraria

Bartolini (ba ´ r-to-lē ´ nē), Lorenzo, a celebrated Italian sculptor, born at Florence about 1778, died 1850. He studied and worked in Paris, and was patronized by Napoleon. On the fall of the Empire he returned to Florence,

where he continued to exercise his profession. Among his greater works may be mentioned his groups of *Charity*, and *Hercules and Lichas*, a colossal bust of Napoleon, and the beautiful monument in the cathedral of Lausanne, erected in memory of Lady Stratford Canning. Bartolini ranks next to Canova among modern Italian sculptors.

Bartolommeo (-mā'ō), Fra, or **Baccio Della Porta** (ba' chō'), Italian painter, born at Florence 1475, died there 1517. He studied painting in Florence, and acquired a more perfect knowledge of art from the works of Leonardo da Vinci. He was an admirer and follower of Savonarola, on whose death he took the Dominican habit, and assumed the name of Fra Bartolommeo. He was the friend of Michael Angelo and Raphael; painted many religious pictures, among them a *Saint Mark* and a *Saint Sebastian*, which are greatly admired. His colouring, in vigour and brilliancy, comes near to that of Titian and Giorgione. His *Holy Family* is in the National Gallery, London.

Bartolozzi (-lot'sē), Francesco, a distinguished engraver, born at Florence in 1725, or, according to others, in 1730, died at Lisbon 1815. In Venice, in Florence, and Milan he etched several pieces on sacred subjects, and then went to London, where he received great encouragement. After forty years' residence in London, he went to Lisbon on the invitation of the Prince Regent of Portugal. He became director of the National Academy at Lisbon, where he remained till his death.

Bar'ton, Andrew, one of Scotland's first great naval commanders; flourished during the reign of James IV, and belonged to a family which for two generations had produced able and successful seamen. In 1497 he commanded the escort which accompanied Perkin Warbeck from Scotland. After doing considerable damage to English shipping, he was killed in an engagement with two ships which had been specially fitted out against him (1512).

Barton, Bernard, known as the Quaker poet, born at Carlisle 1784, died 1849. In 1806 he removed to Woodbridge, in Suffolk, where he was long clerk in a bank. He published *Metrical Effusions* (1812); *Poems by an Amateur* (1818); *Poems* (1820); *Napoleon, and other Poems* (1822); *Poetic Vigils* (1824); *Devotional Verses* (1826); *A New-year's Eve, and other*

Poems (1828); besides many contributions to the annuals and magazines. His poetry, though deficient in force, is pleasing, fluent, and graceful. Barton is chiefly remembered as the friend of Charles Lamb, with whom he began to correspond in 1822.

Barton, Elizabeth, a country girl born in 1506 at Aldington, in Kent (commonly called the Holy Maid or the Nun of Kent), who gained some notoriety in the reign of Henry VIII. She was subject to epileptic fits, and was persuaded by certain priests that she was a prophetess inspired by God. Among other things she prophesied that Henry, if he persisted in his purpose of divorce and second marriage, would not be king for seven months longer, and would die a shameful death, and be succeeded by Catherine's daughter. On arrest the imposture was confessed, and Barton and six others were executed 5th May, 1534.

Barton-upon-Humber, a town of England, in Lincolnshire, on the Humber. It contains two old churches, one of which is an undoubted specimen of Anglo-Saxon architecture. Pop. (1921), 6454.

Bartsch (ba´rch), Karl Friedrich, a German scholar, born in 1832, died in 1888, whose labours have been of immense service in elucidating the older literature and language of his native country as well as in the field of the Romance tongues. Among his publications were editions of the *Nibelungenlied*, *Walther von der Vogelweide*, *Kudrun*, &c.; *Chrestomathie de l'ancien Français*; *Provençalisches Lesebuch*; translations of Burns, of Dante, &c.

Bartsia, a genus of Scrophulariaceæ, green half-parasites upon roots of grasses. Three species are British; *B. alpina* has underground shoots, not unlike those of the allied Toothwort, with fleshy scale-leaves bearing water-secreting glands.

Baru (ba-rö´), a woolly substance used for caulking ships, stuffing cushions, &c., found at the base of the leaves of an East India sago palm.

Baruch (bā´ruk; literally, 'blessed'), a Hebrew scribe, friend and assistant to the prophet Jeremiah. At the captivity, after the destruction of Jerusalem, Jeremiah and Baruch were permitted to remain in Palestine, but were afterwards carried into Egypt, 588 B.C. His subsequent life is unknown. One

of the apocryphal books bears the name of Baruch. The Council of Trent gave it a place in the canon, but its authenticity was not admitted either by the ancient Jews or the early Christian fathers.

Barwood, a dyewood obtained from *Pterocarpus angolensis*, a tall tree of West Africa. It is chiefly used for giving orange-red dyes on cotton yarns. See *Camwood*; *Sandalwood*.

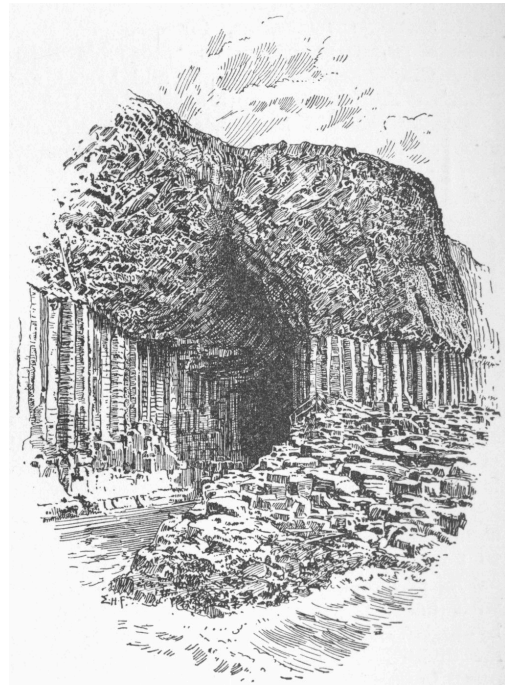
Bary'ta, oxide of barium (BaO) is found in nature associated with sulphuric acid as barium sulphate, heavy-spar or Barytes (BaSO_4), and with carbon dioxide as barium carbonate or witherite (BaCO_3). Baryta is a heavy greyish powder of specific gravity 4.7, obtained by burning barium in oxygen or from barium nitrate or barium carbonate. It has a strong affinity for water, and combines with it in the evolution of heat and the formation of barium hydroxide (Ba(OH)_2). Barium oxide is manufactured in quantity for the preparation of barium dioxide (BaO_2), from which oxygen was at one time obtained. (See *Oxygen*.) Baryta forms white crystalline salts with acids, which are all poisonous with the exception of barium sulphate. Barium hydroxide is soluble in water, forming a strongly-alkaline solution, Baryta water, which is much used in chemical analysis. Barium sulphate is a white substance insoluble in water and in acids, and is the source of most of the barium compounds. Artificially-prepared barium sulphate is used as a pigment, *Permanent White*, also in the paper industry and in vulcanizing rubber. Other salts of barium of importance are barium chloride, barium nitrate, and barium carbonate, used in pyrotechny and in glass manufacture.

Barytes (American Barite), a rhombic mineral consisting of barium sulphate (specific gravity 4.5) occurring in veins and extensive masses as a subsequent deposit in various rocks. Colourless to brownish, the colourless massive examples being in great demand as a substitute for white lead in paint and for other purposes. Barytes is mined in the north of England and in County Cork.

Barytone. See *Baritone*.

Basalt (ba-sa'lt'), a well-known igneous rock occurring in the ancient trap and the recent volcanic series of rocks, but more abundantly in the former. It is a fine-grained, heavy, crystalline rock, consisting of felspar, augite, and

magnetic iron, and sometimes contains a little olivine. Basalt is amorphous, columnar, tabular, or globular. The columnar form is straight or curved, perpendicular or inclined, sometimes nearly horizontal; the diameter of the columns from 3 to 18 inches, sometimes with transverse hemispherical joints, in which the convex part of one is inserted in the concavity of another; and the height from 5 to 150 feet. The forms of the columns generally are pentagonal, hexagonal, or octagonal. When decomposed it is found also in round masses, either spherical or compressed and lenticular. These rounded masses are sometimes composed of concentric layers, with a nucleus, and sometimes of prisms radiating from a centre. Fingal's Cave, in the Island of Staffa, furnishes a remarkable instance of basaltic columns. The pillars of the Giant's Causeway, Ireland, composed of this stone, and exposed to the roughest sea for ages, have their angles as perfect as those at a distance from the waves. Basalt often assumes curious and fantastic forms, as, for example, those masses popularly known as 'Samson's Ribs' at Arthur's Seat, Edinburgh, and 'Lot' and 'Lot's Wife' near the south coast of St. Helena.



Basaltic Columns, Fingal's Cave, Island of Staffa

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Baschi (ba ´ s'kē), Matteo, an Italian Minorite friar of the convent of Montefalcone, founder and first general of the Capuchin branch of the Franciscans. He died at Venice, 1552.

Bas´cinet, or **Bas´net**, a light helmet, sometimes with, but more frequently without, a visor, in general use for English infantry in the reigns of Edward II and III and Richard II.

Base, in architecture, that part of a column which is between the top of the pedestal and the bottom of the shaft; where there is no pedestal, the part between the bottom of the column and the pavement. The term is also

applied to the lower projecting part of the wall of a room, consisting of a plinth and its mouldings.

Base, in chemistry, a term applied to those compound substances which unite with acids to form salts. The most important bases are oxides of metals, and when brought in contact with acids their oxygen combines with the hydrogen of the acid to form water. They are divided into several sections, of which the most important are the alkalies. These substances are the hydrates of the so-called alkaline metals, and may be compared to water in which part of the hydrogen is replaced by a metallic radicle. Potash, for instance, is the hydrate of the metal potassium. The alkalies are readily soluble in water, restore the blue colour to reddened litmus, and give a green with red cabbage, dahlia, and other vegetable blues, and convert the yellow of turmeric into a brownish red. Most of the bases, however, are insoluble in water, and without any effect on vegetable colours. See *Alkali*; *Acids*.

Base, a term in tactics, signifying the original line on which an offensive army forms; or any safe position from which an army takes the field to invade an enemy's country; upon which it depends for its supplies, reinforcements, &c.; to which it sends back its sick and wounded; and upon which it would generally fall back in case of reverse and retreat.

Base-ball, the national game of the United States of America. It holds the position there that is held by cricket in England. It is a scientific development of the old English game of 'rounders', and is played by nine players a side. A diamond-shaped space of ground, 90 feet on the side, is marked out, the corners being the 'bases'. One side takes the field and the other sends a man to bat. When the field side takes its place, the 'pitcher', standing inside the ground near the centre and in front of the batsman, delivers a ball to the batsman, who stands at the 'home base' within a certain marked space, and who tries to drive it out of the reach of the fielders, and far enough out of the field to enable him to run round the bases, which scores a run. If he cannot run round all, he may stop at any one, and may be followed by another batsman. If the ball is caught by an opponent before touching the ground when the batsman is running, or if he is touched by the ball, he is out (also in several other cases), and when three on his side are put out, the field side take the bat. Nine of these innings make a game,

which the highest score wins. The bat is of a cylindrical shape, not more than 2½ inches in diameter nor more than 42 inches long. The ball is about 9 inches in circumference and weighs 5 - 5¼ oz. Base-ball is not a very old game, having attained its present position only since about 1845. Many professional players now engage in it. Attempts have been made to introduce it into Britain and Australia, but with little success.—BIBLIOGRAPHY: *Official Base-ball Guide*, in the Spalding Athletic Library (annually); R. H. Barbour, *The Book of School and College Sports*; A. G. Spalding, *America's National Game*; W. J. Clarke and F. T. Dawson, *Base-ball*.

Basedow (bä'ze-dō), John Bernhard, German educationalist, born in Hamburg 1723, died in 1790. After having gained considerable experience as a teacher, especially at the gymnasium of Altona, he published a number of works dealing with mental and moral philosophy, the teaching of religion and morality, &c., some of which roused a great amount of discussion. His watchword was "Everything according to nature". In 1771 he was called to Dessau by Prince Leopold, and in 1774 took charge of an educational institution in which his views were to receive practical exemplification. This institution, which he called the *Philanthropinum*, was a school free from sectarian bias, and in which the pupils were to be disciplined in all studies—physical, intellectual, and moral. This school led to the establishment of some similar ones, though Basedow retired from it in 1776, not having been very successful in the practical working out of his theories. His place was taken by Joachim Heinrich Campe (1746-1818). He henceforth devoted himself to authorship, writing especially on religious subjects. The chief feature of Basedow's system is the full development of the faculties of the young, in pursuance of the notions of Locke and Rousseau. His name still lives in the history of education, and his efforts were not without result.—BIBLIOGRAPHY: J. C. Meyer, *Leben, Charakter und Schriften Basedows* (2 vols., 1791-2); A. Pinloche, *La Réforme de l'Education en Allemagne au dix-huitième Siècle*; O. H. Lang, *Basedow, His Life and Work*.

Basel (bä'zl; Fr. *Bâle*), a canton and city of Switzerland. The canton borders on Alsace and Baden, has an area of 177 sq. miles, and a pop. of 222,000 (1916), nearly all speaking German. It is divided into two half-cantons, Basel city (Basel-Stadt) and Basel country (Basel-Land). The former

consists of the city and its precincts, the remainder of the canton forming Basel-Land, the capital of which is Liestal. The whole canton belongs to the valley of the Rhine. The soil is generally well cultivated; and the climate in the low grounds allows of the cultivation of the vine and other fruits.—The city of *Basel* is 43 miles N. of Bern, and consists of two parts on opposite sides of the Rhine, and communicating by three bridges, one of them an ancient wooden structure, besides a railway bridge. The older portions are irregularly built with narrow streets, and are now surrounded with pleasant promenades where the old fortifications existed. Basel has an ancient cathedral, founded 1010, containing the tombs of Erasmus and other eminent persons; the fine modern church of St. Elizabeth; town hall (1508); a university, founded by Pius II in 1460; a seminary for missionaries; a bible society; a museum containing the valuable public library, pictures, &c. The industries include silk ribbons (8000 hands), metal articles, tanning, paper, aniline dyes and other chemicals, brewing, &c.; and the position of Basel, a little below where the Rhine becomes navigable and at the terminus of the French and German railways, has made it the emporium of a most important trade. At Basel was signed the treaty of peace between France and Prussia, 5th April, and that between France and Spain, 22nd July, 1795. Pop. (with suburbs), 137,000 (latest estimate).

Basel, Council of, a celebrated œcumenical council of the Church, convoked by Pope Martin V and his successor Eugenius IV. It was opened 14th Dec., 1431, under the presidency of the Cardinal Legate Juliano Cesarini of St. Angelo. The objects of its deliberations were to extirpate heresies (that of the Hussites in particular), to unite all Christian nations under the Catholic Church, to put a stop to wars between Christian princes, and to reform the Church. But its first steps towards a peaceable reconciliation with the Hussites were displeasing to the Pope, who authorized the Cardinal Legate to dissolve the Council. That body opposed the pretensions of the Pope, and, notwithstanding his repeated orders to remove to Italy, continued its deliberations under the protection of the Emperor Sigismund, of the German princes, and of France. On the Pope continuing to issue bulls for its dissolution, the Council commenced a formal process against him, and cited him to appear at its bar. On his refusal to comply with this demand the Council declared him guilty of contumacy, and, after Eugenius had opened a counter-synod at Ferrara, decreed his

suspension from the papal chair (24th Jan., 1438). The removal of Eugenius, however, seemed so impracticable that some prelates, who till then had been the boldest and most influential speakers in the Council, including the Cardinal Legate Juliano, left Basel, and went over to the party of Eugenius. The Archbishop of Arles, Cardinal Louis Allemand, was now made first President of the Council, and directed its proceedings with much vigour. In May, 1439, it declared Eugenius, on account of his disobedience of its decrees, a heretic, and formally deposed him. Excommunicated by Eugenius, they proceeded, in a regular conclave, to elect the Duke Amadeus of Savoy to the papal chair. Felix V—the name he adopted—was acknowledged by only a few princes, cities, and universities. After this the moral power of the Council declined; its last formal session was held 16th May, 1443, though it was not technically dissolved till 7th May, 1449, when it gave in its adhesion to Nicholas V, the successor of Eugenius. The decrees of the Council of Basel are admitted into none of the Roman collections, and are considered of no authority by the Roman lawyers. They are regarded, however, as of authority in points of canon law in France and Germany, as their regulations for the Reformation of the Church have been adopted in the pragmatic sanctions of both countries, and, as far as they regard clerical discipline, have been actually enforced.

Base-line, in surveying, a straight line measured with the utmost precision to form the starting-point of the triangulation of a country or district. See *Geodesy*.

Bash. See *Mining*.

Ba'shan, the name in Scripture for a singularly rich tract of country lying beyond the Jordan between Mount Hermon and the land of Gilead. At the time of the Exodus it was inhabited by the Amorites, who were overpowered by the Israelites, and the land assigned to the half-tribe of Manasseh. The district was, and yet is, famous for its oak forests and its cattle. Remains of ancient cities are common.

Bashaw, or **Basha**. See *Pasha*.

Bashee' Islands, a group of islands in the Chinese Sea between Luzon and Formosa, long. 122° E.; lat. 20° 28' to 20° 55' N. They were discovered by

Dampier in 1687, and belong to the United States. The largest island is Batan, with a population of 8000.

Bashi-Bazooks', irregular troops in the Turkish army. They are mostly Asiatics, and have had to be disarmed several times by the regular troops on account of the barbarities by which they have rendered themselves infamous.

Bash'kirs, a tribe of Finno-Tartar origin, inhabiting the Russian governments of Ufa, Orenburg, Perm, and Samara. They formerly roamed about under their own princes in Southern Siberia, but in 1556 they voluntarily placed themselves under the Russian sceptre. They are nominally Mahommedans, and live by hunting, cattle-rearing, breeding of cattle and horses, and keeping of bees. They are rude and warlike and partially nomadic. They number about 1,000,000.—Cf. Ujfalvy, *Les Bashkirs*.

Bashkirtsev, Marie, Russian painter and authoress, born 1860, died 1884; educated mostly outside of Russia, in France, Germany, and Italy; became an accomplished linguist and musician, and studied art in Paris, attaining high success, but overtaxing her system, with fatal results. She is best known from her journal, an intimate personal record, interesting not only as revealing her own peculiar character and intellectual gifts, but also for notices of the notable personages with whom she came in contact. It has been translated into various languages—into English by Mathilde Blind (1890), who has also published *A Study of Marie Bashkirtsev* (1892). A number of her letters were also published in 1891.—Cf. *The Journal of Marie Bashkirtsev, an Exposure and a Defence*.

Basic Slag, the slag or refuse-matter which is got in making basic steel, and which, from the phosphate of lime it contains, is a valuable fertilizer. See *Manures (Phosphatic)*.

Basic Steel. See *Steel*.

Basi'diomycetes, one of the two sub-classes of the Eumycetes or septate Fungi, including the bulk of the larger and more familiar saprophytic types, such as the Mushroom, Toad-stools, Shelf-fungi, Puff-balls, and Earth-stars, and also the important parasites known as the Rusts. They are characterized

by their principal spores being produced externally, usually in fours, upon an organ called a *basidium*. The basidia are arranged in a continuous layer (hymenium), and are usually massed together upon a specialized fruit-body, of which an ordinary mushroom is a good example. The principal subdivisions of the group, with representative genera, are as follows:—

A. Basidia septate (Proto-Basidiomycetes).

1. Family Uredineæ (Rusts). Basidia transversely septate; teleutospores present; parasites. *Puccinia*, *Phragmidium*, *Melampsora*.
2. Family Auricularineæ. Basidia transversely septate; no teleutospores; saprophytes. *Auricularia*.
3. Family Tremellineæ. Basidia longitudinally septate; saprophytes. *Tremella*.

B. Basidia not septate (Auto-Basidiomycetes).

4. Family Exobasidiineæ. Parasites without fruit-body; basidia exposed on surface of host. *Exobasidium*.
5. Family Dacryomycetinae. Fruit-body soft; hymenium on its surface; saprophytes. *Dacryomyces*, *Calocera*.
6. Group Hymenomycetes (several families). Hymenium generally on gills, in tubes, &c.; saprophytes or, less often, parasites. *Agaricus* *Polyporus*, *Hydnum*.
7. Group Gastromycetes (several families). Hymenium within a closed fruit-body, which does not open until the spores are ripe; saprophytes. *Itthyphallus*, *Lycoperdon*.

The Ustilagineæ (Smuts) are often regarded as Basidiomycetes of a low type, but it is more probable that they are allied to the Unicellular Chytridineæ. The Basidiomycetes are the highest members of the Fungoid alliance; their relations to the lower groups are obscure.

Basidium, the characteristic spore-producing organ of the Basidiomycetes. Typically, e.g. in the Toad-stools, it is a club-shaped structure, produced at its free end into four slender processes, the *sterigmata*, each of which bears

a *basidiospore* at its tip. The young basidium contains two nuclei, which later fuse; the fusion-nucleus then undergoes two successive divisions, involving a reduction of chromosomes, and each of the four resultant nuclei passes through a sterigma into a basidiospore. These latter are thus seen to be carpospores, comparable to those of Red Algæ, and to the ascospores of Ascomycetes. The basidia of Uredineæ (Rusts) and of some other primitive Basidiomycetes are septate, but otherwise agree with the type described above. See *Basidiomycetes*; *Carpospore*; *Rusts*.

Basil. See *Basilus*.

Bas'il, a labiate plant, *Ocimum basilicum*, a native of India, much used in cookery, especially in France, and known more particularly as sweet or common basil. Bush or lesser basil is *O. minimum*; wild basil belongs to a different genus, being the *Calamintha Clinopodium*.

Basil, St., called the *Great*, one of the Greek fathers, was born in 329, and made in 370 Bishop of Cæsarea in Cappadocia, where he died in 379. He was distinguished by his efforts for the regulation of clerical discipline, and, above all, his endeavours for the promotion of monastic life. The Greek Church honours him as one of its most illustrious saints, and celebrates his festival on 1st Jan. The vows of obedience, chastity, and poverty framed by St. Basil are essentially the rules of all the orders of Christendom, although he is particularly the father of the Eastern, as St. Benedict is the patriarch of the Western orders.

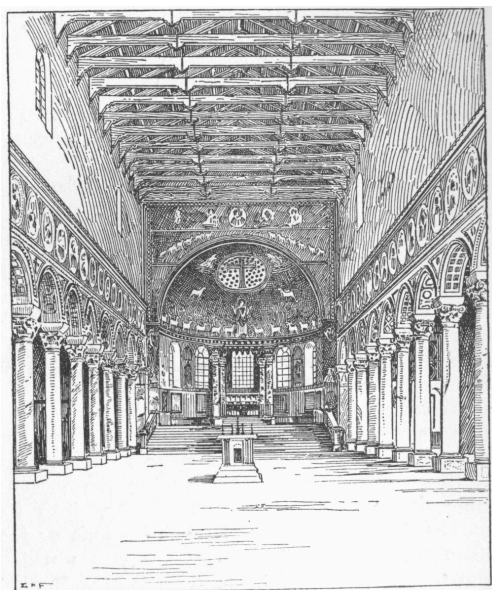
Basilan', the principal island of the Sulu Archipelago, now belonging to the Philippines, off the S.W. extremity of Mindanao, from which it is separated by the Strait of Basilan. It is about 30 miles in length by 20 miles in breadth. Pop. about 8000.

Basile'an Manuscripts, two manuscripts of the Greek New Testament now in the library of Basel. (1) A nearly complete uncial copy of the Gospels of the eighth century; (2) a cursive copy of the whole New Testament except the Apocalypse, tenth century.

Basil'ian Liturgy, that form for celebrating the Eucharist drawn up towards the close of the fourth century by Basil the Great, still used in the Greek Church.

Basilian Monks, monks who strictly follow the rules of St. Basil, chiefly belonging to the Greek Church.

Basil'ica, originally the name applied by the Romans to their public halls, either of justice, of exchange, or other business. The plan of the basilica was usually a rectangle divided into aisles by rows of columns, the middle aisle being the widest, with a semicircular apse at the end, in which the tribunal was placed. The ground-plan of these buildings was generally followed in the early Christian churches, which, therefore, long retained the name of basilica, and it is still applied to some of the churches in Rome by way of distinction, and sometimes to other churches built in imitation of the Roman basilicas.



Basilica di S'Apollinare in Classe,
Ravenna

Basilica'ta, also called **Potenza**, an Italian province, extending north from the Gulf of Taranto, and corresponding pretty closely with the ancient Lucania. Area, 3855 sq. miles; pop. 489,574 (1915).

Basil'icon, a name of several ointments, the chief ingredients of which are wax, pitch, resin, and olive-oil.

Basil'icon Do'ron (the royal gift), the title of a book written by King James I in 1599, containing a collection of precepts of the art of government. It maintains the claim of the king to be sole head of the Church. Printed at Edinburgh, 1603.

Basil'ides (dēz), one of the most famous Gnostic teachers, a native of Alexandria, who lived under the reigns of Adrian and Antoninus Pius about A.D. 120-40. He was well acquainted with Christianity, but mixed it up with the wildest dreams of the Gnostics, peopling the earth and the air with multitudes of *æons*. He was also greatly influenced by Platonism and Zoroastrianism. His disciples (Basilidians) were numerous in Syria, Egypt, Italy, and Gaul, but they are scarcely heard of after the fourth century.

Bas'ilisk, a fabulous creature formerly believed to exist, and variously regarded as a kind of serpent, lizard, or dragon, and sometimes identified with the cockatrice. It inhabited the deserts of Africa, and its breath and even its look was fatal. The name is now applied to a genus of saurian reptiles (*Basiliscus*), belonging to the family *Iguanidæ*;, distinguished by an elevated crest or row of scales, erectable at pleasure, which, like the dorsal fins of some fishes, runs along the whole length of the back and tail. The mitred or hooded basilisk (*B. mitrātus*) is especially remarkable for a membranous bag at the back of the head, of the size of a small hen's egg, which can be inflated with air at pleasure. The other species have such hoods also, but of a less size. To this organ they owe their name, which recalls the basilisk of fable, though in reality they are exceedingly harmless and lively creatures. The *B. amboinensis* is a native of the Indian Archipelago, where it is much used for food. It frequents trees overhanging water, into which it drops when alarmed.

Basil'ius I, a Macedonian, Emperor of the East, born A.D. 820, died 886. He was of obscure origin, but having succeeded in gaining the favour of the Emperor Michael III, he became his colleague in the Empire, 866. After the assassination of Michael, 867, Basilius became emperor. Though he had worked his way to the throne by a series of crimes, he proved an able and equitable sovereign. The versatility, if not the depth, of his intellect is strikingly displayed in his *Exhortations to his Son Leo*, which are still extant.

Basilius II, Emperor of the East, born 958, died 1025. On the death of his father, the Emperor Romanus the Younger, in 963, he was kept out of the succession for twelve years by two usurpers. He began to reign in conjunction with his brother Constantine, 975. His reign was spent in almost perpetual warfare, his most important struggle being that which resulted in the conquest of Bulgaria, 1018.

Ba'sin, in physical geography, the whole tract of country drained by a river and its tributaries. The line dividing one river basin from another is the water-shed, and by tracing the various water-sheds we divide each country into its constituent basins. The basin of a loch or sea consists of the basins of all the rivers which run into it.—In geology a basin is any dipping or disposition of strata towards a common axis or centre, due to upheaval and

subsidence. It is sometimes used almost synonymously with 'formation' to express the deposits lying in a certain cavity or depression in older rocks. The 'Paris basin' and 'London basin' are familiar instances.

Ba'singstoke, a town of England, county of Hants, 18 miles N.N.E. of Winchester. It has a good trade in corn, malt, &c., and now gives its name to one of the parliamentary divisions of the county. Pop. (1921), 12,718.

Bas'kerville, John, celebrated English printer and type-founder, born in 1706, died 1775. He settled at Birmingham as a writing-master, subsequently engaged in the manufacture of japanned works, and in 1750 commenced printing. From his press came highly-prized editions of ancient and modern classics, Bibles, prayer-books, &c., all beautifully-printed works. His first work was a *Virgil*, published in 1757, followed by his famous edition of Milton in 1758.

Basket, a vessel or utensil of wicker-work, made of interwoven osiers or willows, rushes, twigs, grasses, &c. The process of basket-making is very simple, and appears to be well known among the very rudest peoples. The ancient Britons excelled in the art, and their baskets were highly prized in Rome.—Cf. T. Okey, *Introduction to the Art of Basket-making*.

Basking-shark (*Selächē maxima* or *Cetorhīnus maximus*), a species of shark, so named from its habit of basking in the sun at the surface of the water. It reaches the length of 40 feet, and its liver yields a large quantity of oil. It frequents the northern seas, and is known also as the sail-fish or sun-fish.

Basle. See *Basel*.

Basoche. See *Bazoche*.

Basques (ba'sks), or **Biscayans** (in their own language, *Euscaldunac*), a remarkable race of people dwelling partly in the south-west corner of France (Basses-Pyrénées), but mostly in the north of Spain adjacent to the Pyrenees. They are probably descendants of the ancient Iberi, who occupied Spain before the Celts. They preserve their ancient language, former manners, and national dances, and make admirable soldiers, especially in guerrilla warfare. Their language is highly polysynthetic, and stands isolated from other tongues of Europe. There are eight principal dialects,

which are not only distinguished by their pronunciation and grammatical structure, but differ even in their vocabularies. The Basques, who number about 600,000 (450,000 in Spain, and 150,000 in France), occupy in Spain the provinces of Biscay, Guipuzcoa, and Alăva; in France parts of the departments of the Upper and Lower Pyrenees, Ariège, and Upper Garonne. The Basques are very religious and conservative in their religious practices. Ignatius Loyola and Francis Xavier de Navarre were Basques.—BIBLIOGRAPHY: Julien Vinson, *Les Basques et le pays Basque*; *Le Folk-lore du pays Basque*; Michel, *Le pays Basque, sa population, sa langue, ses mœurs, sa littérature, et sa musique*; Van Eys, *Outlines of a Basque Grammar*.

Basra, **Bassora**, or **Basrah**, a city in Lower Mesopotamia, on the west bank of the Shat-el-Arab (the united stream of the Tigris and Euphrates), about 50 miles from its mouth, and nearly 300 miles south-east of Bagdad. It is surrounded by a wall about 10 miles in circuit, from 20 to 25 feet thick; but much of the area enclosed is occupied by gardens, &c. The houses are generally mean. A considerable transit trade is carried on here between the Turkish and Persian dominions and India, and since communication by steamer has been established with Bagdad and Bombay the prosperity of the town has greatly increased. The chief exports are: dates, camels and horses, wool and wheat; imports: coffee, indigo, rice, tissues, &c. The inhabitants are estimated at 80,000; but in the eighteenth century they were said to number 150,000. The substitution of date and wheat cultivation for that of rice has rendered the place much more healthy. The ruins of the ancient and more famous Bassora—founded by Caliph Omar in 636, at one time a centre of Arabic literature and learning and regarded as 'the Athens of the East'—lie about 9 miles south-west of the modern town. The town was occupied by the British on 22nd Nov., 1914. The first through-train from Basra to Bagdad was run in Jan., 1920.

Bas-relief (bă´rē-lēf or bas´rē-lēf) or **Basso-Rilievo**, low-relief, a mode of sculpturing figures on a flat surface, the figures having a very slight relief or projection from the surface. It is distinguished from *haut-relief* (*alto-rilievo*), or high-relief, in which the figures stand sometimes almost entirely free from the ground. Bas-relief work has been described as 'sculptured painting', from the capability of disposing of groups of figures and exhibiting minor adjuncts, as in a painting. The finest specimen of bas-relief

is the frieze around the cella of the Parthenon; large portions of it are to be seen in the British Museum.

Bass (bās; from the It. *basso*, deep, low), in music, the lowest part in the harmony of a musical composition, whether vocal or instrumental. According to some it is the fundamental or most important part, while others regard the melody or highest part in that light. Next to the melody, the bass part is the most striking, the freest and boldest in its movements, and richest in effect.—*Figured bass*, a bass part having the accompanying chords suggested by certain figures written above or below the notes—the most successful system of shorthand scoring at present in use among organists and pianists.—*Fundamental bass*, the lowest note or root of a chord; a bass consisting of a succession of fundamental notes.—*Thorough bass*, the mode or art of expressing chords by means of figures placed over or under a given bass. Figures written over each other indicate that the notes they represent are to be sounded simultaneously, those standing close after each other that they are to be sounded successively. The common chord in its fundamental form is generally left unfigured, and accidentals are indicated by using sharps, naturals, or flats along with the figures.



Bas-relief—Northern frieze of Parthenon

Bass (ba's), the name of a number of fishes of several genera, but originally belonging to a genus of sea-fishes (*Labrax*) of the perch family, distinguished from the true perches by having the tongue covered by small teeth and the preoperculum smooth. *L. lupus*, the only British species, called also sea-dace, and from its voracity sea-wolf, resembles somewhat the salmon in shape, and is much esteemed for the table, weighing about 15 lb. *L. lineatus* (*Roccus lineatus*), or striped bass, an American species, weighing from 25 to 30 lb., is much used for food, and is also known as rock-fish. Both species occasionally ascend rivers, and attempts have been made to cultivate British bass in freshwater ponds with success. Two species of black bass (*Micropterus salmoides* and *M. dolomieu*), American freshwater fishes, are excellent as food and give fine sport to the angler. The

former is often called the large-mouthed black bass, from the size of its mouth. Both make nests and take great care of their eggs and young. The *Centropristis nigricans*, an American sea-fish of the perch family, and weighing 2 to 3 lb., is known as the sea-bass.

Bass (ba's), **The**, a remarkable insular trap-rock, at the mouth of the Firth of Forth, 3 miles from North Berwick, of a circular form, about 1 mile in circumference, rising majestically out of the sea to a height of 313 feet. It pastures a few sheep, and is a great breeding-place of solan geese. During the persecution of the Covenanters its castle, long since demolished, was used as a State prison, in which several eminent Covenanters were confined. It was held from 1691 to 1694 with great courage and pertinacity by twenty Jacobites, who in the end capitulated on highly honourable terms.

Bass. See *Basswood*.

Bassa'no, a commercial city of North Italy, province of Vicenza, on the Brenta, over which is a covered wooden bridge. It has lofty old walls and an old castle, and has various industries and an active trade. Near Bassano, 8th Sept., 1796, Bonaparte defeated the Austrian general Wurmser. Pop. (commune) 17,130.

Bassa'no (from his birth-place; real name **Giacomo da Ponte**), an Italian painter, born 1510, died 1592. He painted historical pieces, landscapes, flowers, &c., and also portraits. He left four sons, who all became painters, Francesco being the most distinguished.

Bassano Dam. See *Dams*.

Bas'saris. See *Cacomistle*.

Bassein (bas-sān'), a town in Lower Burma, province of Pegu, on both banks of the Bassein River, one of the mouths of the Irawadi, and navigable for the largest ships. It has considerable trade, exporting large quantities of rice, and importing coal, salt, cottons, &c. Pop. 37,081.—Bassein District has an area of 4127 sq. miles and a pop. of 9598 (1911).

Bassein (bas-sān'), a town in Hindustan, 28 miles north of Bombay. At the beginning of the eighteenth century it was a fine and wealthy city, with over 60,000 inhabitants; it has now only about 9598 (1911).

Basselin (ba's-lan), Olivier, an old French poet or song-writer, born in the Val-de-Vire, Normandy, about the end of the fourteenth century, died about 1450. His sprightly songs, famous under the name *Vaux-de-Vire*, have given origin and name to the modern Vaudevilles.

Basselisse Tapestry. See *Hautelisse*.

Basses-Alpes (bäs-a'lp; 'Lower Alps'). See *Alpes*.

Basses-Pyrénées (bäs-pē-rā-nā; 'Lower Pyrenees'). See *Pyrénées*.

Bass'et, the name of a game at cards, formerly much played, especially in France. It is very similar to the modern *faro*. See *Lansquenet*.

Basset, or **Basset-hound**, a smooth-haired dog with short crooked legs, rather large head, and large pendulous ears, thus somewhat resembling a bloodhound and a dachshund. It is sometimes used in packs for hunting hares, or in beating and covert work generally.

Basseterre (bäs-tār), two towns in the West Indies.—1. Capital of the Island of St. Christopher's, at the mouth of a small river, on the south side of the island. Trade considerable. Pop. about 9000.—2. The capital of the Island of Guadeloupe. It has no harbour, and the anchorage is unsheltered and exposed to a constant swell. Pop. 8650.

Basset-horn, a musical instrument, now practically obsolete, a sort of clarinet of enlarged dimensions, with a curved and bell-shaped metal end. The compass extends from F below the bass staff to C on the second ledger-line above the treble. Mozart wrote several pieces for the basset-horn, and Beethoven employed it in his *Prometheus* overture.

Bassetlaw, a parliamentary division of the county of Nottingham.

Bassia, a genus of tropical trees found in the East Indies and Australia, nat. ord., Sapotaceæ. The seeds of *B. butyracea*, the Indian butter tree, yield a butter-like substance, which makes good soap. *B. latifolia*, the Mahwa of Bengal, furnishes a valuable hard timber, and has edible flowers, from which a potent spirit is distilled; a gutta-percha is obtained from *B. pallida*.

Bassompierre (bä-son-pyār), Francois de, Marshal of France, distinguished both as a soldier and a statesman, born 1579, died 1646. In 1602 he made

his first campaign against the Duke of Savoy, and he fought with equal distinction in the following year in the imperial army against the Turks. In 1622 Louis XIII appointed him Marshal of France, and became so much attached to him that Luynes, the declared favourite, sent him on embassies to Spain, Switzerland, and England. After his return he became an object of suspicion to Cardinal Richelieu, and was sent to the Bastille in 1631, from which he was not released till 1643, after the death of the cardinal. During his detention he occupied himself with writing his memoirs, which shed much light on the events of that time.

Basso´on, a musical wind-instrument of the reed order, blown with a bent metal mouthpiece, and holed and keyed like the clarinet. Its compass comprehends three octaves rising from B flat below the bass staff. Its diameter at bottom is 3 inches, and for convenience of carriage it is divided into two or more parts, whence its Italian name, *fagotto*, a bundle. It serves for the bass among wood wind-instruments, as hautboys, flutes, &c.

Bassora Gum, an inferior kind of gum resembling gum-arabic.

Bass Rock. See *Bass*.

Bass Strait, a channel beset with islands, which separates Australia from Tasmania, 120 miles broad, discovered by George Bass, a surgeon in the Royal Navy, in 1798.

Basswood, or **Bass**, the American lime tree or linden (*Tilia americana*), a tree common in N. America, yielding a light, soft timber.

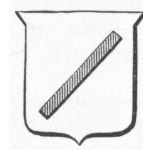
Bast, a term originally applied to fibrous tissue external to the cambium; now used in two senses: (1) for Phloem (q.v.); (2) commercially, for the fibre of the lime or linden. The manufacture of lime-bast into mats, ropes, &c., is in Russia a considerable industry, bast-mats for packing furniture, covering plants, &c., being largely exported.

Bastar´, a feudatory State in Upper Godavari district, Central Provinces of India; area, 13,062 sq. miles; pop. 433,310. Chief town, Jagdalpur; pop. 4294.

Bas´tard, a child begotten and born out of wedlock; an illegitimate child. By the civil and canon laws, and by the law of Scotland, a bastard becomes

legitimate by the intermarriage of the parents at any future time. In some of the United States, bastards can be legitimized by subsequent marriage, in addition to which, however, the father must also recognize the child as his. By the laws of England a child, to be legitimate, must at least be *born* after the lawful marriage; it does not require that the child shall be begotten in wedlock, but it is indispensable that it should be born after marriage, no matter how short the time, the law presuming it to be the child of the husband. The only incapacity of a bastard is that he cannot be heir or next of kin to any one save his own issue. In England the maintenance of a bastard in the first instance devolves on the mother, while in Scotland it is a joint burden upon both parents. The mother is entitled to the custody of the child in preference to the father. See *Legitimation; Illegitimacy*.—
BIBLIOGRAPHY: Bacquet, *Traité de la Bâtardise*; Nicholls and Mackay, *History of the English Poor Law*.

Bastard Bar, more correctly *baton sinister*, the heraldic mark used to indicate illegitimate descent. It is a diminutive of the bend sinister, of which it is one-fourth in width, coupé or cut short at the ends, so as not to touch the corners of the shield.



Bastard Bar

Bastard Fallow. See *Fallow Land*.

Bastard Saffron. See *Safflower*.

Bastia (ba's-tē'a'), the former capital of the Island of Corsica, upon the N.E. coast, 75 miles N.E. of Ajaccio, on a hill slope; badly built, with narrow streets, a strong citadel, and an indifferent harbour; with some manufactures, a considerable trade in hides, soap, wine, oil, pulse, &c. Pop. 29,412.

Bas'tian, Adolf, German traveller and ethnologist, born in 1826, died in 1905. His travels embraced various parts of Europe, the United States, Mexico, Peru, Australia and New Zealand, Southern and Western Africa, Egypt, Arabia, India, South-Eastern Asia, the Asiatic Archipelago, Japan, China, Mongolia, Siberia, &c. His numerous writings throw light on almost every subject connected with ethnology or anthropology, as well as psychology, linguistics, non-Christian religions, geography, &c. Some of his chief works are: *Die Völker des östlichen Asien* (Peoples of Eastern

Asia); *Ethnographische Forschungen*; *Ideale Welten*; *Die Völkerkunde u. der Völkerverkehr*; &c.

Bas'tian, Henry Charlton, English physician and biologist, born at Truro in 1837, died on 17th Nov., 1915. He was educated at Falmouth and at University College, London, where he was assistant curator in the museum from 1860 to 1863. He obtained the degree of M.A. in 1861 from the University of London, graduating subsequently in medicine at the same university (M.B. 1863, M.D. 1866). From 1864 to 1866 he was a medical officer in Broadmoor Criminal Lunatic Asylum, and in 1866 was appointed lecturer on pathology and assistant physician in St. Mary's Hospital. In 1867 he became professor of pathological anatomy in University College, subsequently he was also professor of clinical medicine, and in the period 1887-95 he occupied the chair of medicine and clinical medicine. Apart from numerous contributions to medical and other periodicals, and to Quain's *Dictionary of Medicine*, he published *The Modes of Origin of Lowest Organisms* (1871); *The Beginnings of Life* (1872); *Evolution and the Origin of Life* (1874); *Lectures on Paralysis from Brain Disease* (1875); *The Brain as an Organ of Mind* (1880); which has been translated into French and German; *The Nature and Origin of Living Matter*; *Evolution of Life*; *Origin of Life*; &c.

Bastiat (ba's-tē-a'), Frédéric, French economist, advocate of free-trade, and opponent of protection, born at Bayonne 1801, died at Rome 1850. He became acquainted with Cobden and the English free-traders, whose speeches he translated into French. Whilst combating protectionism, Bastiat was also an opponent of socialism. His chief works are: *Sophismes Économiques* (1846); *Propriété et Loi*; *Justice et Fraternité* (1848); *Protectionisme et Communisme* (1849); *Harmonies Économiques* (1849).

Bastien-Lepage (ba's-tyan'-lè-päzh), Jules, French painter, born 1848, died 1884. He studied at Paris under Cabanel, and about 1874 began to attract some notice as a realistic painter of subjects connected with the country and everyday life, among his first pictures of note being *Song of Spring*, *Portrait of My Grandfather*, and *The First Communion*. Among his most important works are *The Hayfield*, *The Potato Harvest*, *Portrait of Mme Sarah Bernhardt*, *The Beggar*, and *The Forge*.—Cf. Marie Bashkirtsev, *Journal Intime*; Arnic, J. *Bastien-Lepage, lettres et souvenirs*.

Bastille (ba´ s-tēl´), a French name for any strong castle provided with towers, but as a proper name the State prison and citadel of Paris, which was built about 1370 by Charles V. It was ultimately used chiefly for the confinement of persons of rank who had fallen victims to the intrigues of the Court or the caprice of the Government. (See *Cachet, Lettres de.*) The capture of the Bastille by the Parisian mob, 14th July, 1789, was the opening act of the Revolution. On that date the Bastille was surrounded by a tumultuous mob, who first attempted to negotiate with the Governor, Delaunay, but when these negotiations failed, began to attack the fortress. For several hours the mob continued their siege without being able to effect anything more than an entrance into the outer court of the Bastille; but at last the arrival of some of the Royal Guard with a few pieces of artillery forced the Governor to let down the second drawbridge and admit the populace. The Governor was seized, but on the way to the Hotel de Ville he was torn from his captors and put to death. The next day the destruction of the Bastille commenced. Not a vestige of it exists, but its site is marked by a column in the Place de la Bastille.—BIBLIOGRAPHY: François Ravaisson, *Les Archives de la Bastille*; Arnold, *Histoire de la Bastille*; Bingham, *The Bastille*; Funck-Brentano, *The Bastille*; M.J. de Staal, *La Bastille sous la Régence* (edited by Funck-Brentano).

Bastina´do, an Eastern method of corporal punishment, consisting of blows upon the soles of the feet, applied with a bamboo cane.

Bas´tion, in fortification, a large mass of earth, faced with sods, brick, or stones, standing out from a rampart, of which it is a principal part. A bastion consists of two *flanks*, each commanding and defending the adjacent *curtain*, or that portion of the wall extending from one bastion to another, and two *faces* making with each other an acute angle called the *salient angle*, and commanding the outworks and ground before the fortification. The distance between the two flanks is the *gorge*, or entrance into the bastion. The use of the bastion is to bring every point at the foot of the rampart as much as possible under the guns of the place.

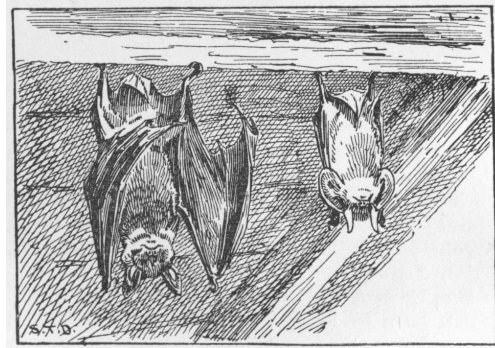
Bast´wick, John, English physician and ecclesiastical controversialist, born in 1593, died 1654. He settled at Colchester, but, instead of confining himself to his profession, entered keenly into theological controversy, and was condemned by the Star Chamber for his books against Prelacy:

Elenchus Religionis Papisticæ, Flagellum Pontificis, and *The Letanie of Dr. J. Bastwick*. With Prynne and Burton he was sentenced to lose his ears in the pillory, to pay a fine of £5000, and to be imprisoned for life. He was released by the Long Parliament, and entered London in triumph along with Prynne and Burton. He appears to have continued his controversies with the Independents and others to the very last.

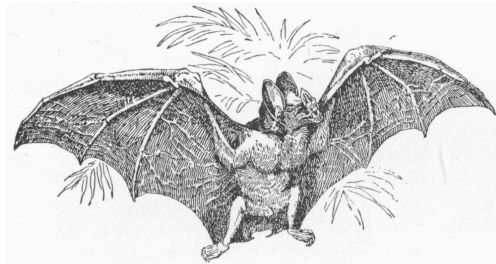
Basu'toland, a native province and British South African possession, situated between the Orange Free State, Natal, Griqualand East, and the Cape Province. The Basutos belong chiefly to the great stem of the Bechuanas, and have made greater advances in civilization than perhaps any other South African race. In 1866 the Basutos, who had lived under a semi-protectorate of the British since 1848, were proclaimed British subjects, their country placed under the government of an agent, and in 1871 it was joined to Cape Colony. In 1879 the attempted enforcement of an Act passed for the disarmament of the native tribes caused a revolt under the chief Moirosi, which the Cape forces were unable to put down. When peace was restored Basutoland was separated from Cape Colony (1884), and is now governed by a Resident Commissioner under the High Commissioner for South Africa. Basutoland has an area of about 11,716 sq. miles, much of it covered with grass, and there is but little wood. The climate is pleasant. The natives keep cattle, sheep, and horses, cultivate the ground, and export grain. The Basuto ponies, which originally were brought from Batavia by the Dutch in the eighteenth century, are famous. It is divided into eight districts, each under a magistrate. Capital, Maseru. Pop. (European), 1396; (native), 404,507.—BIBLIOGRAPHY: J. Widdicombe, *Fourteen Years in Basutoland*; M. Martin, *Basutoland: Its Legends and Customs*; Sir G. Lagden, *The Basutos*; Rev. D. F. Ellenberger, *History of the Basuto*.

Bat, one of the group of wing-handed, flying mammals, having the fore-limb peculiarly modified so as to serve for flight, and constituting the order Cheiroptera. Bats are animals of the twilight and darkness, and are common in temperate and warm regions, but are most numerous and largest in the tropics. All European bats are small, and have a mouse-like skin. The body of the largest British species, *Vespertilio noctūla*, is less than that of a mouse, but its wings stretch about 15 inches. During the day it remains in caverns, in the crevices of ruins, hollow trees, and similar lurking places,

and flits out at evening in search of food, which consists of insects. Several species of the same genus are common in North America. Many bats are remarkable for having a singular nasal cutaneous appendage, bearing in some cases a fancied resemblance to a horse-shoe. Bats may be conveniently divided into two sections—the insectivorous or carnivorous, comprising all European and most African and American species; and the fruit-eating, belonging to tropical Asia and Australia, with several African forms. An Australian fruit-eating bat (*Pteropus edulis*), commonly known as the kalong or flying-fox, is the largest of all the bats; it does much mischief in orchards. At least two species of South American bats are known to suck the blood of other mammals, and thence are called 'vampire-bats'. All the British bats belong to the insectivorous group, the fruit-eating and blood-sucking bats being confined to warmer regions. There are fifteen species on the British list, but of these three are very rare. Among the most noteworthy forms are the greater and the lesser horse-shoe bats, in which the ears are nearly as long as the whole body. As winter approaches, in cold climates bats seek shelter in caverns, vaults, ruinous and deserted buildings, and similar retreats, where they cling together in large clusters, hanging head downwards by the feet, and remain in a torpid condition until the returning spring recalls them to active exertions. Bats generally bring forth two young. The parent shows a strong degree of attachment to her offspring, and, when they are captured, will follow them, and even submit to captivity herself rather than forsake her charge.



Bats Asleep
Left—Serotine (*Vespertilio serotinus*).
Right—Long-eared bat (*Plecotus auritus*).



Bataille, Henry, French poet and dramatist, born at Nîmes in 1872. His first volume of poems, *La Chambre Blanche*, appeared in 1895. His other works include *Le Beau Rivage*, *Maman Colibri*, *La Marche Nuptiale*, *La Femme Nue*, *La Divine Tragédie*, &c.

Batalha (ba'-ta'l'ya'), a village in Portugal, 69 miles north of Lisbon, with a renowned convent of Dominicans, a splendid building.

Batan'gas, a town of the Philippines, in the Island of Luzon, capital of a province of same name, 58 miles south of Manila. Pop. 33,131.

Bata'tas. See *Sweet-potato*.

Bat'avi. See *Batavians*.

Bata'via, a city and seaport of Java, on the north coast of the island, the capital of the Dutch East Indies. It is situated on a wide, deep bay, the principal warehouses and offices of the Europeans, the Java Bank, the exchange, &c., being in the old town, which is built on a low, marshy plain near the sea, intersected with canals and very unhealthy; while the Europeans reside in a new and much healthier quarter. Batavia has a large trade, sugar being the chief export. It was founded by the Dutch in 1619, and attained its greatest prosperity in the beginning of the eighteenth century. Its inhabitants are chiefly Malay, with a considerable admixture of Chinese and a small number of Europeans. Pop. 234,697 (1918).

Batavian Republic. See *Belgium; Netherlands*.

Batavians, an old German nation which inhabited a part of the present Holland, especially the island called *Batavia*, formed by that branch of the Rhine which empties itself into the sea near Leyden, together with the Waal and the Meuse. Tacitus asserts that they were a branch of the Catti. They were subdued by Germanicus, and were granted special privileges for their faithful services to the Romans, but revolted under Vespasian. They were, however, again subdued by Trajan and Adrian, and at the end of the third century the Salian Franks obtained possession of the Island of Batavia.

Batchian. See *Bachian*.

Bates, Henry Walter, traveller and naturalist, born at Leicester 1825, learned the hosiery trade, but in 1847 went to the Amazon with Russel Wallace, and remained there for eleven years studying the natural history of the region. Returning to England, he published in 1862 his *Naturalist on the River Amazons*, a work regarded as a classic. He was assistant secretary to the Royal Geographical Society from 1864 till his death in 1892. A large part of his collections is in the British Museum.

Bath (ba`th), a city of England, in Somersetshire, on the Avon, which is navigable for barges from Bristol; is beautifully placed among the hills, and the houses are built of freestone, obtained from the neighbourhood. The Abbey Church, dating from the fifteenth century, ranks as one of the finest specimens of Perpendicular Gothic architecture. Bath is remarkable for its medicinal waters, the four principal springs yielding no less than 184,000 gallons of water a day; and the baths are both handsome and roomy. The temperature of the springs varies from 109° to 117° F. They contain carbonic acid, chloride of sodium and of magnesium, sulphate of soda, carbonate and sulphate of lime, &c. Bath was founded by the Romans, and called by them *Aquæ Solis* (Waters of the Sun). Amongst the Roman remains discovered there have been some fine baths. The height of its prosperity was reached, however, in the eighteenth century, when Beau Nash was leader of the fashion and master of its ceremonies. Since then, though it still attracts large numbers of visitors, it has become the resort of valetudinarians chiefly. Jointly with Wells it is the head of a diocese, and since 1918 returns one member to the House of Commons. Pop. (1921), 68,648.

Bath, a town, United States, Maine, on the west side and at the head of the winter navigation of the Kennebec, 12 miles from the sea. Chief industries: shipbuilding and allied crafts. Pop. 9396.

Bath, the immersion of the body in water, or an apparatus for this purpose.

Ancient Baths.—The use of the bath as an institution, apart from occasional immersion in rivers or the sea, is, as might be anticipated, an exceedingly old custom. Homer mentions the bath as one of the first refreshments offered to a guest; thus, when Ulysses enters the palace of Circe, a bath is prepared for him, and he is anointed after it with costly perfumes. No

representation, however, of a bath as we understand it is given upon the Greek vases, bathers being represented either simply washing at an elevated basin, or having water poured over them from above. In later times, rooms, both public and private, were built expressly for bathing, the public baths of the Greeks being mostly connected with the gymnasia. Apparently, by an inversion of the later practice, it was customary in the Homeric epoch to take first a cold and then a hot bath; but the Lacedæmonians substituted the hot-air sudorific bath, as less enervating than warm water, and in Athens at the time of Demosthenes and Socrates the warm bath was considered by the more rigorous to be an effeminate custom. The fullest details we have with respect to the bathing of the ancients apply to its luxurious development under the Romans. Their bathing establishments consisted of four main sections: the undressing room, with an adjoining chamber in which the bathers were anointed; a cold room with provision for a cold bath; a room heated moderately to serve as a preparation for the highest and lowest temperatures; and the sweating-room, at one extremity of which was a vapour-bath and at the other an ordinary hot bath. After going through the entire course both the Greeks and Romans made use of strigils or scrapers, either of horn or metal, to remove perspiration, oil, and impurities from the skin. Connected with the baths were walks, covered race-grounds, tennis-courts, and gardens, the whole, both in the external and internal decorations, being frequently on a palatial scale. The group of the *Laocoon* and the *Farnese Hercules* were both found in the ruins of Roman baths.

Modern Baths.—With respect to modern baths, that commonly in use in Russia consists of a single hall, built of wood, in the midst of which is a powerful metal oven, covered with heated stones, and surrounded with broad benches, on which the bathers take their places. Cold water is then poured upon the heated stones, and a thick, hot steam rises, which causes the sweat to issue from the whole body. The bather is then gently whipped with wet birch rods, rubbed with soap, and washed with luke-warm and cold water; of the latter, some pailfuls are poured over his head; or else he leaps, immediately after this sweating-bath, into a river or pond, or rolls in the snow. The Turks, by their religion, are obliged to make repeated ablutions daily, and for this purpose there is, in every city, a public bath connected with a mosque. A favourite bath among them, however, is a modification of the hot-air sudorific bath of the ancients introduced under

the name of 'Turkish' into other than Mahomedan countries. A regular accompaniment of this bath, when properly given, is the operation known as 'kneading', generally performed at the close of the sweating process, after the final rubbing of the bather with soap, and consisting in a systematic pressing and squeezing of the whole body, stretching the limbs, and manipulating all the joints as well as the fleshy and muscular parts. Public baths are now common in Europe, but the first English public baths and wash-houses of the kind now common in all cities were established in Liverpool and near the London docks in 1844. In 1846 an Act was passed for their encouragement, and a series of statutes, known as 'The Baths and Wash-houses Acts 1846 to 1896', followed. The establishment of cheap swimming-baths was authorized in 1878.

The principal natural warm baths in England are at Bath, in Somersetshire (the hottest), and Buxton and Matlock, in Derbyshire. The temperature of the Bath springs ranges from 109° to 117° F., while that of the Buxton and Matlock waters scarcely exceeds 82°. The baths of Harrogate, which are strongly impregnated with sulphuretted hydrogen gas, are also of great repute for the cure of obstinate cutaneous diseases, indurations of the glands, &c. The most celebrated natural hot baths in Europe are those of Aix-la-Chapelle, and the various Baden, in Germany; Toeplitz, in Bohemia; Bagnières, Baréges, and Dax, in the south of France; and Spa, in Belgium. Besides the various kinds of water-bath with or without medication or natural mineral ingredients, there are also milk, oil, wine, earth, sand, mud, and electric baths, smoke-baths and gas-baths; but these are as a rule only indulged in after specific prescription.

The practice of bathing as a method of cure in cases of disease falls under the head of hydropathy; but even when it is employed simply for pleasure or purification due regard should be paid to the physiological condition of the bather. In many cases cold bathing should be avoided altogether, especially by those who have any tendency to spitting of blood or consumption, by gouty people, or by those who have any latent visceral disease or apoplectic tendency. Wherever the bath is followed by shivering instead of by a healthy reactionary glow, it is undesirable; and a cold bath in the morning after any debauchery or excess in eating or drinking on the previous evening is exceedingly imprudent. Delicate persons and children ought not to bathe in the sea before ten or eleven o'clock in the morning,

and in no case should bathing be indulged in after a long fast. In cold streams and rivers additional precautions should be taken, the cold plunge, when heated or fatigued, being frequently attended with fatal results. Even warm baths are not wholly free from danger; apoplexy and death having been known to follow a hot bath when entered with a full stomach. As a rule the temperature should not exceed 105°, and they should not be too long continued. Frequent indulgence in them has an enervating effect, though the majority of people need as yet no renewal of Hadrian's prohibitive legislation in this matter. See also *Douches*; *Thermæ*.—BIBLIOGRAPHY: W. A. Becker, *Gallus*; J. Farrar, *Baths and Bathing*; W. P. Gerhard, *Modern Baths and Bath Houses*; R. F. Fox, *Principles and Practice of Medical Hydrology*.

Bath, Knights of the, an order of England, supposed to have been instituted by Henry IV on the day of his coronation, but allowed to lapse after the reign of Charles II till 1725, when George I revived it as a military order. By the book of statutes then prepared the number of knights was limited to the sovereign and thirty-seven knights companions; but the limits of the order were greatly extended in 1815, and again in 1847, when it was opened to civilians. It now consists of three classes, each subdivided into (1) military members, (2) civil members, and (3) honorary members, consisting of foreign princes and officers. The first class consists of Knights of the Grand Cross (G.C.B.); the second of Knights Commanders (K.C.B.); and the third of Companions (C.B.). The Dean of Westminster is dean of the order. In 1917 the Companions of the order were given the privileges and precedence of Commanders, the badge being now worn round the neck. In 1918 the statutes were amended in order to admit officers of the Royal Air Force to the military division. The ribbon of the order is crimson; the military badge a gold cross of eight points, with the lion of England between the four principal angles, and having in a circle in the centre the rose, thistle, and shamrock between three imperial crowns; motto: *Tria juncta in uno*. Stars are worn by the two first classes, with additional motto, *Ich dien*, for the military members only. See *Knighthood*.

Bath-brick, a preparation of siliceous earth found in the River Parret, in Somersetshire, in the form of a solid brick, used for cleaning knives, &c. See *Sand*.

Bath'gate, a town, Scotland, Linlithgowshire, pleasantly situated, and having in the vicinity the paraffin-works known as Young's, a paper-mill and other works, and coal and iron-stone mines. Pop. (1921), 8504.

Batholite, a mass of igneous rock having no visible floor, and presumed to pass down into some region of the earth's crust where molten rocks at one time prevailed over all others. Granite blocks, like those of the Cairngorm Mountains in Scotland, or the long ridge of the Leinster Chain, which have intruded into their surroundings, are regarded as batholites the crests of which have become exposed by denudation. A batholite in its original molten condition may exist under any large area of volcanic activity, and the similarity of the igneous rocks sent to the surface over broad regions of the earth, constituting what are called 'petrographical provinces', points to the existence of very considerable continuous rock-cauldrons in the crust.

Bathom'eter, an instrument, invented by C. Williams Siemens, for measuring the depth of sea beneath a vessel without casting a line. It is based upon the fact that the attraction exerted upon any given mass of matter on the ship is less when she is afloat than ashore, because of the less density of sea-water as compared with that of earth or rock.

Bathonian Stage, a division of the middle Jurassic series of rocks, including the beds between the Inferior Oolite and the Oxford Clay. The name is derived from its excellent development near Bath, in Somersetshire, and the most important member is the Oolitic limestone known as Bath-stone (q.v.). The Bathonian stage includes the following beds in ascending order:—

1. *Stonesfield Slate*, named from a village near Woodstock; a flaggy limestone used for roofing, with remains of ferns and cycads washed in from the Jurassic land, and of some of the earliest-known mammals, allies of the monotremes or the marsupials. 2. The *Great* or *Bath Oolite*, with marine fossils, and also remains of the large reptiles known as dinosaurs. 3. *Bradford Clay*. 4. *Forest Marble*. 5. *Cornbrash*, so called from its 'brashy' or rubbly nature, an earthy oolite yielding fair land for corn.

Bathori (bă'to-rē), or **Bathory**, a Hungarian family, which gave Transylvania five princes, and Poland one of its greatest kings. The more important members were:—1. Stephen, born in 1532, elected Prince of

Transylvania in 1571, on the death of Zapolya, and in 1575 King of Poland. He accomplished many internal reforms, recovered the Polish territories in possession of the Czar of Moscovy, and reigned prosperously till his death in 1586.—2. Sigismund, nephew of Stephen, educated by the Jesuits, became Prince of Transylvania in 1581, shook off the Ottoman yoke, and had begun to give hopes of reigning gloriously when he resigned his dominions to the Emperor Rudolph II, in return for two principalities in Silesia, a cardinal's hat, and a pension. Availing himself, however, of an invitation by the Transylvanians, he returned, and placed himself under the protection of the Porte, but was defeated by the Imperialists in every battle, and finally sent to Prague, where he died almost forgotten in 1613.—3. Elizabeth, niece of Stephen, King of Poland, and wife of Count Nadasdy, of Hungary. She is said to have bathed in the blood of 600 young girls in the hope of renewing her youth, and to have committed other enormities. She was afterwards seized and confined till her death in 1614.

Bat-horse. See *Batman*.

Ba'thos, a Greek word meaning depth, now used to signify a ludicrous sinking from the elevated to the mean in writing or speech. First used in this sense by Pope in *The Art of Sinking in Poetry* (*Scriblerus Papers*, 1729).

Bath-stone (also called **Bath-oolite** and **Roe-stone**, from the small rounded grains of which it is composed), a species of English limestone. It is extensively worked near Bath for building purposes. When just quarried it is soft; but though it soon becomes hard on exposure to the atmosphere, and is of handsome appearance, it is not very durable.

Bath'urst, a British settlement on the west coast of Africa, on the Island of St. Mary's, near the mouth of the Gambia, with a trade in gum, bees-wax, hides, ivory, gold, rice, cotton, and palm-oil. Pop. 8000.

Bathurst, a town in the western district of New South Wales, on the Macquarie River, with wide, well-laid-out streets at right angles, and a central square, tanneries, railway workshops, breweries, flour-mills, and other industries. Pop. 9200.

Bathurst, Allen Bathurst, First Earl, a distinguished statesman in Queen Anne's reign; born 1684. He took part with Harley and St. John in opposing

the influence of Marlborough, was raised to the peerage in 1711, impeached the promoters of the South Sea scheme, opposed the Bill against Atterbury, and was a leading antagonist of Walpole. He was created earl in 1772. His name is also associated with those of the leading writers and wits of the day. Died 1775.

Bathurst, Henry Bathurst, Third Earl, son of the second earl, a prominent Tory statesman, after whom various capes, islands, and districts were named. Born 1762; in 1807, President of Board of Trade; in 1809, Secretary for Foreign Affairs; and in 1812, Secretary for the Colonies, a post held by him for sixteen years. He was also President of the Council under Wellington, 1828-30. Died 1834.

Bathurst Island, on the North Australian coast, belonging to S. Australia, separated from Melville Island by a narrow strait; triangular in shape, with a wooded area of about 1000 sq. miles.—Also an island in the Arctic Ocean discovered by Parry, E. of Cornwallis and W. of Melville Island, 76° N., 100° W.

Bathyb'ius (Gr. *bathys*, deep, *bios*, life), the name given by Huxley to what was regarded as granular slimy masses of animal matter found covering the sea-bottom at great depths.

Batiste (ba-tēst'), a fine linen cloth made in Flanders and Picardy, named after its inventor, Jean Baptiste of Cambrai.

Bat'ley, a municipal and parliamentary borough of England, West Riding of York, about a mile from Dewsbury; principal manufactures: heavy woollen cloths, such as pilot, beaver, police, army, and frieze cloths, flushings, and blankets. Batley and Morley unite in returning one member to Parliament. Pop. 36,151.

Batman (bat'man or bā'man; from Fr. *bât*, a pack-saddle), in the British army, a person allowed by the Government to every company of a regiment on foreign service. His duty is to take charge of the cooking utensils, &c., of the company, and he has a bat-horse to convey these utensils from place to place.

Ba'ton, a short staff or truncheon, in some cases used as an official badge, as that of a field-marshal. The conductor of an orchestra has a baton for the

purpose of directing the performers as to time, &c. In heraldry, what is usually called the 'bastard bar', or 'bar sinister', is properly a baton sinister. See *Bastard Bar*.

Bat'on Rouge (rözh), the capital of Louisiana, United States, on the left bank of the Mississippi, with an arsenal, barracks, military hospital, State house, State university, &c. On 5th Aug., 1862, the Confederates under General Breckenridge suffered a severe defeat before it. Pop. (1920), 21,782.

Batoum, or **Batum** (ba'-töm'), a port on the east coast of the Black Sea, acquired by Russia by the Treaty of Berlin, on condition that its fortifications were dismantled and it was thrown open as a free port. It rapidly grew to be the main outlet for Transcaucasia; its harbour was enlarged for commercial reasons; an arsenal was built outside it; it was connected by a military road with Kars; and finally, in July, 1886, the Russian Government declared it to be a free port no longer. Its importance as a naval and military station to Russia was unquestionably great; it is of most importance commercially as an outlet for the mineral oil of Baku. The water is of great depth close inshore, and the shipping lies under protection of the overhanging cliffs of the Gouriél Mountains. Batoum was abandoned by the Bolshevik Government in March, 1918, and occupied by the Turks on 15th April of the same year. By the terms of the armistice concluded with Turkey in Nov., 1918, the Allies occupied the port, and in Nov., 1919, a military governor was appointed by the British Government. On 9th July, 1920, Batoum was evacuated by the British, and occupied by Georgian troops. Pop. 46,000.—The province of *Batoum* has an area of 2693 sq. miles. Pop. 186,000.

Batrachians (ba-trā'ki-anz; Gr. *batrachōs*, a frog), the fourth order in Cuvier's arrangement of the class Reptilia, comprising frogs, toads, newts, salamanders, and sirens. The term is now sometimes employed as synonymous with amphibia, but is more usually restricted to the order Anura or tailless amphibia. See *Amphibia*.

Batshian. See *Bachian*.

Bat'ta, an allowance which military officers in India receive in addition to their pay. It was originally given only when the officers were on the march

or in the field, but now half batta is paid when troops are in cantonments.

Battal'ion, the tactical unit of command in infantry, supposed to be of the maximum strength to be efficiently handled by one officer in chief command, with others under him. In most armies it is about 1000 to 1100 men. The former is about the strength of a battalion in the British army, in which battalions now correspond to what were formerly regiments, the chief commanding officer (in actual command) being the lieutenant-colonel. See *Army*.

Battalion of Death, the legion of Russian women and girls, belonging to all classes, organized in 1917 and commanded by the famous revolutionary Madame Botshkalova. The battalion took part in several engagements.

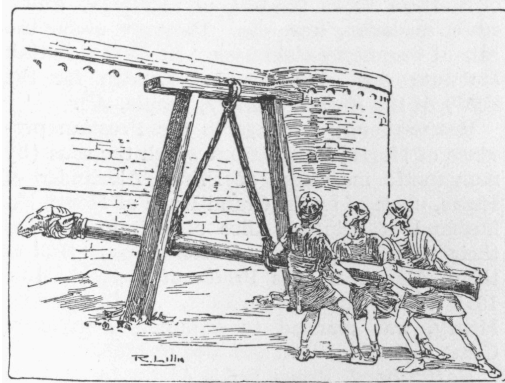
Bat'tas, or **Battaks**, a people belonging to the Malayan race inhabiting the valleys and plateaus of the mountains that extend longitudinally through the Island of Sumatra. They practise agriculture and cattle-rearing, and are skillful in various handicrafts; they have also a written literature and an alphabet of their own, their books treating of astrology, witchcraft, medicine, war, &c. They are under the rule of hereditary chieftains. In 1908 a Battak Institute was established at Leyden for the study of the Battak country, people, &c.

Bat'tenberg, a village in the Prussian province of Hesse-Nassau, from which the sons (bymorganatic marriage) of Prince Alexander of Hesse, uncle of Louis, Grand-Duke of Hesse, the husband of Princess Alice of Britain, derived their titles of princes of Battenberg. One of them, Alexander, was Prince of Bulgaria from 1879 to 1886, and died in 1893. Another, Prince Henry, was married to Princess Beatrice of Great Britain in 1885, and died in 1896.

Battenberg, Prince Louis Alexander of, born 24th May, 1854, in Graz, son of Prince Alexander of Hesse. He became a naturalized British subject, entered the navy in 1868, and in 1884 married Princess Victoria, eldest daughter of Princess Alice Maud, Grand-Duchess of Hesse-Darmstadt, and second daughter of Queen Victoria. He became rear-admiral in 1904, was second admiral of the Mediterranean fleet from 1906 to 1908, commanded the Atlantic fleet from 1908 to 1910, the 3rd and 4th divisions of the Home fleet in 1911, and was appointed Second Sea Lord in Nov., 1911, and First

Sea Lord in Dec., 1912, in succession to Sir Francis Bridgeman. In October, 1914, however, in consequence of a campaign against alien enemies which culminated in an attack in the *Globe* on the First Sea Lord, he tendered his resignation to Mr. Winston Churchill, then First Lord of the Admiralty, on the ground that his birth and parentage in some respects impaired his usefulness. In 1917 the king determined that those princes of his family who were his subjects and bore German names and titles should relinquish those titles and adopt British surnames, and Prince Louis adopted the title of Marquess of Milford Haven, and the surname of Mountbatten. He was the author of *Men-of-war Names, their Meaning and Origin* (1908). He was made a Privy Councillor in 1914, and was both a Civil and a Military Knight Grand Cross of the Bath. He died 11th Sept., 1921.

Battering-ram, an engine for battering down the walls of besieged places. The ancients employed two different engines of this kind—one suspended in a frame, the other movable on wheels or rollers. They consisted of a beam or spar with a massive metal head, and were set in motion either by a direct application of manual force or by means of cords passing over pulleys. Some are said to have been 120 feet or more in length, and to have been worked by 100 men. One is described as being 180 feet long, and having a head weighing $1\frac{1}{2}$ tons. Smaller ones, manned by three or four soldiers, were also frequently employed. They were generally covered with a roof or screen for the protection of the workers. The battering-ram is mentioned by Ezekiel (iv, 2; xxi, 22), and the Romans learnt the use of it from the Greeks.



Roman Battering-ram

Bat'tersea, a municipal and parliamentary borough of London, in Surrey, in a low situation on the south bank of the Thames, nearly opposite Chelsea, with a fine public park extending over 185 acres. The district is associated with the names of Pope and Bolingbroke, and with the Wellington-Winchelsea duel. Battersea returns two members to Parliament. Pop. (municipal borough), (1921), 167,693.

Bat'tery, (1) any number of guns grouped for action in one place and under one control; (2) the tactical and administrative unit of the Royal Regiment of Artillery; (3) an entrenched work constructed as a position for guns in siege warfare.

The Royal Regiment of Artillery is organized in batteries of horse, field, heavy, siege, and mountain artillery. These differ in the weight and type of gun with which they have to do, but the general principles of organization are the same. (See *Army*.) Nominally a battery consists of six guns with their attendant ammunition wagons (12) and certain other transport. A major commands the battery, and has a captain and three subalterns to assist him. For all purposes a battery is permanently organized in three sections, each of two guns and four wagons, under subaltern officers; while in action it is divided into two parts, (a) the fighting battery of six guns and six wagons; (b) the wagon line of the remaining wagons, under command of the captain. The change from march formation to that necessary for firing is termed 'coming into action'.

Horse, field, and some heavy batteries are horse-drawn (six or eight horses to a gun), while other heavy batteries and siege-guns are drawn by mechanical transport or mounted on railway platforms. Mountain batteries are organized for pack transport only—either mules or carriers—all the material being specially constructed to this end. Among combinations with the word battery come the following: *battery commander*, the officer commanding a battery; *battery leader*, the senior subaltern, who receives and carries out the commander's orders in action; *battery-fire*, each gun of a battery firing in turn: as opposed to gun-fire, where each gun fires independently. See *Artillery*.

Battery, in criminal law, an assault by beating or wounding another. The least touching or meddling with the person of another against his will may be held to constitute a battery.

Battery, in electricity and galvanism, a combination of several jars or metallic plates. See *Daniell's Cell*; *Electric Battery*; *Secondary Cell*.

Batthyanyi (ba't-ya'n'yē), one of the oldest and most celebrated Hungarian families, traceable as far back as the Magyar invasion of Pannonia in the ninth century. Among later bearers of the name have been—Count Casmir

Batthyanyi, who was associated with Kossuth, was Minister of Foreign Affairs in Hungary during the insurrection of 1849, and died in Paris 1854; Count Louis Batthyanyi, born 1809, of another branch of the family, was leader of the opposition in the Hungarian Diet until the breaking out of the commotions of 1848, when he took an active part in promoting the national cause; but on the entry of Windischgrätz into Pesth he was arrested and shot, 1849.

Battle, a combat between two armies. In ancient times and the Middle Ages the battleground was often chosen by agreement, and the battle was a mere trial of strength, a duel *en gros*; and as the armies of the ancients were imperfectly organized, and the combatants fought very little at a distance, after the battle had begun manœuvres were much more difficult, and troops almost entirely beyond the control of the general. Under these circumstances the battle depended almost wholly upon the previous arrangements and the valour of the troops. In modern times, however, the finest combinations, the most ingenious manœuvres, are rendered possible by the better organization of the armies, and it is the skill of the general rather than the courage of the soldier that now determines the event of a battle. Battles are distinguished as *offensive* or *defensive* on either side, but there is a natural and ready transition from one method to the other. The purely defensive attitude is condemned by tacticians except in cases where the only object desirable is to maintain a position of vital consequence, the weight of precedent being in favour of the dash and momentum of an attacking force even where opposed to superior forces. Where the greatest generals have acted upon the defensive, it has almost always been with the desire to develop an opportunity to pass to the offensive, and, having discovered their opponent's hand, to marshal against the enemy, exhausted with attack, the whole strength of their resources. Napoleon won more than one great victory by this method, and Wellington's reputation was largely based upon his skill in defensive-offensive operations. Tacticians have divided a battle into three periods, which are disposition, combat, and the decisive moment. In some measure they require distinct qualities in a commander, the intellect which can plot a disposition being by no means always of the prompt judgment passing to instant action which avails itself of the crucial moment to crush an enemy. Cooperation and mutual support of all arms and services is essential to success in a modern battle, which

may occupy many days or even weeks and embrace a large area of country, e.g. the battle of the Marne, the battle of the Somme. This co-operation is ensured by the most careful and detailed arrangements being made beforehand, such arrangements even including the rehearsal of more important phases over ground prepared to represent the country over which the operations will eventually take place. The modern battle may be, and probably will be, a continuation of many operations—*offensive* and *defensive*—carried out by distinct and separate formations. The general scope of the operations as a whole will be planned by the supreme commander, while the choice of the methods to be used is left to the commander of the formations affected. See *Strategy*; *Tactics*.

Battle, a town in Sussex, so named from the battle of Hastings having been fought there. An abbey built by William the Norman has disappeared, but important remains of a later building exist on the same site; and there is an old church of great interest. Pop. (1921), 2891.

Battle (or Battel), Wager of, an obsolete method, according to English law, of deciding civil or criminal cases by personal combat between the parties or their champions in presence of the court. A woman, a priest, a peer, or a person physically incapable of fighting could refuse such a trial. It was not abolished till 1819 (by the Statute of 59 George III, c. 46), but had long previously been in abeyance. See *Ordeal*.

Battle-axe, a weapon much used in war in the early part of the Middle Ages among knights. It is a weapon which affords hardly any guard, and the heavier the blow given with it the more the fighter is exposed; but its use was to some extent necessitated by the resistance of iron armour to all but heavy blows. In England and Scotland the battle-axe was much employed, the Lochaber-axe remaining a formidable implement of destruction in the hands of the Highlanders to a recent period.

Battle Creek, a town of the United States, in Michigan, at the junction of the Kalamazoo and Battle Creek, with a college, and manufactures of agricultural implements, &c. Pop. 25,267.

Battleford, a trading town of Canada, province of Saskatchewan, at the junction of the Battle and Saskatchewan (North Fork). Pop. 1335.

Bat'tlement, a notched or indented parapet of a fortification, formed by a series of raised parts called *cops* or *merlons*, separated by openings called *crenelles* or *embrasures*, the soldier sheltering himself behind the merlon, while he fires through the embrasure. Battlements were originally military, but were afterwards used freely in ecclesiastical and civil buildings by way of ornament, on parapets, cornices, tabernacle work, &c.

Battle-piece, a painting representing a battle. Some of the greatest pieces of this kind are the *Battle of Constantine*, of which the cartoons were drawn by Raphael, and which was executed by Giulio Romano; Lebrun's *Battles of Alexander*; and the *Battles of the Amazons*, by Rubens.

Battue (ba-tü'), a method of killing game by having persons to beat a wood, copse, or other cover, and so drive the animals (pheasants, hares, &c.) towards the spot where sportsmen are stationed to shoot them.

Battus, legendary founder of the Greek colony of Cyrene, in Libya, about 650 B.C. There were eight rulers of the family founded by him, bearing alternately the names Battus and Arcesilaus.

Batu Khan, Mongol ruler of the Western conquests of his grandfather, Genghis Khan, from 1224 to 1255. He overran Russia, Poland, Hungary, and Dalmatia, holding Russia for ten years.

Batum. See *Batoum*.

Baudelaire (bōd-lār), Charles Pierre, French poet, born 1821. His first work of importance was a series of translations from Poe, ranking among the most perfect translations in any literature. A volume of poems, *Les Fleurs du Mal* (1857), established his reputation as a leader of the Romanticists, though the police thought it necessary to deodorize them. Of a higher tone were his *Petits Poèmes en Prose*; followed in 1859 by a monograph on Théophile Gautier, in 1860 by *Les Paradis Artificiels* (opium and hashish studies), and in 1861 by *Wagner et Tannhäuser à Paris*. He died in 1867. The best English rendering of the *Fleurs du Mal* is by A. Symonds (1905).—
BIBLIOGRAPHY: Henry James, *French Poets and Novelists*; Asselineau, *Charles Baudelaire, sa vie et son œuvre*; F. Gautier, *Charles Baudelaire*.

Baudry (bō-drē), Paul Jacques Aimé, a prominent modern French painter, born in 1828, died Jan., 1886. He took the grand prix de Rome in 1850, and

exhibited many important works, of which the better known are his *Charlotte Corday* and *La Perle et la Vague*. The decoration of the foyer of the New Opera House at Paris was entrusted to him—an enormous work, occupying a total surface of 500 sq. metres, but admirably accomplished by him in eight years.

Bauer (bou´er), Bruno, German philosopher, historian, and Biblical critic of the rational school, born 1809, died 1882. Wrote *Critique of the Gospel of John* (1840); *Critique of the Synoptic Gospels* (1840); *History of the French Revolution to the Founding of the Republic* (1847); *History of Germany during the French Revolution and the Rule of Napoleon* (1846); *Critique of the Gospels* (1850-1); *Critique of the Pauline Epistles* (1850); *Philo, Strauss, Renan, and Primitive Christianity* (1874); *Christ and the Cæsars* (1877), &c.

Bauhin (bō-an), Gaspard, born at Basel in 1560; in 1580 elected to the Greek chair at Basel, and in 1589 to that of anatomy and botany. He died in 1624. His fame rests chiefly on his *Pinax Theatri Botanici* and *Theatrum Botanicum*. Linnæus gave his name to a genus of plants.

Bauhin´ia, a genus of plants, ord. Leguminosæ, usually lianes, found in the woods of hot countries, and often stretching from tree to tree like cables. Many are showy and interesting. The bark of *B. variegāta* is used in tanning; the bast fibres of some Indian species are made into ropes and twine.

Baumgarten (boum´ga´r-tn), Alexander Gottlieb, a German philosopher, born in 1714 at Berlin; in 1740 was made professor of philosophy at Frankfort-on-the-Oder, and died there in 1762. He is the founder of æsthetics as a science, and the inventor of this name. His ideas were first developed in his *De Nonnullis ad Poema pertinentibus* (1735), and afterwards in the two volumes of his uncompleted *Æsthetica*, published 1750-8.

Baur (bour), Ferdinand Christian, German theologian, founder of the 'Tübingen School' of theologians and Hegelian thinkers, born in 1792. The publication of his first work, *Symbolism and Mythology, or The Natural Religion of Antiquity*, in 1824-5, led to his appointment as professor in the evangelical faculty of Tübingen University, a position occupied by him till

his death in 1860. His chief works in the department of the history of Christian dogma are: *The Christian Gnosis, or The Christian Philosophy of Religion* (1835); *The Christian Doctrine of the Atonement* (1838); *The Christian Doctrine of the Trinity and the Incarnation* (1841-3); *The Compendium of and Lectures on the History of Christian Dogmas* (1847, 1865). To the department of New Testament criticism and the early history of Christianity belong the so-called *Pastoral Epistles of the Apostle Paul* (1835); *Paul the Apostle of Jesus Christ* (1845); *Critical Inquiries Concerning the Canonic Gospels* (1847); *A History of Christian Doctrine to the End of the Eighteenth Century* (1853-63). Baur's views in regard to the Church of the earliest times and the New Testament Scriptures have been very influential. He saw different and opposing tendencies at work in the Church of apostolic times, and believed that the New Testament mainly took form in the second century, the only genuine writings previous to A.D. 70 being the four great Pauline epistles and *Revelation*.

Bautzen (bout'sen), or **Budissin**, a German town in Saxony, upon a height on the right bank of the Spree, with some old and interesting buildings. The inhabitants are mostly Lutheran, and both Catholics and Protestants worship in the same cathedral. Chief manufactures: woollen goods, paper, gunpowder, machines. Napoleon defeated the united armies of the Russians and the Prussians at Bautzen on the 21st May, 1813. Pop. 32,754.

Bauxite (ba'k'sīt), a rock resembling clay, but consisting of hydrated aluminium oxides, with, in good examples, a silica content of less than 2 per cent. Bauxite is mined at Les Baux in Provence, in Arkansas, and other places, including County Antrim, where it occurs between basaltic lavas. It is the source of commercial aluminium, and is also used for bricks for high-temperature furnaces, and in alum manufacture. Like laterite (q.v.), it results from tropical conditions of weathering acting on a variety of silicate rocks.

Bava'ria (Ger. *Bayern*; Fr. *Bavière*), previous to the changes caused by the European War (1914-8), a kingdom in the south of Germany, the second largest State of the *Reich*, composed of two isolated portions, the larger comprising about eleven-twelfths of the monarchy, having Würtemberg, Baden, &c., on the west, while the smaller portion, the Pfalz or Palatinate, is separated from the other by Würtemberg and Baden, and lies west of the Rhine; total area, 30,346 sq. miles. The main political divisions are: Upper

Bavaria (pop. 1,532,065; chief town, Munich, the capital of the country, pop. 596,467); Lower Bavaria (724,331); Palatinate (937,085); Upper Palatinate and Regensburg (600,284); Upper Franconia (661,862); Middle Franconia (930,868); Lower Franconia and Aschaffenburg (710,943); Schwaben and Neuburg (789,853); total in (1919), 7,150,146. Next towns after Munich are Nürnberg, Augsburg, Würzburg, and Ratisbon (Regensburg). The main portion of the country is hilly; in the south, where it belongs to the Alps, it is mountainous; but north of the Alps and south of the Danube, which flows east through the country from Ulm to Passau, there is a considerable plateau, averaging about 1600 feet above the sea-level. The south frontier is formed by a branch of the Noric Alps, offsets from which project far into the plateau; principal peaks: the Zugspitze, 10,394 feet, and the Watzmann, 9470 feet. The highest summits on the Bohemian (Austrian) frontier, belonging to the Böhmerwald Mountains, are the Rachel, 5102 feet, and the Arber, 5185 feet. Ranges of less elevation bordering on or belonging to the country are the Fichtelgebirge in the north-east, the Frankenwald, Rhöngebirge, and Spessart in the north, and the Steigerwald and Franconian Jura in the middle. The Palatinate is traversed by the north extremity of the Vosges Mountains, the highest peak being the Königstuhl, 2162 feet. The greater part of the country belongs to the basin of the Danube, which is navigable, its tributaries on the south being the Iller, Lech, Isar, and Inn; on the north, the Wornitz, Altmühl, Nab, and Regen. The northern portion belongs to the basin of the Main, which receives the Regnitz and Saale, and is a tributary of the Rhine. The Palatinate has only small streams that flow into its boundary river the Rhine. The chief lakes of Bavaria are all on the higher part of the south plateau; the smaller within the range of the Alps. The Ammer-See is about 10 miles long by 2½ broad, 1736 feet above the sea; the Würm-See or Starnberger-See, about 12 miles long by 3 broad, 1899 feet; and Chiem-See, 9 miles long by 9 to 4 broad, 1651 feet. The climate in general is temperate and healthy, though somewhat colder than the other South German States; yearly average about 47° F.

As regards soil, Bavaria is one of the most fertile countries in Germany, producing the various cereals in abundance, the best hops in Germany, fruit, wine, tobacco, &c., and having extensive forests. Lower Franconia (the Main valley) and the Palatinate are the great vine-growing districts. The

celebrated Steinwein and Leistenwein are the produce of the slopes of the Steinberg and Marienberg at Würzburg (on the Main). The forests of Bavaria, chiefly fir and pine, yield a large revenue, much timber being annually exported, together with potash, tar, turpentine, &c. The principal mineral products are salt, coal, and iron, some of the mining works belonging to the State. The minerals worked include copper, quicksilver, manganese, cobalt, porcelain clay, alabaster, graphite. Large numbers of horses and cattle are reared, as also sheep and swine. The return of cattle in 1917 showed 272,667 horses, 3,896,702 cattle, and 616,464 sheep. The manufactures are individually mostly on a small scale. The principal articles manufactured are linens, woollens, cottons, leather, paper, glass, earthen and iron ware, jewellery, &c. The optical and mathematical instruments made are excellent. A most important branch of industry is the brewing of beer, for which there are upwards of 7000 establishments, producing over 300 millions of gallons a year. The distilleries produce over 4 millions of gallons of alcohol. A number of the people maintain themselves by the manufacture of articles in wood, and by felling and hewing timber. The trade of Bavaria in home produce exported and goods imported for consumption in the country is not large. Principal exports: corn, timber, wine, cattle, glass, hops, fruit, beer, wooden wares, &c. The chief imports are sugar, coffee, cotton, rice, spices, dye-stuffs, silk and silk goods, lead, &c. From its position Bavaria has a considerable transit trade. The König Ludwig Canal connects the Main at Bamberg with the Altmühl a short distance above its embouchure in the Danube, thus establishing water communication between the North Sea and the Black Sea. The railway system has a total mileage of 5290, mostly belonging to the State.

Education is in a less satisfactory condition than in most German States. There are about 7500 elementary schools, at which attendance is compulsory up to fourteen years of age. There are three universities, two of which (Munich and Würzburg) are Roman Catholic, and one (Erlangen) Protestant. In art Bavaria is best known as the home of the Nürnberg school, founded about the middle of the sixteenth century by Albert Dürer. Hans Holbein is also claimed as a Bavarian; and to these have to be added the eminent sculptors Kraft and Vischer, both born about the middle of the fifteenth century. The restoration of the reputation of Bavaria in art was chiefly the work of Ludwig I, under whom the capital became one of the

most prominent seats of the fine arts in Europe. The religion of the State is Roman Catholicism, over 70 per cent of the population being Roman Catholics. All citizens, whatever their creed, possess the same civil and political rights. The dioceses of Bavaria comprise two Roman Catholic archbishoprics, Munich and Bamberg; and six bishoprics, Augsburg, Ratisbon, Eichstädt, Passau, Würzburg, and Spire.

For over a century Bavaria was a kingdom, and the crown hereditary in the male line of the descendants of the ancient Counts of Wittelsbach. The executive power was vested in the king, but his ministers were responsible for his acts. The legislature consisted of two chambers—one of senators, composed of princes of the royal family, the great officers of the State, the two archbishops, the heads of certain noble families, and certain members appointed by the Crown; the other of deputies, 163 in number, elected by direct and secret vote, and calculated at the rate of 1 for every 38,000 inhabitants. The lower chamber was elected for six years. The Budget for each of the years 1916 and 1917 balanced at £12,183,411; for 1918 and 1919 at £42,639,595; the public debt on 1st July, 1917, was £128,719,000, £98,179,000 being railway debt. The army (peace footing, 72,000) was raised by conscription—every man being liable to serve from the 1st of January of the year in which he completes his twentieth year. In time of peace the army was under the command of the King of Bavaria, but in time of war under that of the Emperor of Germany, as commander-in-chief of the whole German army.

As a result of the revolutionary movement in Nov., 1918, the Bavarian dynasty was deposed and a republican form of government set up. The provisional constitution, made public on 7th Jan., 1919, established a Socialist Republic in Bavaria. The supreme power lies with the people; the Diet consists of one chamber; the suffrage is universal; the privileges of birth and caste are abolished. The supreme executive power is exercised by the ministry as a whole. The Church is separated from the State, and all religious associations have equal rights.

History.—The Bavarians take their name from the Boii, a Celtic tribe whose territory was occupied by a confederation of Germanic tribes, called after their predecessors Boiarii. These were made tributary first to the Ostrogoths, and then to the Franks; and on the death of Charlemagne his

successors governed the country by lieutenants with the title of margrave, afterwards converted (in 921) into that of duke. In 1070 Bavaria passed to the family of the Guelphs, and in 1180 by imperial grant to Otho, Count of Wittelsbach, founder of the dynasty reigning until 1918. In 1623 the reigning duke was made one of the Electors of the empire. Elector Maximilian II joined in the War of the Spanish Succession on the side of France, and this led, after the battle of Blenheim, 1704, to the loss of his dominions for the next ten years. His son, Charles Albert, likewise lost his dominions for a time to Austria, but they were all recovered again by Charles's son, Maximilian III (1745). In the wars following the French revolution Bavaria was in a difficult position between France and Austria, but in the end joined Napoleon, from whom its Elector Maximilian IV received the title of king (1805), a title afterwards confirmed by the treaties of 1814 and 1815. King Maximilian I was succeeded by his son, Ludwig (or Louis) I, under whom various circumstances helped to quicken a desire for political change. Reform being refused, tumults arose in 1848, and Ludwig resigned in favour of his son, Maximilian II, under whom certain modifications of the constitution were carried out. At his death in 1864 he was succeeded by Ludwig II. In the war of 1866 Bavaria sided with Austria, and was compelled to cede a small portion of its territory to Prussia, and to pay a war indemnity of £2,500,000. Soon after Bavaria entered into an alliance with Prussia, and in 1867 joined the Zollverein. In the Franco-German War of 1870-1 the Bavarians took a prominent part, and it was at the request of the King of Bavaria, on behalf of all the other princes and the senates of the free cities of Germany, that the King of Prussia agreed to accept the title of Emperor of Germany. From Jan., 1871, to 1918 Bavaria formed part of the German Empire, and was represented in the Bundesrath by six, and in the Reichstag by forty-eight members. The eccentricity early displayed by Ludwig II developed to such an extent that in June, 1886, he was placed under control, and a regency established under Prince Luitpold (Leopold). The change was almost immediately followed by the suicide of the king, and as Prince Otto, the brother and heir of the late king, was insane, the regency was continued. Prince Luitpold died on 12th Dec., 1912, and his son, Ludwig, became regent. On 5th Nov., 1913, the prince was proclaimed king, under the name of Ludwig III, in succession to his cousin, King Otto, who was declared incapable of ruling. King Otto died on 12th Oct., 1916. Ludwig III abdicated in Nov., 1918, and the Bavarian

Government was taken over by a cabinet under the leadership of Kurt Eisner. Eisner was assassinated in Feb., 1919, and a struggle ensued between the moderate Socialists and the extremists of the Left. The moderate party, however, returned to power in May, 1919.—BIBLIOGRAPHY: S. Riezler, *Geschichte Bayerns* (4 vols.); R. Piloty, *Die Verfassungsgeschichte des Königreichs Bayern*; Götz, *Geographisch-historisches Handbuch von Bayern*; M. Docberl, *Entwicklungsgeschichte Bayerns*; C. P. Higby, *Religious Policy of the Bavarian Government during the Napoleonic Period*.

Baxter, Richard, the most eminent of the English nonconforming divines of the seventeenth century, born in Rowton, Shropshire, 1615; ordained in 1638; parish minister of Kidderminster in 1640. The imposition of the oath of universal approbation of the doctrine and discipline of the Church of England (the *et cætera oath*) detached him from the Establishment. After the battle of Naseby he accepted the chaplaincy of Colonel Whalley's regiment. He can scarcely be said, however, to have separated as yet in spirit from the Establishment. He upheld the monarchy, condemned the execution of the king and the election of Cromwell, preached against the Covenant and against separatists and sectaries, but his piety won him the respect of all parties. At the Restoration he became king's chaplain, but declined the bishopric of Hereford, and on the passage of the Act of Uniformity threw in his lot entirely with the nonconformists. In 1685 he was arrested, refused a hearing by Jeffreys, and imprisoned. After his release he lived in retirement till his death in 1691. He left about 150 treatises, of which his *Saints' Everlasting Rest*, *Call to the Unconverted*, and *Reformed Pastor* have been the most popular.—BIBLIOGRAPHY: William Orme, *Life and Times of Richard Baxter*; J. Stalker, *Lecture on Baxter in The Evangelical Succession*; M'Adam Muir, *Religious Writers of England*; Currier, *Nine Great Preachers*.

Baxterians, followers of Baxter in respect of his attempted compromise between Calvinism and Arminianism. They reject the doctrine of reprobation, admit a universal potential salvation, becoming actual in the case of the elect, and assert the possibility of falling from grace. The Baxterians never formed a sect or even a school, but were men of independent minds, distinguishing between essentials and non-essentials: in things necessary, unity; in things doubtful, liberty; in all things, charity. The

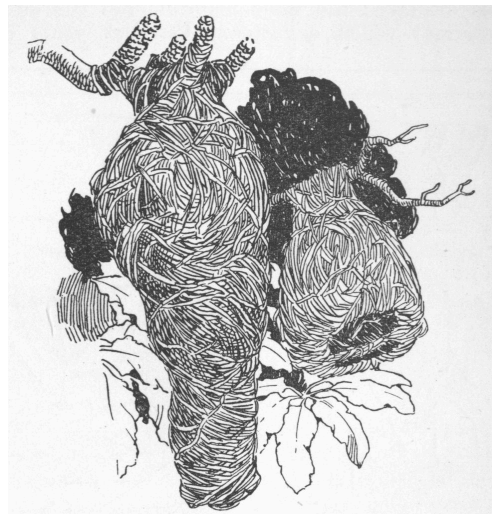
two most eminent Baxterians were Dr. Isaac Watts and Dr. Philip Doddridge.—Cf. M'Adam Muir, *Religious Writers of England*.

Bay, the laurel tree, noble laurel, or sweet-bay (*Laurus nobilis*); but the term is loosely given to many trees and shrubs resembling this. A fatty or fixed oil (used in veterinary medicine) and also a volatile oil are obtained from the berries, but what is called 'bayberry oil' is also obtained from the genus *Myrica* or candleberry. In N. America the fragrant-flowered *Magnolia glauca* is called sweet-bay, the red-bay being *Laurus carolinensis*, the loblolly-bay *Gordonia lasianthus*. See *Laurel*.

Bay, in geography, an indentation of some size into the shore of a sea or lake, generally said to be one with a wider entrance than a gulf.

Bay, in architecture, a term applied to a recessed division or compartment of a building, as that marked off by buttresses or pillars.

Ba'ya, the weaver-bird (*Ploceus philippinus*), an interesting East Indian passerine bird, somewhat like the bullfinch. Its nest resembles a bottle, and is suspended from the branch of a tree. The entrance is from beneath, and there are two chambers, one for the male, the other for the female. The baya is easily tamed, and will fetch and carry at command.



Nests of Baya (*Ploceus philippinus*)

Bayaderes (bā-a-dērz'), the general European name for the dancing and singing girls of India, some of whom are attached to the service of the Hindu temples, while others travel about and dance at entertainments for hire. Those in the service of the temples are generally devoted to this profession (including that of prostitution) from their childhood.

Bayamo (ba'-yā'mō), or **St. Salvador**, a town in the east of Cuba, near the Cauto. Pop. 18,180.

Bayard (ba'-yär), Pierre du Terrail, Seigneur de, the *Chevalier sans peur et sans reproche*, born in 1476 in Castle Bayard, near Grenoble, in Southern France. At the age of eighteen he accompanied Charles VIII to Italy, and in the battle of Fornovo took a standard. At the beginning of the reign of Louis XII, in a battle near Milan, he entered the city at the heels of the fugitives, and was taken prisoner, but released by Ludovico Sforza without ransom. In Apulia he killed his calumniator, Sotomayor, and afterwards defended a bridge over the Garigliano singly against the Spaniards, receiving for this exploit as a coat of arms a porcupine, with the motto *Vires agminis unus habet* ('one has the strength of a band'). He distinguished himself equally against the Genoese and the Venetians, and, when Julius II declared himself against France, went to the assistance of the Duke of Ferrara. He was severely wounded at the assault of Brescia, but returned, as soon as cured, to the camp of Gaston de Foix, before Ravenna, and after new exploits was again dangerously wounded in the retreat from Pavia. In the war commenced by Ferdinand the Catholic he displayed the same heroism, and the fatal reverses which embittered the last years of Louis XII only added to the personal glory of Bayard. When Francis I ascended the throne he sent Bayard into Dauphiné to open a passage over the Alps and through Piedmont. Prosper Colonna lay in wait for him, but was made prisoner by Bayard, who immediately after further distinguished himself in the battle of Marignano. After his defence of Mézières against the invading army of Charles V he was saluted in Paris as the saviour of his country, receiving the honour paid to a prince of the blood. His presence reduced the revolted Genoese to obedience, but failed to prevent the expulsion of the French after the capture of Lodi. In the retreat the safety of the army was committed to Bayard, who, however, was mortally wounded by a stone from a blunderbuss in protecting the passage of the Sesia. He kissed the cross of his sword, confessed to his squire, and died, 30th April, 1524. He was buried in a church of the Minorites, at Grenoble.—BIBLIOGRAPHY: "Le Loyal Serviteur" (supposed to be Jacques de Mailles), *La très joyeuse ... histoire ... des faiz, gestes, triumphes et prouesses du bon chevalier sans paour et sans reproche, le gentil seigneur de Bayart* (printed in 1527); modern edition of this work by M. J. Roman; A. de Terrebasse, *Histoire de Pierre Terrail, Seigneur de Bayart*.

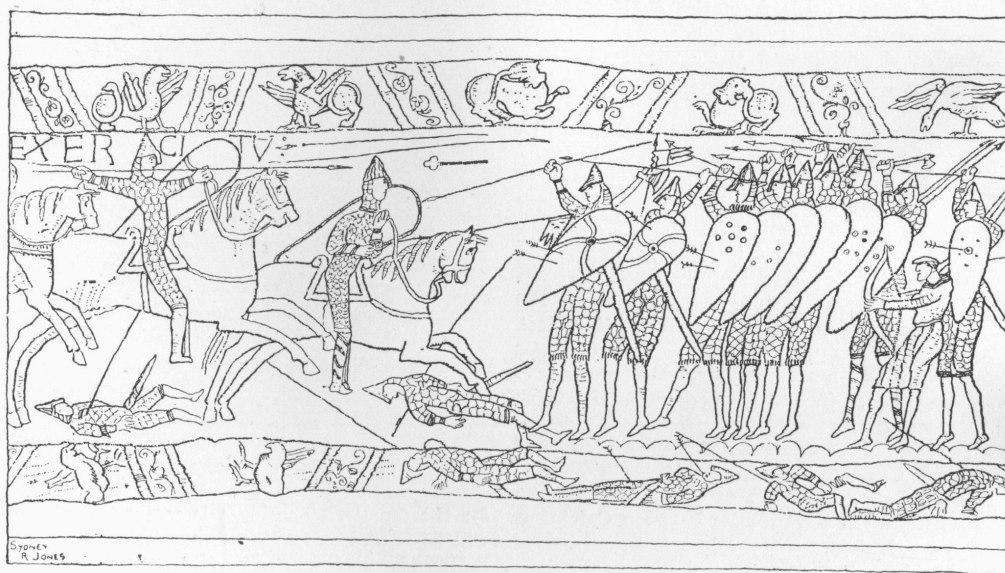
Bayazid', a ruined town in Armenia, 140 miles S.E. of Erzerum, S.W. of Mount Ararat, at various times the scene of warlike operations, and twice occupied by the Russians in 1877. Total pop. 4000.

Bayazid. See *Bajazet*.

Bayberry. See *Bay* (tree) and *Candleberry*.

Bay City, an American city, Michigan, on the east side of Saginaw River, near its mouth in Saginaw Bay, Lake Huron. Chief articles of trade, lumber and salt. Pop. (1920), 47,554.

Bayeux, (ba'-yeu) an ancient town, France, department Calvados, 16 miles N.W. of Caen, with manufactures of lace, calico, and porcelain. In its cathedral, said to be the oldest in Normandy, the famous Bayeux tapestry was preserved for a long time. Pop. 7638.



A section of the Bayeux Tapestry showing the Norman cavalry attacking the English footmen at the Battle of Hastings

Bayeux Tapestry, so called because it was originally found in the cathedral of Bayeux, in the public library of which town it is still preserved. Formerly known as the 'Toile de St. Jean', it is supposed to have been worked by Matilda, Queen of William the Conqueror, and to have been presented by Odo, Bishop of Bayeux, the half-brother of William, to the church in which it was found. It is 231 feet in length and 20 inches in breadth, and is divided

into seventy-two compartments, the subject of each scene being indicated by a Latin inscription. These scenes give a pictorial history of the invasion and conquest of England by the Normans, beginning with Harold's visit to the Norman Court, and ending with his death at Hastings.—BIBLIOGRAPHY: Bolton Corney, *Researches and Conjectures on the Bayeux Tapestry*; Jules Comte, *Tapisserie de Bayeux*; F. R. Fowke, *The Bayeux Tapestry*.

Bay Islands, an island group, Bay of Honduras, off N. coast of State of Honduras, incorporated as a British colony in 1852, and ceded to Honduras in 1856. The largest is Ruatan, 30 miles long. Pop. about 5000.

Bayle (bāl), Pierre, French critic and miscellaneous writer, the son of a Calvinist preacher, born at Carlat (Languedoc) in 1647, died at Rotterdam 1706. A statue in his honour was erected at Pamiers in 1906. He studied at Toulouse, and was employed for some time as a private tutor at Geneva and Rouen. He went to Paris in 1674, and soon after was appointed professor of philosophy at Sedan. Six years after he removed to Rotterdam, where he filled a similar chair. The appearance of a comet, in 1680, which occasioned an almost universal alarm, induced him to publish, in 1682, his *Pensées Diverses sur la Comète*, a work full of learning, in which he discussed various subjects of metaphysics, morals, theology, history, and politics. It was followed by his *Critique Générale de l'Histoire du Calvinisme de Maimbourg*. This work excited the jealousy of his colleague, the theologian Jurieu, and involved Bayle in many disputes. In 1684 he undertook a periodical work, *Nouvelles de la République des Lettres*, containing notices of new books in theology, philosophy, history, and general literature. This publication, which lasted for three years, added much to his reputation as a philosophical critic. In 1693 Jurieu succeeded in inducing the magistrates of Rotterdam to remove Bayle from his office. He now devoted all his attention to the composition of his *Dictionnaire Historique et Critique*, which he first published in 1696, in 2 vols. fol. It is a vast storehouse of facts, discussions, and opinions, and though it was publicly censured by the Rotterdam consistory for its frequent impurities, its pervading scepticism, and tacit atheism, it long remained a favourite book both with literary men and with men of the world. The articles in his dictionary, in themselves, are generally of little value, and serve only as a pretext for the notes, in which the author displays, at the same time, his learning and the power of his logic. The best editions are that of 1740, in 4 vols. fol. (Amsterdam and

Leyden), and that in 16 vols., published in 1820-4 at Paris.—BIBLIOGRAPHY: L. A. Feuerbach, *Pierre Bayle*; F. Brunetière, *Études Critiques* (5th series); A. Cazes, *Pierre Bayle, sa vie, ses idées, son influence, son œuvre*.

Bay-leaf, the leaf of the sweet-bay or laurel tree (*Laurus nobilis*). These leaves are aromatic, and are used in cookery and confectionery. See *Bay*.

Baylen (bī-len´). Same as *Bailen*.

Bayly (bā´li), Thomas Haynes, English poet, novelist, dramatist, and miscellaneous writer, born 1797, died 1839. He was educated at Oxford, and intended for the Church. He wrote thirty-six pieces for the stage, most of which were successful; several novels: *The Aylmers*, *A Legend of Killarney*, &c.; and numerous songs. As a song-writer he was most prolific and most popular. *The Soldier's Tear*, *We met—'twas in a Crowd*, were the best known.

Bay Mahogany, that variety of mahogany exported from Honduras. It is softer and less finely marked than the variety known as Spanish mahogany, but is the largest and most abundant kind.

Baynes (bānz), Thomas Spencer, LL.D., born at Wellington, Somerset, in 1823, died suddenly in London, 1887. He studied under Sir William Hamilton at Edinburgh, and acted as his class assistant from 1851 to 1855. From 1857 to 1863 he was resident in London, where he acted as examiner in logic and mental philosophy in the University of London, and as assistant editor of the *Daily News*. In 1864 he was appointed to the chair of logic, rhetoric, and metaphysics in St. Andrews University, a post he held till his death. In 1873, when he became editor of the ninth edition of the *Encyclopædia Britannica*, his wide acquaintance with men of letters and learning assisted him greatly in the selection of suitable contributors. He translated the *Port Royal Logic* (see *Arnauld*), and was a frequent contributor to the principal reviews and literary journals.

Bay of Islands, a large, deep, and safe harbour on the N.E. coast of the N. Island of New Zealand. On it is Kororarika, the first European settlement in New Zealand.—Also a large bay formed by the Gulf of St. Lawrence, on the west coast of Newfoundland. See *New Zealand*.

Bayonet (Fr. *baïonnette*), a straight, sharp-pointed weapon, invented by a Frenchman (probably a native of Bayonne) in the seventeenth century. A bayonet was originally screwed into the muzzle of the firelock, converting it into a short pike. This system, which practically precluded any further use of the firelock as such, was later improved by a Scotchman, who fitted the weapon with a socket and bolt, enabling it to be placed over the muzzle of the musket instead of being screwed into it. The firelock could then be used for both missile and shock action, according to requirements. The bayonet replaced the pike in the English army after the revolution of 1690. The length, shape, and weight of the bayonet have varied considerably at different periods and in different countries; that at present in use in the British army is a short sword-shaped weapon, 1 foot 6 inches long and weighing about 1 pound. In war it is considered that an attack with the bayonet or the immediate threat of one is a necessary preliminary to a successful assault.

Bayonne (ba'-yon), a well-built fortified town, the largest in the French department Basses-Pyrénées, at the confluence of the Nive and the Adour, about 2 miles from their mouth in the Bay of Biscay; with a citadel commanding the harbour and city, a cathedral—a beautiful ancient building—shipbuilding and other industries, and a considerable trade. Among the lower class the Basque language is spoken. Catherine de' Medici had an important interview with the Duke of Alva in Bayonne, June, 1565, at which it is said the massacre of St. Bartholomew was arranged. It was also the scene of the abdication of Charles IV of Spain in favour of Napoleon (1808). In 1814 the British forced the passage of the Nive and invested the town, from which the French made a desperate but unsuccessful sortie. Pop. 27,886.

Bayonne City, a suburb of New York, in Hudson County, New Jersey. Pop. 76,754.

Bayou (bā-yö'), in the southern States of North America, a stream which flows from a lake or other stream: frequently used as synonymous with creek or tidal channel.

Bayreuth (bī'roit). See *Baireuth*.

Bay Rum, a spirit obtained by distilling the leaves of *Myrica acris*, or other West Indian trees of the same genus. It is used for toilet purposes, and as a liniment in rheumatic affections.

Bay-salt, a general term for coarse-grained salt, but properly applied to salt obtained by spontaneous or natural evaporation of sea-water in large shallow tanks or *bays*.

Bay-window, a window forming a recess or bay in a room, projecting outwards, and rising from the ground or basement on a plan rectangular, semi-octagonal, or semi-hexagonal, but always straight-sided. One of the finest examples of bay-windows is that of the banqueting-hall at Hampton Court. The term is, however, also often employed to designate a *bow-window*, which more properly forms the segment of a circle, and an *oriel-window*, which is supported on a kind of bracket, and is usually on the first floor.



Bay-window at Bramhall Hall, Cheshire

Baza (bā'tha'), an old town of Spain, Andalusia, province of Granada, formerly a large and flourishing city. In 1810 the French, under Marshal Soult, here defeated the Spaniards under Generals Blake and Freire. Pop. 15,964.

Bazaar. See *Bazar*.

Bazaine (ba'-zān), François Achille, French general, born 1811, died 1888. He entered the army as a private soldier in 1831, and served in Algeria with distinction, gaining the cross of the Legion of Honour, and rising to the rank of lieutenant. He next went to Spain and fought in the Foreign Legion against the Carlists; and in 1839 returned to Algeria, where he eventually held the rank of colonel (1850). He was next engaged in the Crimean War, being at first commander of a brigade and then general of division, leading the French troops sent to attack the fortress of Kinburn (1855). He did good

service also in the Italian war of 1859, being actively engaged in the battle of Solferino. His military reputation was increased by the part he took in the Mexican expedition (1862-4), in which he led the first division under Forey, and when this general was recalled became commander-in-chief of the French forces in Mexico and marshal of France. His loyalty, however, to the cause of the Emperor Maximilian was very doubtful. In 1870 he took command of the army of the Rhine, or the third army corps, in the Franco-German War, collected a very large army in the neighbourhood of Metz, and had the intention of joining his forces with those of MacMahon at Châlons. He found this impossible, however, especially after Gravelotte, and was forced into Metz, where he capitulated after a seven weeks' siege, with an army of 175,000 men. For this act he was tried by court-martial in 1871, found guilty of treason, and condemned to death. The sentence was commuted to twenty years' seclusion in the Isle of St. Marguerite, from which he escaped in 1874, and retired to Spain. He published *Épisodes de la Guerre de 1870*.—BIBLIOGRAPHY: La Brugère, *L'Affaire Bazaine*; Comte d'Herisson, *La Légende de Metz*.

Bazar´, or **Bazaar´**, in the East an exchange, market-place, or place where goods are exposed for sale, usually consisting of small shops or stalls in a narrow street or series of streets. These bazar-streets are frequently shaded by a light material laid from roof to roof, and sometimes are arched over. Marts for the sale of miscellaneous articles, chiefly fancy goods, are now to be found in most European cities, bearing the name of *bazars*. The term bazar is popularly applied to a sale of miscellaneous articles, mostly of fancy work, and contributed gratuitously, in furtherance of some charitable or other purpose.

Bazar´jik, now officially called Dobritsh (q.v.).

Bazigars´, a tribe of Indians dispersed throughout the whole of Hindustan mostly in wandering tribes. They are divided into seven castes; their chief occupation is that of jugglers, acrobats, and tumblers, in which both males and females are equally skilful. They present many features analogous to the gipsies of Europe, and like these each clan has its king.

Bazin, René, French novelist, born 1853. For some time he was professor of law in the Catholic University of Angers. His literary reputation was

established by *Une Tache d'Encre*. His other novels include *La Terre qui meurt*, *De Toute son Âme*, *Les Oberlé*, *Les Noellet*, *Davidée Birot*, all of which have been translated into English by Dr. A. S. Rappoport. He has also written *Questions littéraires et sociales*, &c. Bazin was elected a member of the Académie Française in 1903.

Bazoche (ba'-zosh'), or **Basoche** (a corruption of *Basilica*), a brotherhood formed by the clerks of the Parliament of Paris at the time it ceased to be the grand council of the French king. They had a king, chancellor, and other dignitaries; and certain privileges were granted them by Philip the Fair early in the fourteenth century, as also by subsequent monarchs. They had an annual festival, having as a principal feature dramatic performances in which satirical allusions were freely made to passing events. These farces or satires were frequently interdicted, but they had a considerable influence on the French drama. The Bazoche took an active part in the French Revolution, but the order was suppressed by the decree of 13th Feb., 1791.

Bazzi (ba't'sē). See *Sodoma*.

Bdellium (del'i-um), an aromatic gum resin brought chiefly from Africa and India, in pieces of different sizes and figures, externally of a dark reddish brown, internally clear, and not unlike glue. To the taste it is slightly bitterish and pungent; its odour is agreeable. It is used as a perfume and a medicine, being a weak deobstruent. Indian bdellium is the produce of *Balsamodendron Roxburghii*; African of *B. africanum*; Egyptian bdellium is obtained from the doum palm; and Sicilian is produced by *Daucus gummifer*, a species of the genus to which the carrot belongs. The bdellium mentioned in *Gen.* ii, 12, was apparently a precious stone, perhaps a pearl. See *Myrrh*.

Beaches, Raised, a term applied to those long, terraced, level pieces of land, consisting of sand and gravel, and containing marine shells, now, it may be, a considerable distance above and away from the sea, but bearing sufficient evidences of having been at one time sea-beaches. In Scotland such a terrace has been traced extensively along the coasts at about 25 feet above the present sea-level.

Beachy Head, a promontory in the south of England, on the coast of Sussex, rising 575 feet above sea-level, with a revolving light, visible in

clear weather from a distance of 28 miles. A naval battle took place here, 30th June, 1690, in which a French fleet under Tourville defeated an English and Dutch combined fleet under Lord Torrington.

Beacon (bē'kon), an object visible at some distance, and serving to notify the presence of danger; commonly applied to a fire-signal set on a height to spread the news of hostile invasion or other great event; and also applied to a mark or object of some kind placed conspicuously on a coast or over a rock or shoal at sea for the guidance of vessels, often an iron structure of considerable height.

Beaconsfield (bē'konz-fēld), a village of Buckinghamshire, the parish church of which contains the remains of Edmund Burke, whose seat was in the neighbourhood; while a marble monument to the poet Waller, who owned the manor, is in the churchyard. It gave the title of earl to the English statesman and novelist Benjamin Disraeli. Pop. (1921), 3642.

Beaconsfield, formerly known as Du Toit's Pan, a town of Cape Province about a mile to the east of Kimberley, of which it forms a suburb. Pop. 20,364.

Beaconsfield, Benjamin Disraeli, Earl of, an eminent English statesman and novelist, of Jewish extraction; eldest son of Isaac D'Israeli, author of the *Curiosities of Literature*; born in London in 1804, died there in 1881, buried at Hughenden. He attended for a time a private school, and was first destined for the law, but showing a decided taste for literature he was allowed to follow his inclination. In 1826 he published *Vivian Grey*, his first novel; and subsequently travelled for some time, visiting Italy, Greece, Turkey, and Syria, and gaining experiences which were afterwards reproduced in his books. His travels and impressions are embodied in a volume of letters addressed to his sister and his father. In 1831 another novel, *The Young Duke*, came from his pen. It was followed at short intervals by *Contarini Fleming*, *Alroy*, *Henrietta Temple*, *Venetia*, *The Revolutionary Epic* (a poem), &c. In 1832, and on two subsequent occasions, he appeared as candidate for the representation of High Wycombe, with a programme which included vote by ballot and triennial parliaments, but was unsuccessful. His political opinions gradually changed: in 1835 he unsuccessfully contested Taunton as a Tory. In 1837 he

gained an entrance to the House of Commons, being elected for Maidstone. His first speech in the House was treated with ridicule; but he finished with the prophetic declaration that the time would come when they would hear him. During his first years in Parliament he was a supporter of Peel; but when Peel pledged himself to abolish the corn-laws, Disraeli became the leader of the protectionists. About this time he became a leader of what was known as the 'Young England' party, the most prominent characteristic of which was a sort of sentimental advocacy of feudalism. This spirit showed itself in his two novels of *Coningsby* and *Sybil*, published respectively in 1844 and 1845. Having acquired the manor of Hughenden, in Buckinghamshire, he was in 1847 elected for this county, and he retained his seat till raised to the peerage nearly thirty years later. His first appointment to office was in 1852, when he became Chancellor of the Exchequer under Lord Derby. The following year, however, the ministry was defeated. He remained out of office till 1858, when he again became Chancellor of the Exchequer, and brought in a Reform Bill which wrecked the Government. During the time the Palmerston Government was in office Disraeli led the opposition in the Lower House with conspicuous ability and courage. In 1866 the Liberals resigned, and Derby and Disraeli came into power, the latter being again Chancellor of the Exchequer. They immediately brought in, and carried, after a violent and bitter struggle, a Reform Bill on the basis of household suffrage. In 1868 he became Premier on the resignation of Lord Derby, but his tenure of office was short. In 1874 he again became Prime Minister with a strong Conservative majority, and he remained in power for six years. This period was marked by his elevation to the peerage in 1876 as Earl of Beaconsfield, and by the prominent part he took in regard to the Eastern question and the conclusion of the Treaty of Berlin in 1878. In 1880 Parliament was rather suddenly dissolved, and, the new Parliament showing an overwhelming Liberal majority, he resigned office, though he still retained the leadership of his party. Within a few months of his death the publication of a novel called *Endymion* (his last, *Lothair*, had been published ten years before) showed that his intellect was still vigorous. Among others of his writings, besides those already mentioned, are: *A Vindication of the English Constitution* (1834); *Alarcos, a Tragedy* (1839); and *Lord George Bentinck, a Political Biography* (1852).—BIBLIOGRAPHY: *Selected Speeches of the Earl of Beaconsfield* (edited by T. E. Kebbel); T. Martin, *Memorials of Lord*

Beaconsfield; the article *Disraeli*, by T. E. Kebbel, in *Dictionary of National Biography*; Sir William Fraser, *Disraeli and his Day*; Sichel, *Disraeli*; the best work, however, is W. F. Monypenny's *Life of Disraeli, Earl of Beaconsfield* (continued by G. E. Buckle).

Bead (bēd), originally a prayer; then a small perforated ball of gold, pearl, amber, glass, or the like, to be strung on a thread, and used in a rosary by Roman Catholics in numbering their prayers, one bead being passed at the end of each ejaculation or short prayer; finally any such small ornamental body. The use of beads among pagans is of greater antiquity than their Christian use, but there is no evidence to show that the latter is derived from the former. Glass beads are now the most common sort; they form a considerable item in the African trade.—In architecture and joinery the bead is a small round moulding. It is of frequent occurrence in architecture, particularly in the classical styles, and is used in picture-frames and other objects carved in wood.—*St. Cuthbert's Beads*, the popular name of the detached and perforated joints of encrinites.

Beadle (bē'dl), an officer in a university, whose chief business is to walk with a mace in a public procession: also, a parish officer whose business is to punish petty offenders, and a church officer (chiefly, although not exclusively, in Scotland) with various subordinate duties, as waiting on the clergyman, keeping order in church, attending meetings of vestry or session, &c. The primary meaning of the word (from O.E. *beodan*, to offer, announce) appears to be 'herald', one who announces or proclaims something.

Bead-snake (*Elaps fulvus*), a beautiful snake of North America, inhabiting cultivated grounds, especially plantations of the sweet-potato, and burrowing in the ground. It is finely marked with yellow, carmine, and black. Though it possesses poison-fangs it never seems to use them.

Beagle (bē'gl), a small hound, resembling a foxhound or harrier, and used to hunt rabbits and hares, being often kept in packs. The beagle is smaller than the harrier, compactly built, smooth-haired, and with pendulous ears. The smallest of them are little larger than the lap-dog.

Beale, Lionel Smith, English physician and biologist, born in 1828, died in 1906. He was educated at King's College School and King's College,

London, and was a professor in the medical department of the latter institution for forty-three years, finally holding the chair of the principles and practice of medicine. His numerous published works treat of medical, anatomical, physiological, and biological subjects; the microscope; various questions of morality, &c. They include *How to Work with the Microscope*, *The Structure of the Tissues*, *Protoplasm*, *Disease Germs*, *Life Theories and Religious Thought*, *The Mystery of Life*, *Bioplasm*, *On Slight Ailments*, *Religio Medici*, *Religio Scientiæ*, *Religio Vitæ*, &c.

Beam, a long, straight, and strong piece of wood, iron, or steel, especially when holding an important place in some structure, and serving for support or consolidation; often equivalent to *girder*. In a balance it is the part from the ends of which the scales are suspended. In a loom it is a cylindrical piece of wood on which weavers wind the warp before weaving; also, the cylinder on which the cloth is rolled as it is woven. In a ship, one of the strong transverse pieces stretching across from one side to the other to support the decks and retain the sides at their proper distance: hence a ship is said to be 'on her beam ends' when lying over on her side. For calculations relating to beams, see Morley's *Strength of Materials* and *Theory of Structures*.

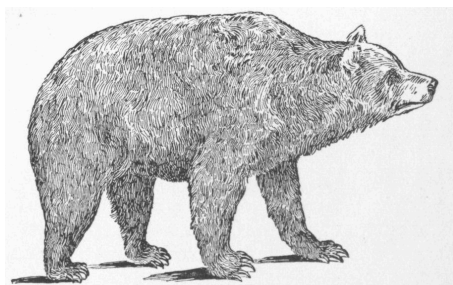
Beam Tree (*Pyrus aria*), a tree of the same genus as the apple, mountain-ash, and service tree found throughout Britain, having berries that are edible when quite mellow, and yielding a hard and fine-grained wood.

Bean, a name given to several kinds of leguminous seeds and the plants producing them, probably originally belonging to Asia. They belong to several genera, particularly to *Vicia*, garden and field bean; *Phaseolus*, French or kidney bean; and *Dolichos*, tropical bean. The common bean (*Vicia Faba*) is cultivated both in fields and gardens as food for man and beast. Beans were believed by some of the ancients to contain the souls of their ancestors, and Pythagoras would not eat beans for this reason. One of the bean family still retains the name of the Pythagorean bean. Beans were introduced by the Moors into Spain, whence they came to France and later to England. It is possible, however, that they were brought to Britain by the Romans. They are now largely imported from Egypt, Turkey, Greece, and the Netherlands. There are many varieties, as the Mazagan, the Windsor, the long-pod, &c., in gardens, and the horse or tick bean in fields. The soil that

best suits is a good strong clay. The seed of the Windsor is fully an inch in diameter; the horse bean is much less, often not much more than half an inch in length and three-eighths of an inch in diameter. Beans are very nutritious, containing nearly 50 per cent of digestible carbohydrate and 25 per cent of nitrogenous matter. The bean is an annual, from 2 to 4 feet high. The flowers are beautiful and fragrant. The *kidney bean*, *French bean*, or *haricot* is the *Phaseolus vulgāris*, a well-known culinary vegetable. There are two principal varieties, annual dwarfs and runners. The beans cultivated in America and largely used as articles of food belong to the genus *Phaseolus*. The *scarlet-runner bean* (*Phaseolus multiflorus*), a native of Mexico, is cultivated on account of its long rough pods and its scarlet flowers.

Bean-goose (*Anser segētum*), a species of wild goose, a migratory bird which arrives in Britain in autumn and retires to the north in the end of April, though some few remain to breed. Being rather less in size than the common wild goose, it is sometimes called the *small grey goose*.

Bean-king, the person chosen king in Twelfth Night festivities in virtue of having got the piece of cake containing the bean buried in the cake for this purpose.



Brown or Black Bear (*Ursus arctos*)

Bear, the name of several large plantigrade carnivorous mammals of the genus *Ursus*. The teeth are forty-two in number, as in the dog, but there is no carnassial or sectorial tooth, and the molars have a more tubercular character than in other carnivores. The eyes have a nictitating membrane, the nose is prominent and mobile, and the tail very short. The true bears are about ten in number, natives chiefly of Europe, Asia, and N. America. They generally lie dormant in their den during the winter months. The brown or

black bear of Europe is the *Ursus arctos*. It is a native of almost all the northern parts of Europe and Asia, and was at one time common in the British Islands. It feeds on fruits, roots, honey, ants, and, in case of need, on mammals. It sometimes reaches the length of 7 feet, the largest specimens being found farthest to the north. It lives solitarily. The American black bear is the *U. americanus*, with black shining fur, and rarely above 5 feet in length. It is a



The Polar or White Bear (*Ursus maritimus*)

great climber, is less dangerous than the brown bear, and is hunted for its fur and flesh. It is very amusing in captivity. The grizzly bear (*U. ferox* or *horribilis*) is an inhabitant of the Rocky Mountains; it is a ferocious animal, sometimes 9 feet in length, and has a bulky and unwieldy form, but is nevertheless capable of great rapidity of motion. The extinct cave-bear (*U. spelæus*) seems to have been closely akin to the grizzly. The Siberian bear (*U. collāris*) is perhaps a variety of the brown bear. The polar or white bear (*U. maritimus*) is an animal possessed of great strength and fierceness. It lives in the polar regions, frequents the sea, feeds on fish, seals, &c., and usually is 7 to 8 feet in length. The Malayan or coco-nut palm bear (*U. malayānus*) inhabits Cochin-China, Nepaul, the Sunda Islands, &c., lives exclusively on vegetable food, and is an expert climber. It is called also sun-bear and bruang. The Indian black bear or sloth-bear of India and Ceylon (*U. labiātus*) is reputed to be a fierce and dangerous animal.

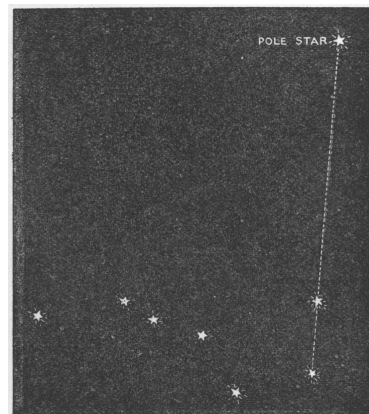
Bear, or **Bere**, a species of barley (*Hordĕum hexastichum*), having six rows in the ear, cultivated in Scotland and the north of England.

Bear, **Great** and **Little**, the popular names of two constellations in the northern hemisphere. The Great Bear (*Ursa Major*) is situated near the pole. It is remarkable for its well-known seven stars, by two of which, called the Pointers, the pole-star is always readily found. These seven stars are popularly called the *Wagon*, *Charles's Wain*, or the *Plough*. The Little Bear (*Ursa Minor*) is the constellation which contains the pole-star. This constellation has seven stars placed together in a manner resembling those in the Great Bear.

Bear-baiting, the sport of baiting bears with dogs, formerly one of the established amusements, not only of the common people, but of the nobility and even royalty itself. The places where bears were publicly baited were called bear-gardens. Butler gives a description of bear-baiting in his *Hudibras*. Bear-baiting was prohibited by Act of Parliament in 1835.

Bearberry (*Arctostaphylos uva-ursi*), an evergreen shrub of the heath family growing on the barren moors of Scotland, Northern Europe, Siberia, and N. America. The leaves, under the name of *uva ursi*, are used in medicine as an astringent and tonic.

Beard, the hair round the chin, on the cheeks, and the upper lip, which is a distinction of the male sex and of manhood. It differs from the hair on the head by its greater hardness and its form. Some nations have hardly any, others a great profusion. The latter generally consider it as a great ornament; the former pluck it out; as, for instance, the American Indians. The beard has often been considered as a mark of the sage and the priest. Moses forbade the Jews to shave their beards. With the ancient Germans the cutting off another's beard was a high offence. Even now the beard is regarded as a mark of great dignity among many nations in the East, as the Turks. Alexander the Great introduced shaving among the Greeks, by ordering his soldiers to wear no beards; among the Romans it was introduced in 296 B.C. The custom of shaving came into use in modern times during the reigns of Louis XIII and XIV of France, both of whom ascended the throne without a beard. Till then fashion sanctioned divers forms of moustaches and beards. In the reign of Henry VIII of England, the authorities of Lincoln's Inn prohibited wearers of beards from sitting at the great table unless they paid double commons. Taxation of beards was introduced in the reign of Elizabeth, beards of above a fortnight's growth being subject to a yearly tax of 3s. 4d. This impost was copied by Peter the Great in Russia. It is only in comparatively recent times that beards and moustaches have again become common.—Cf. Philippe, *Histoire philosophique, politique, et religieuse de la barbe*.



The Constellation of the Great Bear. The two stars on the right are the Pointers

Beard-moss (*Usnea barbāta*), a lichen of grey colour, forming a shaggy coat on many forest trees.

Beardsley, Aubrey Vincent, artist in black and white, born 1872, died 1898; took up art as a profession at the age of nineteen, and executed a large number of drawings for books and periodicals, showing great technical skill, originality, and disregard of conventionality, with sometimes a tendency towards the repulsive or morbid. Consumption had marked him as its victim from the first. Collections of his drawings were published under the titles *A Book of Fifty Drawings*, *The Early Work of Aubrey Beardsley*, *The Late Work of Aubrey Beardsley*, &c. *Under the Hill*, with his letters and poems, was published in 1904. Cf. G. Derry, *An Aubrey Beardsley Scrap-book*.

Bearer Securities. See *Stock Exchange*.

Bearing, the direction or point of the compass in which an object is seen, or the situation of one object in regard to another, with reference to the points of the compass. Thus, if from a certain situation an object is seen in the direction of north-east, the *bearing* of the object is said to be N.E. from the situation.—*To take bearings*, to ascertain on what point of the compass objects lie.

Bear Lake, Great, an extensive sheet of fresh water in the North-West Territory of Canada, between about 65° and 67° 32' N. lat.; and under the 120th degree of W. long.; of irregular shape; area about 14,000 sq. miles. The water is very clear and the lake abounds in fish.—*Bear Lake River*, the outlet at the S.W. extremity of Great Bear Lake, runs S.W. for 70 miles and joins the Mackenzie River.

Béarn (bā-a' r), one of the provinces into which France was formerly divided, now chiefly included in the department of Lower Pyrénées. Pau is the chief town (pop. 35,000). There is a peculiar and well-marked dialect—the Béarnese—spoken in this district, which has much more affinity with Spanish than with French.

Bear-pit, a deep, open pit with perpendicular walls, built in a zoological garden for keeping bears, and having in the centre a pole on which they may exercise their climbing powers.

Bear River, a river of the United States, 400 miles long; rises in the north of Utah, and flows northward into Idaho; turns abruptly southward, re-enters Utah, and empties into Great Salt Lake.

Bears and Bulls. See *Stock Exchange*.

Bear's-grease, the fat of bears, esteemed as of great efficacy in nourishing and promoting the growth of hair. The unguents sold under this name, however, are in a great measure made of hog's lard or veal fat, or a mixture of both, scented and slightly coloured.

Beas, river of India. See *Bias*.

Beat, in acoustics and music, the beating or pulsation resulting from the joint vibrations of two sounds of the same strength, and all but in unison. Also a short shake or transient grace-note struck immediately before the note it is intended to ornament. The word is also used for the movement of the hand or baton, by which the rhythm of a piece of music is indicated, and by which a conductor ensures perfect agreement in tempo and accent on the part of the orchestra or chorus.

Beath (bēth), a populous parish of Fifeshire, Scotland, containing the towns of Cowdenbeath, Kelty, and Hill of Beath. Pop. 24,350, (Cowdenbeath having 7908).

Beatification, in the Roman Catholic Church, an act by which the Pope declares a person beatified or blessed after his death. It is the first step to canonization, that is, the raising one to the honour and dignity of a saint. No person can be beatified till fifty years after his or her death. All certificates or attestations of virtues and miracles, the necessary qualifications for saintship, are examined by the Congregation of Rites. This examination often continues for several years; after which his Holiness decrees the beatification, and the corpse and relics of the future saint are exposed to the veneration of all good Christians. The present custom dates from a Bull of Urban VIII in 1634, although local veneration may be traced back to the earliest Christian ages.—In the Orthodox Eastern Church beatification is not distinguished from canonization.

Beating the Bounds, the periodical survey or perambulation by which the boundaries of parishes in England are preserved. It was the custom in some

places that the clergyman of the parish, with the parochial officers and the boys of the parish school, should march to the boundaries, where the boys were struck with willow rods. A similar ceremony in Scotland was called *riding the marches*.

Bea'ton, David, Archbishop of St. Andrews, and cardinal, born 1494. Pope Paul III raised him to the rank of cardinal in Dec., 1538. On the death of his uncle, Archbishop James Beaton, he succeeded him in the see of St. Andrews in 1539. After the accession of Mary he became Chancellor of Scotland, and distinguished himself by his zeal in persecuting members of the Reformed party, among the rest the famous Protestant preacher George Wishart, whose sufferings at the stake he viewed from his window with apparent exultation. At length a conspiracy was formed against him, and he was assassinated at his own castle of St. Andrews, on the 29th May, 1546. His private character was marked by pride, cruelty, and licentiousness.

Beatrice Portinari (bā-a ' -trē'chā por-tē-nā'rē), the "glorious lady" of Dante, born about 1266, died 1290; the daughter of a wealthy citizen of Florence, and wife of Simone de Bardi. She was but eight years of age, and Dante nine, when he met her first at the house of her father. He altogether saw her only once or twice, and she probably knew little of him. The story of his love is recounted in the *Vita Nuova*, which was mostly written after her death.

Beattie (bē'ti), James, a Scottish poet and philosophical writer, born at Laurencekirk, Kincardineshire, in 1735, died at Aberdeen 1803. He studied at Marischal College, Aberdeen, for four years, and received the M.A. degree. In 1753 he was appointed schoolmaster at Fordoun, a few miles from his native place; from whence he obtained a mastership in the Grammar School of Aberdeen, and ultimately was installed professor of moral philosophy and logic in Marischal College. In 1760 he published a volume of poems, which he subsequently endeavoured to buy up, considering them unworthy of him. In 1765 he published a poem, *The Judgment of Paris*, and in 1770 his celebrated *Essay on the Nature and Immutability of Truth*, attacking Helvetius and Hume and advocating what was afterwards called the doctrine of Common Sense, for which the University of Oxford conferred on him the degree of D.C.L., and George III honoured him, when on a visit to London, with a private conference and a

pension. He next published in 1771 the first book of his poem *The Minstrel*, and in 1774 the second; this is the only work by which he is now remembered. In 1776 he published dissertations on *Poetry and Music, Laughter and Ludicrous Composition, &c.*; in 1783 *Dissertations, Moral and Critical*; in 1786 *Evidences of the Christian Religion*; and in 1790-3 *Elements of Moral Science*. His closing years were darkened by the death of his two sons.—BIBLIOGRAPHY: Margaret Forbes, *Beattie and His Friends*; A. Mackie, *James Beattie, the Minstrel: Some Unpublished Letters*.

Beattie, William, M.D., Scottish physician, poet, and miscellaneous writer, born in 1793, died at London 1875. He was author of the standard *Life of Thomas Campbell*, whose intimate friend he was; published several poems, including *John Huss*, *The Heliotrope*, and *Polynesia*; wrote a series of descriptive and historical works, beautifully illustrated by his friend and fellow-traveller, W. H. Bartlett, on Switzerland, Scotland, &c. He had a very extensive and lucrative medical practice.

Beatty, Admiral of the Fleet, Earl, born in 1871. He entered the navy in 1884, became Commander in 1898, Rear-Admiral in 1910, and Vice-Admiral in 1915. He served in the Sudan from 1896 to 1897, and in China in 1900. In 1912 he was Naval Secretary to the First Lord of the Admiralty, and from 1912 to 1916 commanded the First Battle Cruiser Squadron. He distinguished himself in the battle of Jutland in 1916, and until 1919, when he succeeded Admiral Sir Rosslyn Wemyss (now Lord Wester Wemyss) as First Sea Lord, he was in command of the Grand Fleet. When the entire German High Seas Fleet left port on 31st May, 1916, and steamed up the west coast of Denmark, it was quickly sighted by the British scouts, so that the main British fleet immediately steamed out from its base to engage the enemy. The British Battle Cruiser Squadron, under the command of Earl Beatty (then Sir David) was nearer the scene than the main battle fleet under Sir John Jellicoe (Viscount Jellicoe of Scapa since 1918). The admiral sighted the enemy fleet north-west of the Horn Reef, and about 3.30 in the afternoon the ships engaged. The Jutland battle lasted only a few hours. At the end of Nov., 1916, Admiral Sir John Jellicoe was made First Sea Lord of the Admiralty, and Vice-Admiral Sir David Beatty took over the supreme command of the Grand Fleet.

He is a D.S.O., an O.M., a G.C.B., a G.C.V.O., and a Grand Officer of the French Legion of Honour. He was knighted in 1914, and received a grant of £100,000 in Aug., 1919, when he was created an earl. He was elected Lord Rector of Edinburgh University in 1917.

Beaucaire (bō-kār), a small commercial city of Southern France, department Gard, on the Rhone opposite Tarascon, with which it communicates by a fine suspension-bridge. It is chiefly famous for its great fair (founded in 1217), held yearly from the 21st to the 28th July. Pop. 8488.

Beauchamp (bō-shän), Alphonse de, French historian and publicist, born at Monaco 1767, died at Paris 1832. Under the Directory he had the surveillance of the press, a position which supplied him with materials for his *History of La Vendée*. He contributed to the *Moniteur* and the *Gazette de France*. Among his chief works are the *History of the Conquest of Peru*, the *History of Brazil*, and the *Life of Louis XVIII*. The *Mémoires de Fouché* is also with good reason ascribed to him.

Beaufort, (bō'fort), Henry, cardinal, natural son of John of Gaunt and half-brother of Henry IV, King of England, born 1377, died 1447; was made Bishop of Lincoln, whence he was translated to Winchester. He repeatedly filled the office of Lord Chancellor, and took part in all the most important political movements of his times.

Beaufort Scale. See *Wind Scale; Navigation*.

Beaufort West, a town of Cape Province, capital of the division of that name, 339 miles by railway north-east of Cape Town, well built and well supplied with water, though in a district with a small rainfall, and chiefly yielding wool. It is a popular health resort. Pop. 4530.

Beaugency (bō-zha'n'-sē), an ancient town, France, department Loiret, on the Loire, of some historical interest. General Chanzy was defeated there by the Grand-Duke of Mecklenburg, 7th-8th Dec., 1870. Pop. 3532.

Beauharnais (bō-a'r-nā), Alexandre, Viscount, was born in 1760 in Martinique. He married Joséphine Tascher de la Pagerie, who was afterwards the wife of Napoleon. At the breaking out of the French Revolution he was chosen a member of the National Assembly, of which he

was for some time president. In 1792 he was general of the army of the Rhine. He was falsely accused of having promoted the surrender of Mainz, and was sentenced to death 23rd July, 1794.

Beauharnais, Eugène de, Duke of Leuchtenberg, Prince of Eichstädt, and Viceroy of Italy during the reign of Napoleon, was born 1781, died at Munich 1824. He was the son of Alexandre Beauharnais and Joséphine, afterwards wife of Napoleon and Empress of France. After his father's death he joined Hoche in La Vendée, and subsequently studied for a time in Paris. He accompanied Napoleon to Egypt in 1798, rose rapidly in the army, was appointed Viceroy of Italy in 1805, and married a daughter of the King of Bavaria in 1806. He administered the government of Italy with great prudence and moderation, and was much beloved by his subjects. In the Russian campaign he commanded the third *corps d'armée*, and greatly distinguished himself. To him and to Ney France was mainly indebted for the preservation of the remains of her army during the retreat from Moscow. After the battle of Lützen of 2nd May, 1813, where, by surrounding the right wing of the enemy, he decided the fate of the day, he went to Italy, which he defended against the Austrians until the deposition of Napoleon. After the fall of Napoleon he concluded an armistice, by which he delivered Lombardy and all Upper Italy to the Austrians. He then went immediately to Paris, and thence to his father-in-law at Munich, where he afterwards resided.—His sister, Hortense Eugénie, Queen of Holland, was born in 1783, died in 1837. She became Queen of Holland by marrying Louis Bonaparte, and after Louis's abdication of the throne she lived apart from him. She wrote several excellent songs, and composed some deservedly popular airs, among others the well-known *Partant pour la Syrie*. Napoleon III was her third and youngest son.

Beaully (bū'li), a small seaport of Scotland, Inverness-shire, near the mouth of the River Beaully, which enters the Beaully Firth, a sea loch branching off from Inverness Firth, with interesting ruins of an old priory. Pop. 882.

Beaumarchais (bō-mär-shā), Pierre Augustin Caron de, a French wit and dramatist, was born at Paris in 1732, died 1799. He was the son of a watchmaker named Caron, whose trade he practised for a time. He early gave striking proofs of his mechanical and also of his musical talents; attained proficiency as a player on the guitar and harp, and was appointed

harp-master to the daughters of Louis XV. By a rich marriage (after which he added 'de Beaumarchais' to his name) he laid the foundation of the immense wealth which he afterwards accumulated by his speculations, and which was also increased by a second marriage. In the meantime he occupied himself with literature, and published two dramas—*Eugénie* in 1767 and *Les Deux Amis* in 1770. He first really distinguished himself by his *Mémoires* (Paris, 1774), or statements in connection with a lawsuit, which by their wit, satire, and liveliness entertained all France. The *Barber of Seville* (1775) and the *Marriage of Figaro* (1784) have given him a permanent reputation. His last work was *Mes Six Époques*, in which he relates the dangers to which he was exposed in the revolution. He lost about a million livres by his edition of the works of Voltaire (1785), and still more at the end of 1792 by his attempt to provide the French army with 60,000 muskets. He was a singular instance of versatility of talent, being at once an artist, politician, projector, merchant, and dramatist.—BIBLIOGRAPHY: L. de Loménie, *Beaumarchais et son temps* (English translation by H. S. Edwards); Gudin de la Brenellerie, *Histoire de Beaumarchais*.

Beumaris (bō-ma'ris), a seaport town, North Wales, Isle of Anglesey, on the Menai Strait. It is a favourite watering-place, and contains the remains of a castle built by Edward I about 1295. Pop. (1921), 1839.

Beaumont, a city of the United States, in Eastern Texas, in a region rich in cotton, timber, and petroleum, and an important railway centre. Pop. 28,851.

Beaumont (bō'mont), Francis, and **Fletcher**, John, two eminent English dramatic writers, contemporaries of Shakespeare, and the most famous of literary partners. The former, son of a common-pleas judge, was born at Grace-Dieu, in Leicestershire, in 1584, died in 1616, and was buried in Westminster Abbey. At the age of sixteen he published a translation, in verse, of Ovid's *Fable of Salmacis and Hermaphroditus*, and before nineteen became the friend of Ben Jonson. With Fletcher also he was early on terms of friendship. He married Ursula, daughter of Henry Isley of Sundridge, in Kent, by whom he left two daughters.—*John Fletcher* was born at Rye, Sussex, in 1579. His father was successively Dean of Peterborough, Bishop of Bristol, Worcester, and London. *The Woman Hater*, produced in 1606-7, is the earliest work known to exist in which he

had a hand. It does not appear that he was ever married. He died in London of the plague, Aug., 1625, and was buried at St. Saviour's, Southwark. The friendship of Beaumont and Fletcher, like their literary partnership, was singularly close; they lived in the same house, and are said to have even had their clothes in common. The works that pass under their names consist of over fifty plays, a masque, and some minor poems. It is believed that all the minor poems except one were written by Beaumont. After the death of Beaumont, Fletcher continued to write plays alone or with other dramatists. It is now difficult, if not indeed impossible, to determine with certainty the respective shares of the two poets in the plays passing under their names. According to the testimony of some of their contemporaries, Beaumont possessed the deeper and more thoughtful genius, Fletcher the gayer and more idyllic. *Four Plays in One*, *Wit at Several Weapons*, *Thierry and Theodore*, *Maid's Tragedy*, *Philaster*, *King and no King*, *Knight of the Burning Pestle*, *Cupid's Revenge*, *Little French Lawyer*, *Scornful Lady*, *Coxcomb*, and *Laws of Candy* have been assigned to Beaumont and Fletcher conjointly. To Beaumont alone—*The Masque of the Inner Temple and Gray's Inn*. To Fletcher alone—*The Faithful Shepherdess*, *Woman Hater*, *Loyal Subject*, *Mad Lover*, *Valentinian*, *Double Marriage*, *Humorous Lieutenant*, *Island Princess*, *Pilgrim*, *Wild-geese Chase*, *Spanish Curate*, *Beggar's Bush*, *Rule a Wife and Have a Wife*, *Fair Maid of the Inn*, &c. To Fletcher and Rowley—*Queen of Corinth*, *Maid of the Mill*, and *Bloody Brother*. To Fletcher and Massinger—*False One* and *Very Woman*. To Fletcher and Shirley—*Noble Gentleman*, *Night-walker*, and *Love's Pilgrimage*. To Fletcher and Shakespeare—*Two Noble Kinsmen*.—BIBLIOGRAPHY: G. C. Macaulay, *Francis Beaumont, a Critical Study*; Sir A. W. Ward, *History of English Dramatic Literature* (vol. ii).

Beaumont, Sir George, born of an ancient family in Leicestershire in 1753, died 1827. He possessed considerable skill as a landscape-painter, but was noted more especially as a munificent patron of the arts. The establishment of the National Gallery was mainly owing to his exertions. He was a friend of Wordsworth, who dedicated to him the 1815 edition of his *Poems*.

Beaumont, Sir John, born 1582, died 1627, brother of Francis Beaumont the dramatist; was author of *Bosworth Field*, an historical poem, and various sacred and other poems. A poem in eight books, called *The Crown of Thorns*, has been lost.

Beaumont, Joseph, D.D., born 1615, died 1699; descended from an old Leicestershire family. In 1663 he became master of Peterhouse, Cambridge. He wrote *Psyche, or Love's Mystery*, a poem once very popular, and an attack on Henry More's *Mystery of Godliness*, for which he received the thanks of the university.

Beaumont, William, M.D., an American surgeon, born 1785, died 1853. His experiments on digestion with the Canadian St. Martin, who lived for years after receiving a gunshot wound in the stomach which left an aperture of about two inches in diameter, were of great importance to physiological science.

Beaumont-Hamel, village of France, department Somme, the scene of fierce fighting in 1918. See *Somme, Battles of the*.

Beaune (bōn), a town, France, department Côte d'Or, 23 miles s.s.w. of Dijon, well built, with handsome church, public library, museum, &c., and a trade in the fine Burgundy and other wines of the district. Pop. 13,409.

Beaune (bōn), Florimond, a distinguished mathematician and friend of Descartes, born at Blois 1601, died at the same place 1652. He may be regarded as the originator of the integral calculus.

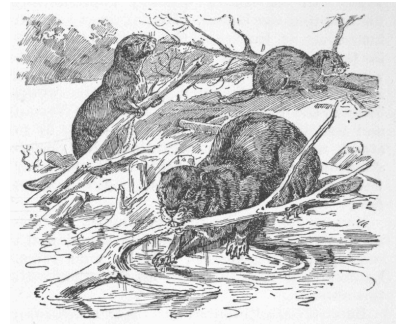
Beauregard (bō'rè-ga 'rd), Pierre Gustavus Toutant, a general of the Confederate troops in the American Civil War, born in 1818 near New Orleans. He studied at the military academy, West Point, and left it as artillery lieutenant in 1838. He served in the Mexican War, and on the outbreak of the Civil War joined the Confederates. He commanded at the bombardment of Fort Sumter, gained the battle of Bull Run, lost that of Shiloh, assisted in the defence of Charleston, and aided Lee in that of Richmond. He was the author of *Principles and Maxims of the Art of War* (1863). He died in 1893.

Beausobre (bō-sō-br), Isaac, born in 1659 at Niort, in France, died at Berlin 1738. In 1683 he became Protestant minister of Chatillon-sur-Indre, but was compelled by persecution to go into exile in 1685. In 1694 he became minister to French Protestants at Berlin. He enjoyed much of the favour both of Frederick William I and of the Crown Prince, afterwards Frederick

the Great, and died in 1738. His most remarkable work is the *Histoire Critique de Manichée et du Manichéisme* (1734).

Beauvais (bō-vā; ancient, *Bellovacum*), a town, France, capital of the department of Oise, at the confluence of the Avelon with the Thérain, 43 miles north of Paris, poorly built, but with some fine edifices, the choir of the uncompleted cathedral being one of the finest specimens of Gothic architecture in France. In 1472 Beauvais resisted an army of 80,000 Burgundians under Charles the Bold. On this occasion the women particularly distinguished themselves, and one of them, Jeanne Lainé, called La Hachette, seeing a soldier planting a standard on the wall, seized it and hurled him to the ground. The banner is preserved in the town hall, and an annual procession of young girls commemorates the deed. Manufactures: tapestry and carpets, trimmings, woollen cloth, cottons, &c. Pop. 20,250.

Beaver, the largest rodent quadruped now existing in the northern hemisphere, about 2 feet in length exclusive of the tail, genus *Castor* (*C. fiber*), is now found in considerable numbers only in North America, living in colonies, but occurring solitary in Central Europe and Asia. It has short ears, a blunt nose, small fore-feet, large webbed hind-feet, with a flat ovate tail covered with scales on its upper surface. It is valued for its fur, which used to be largely employed in the manufacture of hats, but for which silk is now for the most part substituted, and for an odoriferous secretion named castor, at one time in high repute, and still largely used in some parts of the world as an antispasmodic medicine. The food of the beaver consists of the bark of trees, leaves, roots, and berries. Their favourite haunts are rivers and lakes which are bordered by forests. In winter they live in houses, which are 3 to 4 feet high, are built on the water's edge, and, being substantial structures with the entrance under water, afford them protection from wolves and other wild animals. These dwellings are called beaver 'lodges', and accommodate a single family. They also live in burrows. They can gnaw through large trees with their strong teeth, this being done partly to obtain food, partly to get materials for houses or dam-building. When they find a stream not sufficiently deep for their purpose, they throw across it a dam constructed with great ingenuity of wood, stones, and mud. Beavers were fairly numerous in Britain in very early times, at least in Scotland and Wales. Their existence has been recorded in Wales till the thirteenth century, and in Scotland to a later date, but they are now extinct in Britain. A colony was some years ago introduced into the Island of Bute, but they died out again.—BIBLIOGRAPHY: Harting, *British Animals Extinct within Historic Times*; E. A. Mills, *In Beaver World*; A. A. R. Dugmore, *The Romance of the Beaver*.



Beavers (*Castor fiber*)

Beaver (from Fr. *bavière*, a child's bib, from *bave*, saliva), the movable face-guard of a helmet, so fitted on as to be raised and lowered at pleasure.

Beaver Falls, a town of the United States, Pennsylvania, on Beaver River, 30 miles north-west of Pittsburg, in the coal and natural-gas region. Pop.

12,191.

Beaver-rat (*Hydromys chrysogaster*), a Tasmanian rodent quadruped, inhabiting the banks both of salt and fresh waters. They are admirable swimmers and divers, and exceedingly shy.

Bebée'ru. See *Greenheart*.

Bebel (bā'bl), Ferdinand August, German Socialist, born in 1840, died in 1913. He became a master turner in Leipzig, was elected to the Reichstag or Diet of the new German Empire in 1871, in which he was prominent until his death. He opposed the leadership of Prussia in Germany and the establishment of the empire, and showed himself favourable to the Paris Commune and the International. Found guilty of treasonable practices, he was condemned to two years' imprisonment in a fortress in 1872, with six months' ordinary imprisonment for insulting the Kaiser. His influence kept on increasing as leader of the Social Democrats in Germany and in the German Parliament, where he spoke strongly against militarism and the emperor's naval policy. His works include *Die Frau und der Socialismus* (Woman and Socialism), in which he went so far as to attack marriage as an institution; *Die Socialdemokratic und das allgemeine wahlrecht*; *Für Volkswehr gegen Militarismus*; *Aus meinem Leben*; &c.

Bec, a celebrated abbey of France, in Normandy, near Brionne, founded in the eleventh century, now represented only by some ruins. Lanfranc and Anselm were both connected with this abbey.

Beccafi'co, a European bird (*Sylvia hortensis*), the garden-warbler. These birds are much esteemed as dainties in the autumn, when they have fattened on figs and grapes.

Beccafu'mi, Domen'ico, Italian painter, born near Sienna in the latter half of the fifteenth century, enriched the churches of Sienna with many noble frescoes and other paintings. He drew and coloured well, and possessed strong inventive powers. He died at Sienna in 1551, and was buried in its cathedral.

Beccaria (bek-a'-rē'a'), Cesare Bonesana, Marchese di, Italian economist and writer on penal laws, born 1738, died 1794. He is principally known from his treatise, *On Crimes and Punishments*, which was speedily

translated into various languages, and to which many of the reforms in the penal codes of the principal European nations are traceable. He became professor of political economy at Milan, where he died.

Beccaria (bek-a´-rē´a´), Giovanni Battista, an Italian natural philosopher, born 1716, died 1781; was appointed professor of experimental physics at Turin, 1748; author of a treatise on *Natural and Artificial Electricity, Letters on Electricity, &c.* He contributed several articles to the *Transactions of the Royal Society of London*, and was commissioned in 1759 to measure an arc of the meridian in the neighbourhood of Turin.

Beccles (bek´lz), a municipal borough in Suffolk, England, 33 miles N.N.E. from Ipswich, on the right bank of the Waveney; has a fine church of the fourteenth century, and a good trade coastwise. Pop. (1921), 7077.

Becelære, town of Belgium, province of West Flanders. It was the scene of fighting in the third battle of Ypres (1917), and in the battle of Flanders (1918).

Becerra (be-ther´a´), Gaspar, Spanish painter and sculptor, born 1520, died 1570. He studied under Michel Angelo at Rome, and is credited with the chief share in the establishment of the fine arts in Spain.

Beche (bāsh), Sir Henry de la, an English geologist, born 1796, died 1855. He founded the geological survey of Great Britain, which was soon undertaken by the Government, De la Beche being appointed director-general. He also founded the Jermyn Street Museum of Economic or Practical Geology, and the School of Mines. His principal works are: *Geology of Jamaica, Classification of European Rocks, Geological Manual, Researches in Theoretical Geology, Geology of Cornwall, Devon, and West Somerset, &c.*

Bêche-de-Mer (bāsh-dē-mār). See *Trepang*.

Becher (beh´er), Johann Joachim, German chemist, born in 1625 or 1635, died in London 1682. He became a professor at Mainz; was elected a member of the Imperial Council at Vienna, 1660, but fell into disgrace and subsequently resided in various parts of Germany, Holland, Italy, Sweden, and Great Britain. His chief work, *Physica Subterranea*, containing many of the fanciful theories of the alchemists, was published in 1669, and enlarged

in 1681. Among his other works are *Laboratorium portabile* and *Alphabetum minerale*.

Bechstein (beh'stīn), Johann Matthäus, German naturalist, born in 1757, died in 1822. He wrote a popular *Natural History of Germany*, and various works on forestry, in which subject his labours were highly valuable. In Britain he is best known by a treatise on cage birds.

Bechuanas, or **Betchuanas** (bech-wan'az), a widely-spread race of people inhabiting the central region of South Africa north of Cape Province. They belong to the great Kaffir stem, and are divided into tribal sections. They live chiefly by husbandry and cattle-rearing, and they work with some skill in iron, copper, ivory, and skins. They were led to seek British protection owing to the encroachments of the Boers. The southern portion of their territory was first placed under British protection in 1885, and subsequently the whole Bechuana country up to the Zambezi was annexed. In 1895 the southern portion (then a Crown colony) was united to the Cape Colony; the remainder is still a protectorate partly under native chiefs. The headquarters of the British administration are at Mafeking in the Cape Province. Bechuanaland Protectorate comprises the territory lying between the Molopo River on the south and the Zambezi on the north, and extending from the Transvaal Province and Matabeleland on the east to South-West Africa; area, 275,000 sq. miles; pop. 125,350, of whom about 1700 are Europeans. It is generally flat or slightly undulating, and is essentially a grass country, the grasses being nutritious and standing drought well. Surface water is scarce, but there is underground water, besides swampy lakes such as Ngami. Some parts are wooded and well watered. Gold, coal, and copper have been found. There is railway connection both with south and north.—Cf. G. W. Stow, *The Native Races of South Africa*.

Beck Case, The, a celebrated case of mistaken identity, which involved a grave miscarriage of justice. Adolf Beck, a Swede, had been sentenced in 1890 to seven years' penal servitude for a series of mean frauds committed against some women, and in 1904 he was again convicted and sentenced for a similar offence. Soon after his second trial a Jew named John Smith, who had been convicted of fraudulency in 1877, was arrested for similar frauds committed while Beck was in custody. The resemblance between the two men was remarkable, and it was eventually discovered that Beck was

entirely innocent. He received two King's pardons, and the sum of £5000 from the Home Office as compensation. A commission which considered the case drew attention in its report to the carelessness of the police in identifying Beck with Smith, though the latter was a Jew; and stated that further legal training was desirable for the subordinate officials in the Home Office. Beck died in poverty in 1909. Cf. J. Kempster, *Perversion of Justice as exhibited in the Beck Case*.

Beckenham, a suburban locality on the south of London, in West Kent, now forming an urban district, lying south of Sydenham. Pop. 33,350.

Beck'er, Wilhelm Adolf, German archæologist, born at Dresden 1796, died at Meissen 1846. In 1828 he became a teacher at Meissen, in 1837 was appointed extraordinary professor of classical archæology at Leipzig, and in 1842 ordinary professor. Best-known works: *Gallus, or Roman Scenes of the Time of Augustus*; and *Charicles, or Illustrations of the Life of the Ancient Greeks*, which wonderfully reproduce the social life of old Rome and Greece.

Beck'et, Thomas (the form À Becket is also common), Archbishop of Canterbury, born in London 1117 or 1119, assassinated in Canterbury Cathedral, 29th Dec., 1170. He was educated at Oxford and Paris, and was sent, by the favour of Theobald, Archbishop of Canterbury, to study civil law at Bologna in Italy, and on his return made Archdeacon of Canterbury and Provost of Beverley. In 1158 Henry II appointed him High Chancellor and preceptor to his son, Prince Henry—the first instance after the Conquest of a high office being filled by a native Englishman. At this period he was a complete courtier, conforming in every respect to the humour of the king. He was, in fact, the king's prime companion, held splendid levees, and courted popular applause. On the death of Theobald, 1162, he was consecrated archbishop, when he affected an extraordinary austerity of character, and appeared as a zealous champion of the Church against the aggressions of the king, whose policy was to have the clergy in subordination to the civil power. Becket was forced to assent to the Constitutions of Clarendon, but a series of bitter conflicts with the king followed, ending in Becket's flight to France, when he appealed to the Pope, by whom he was supported. After much negotiation a sort of reconciliation took place in 1170, and Becket returned to England, resumed his office, and

renewed his defiance of the royal authority. A rash hint from the king induced four barons, Reginald Fitz-Urse, William de Tracy, Hugh de Morville, and Richard Breto, to go to Canterbury and murder the archbishop while at vespers in the cathedral. He was canonized in 1172 by Pope Alexander III, and the splendid shrine erected at Canterbury for his remains was, for three centuries, a favourite place of pilgrimage.—BIBLIOGRAPHY: J. C. Robertson and J. B. Sheppard, *Materials for the History of Archbishop Becket* (Rolls Series); Canon Morris, *Life and Martyrdom of St. Thomas Becket*; Lhuillier, *Saint Thomas de Cantorbéry*; Abbott, *St. Thomas of Canterbury, His Death and His Miracles*.

Beckett, Gilbert Abbot À. See *À Beckett*.

Beck'ford, William, an English writer famous in his time for his immense wealth and his eccentricities. He was born at Fonthill, his father's estate in Wiltshire, in 1759. In 1770 the death of his father left him in the possession of £1,000,000 of money, and an income of £100,000 a year. He travelled much, and for some time lived in Portugal. He expended an enormous sum in building and rebuilding Fonthill Abbey, near Salisbury, which he filled with rare and expensive works of art. Here he lived in seclusion for twenty years. In 1822 the abbey and greater part of its contents were sold, and he retired to Bath, where, with a much-diminished fortune, but one amply sufficient, he lived till 1844. His literary fame rests upon his Eastern tale *The History of the Caliph Vathek*, which he wrote in French, and a translation of which by the Rev. Samuel Henley appeared at London in 1786. The tale is still much read, and was highly commended by Lord Byron. He had two daughters, one of whom became Duchess of Hamilton, and brought his valuable library to this family.—*William Beckford*, his father, a London merchant and West Indian proprietor, was famous for a spirited speech made to George III when Lord Mayor of London.—BIBLIOGRAPHY: Cyrus Redding, *Memoir*; R. Garnett, *Vathek* (with a critical essay); Melville, *The Life and Letters of William Beckford of Fonthill*.

Beckmann, Johann, German writer on the industrial arts and agriculture, born 1739, died 1811. He was for a short time professor of physics and natural history at Petrograd, and afterwards for almost forty-five years professor of philosophy and economy in Göttingen. His *History of*

Inventions is well known in the English translation of it. This work entitled the author to be regarded as the founder of scientific technology.

Beckx (beks), Pierre Jean, general of the order of Jesuits, born near Louvain, Belgium, 1795, died 1887. The success of the Jesuits, especially in non-Catholic countries, was greatly due to his tact and energy.

Becquerel (bek-rel), Antoine César, French physicist, born 1788, died 1878. He served as an officer of engineers, and retired in 1815, after which he devoted himself to the study of electricity, especially electro-chemistry. In 1837 he was awarded the Copley medal of the Royal Society of London. He refuted the 'theory of contact' by which Volta explained the action of his pile or battery. Becquerel may be considered one of the creators of electro-chemistry.—His son, Alexandre Edmond (1820-91), was associated with him in much of his work. He was the author of a work *La Lumière, ses causes et ses effets* (1867-8).

Becquerel Rays (bek-rel), the rays given out by radium and other 'radioactive' substances, so named from their discoverer, the French physicist, Henri Becquerel (born 1852, died 1908), son of Alexandre Edmond Becquerel. They were first detected in 1896, as proceeding from uranium salts; and it is only by uranium, thorium, radium, and one or two other elements that they are emitted, these bodies giving them out spontaneously and without any apparent loss of radioactive power or change of any kind. The Becquerel rays are invisible, and only known by their effects, which are of various kinds: thus, like the Röntgen rays, they blacken a photographic plate, even after passing through glass or other intervening substances; they cause a number of different substances to give out a fluorescent light, and they render air a conductor of electricity. Like the Röntgen rays, they act strongly on the human skin. They consist of a mixture of α -, β -, and γ -rays. See *Radio-activity*; *Radium*.

Becse (bech'e), **Old**, a town of Hungary, 48 miles s. of Szegedin, on the right bank of the Theiss. Pop. 19,000.—**New Becse**, a market-town on the left bank of the Theiss, 5 miles E. of Old Becse. Pop. 7725, or, with the immediately adjoining village of Franyova, about 15,000. Both towns carry on an extensive trade in grain.

Becskerek (bech'ke-rek), two towns of South Hungary.—*Great Becskerek*, now in Yugo-Slavia, is on the Bega, 45 miles N. of Belgrade, with which it communicates by the Bega Canal. Trade in cattle and agricultural produce. Pop. 26,407.—*Little Becskerek*, now belonging to Roumania, is 11 miles by railway from Temesvar. Pop. 3660.

Bed, Bedstead, an article of furniture to sleep or rest on. The term *bed* properly is applied to a large flat bag filled with feathers, down, wool, or other soft material, and also to a mattress supported on spiral springs or form of elastic chains or wirework which is raised from the ground on a bedstead. The term, however, sometimes includes the bedstead or frame for supporting the bed. The forms of beds are necessarily very various—every period and country having its own form of bed. Air-beds and water-beds are much used by invalids.

Bed, in geology, a layer or stratum, usually a stratum of considerable thickness.

Beda. See *Bede*.

Bédarieux (bā-där-i-eu), a thriving town, Southern France, department Hérault, situated on the Orb. Pop. 6186.

Bed-bug. See *Bug*.

Bed-chamber, Lords of the, officers of the royal household of Britain under the Groom of the Stole. They are twelve in number, and wait a week each in turn. In the case of a queen regnant these posts are occupied by ladies, called *Ladies of the Bed-chamber*.

Beddoes (bed'ōz), Thomas, physician and author, born 1760; educated at Oxford, London, and Edinburgh. After taking his doctor's degree, and visiting Paris, he was appointed professor of chemistry at Oxford. There he published some excellent chemical treatises, and essays upon such subjects as the calculus, sea-scurvy, consumption, catarrh, and fever. His expressed sympathy with the French revolutionists led to his retirement from his professorship in 1792, soon after which he published his *Observations on the Nature of Demonstrative Evidence*, and the exceedingly popular *History of Isaac Jenkins*. In 1794 he married a sister of Maria Edgeworth; and in 1798, with the pecuniary aid of Wedgwood, opened a pneumatic institution

for curing phthisical and other diseases by inhalation of gases. It speedily became an ordinary hospital, but was noteworthy as connected with the discovery of the properties of nitrous oxide, and as having been superintended by the young Humphry Davy. Beddoes' essays *On Consumption* (1779) and *On Fever* (1807), and his *Hygeia* (3 vols., 1807) had a high contemporary repute. He died in 1808.

Beddoes, Thomas Lovell, poet, son of above, born 1803, died 1849; published *The Bride's Tragedy* while a student at Oxford, studied medicine, and lived long abroad. His work was largely fragmentary, but his posthumous *Death's Jest-book, or The Fool's Tragedy* (1850), received the high praise of such judges as Landor and Browning. His *Poems*, with memoir, appeared in 1851.

Bede, **Beda**, or **Bæda**, known as the Venerable, English historian and theologian, born in 672 or 673 in the neighbourhood of Monkwearmouth, County Durham; educated at St. Peter's Monastery, Wearmouth; took deacon's orders in his nineteenth year at St. Paul's Monastery, Jarrow, and was ordained priest at thirty by John of Beverley, Bishop of Hexham. His life was spent in studious seclusion, the chief events in it being the production of homilies, hymns, lives of saints, commentaries, and works in history, chronology, grammar, &c. He was the most learned Englishman of his day, and in some sense the father of English history, his most important work being his *Historia Ecclesiastica Gentis Anglorum* (or *Ecclesiastical History of England*), afterwards translated by King Alfred into Anglo-Saxon. Besides his familiarity with Latin, he knew Greek and had some acquaintance with Hebrew. Most of his writings were on scriptural and ecclesiastical subjects, but he also wrote on chronology, physical science, grammar, &c., and had considerable ability in the writing of Latin verse. He died in 735, an interesting record of his closing days being preserved in a letter by his pupil Cuthbert. His body was after a lapse of time removed from Jarrow church to Durham, but of the shrine which formerly enclosed them only the Latin inscription remains, ending with the verse—'Hac sunt in fossa Bedæ venerabilis ossa'. On 13th Nov., 1899, Leo XIII decreed that the feast of the Venerable Bede should be celebrated in the Church on 27th May. An edition of the whole works of Bede (12 vols.), with an English translation, was prepared by Dr. Giles (1843-4).—BIBLIOGRAPHY: Gehle. *De*

Bedæ Venerabilis Vita et Scriptis; Browne, *The Venerable Bede*; C. Plummer's introduction to his edition of the *Historia Ecclesiastica*.

Bedeguar, or **Bedegar** (bed'e-gär), a spongy excrescence or gall, sometimes termed sweet-brier sponge, found on various species of roses, and produced by several insects as receptacles for their eggs, especially by the *Cynips rosæ*, It was once thought to be a diuretic and vermifuge.

Bedell, William, a celebrated Irish bishop, born in Essex in 1570. In 1604 he went to Venice as chaplain to Sir Henry Wotton, and remained eight years. After holding the living of Horingsheath from 1615-27, he became Provost of Trinity College, Dublin, and in 1629 Bishop of Kilmore and Ardagh, though he resigned the latter of the united sees in 1630. He set himself to reform abuses and promote the spread of Protestantism, procured the translation of the Old Testament into Irish, and by his tact and wisdom conciliated the adherents of both creeds. He underwent a brief imprisonment on the breaking out of the rebellion in 1641, and died in the year following. His biography was written by Bishop Burnet.

Be'der Ware. See *Bidery*.

Bed'ford, a municipal and, until 1918, a parliamentary borough, England, county town of Bedfordshire, on the Ouse. The chief buildings are the law courts, a range of public schools, a large infirmary, county jail, &c., and the churches. The town is rich in charities and educational institutions, the most prominent being the Bedford Charity, embracing grammar and other schools, and richly endowed. There is an extensive manufactory of agricultural implements; lace is also made, and there is a good trade. John Bunyan was born at Elstow, a village near the town, and it was at Bedford that he lived, preached, and was imprisoned. Bedford gives its name to a parliamentary division of the county. Pop. 40,247.—*Bedfordshire* or *Beds*, the county, is bounded by Northampton, Bucks, Herts, Cambridge, and Huntingdon; area, 302,942 acres, of which 260,000 are under tillage or in permanent pasture. Chalk hills, forming a portion of the Chilterns, cross it on the south; north of this is a belt of sand; the soil of the vale of Bedford, consisting mostly of clay and loam, is very fertile; and the meadows on the Ouse, Ivel, and other streams furnish rich pasturage. Two-thirds of the soil is under tillage. Besides the usual cereal and other crops, culinary

vegetables are extensively cultivated for the London market. Principal manufactures: agricultural implements, and straw-plait for hats, which is made up principally at Dunstable and Luton. The county returns three members to Parliament (divisions: Bedford, Luton, Midlands). Pop. (1921), 206,478.

Bedford, John, Duke of, one of the younger sons of Henry IV, King of England; famous as a statesman and a warrior. He defeated the French fleet in 1416, commanded an expedition to Scotland in 1417, and was lieutenant of England during the absence of Henry V in France. On the king's death he became Regent of France, and for several years his policy was as successful as it was able and vigorous, the victory of Verneuil in 1424 attesting his generalship. The greatest stain on his memory is his execution of the Maid of Orleans (Joan of Arc) in 1431. He died in 1435 at Rouen, and was buried in the cathedral of that city.

Bedford Level, a large tract of marshy land in England, of about 400,000 acres total area, comprising 63,000 in Norfolk, 30,000 in Suffolk, 50,000 in Huntingdon, the Peterborough fen in Northampton, the Holland district in Lincolnshire, and most of the Isle of Ely in Cambridge. It derives its name from Francis, Earl of Bedford, who in 1634 made an agreement with Charles I for the drainage of the Level, in consideration of receiving 95,000 acres of the reclaimed land. A great part of the Level is under cultivation, and produces grain, flax, and cole-seed; the remainder yields a winter harvest of wild-fowl for the London market. See *Fenland*.

Bedfordshire Regiment, The, raised in 1688 on the approach of the Prince of Orange, suffered severely at Blenheim, and distinguished itself in a marked manner at the siege of Lille. In 1797 it went to Scotland and there recruited 1000 'parish boys' under sixteen years old, who later made excellent soldiers. The regiment shared in the Chitral and South African (1900-2) campaigns; on going to the front in 1914, sustained heavy losses at Givenchy, and distinguished itself at Ypres and the Aisne.

Bedivere, the last knight of the Round Table (q.v.). See *Morte d'Arthur*.

Bed'lam, a corruption of Bethlehem (Hospital), the name of a religious house in London, converted, after the general suppression by Henry VIII, into a hospital for lunatics. The original Bedlam stood in Bishopsgate

Street; its modern successor in St. George's Fields was opened in 1815, having been built on the site of the notorious tavern called 'The Dog and the Duck'. The lunatics were at one time treated as little better than wild beasts, and hence Bedlam came to be typical of any scene of wild confusion. The average number of patients is about 300.

Bed'lington, a town and urban district of England, in Northumberland, near the mouth of the Blyth, and not far from the seaport of Blyth, with collieries, ironworks, &c. Pop. 25,440.

Bedlington Terrier, an English Dog, deriving its name from Bedlington, in Northumberland, having first become well known as a favourite among the miners of that place. It is a dog of moderate size, head rather long, with a light, silky tuft on top, ears hanging close to the cheeks, legs moderately long and strong, tail tapering to the point, which is almost bare; colour, dark blue, blue and tan, liver, liver and tan, sandy, or sandy and tan; courageous, intelligent, and generally useful.

Bedlis. See *Betlis*.

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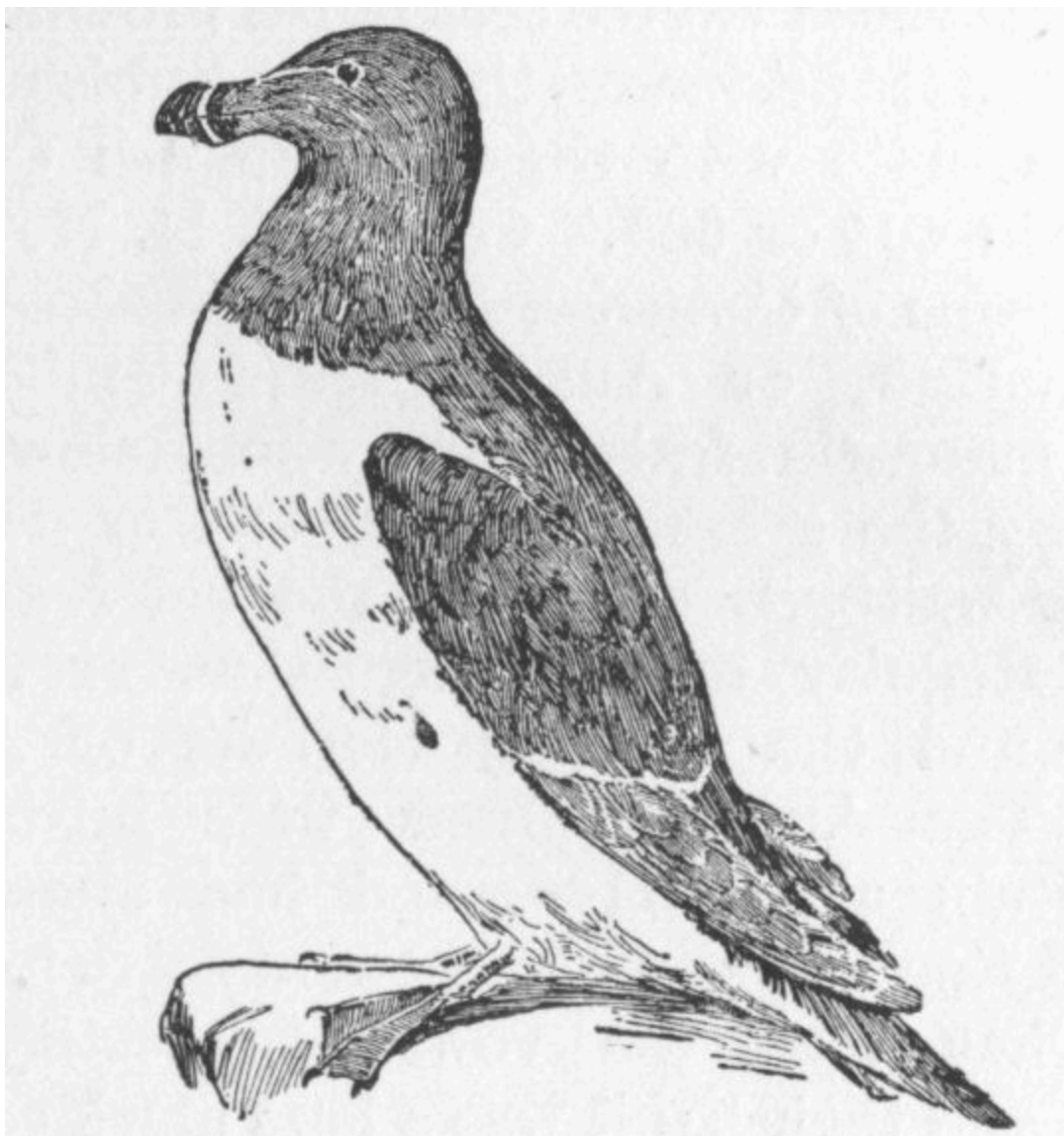
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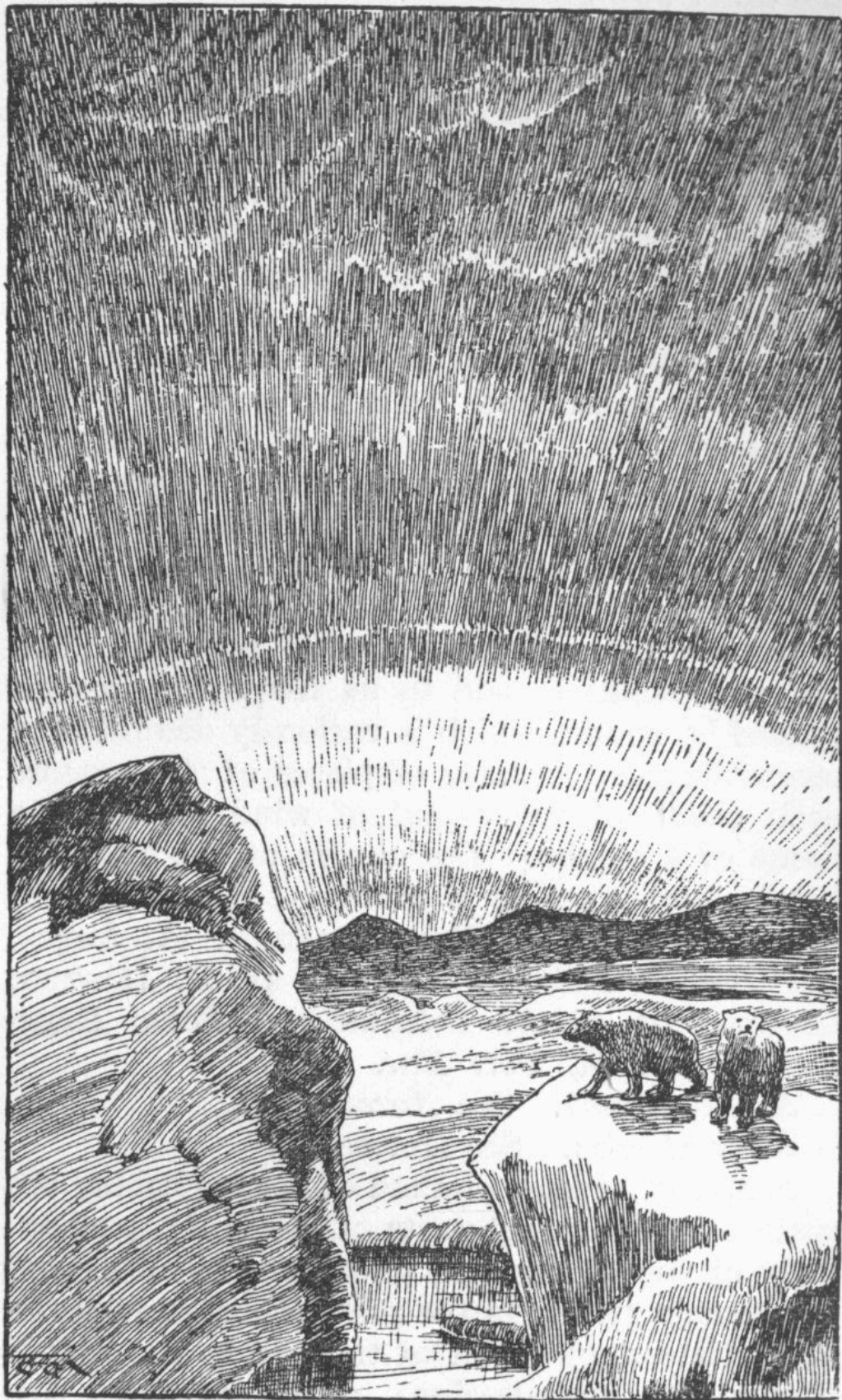
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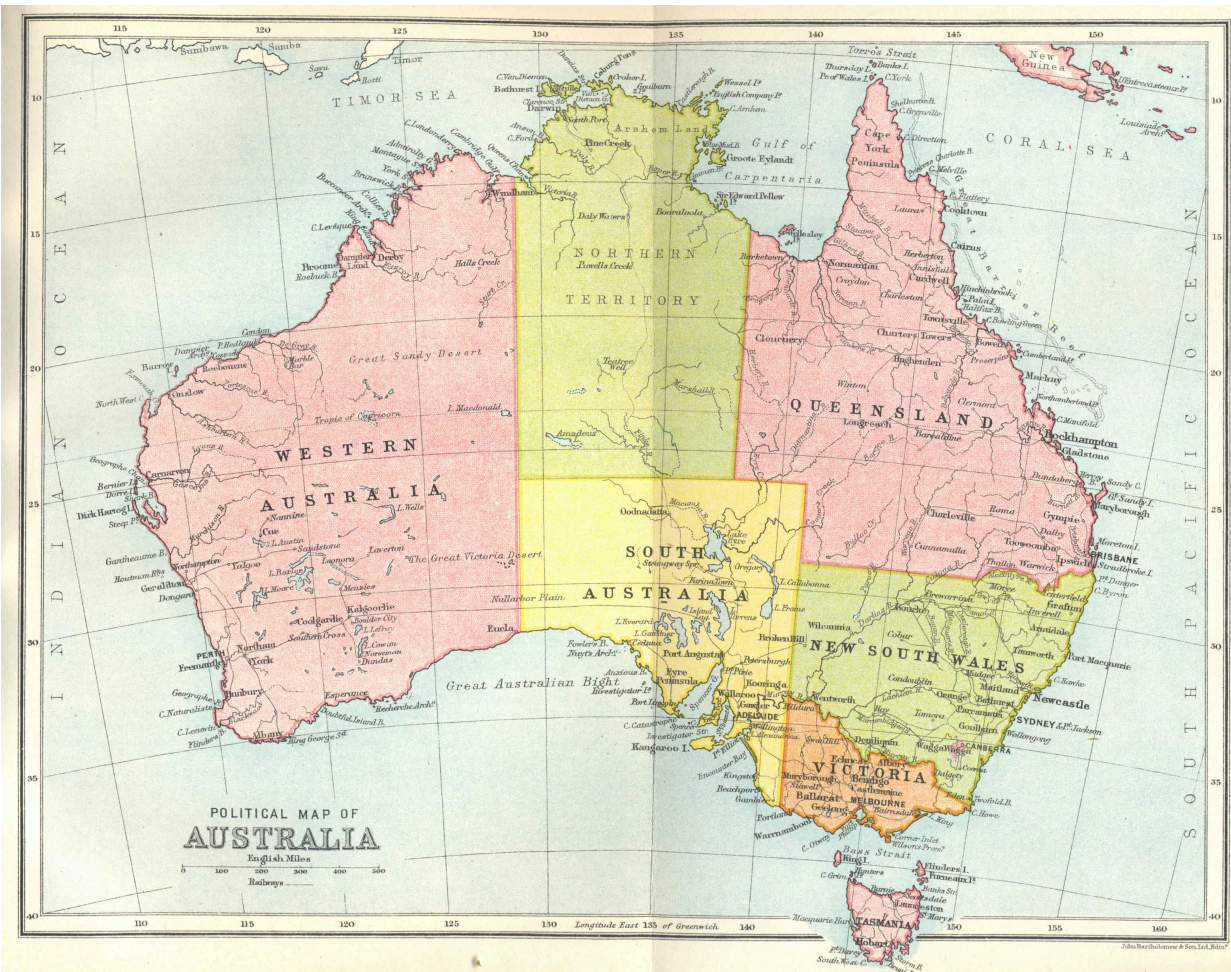
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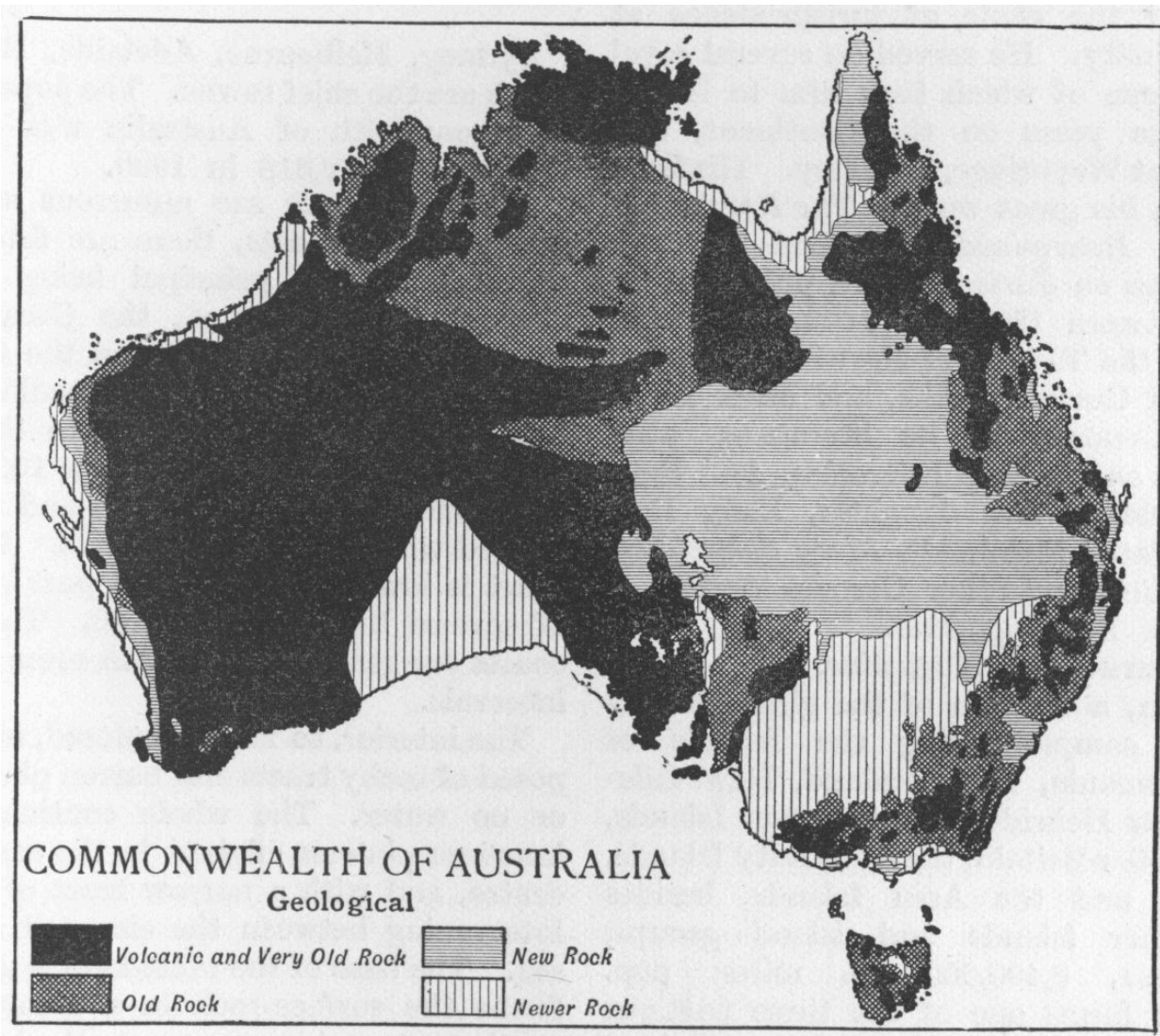
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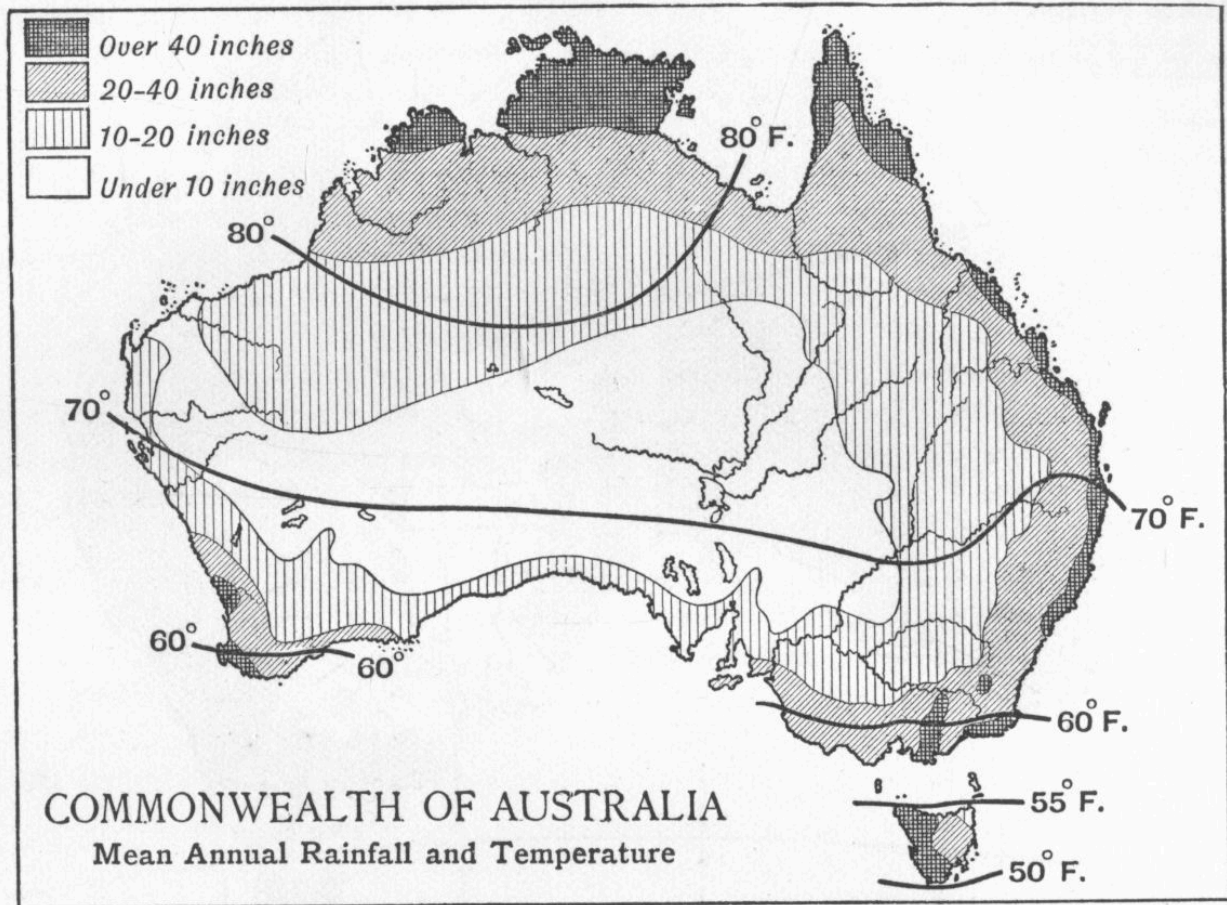
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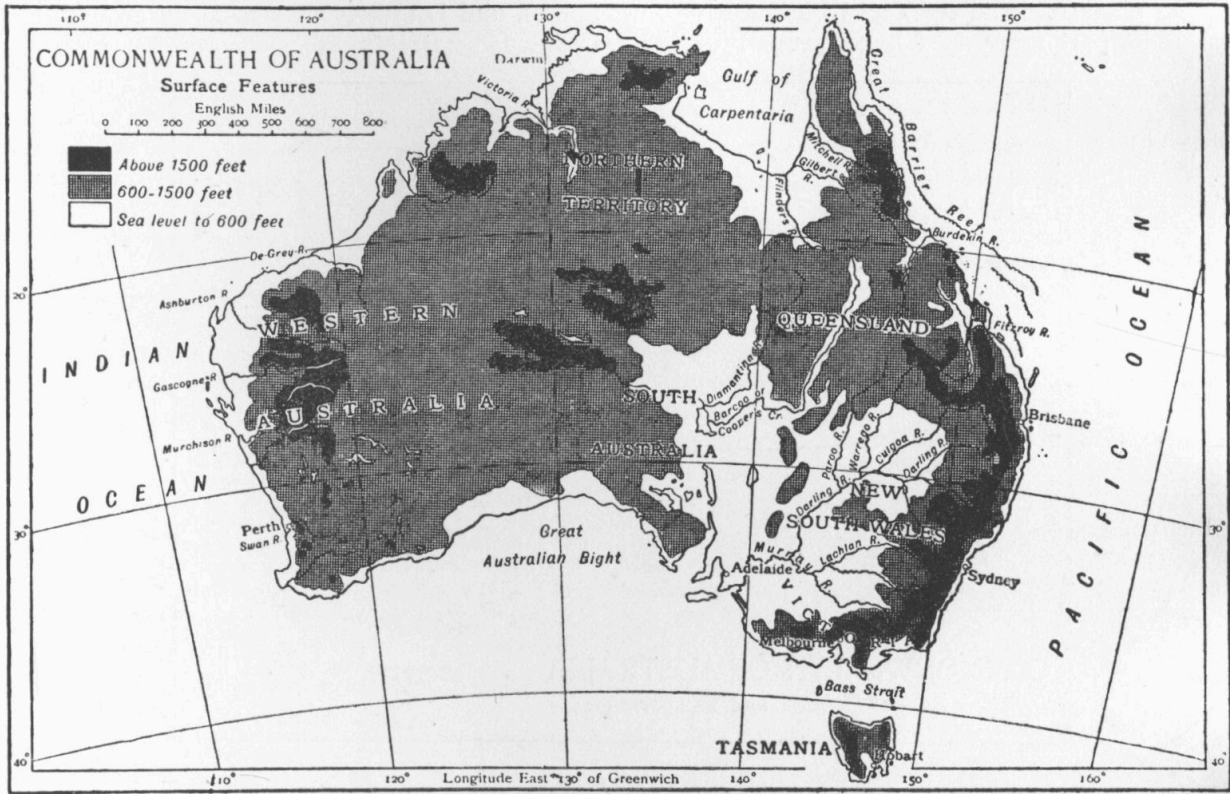
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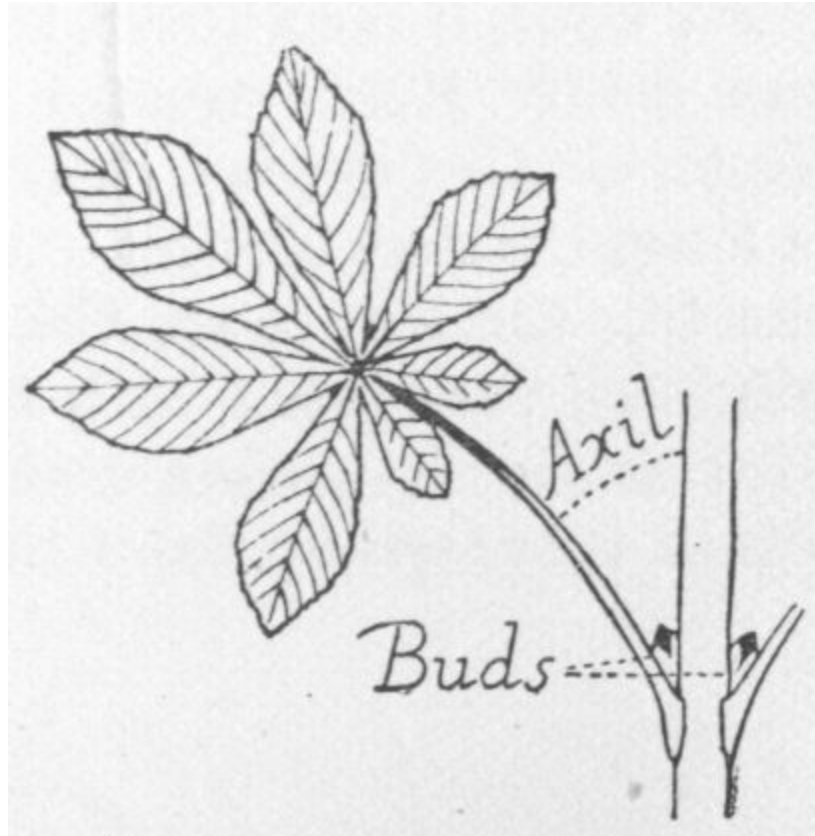
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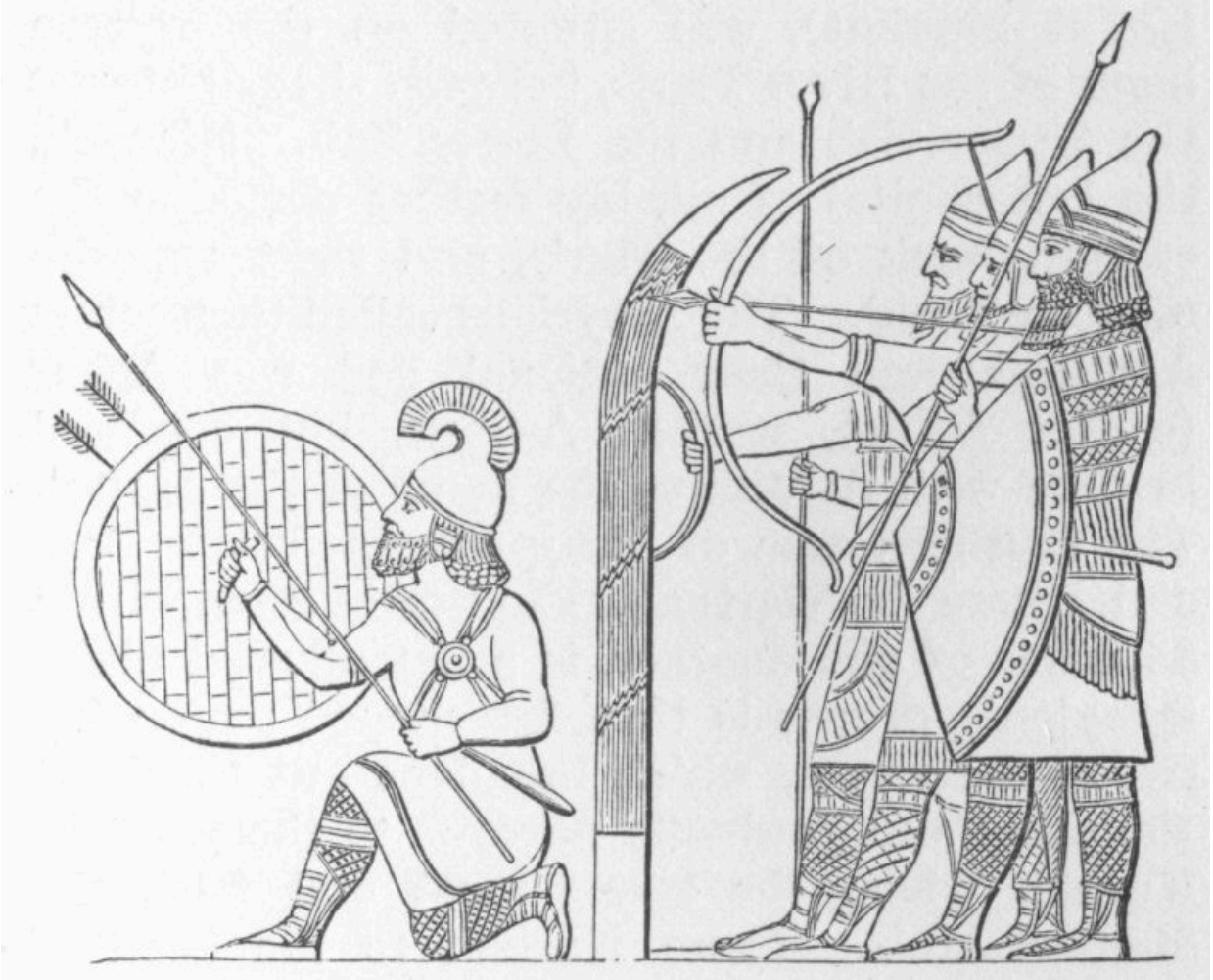
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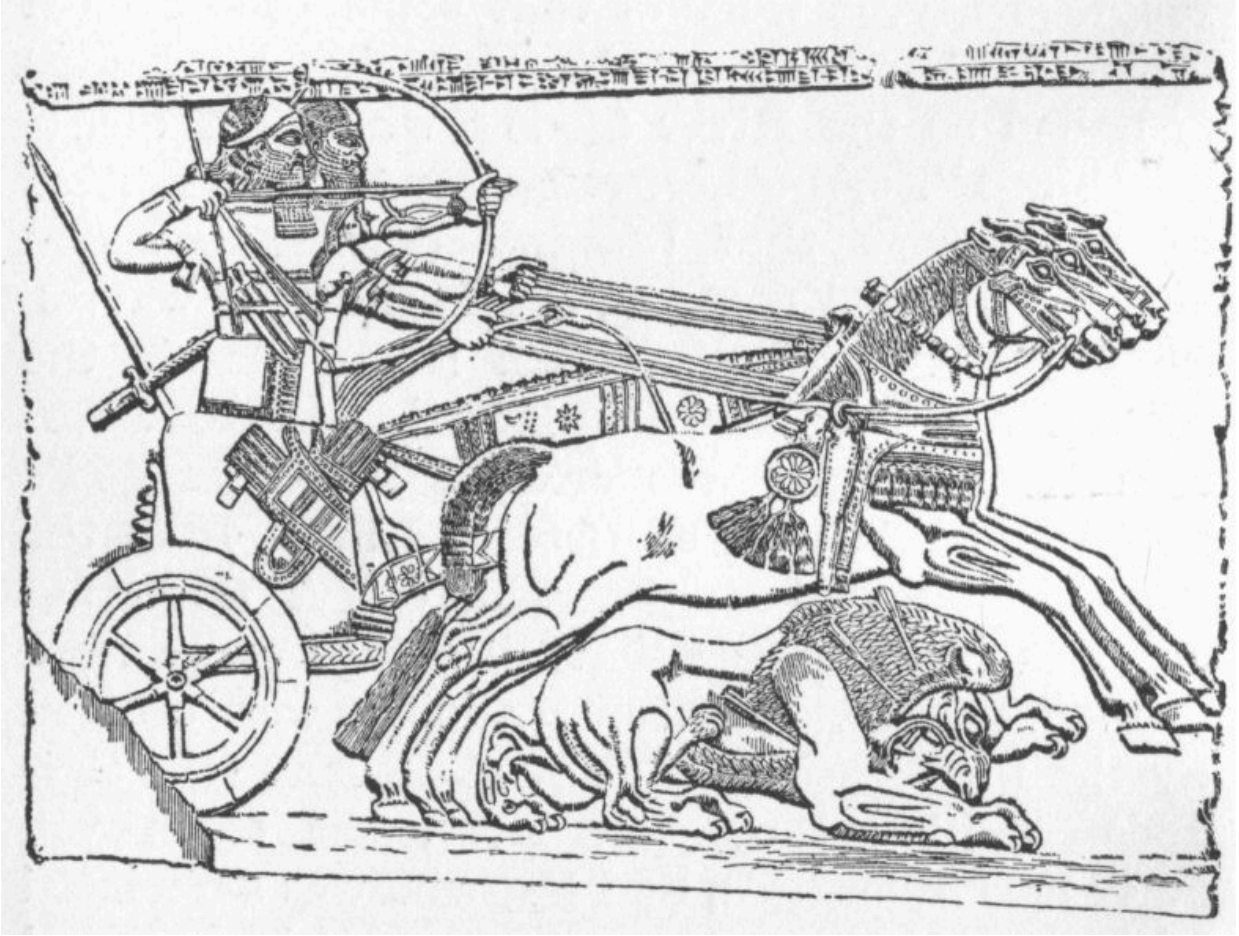
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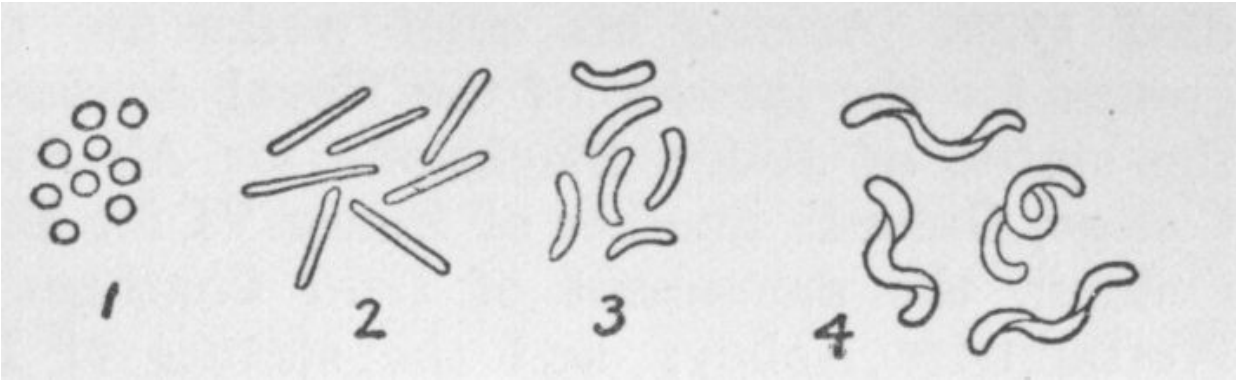
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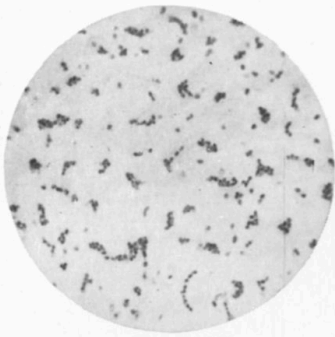
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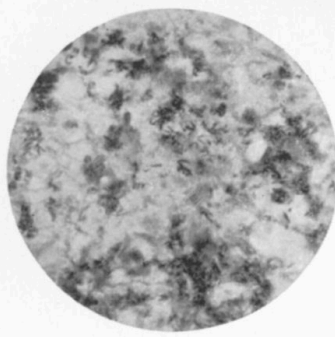
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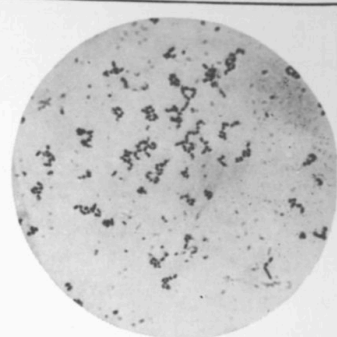
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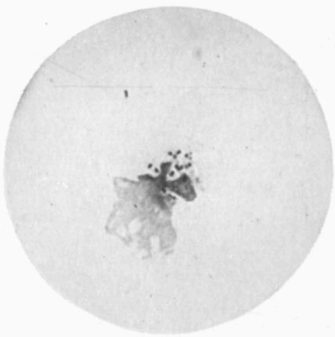
Pneumococcus.



Diphtheria.



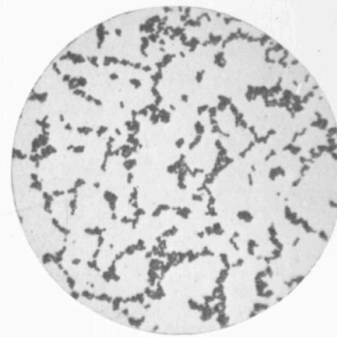
Nitrifying Bacteria of Soil.



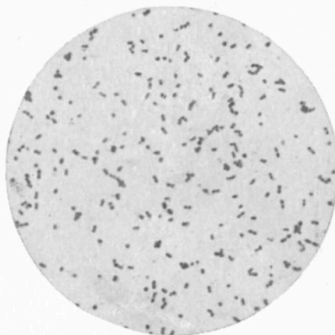
Gonococcus.



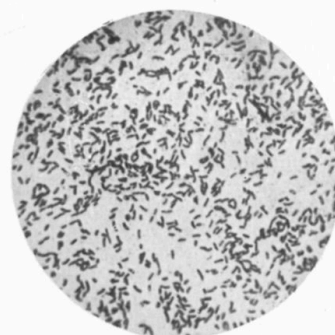
Tetanus.



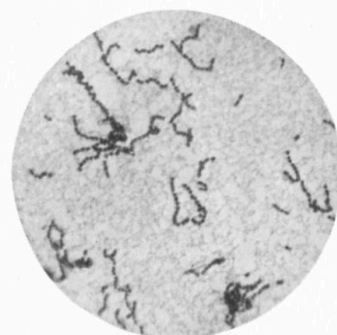
Staphylococcus aureus.



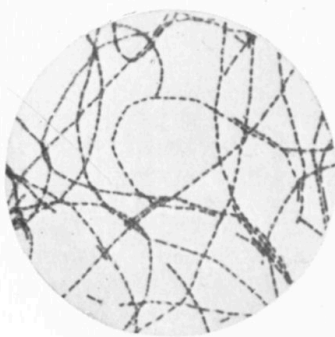
Diplococcus Meningitidis.



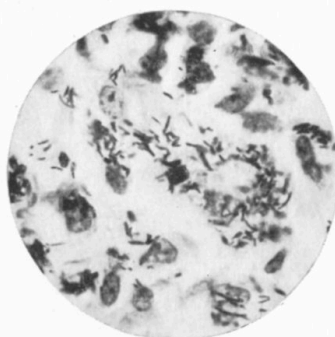
Asiatic Cholera.



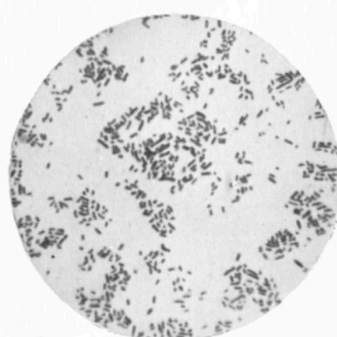
Streptococcus erysipielatis.



Anthrax.



Tubercle Bacilli in Human Lung.

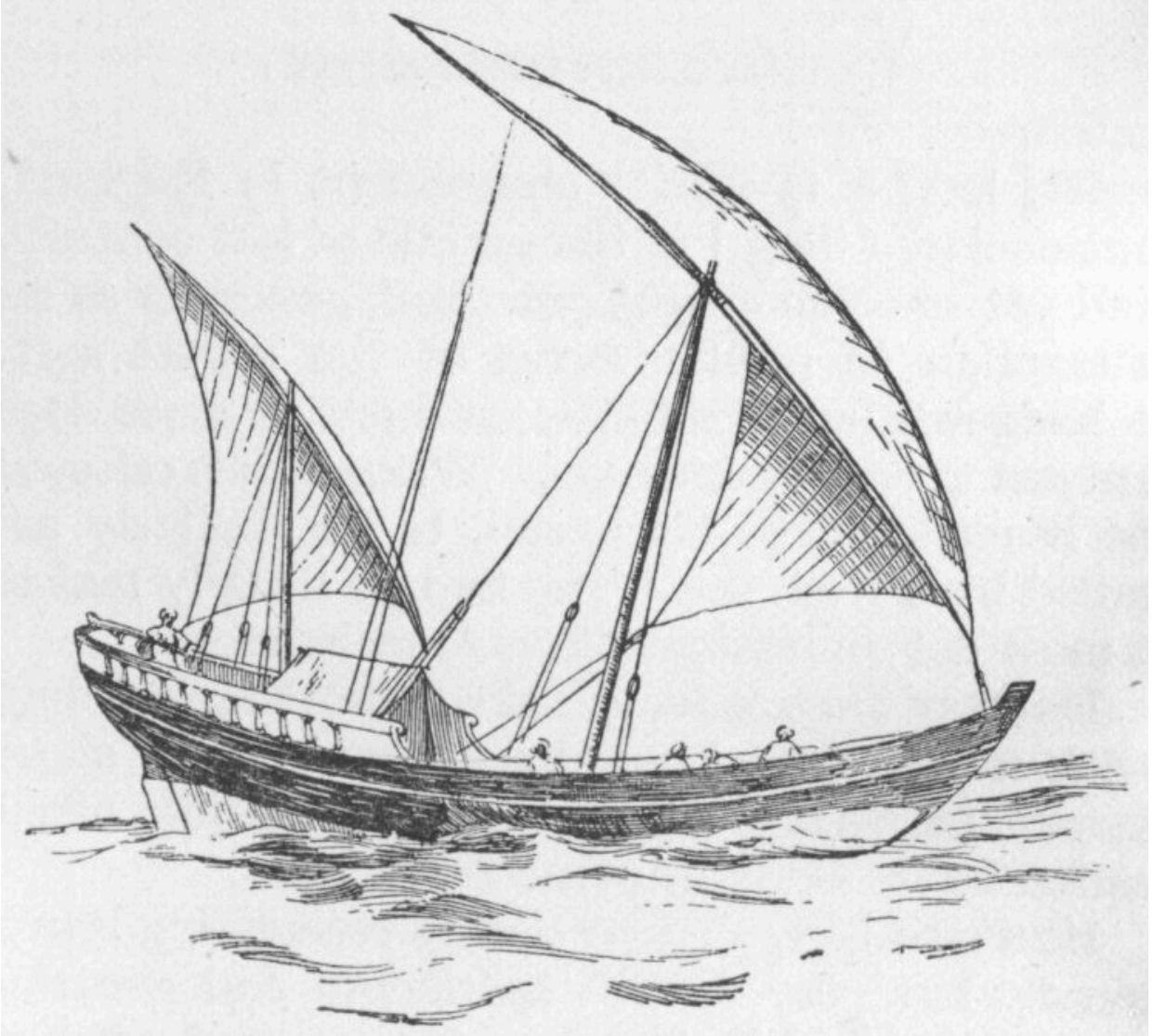


Typhoid.

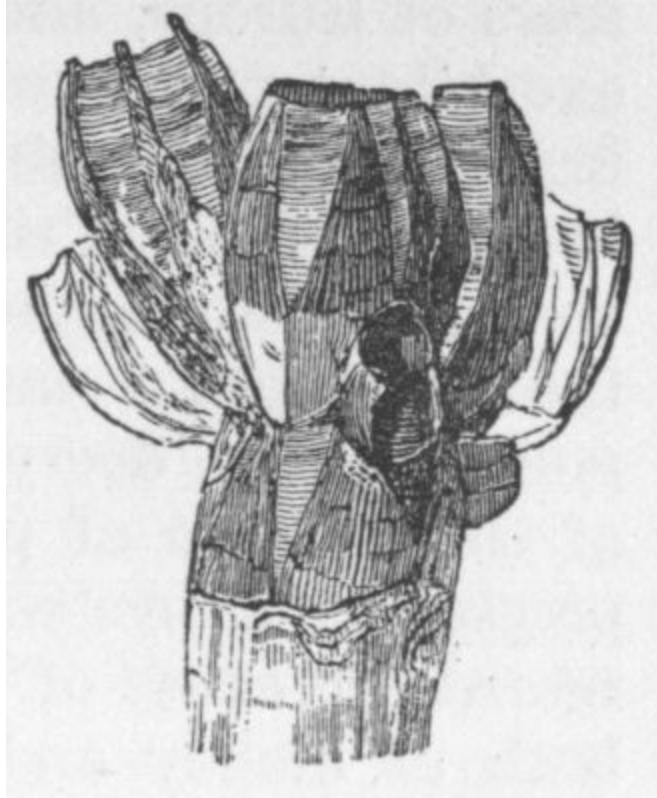
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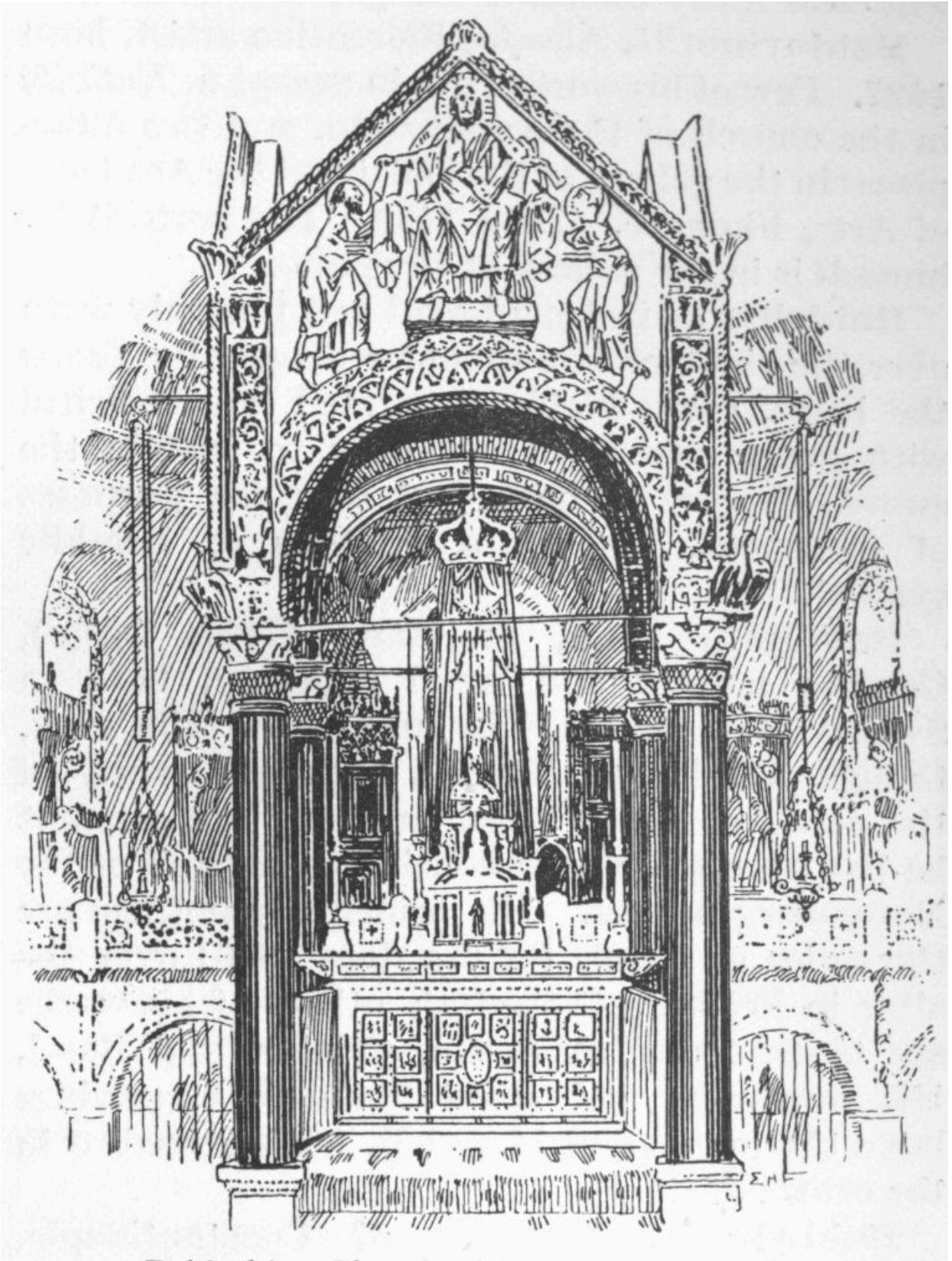
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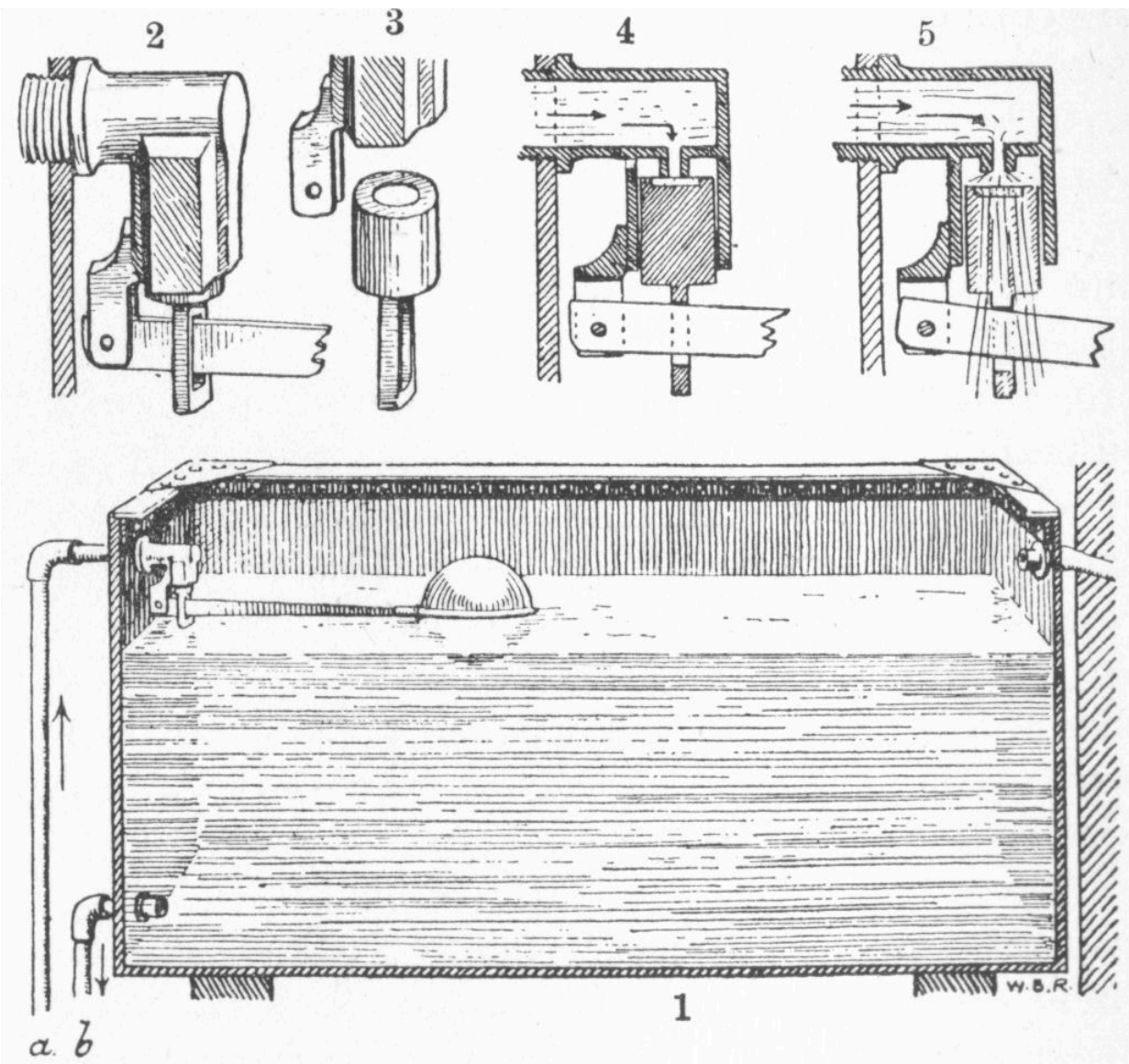
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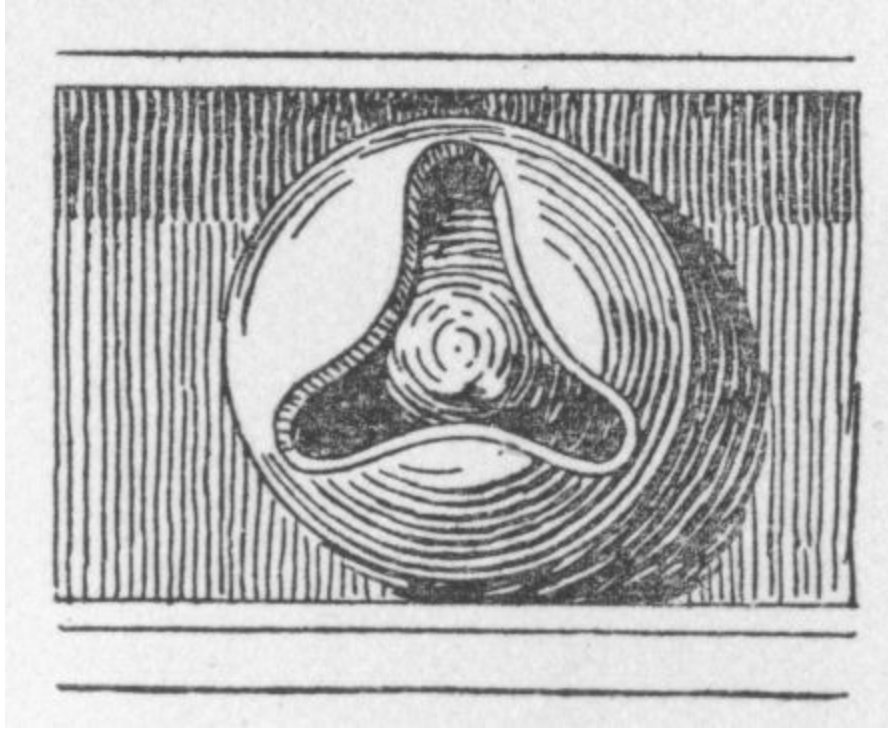
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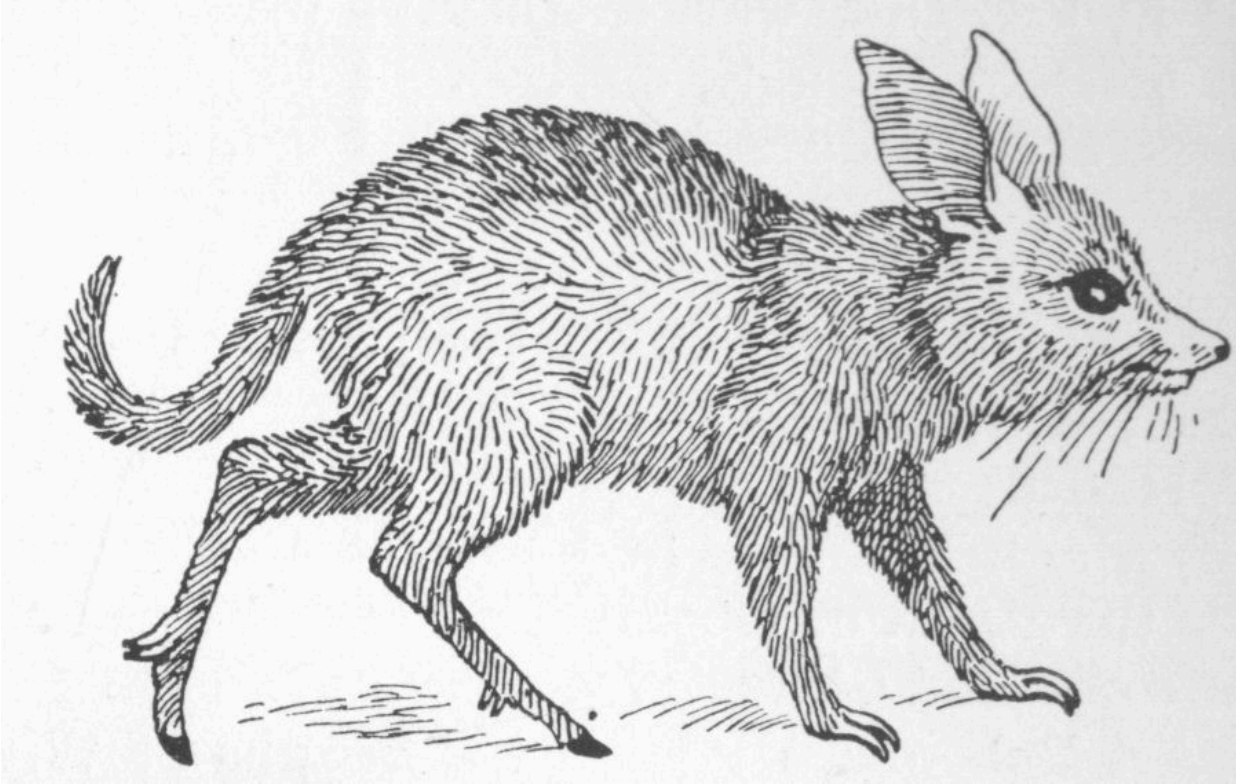
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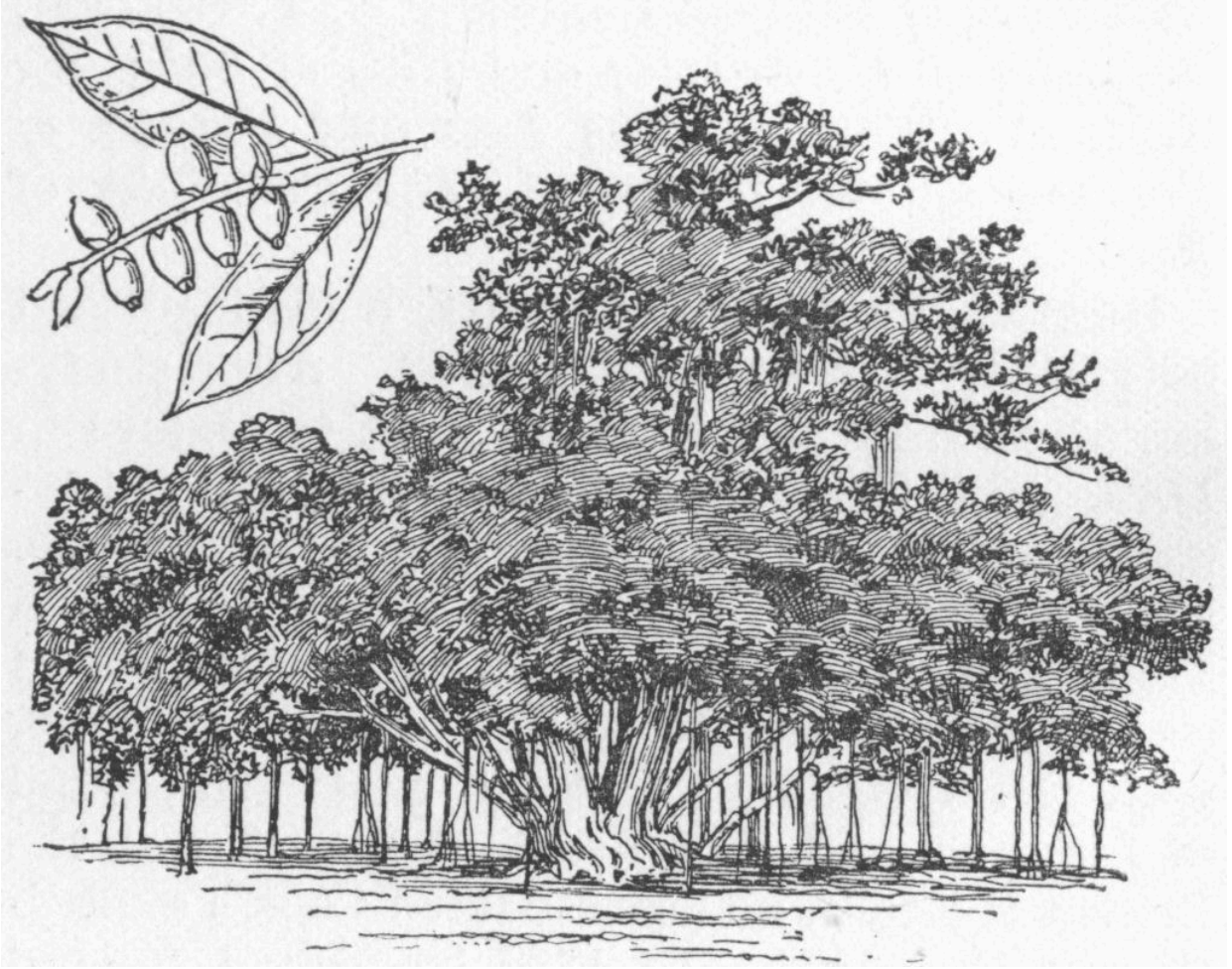
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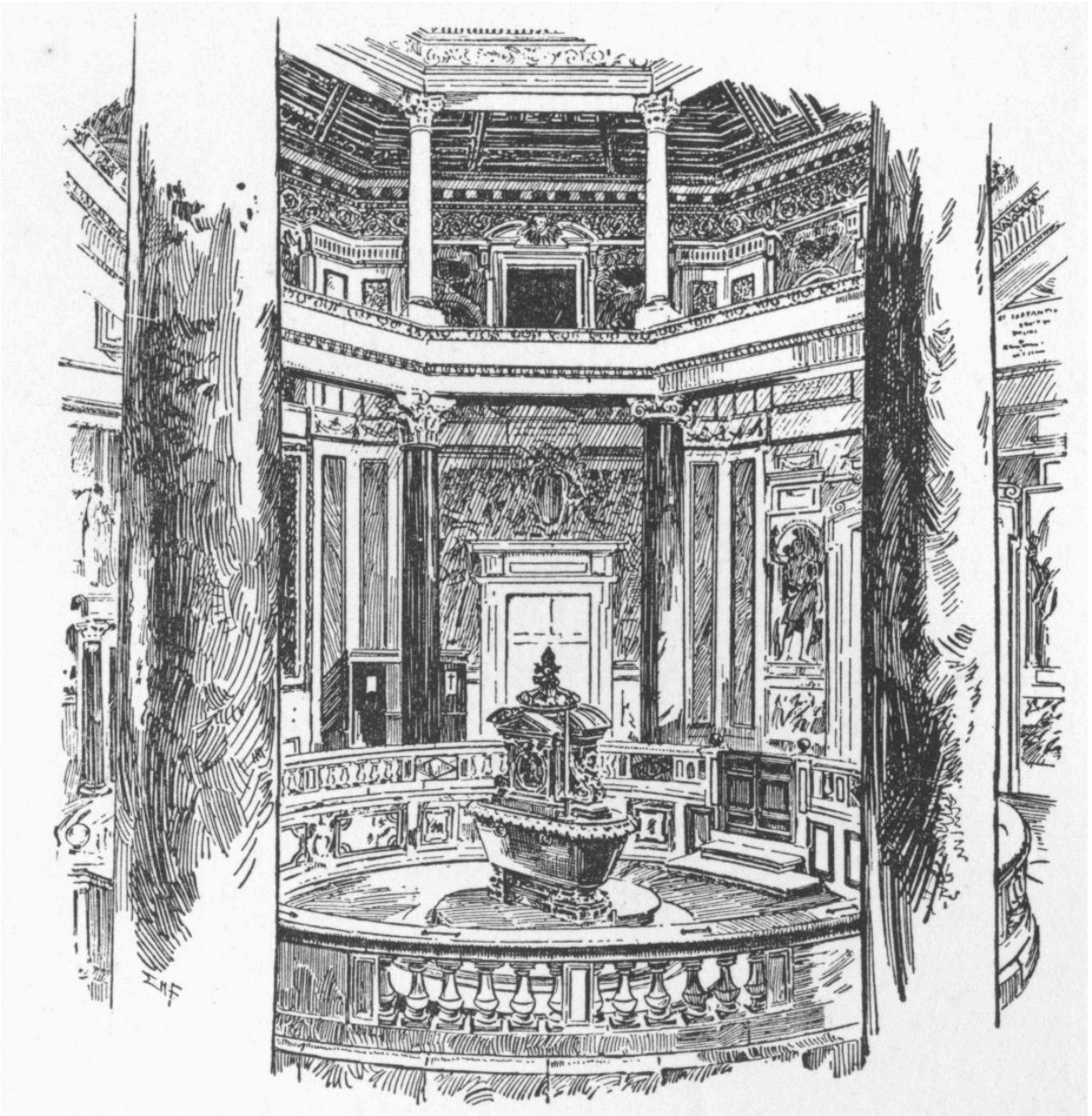
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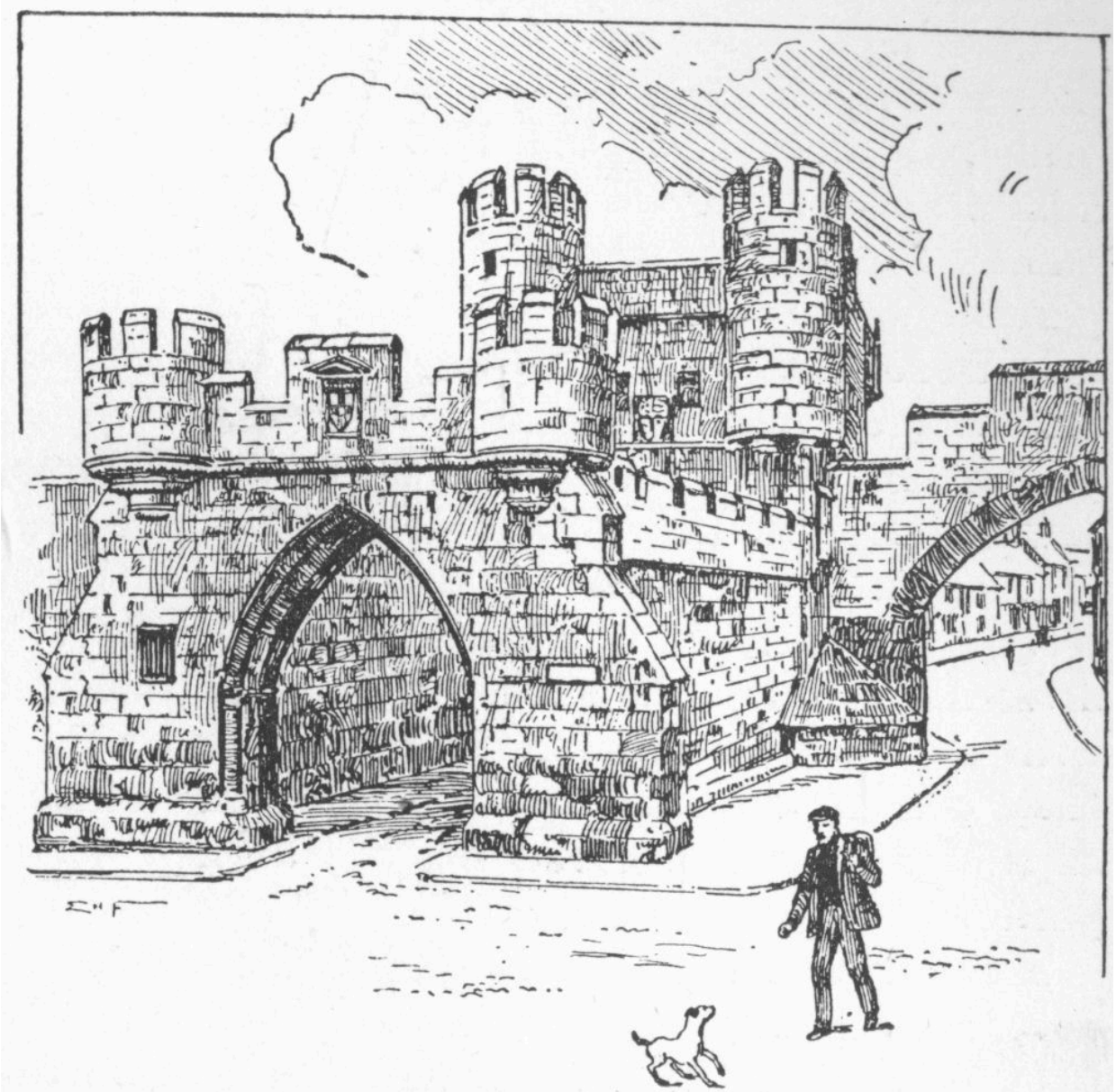
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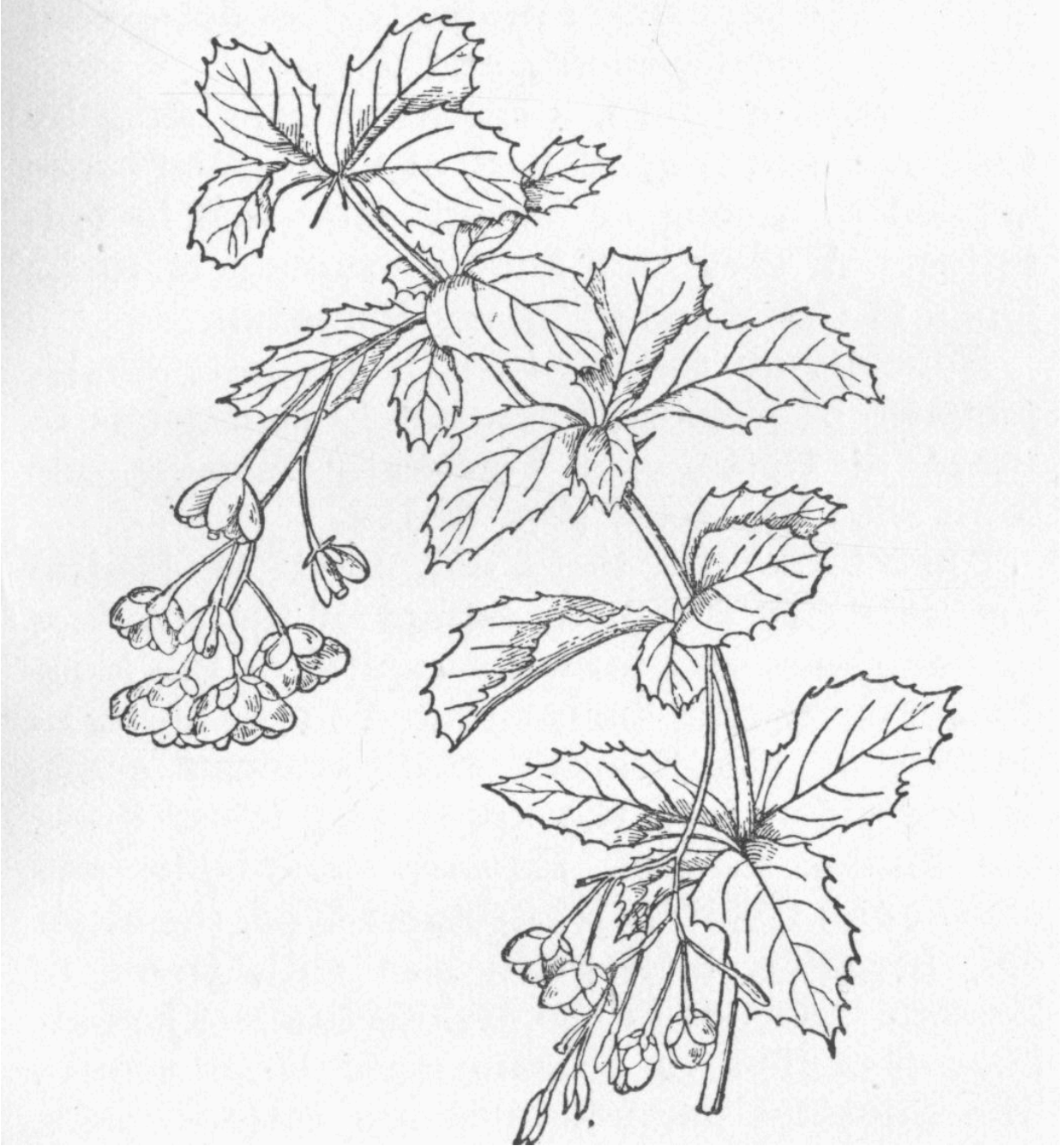
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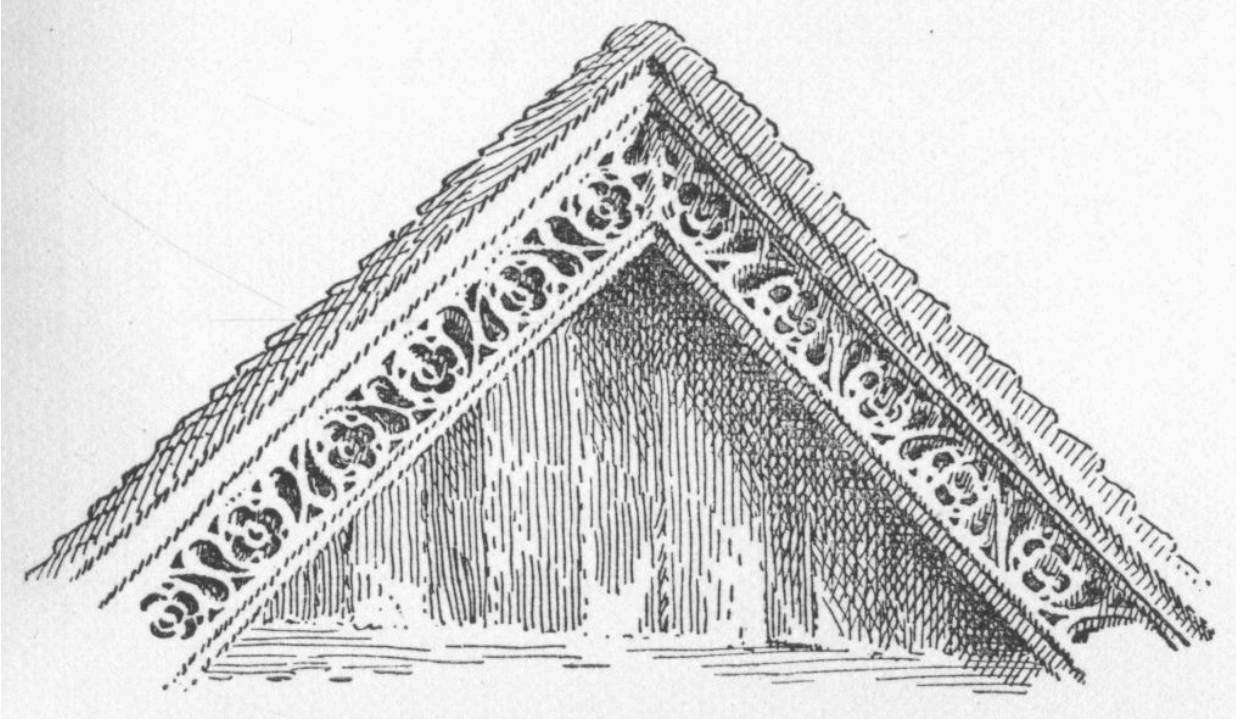
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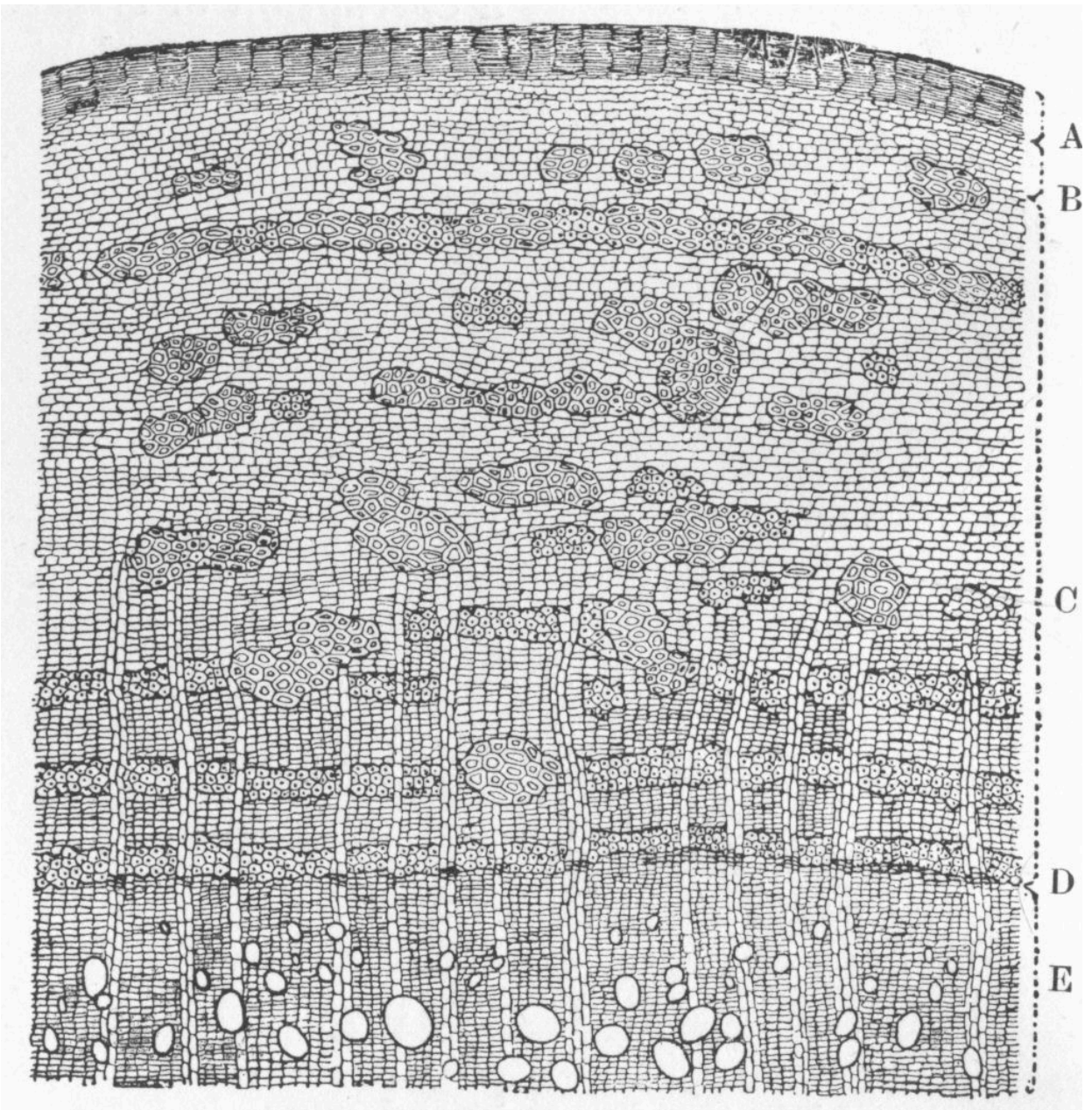
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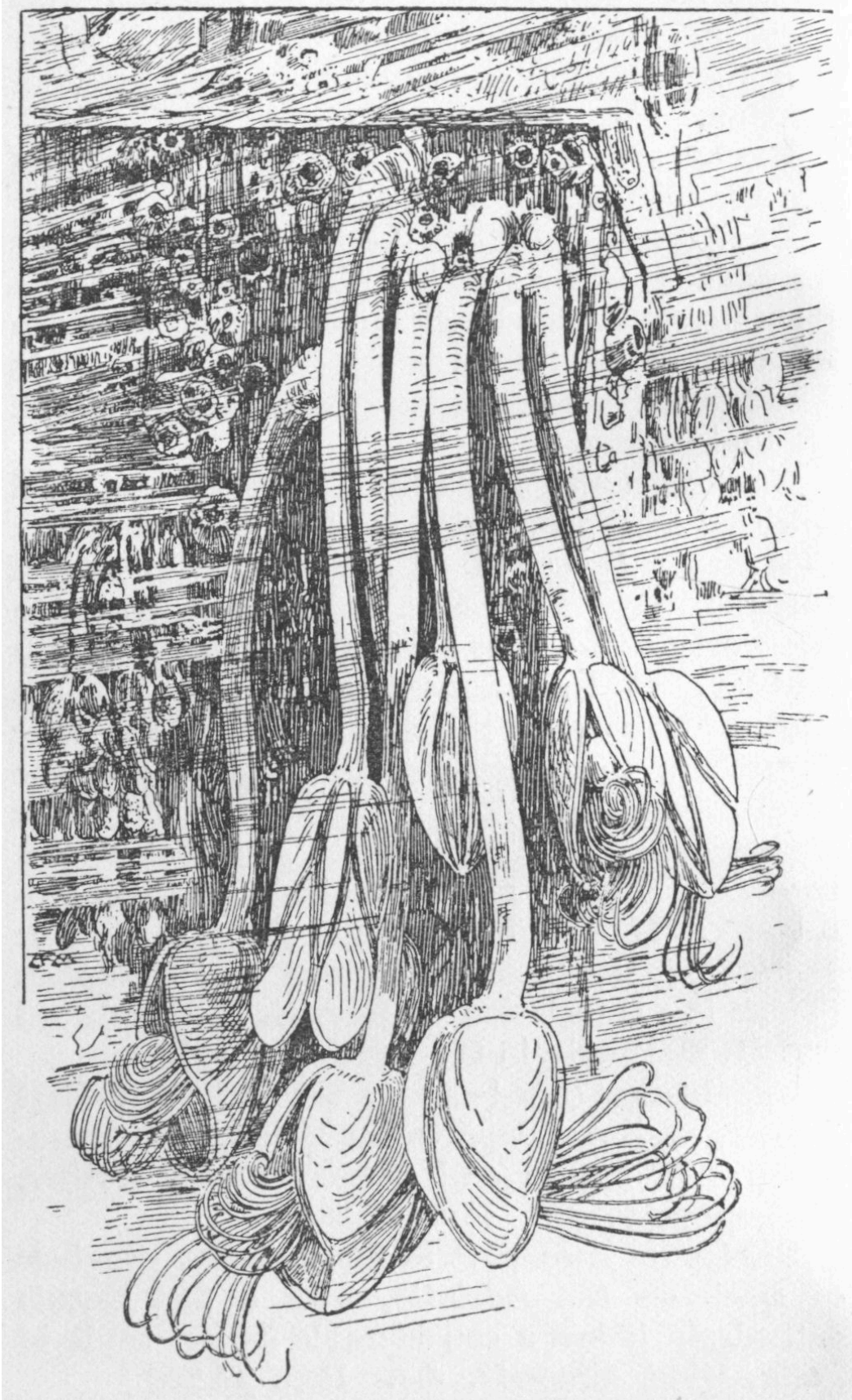
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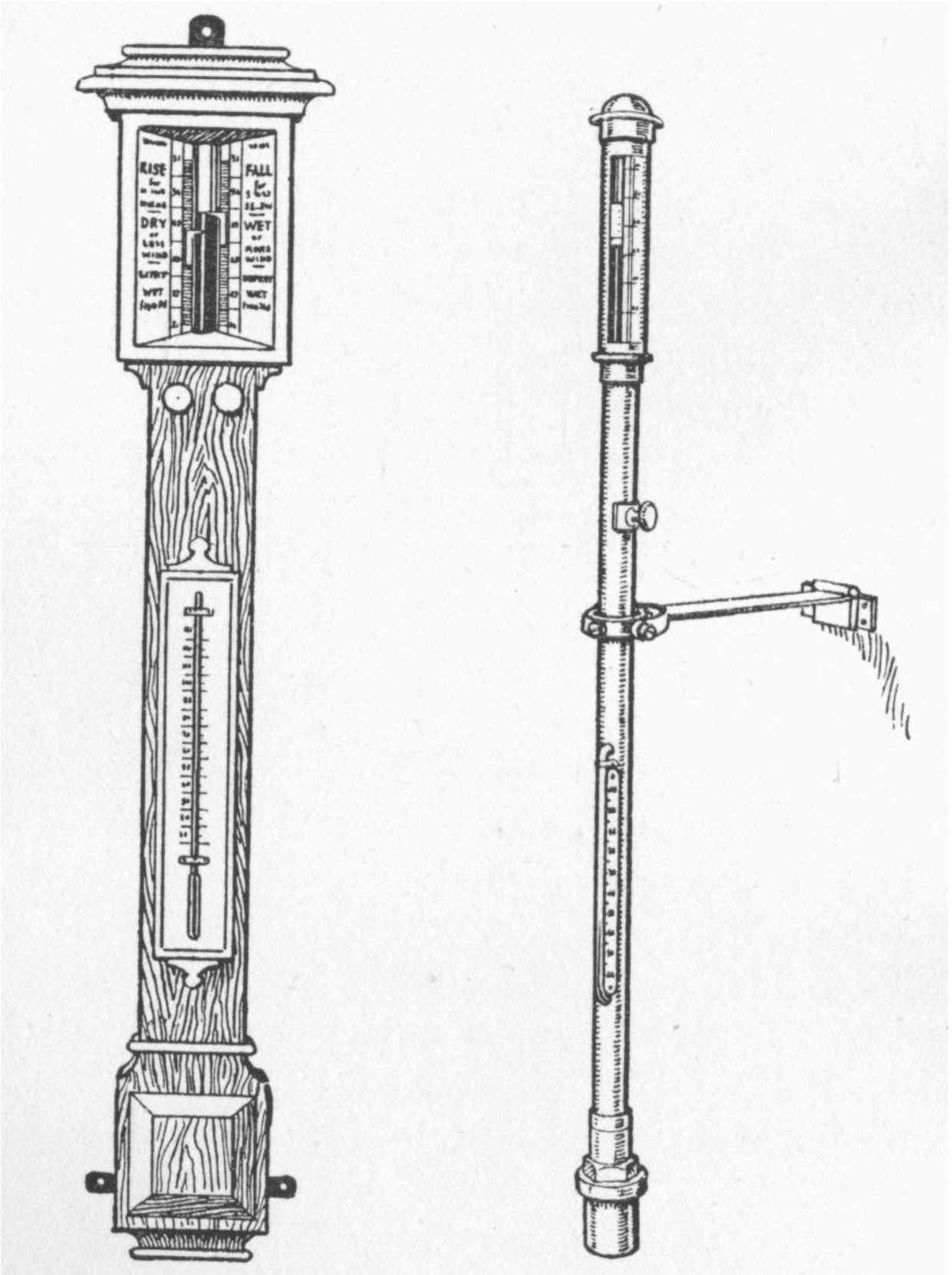
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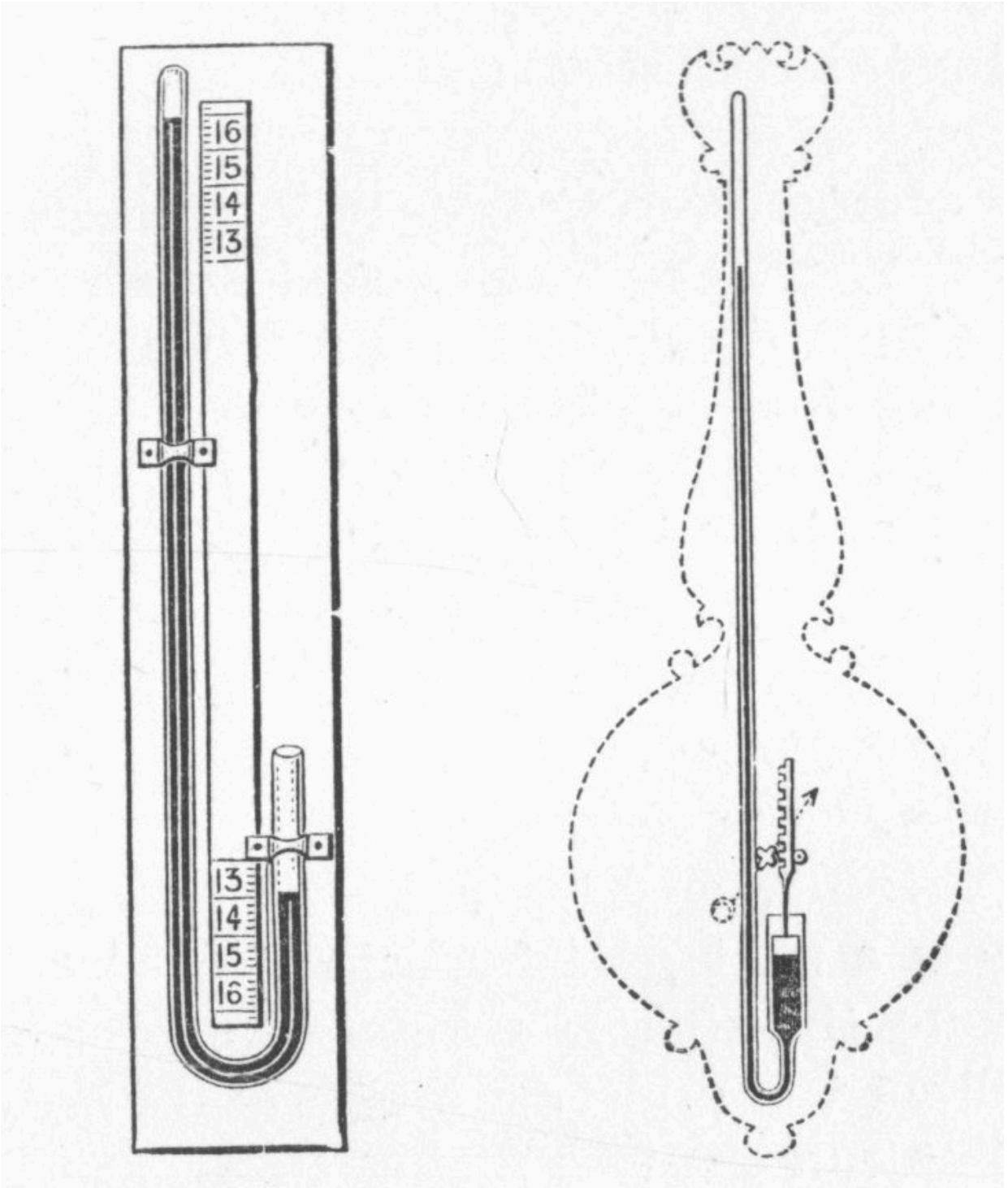
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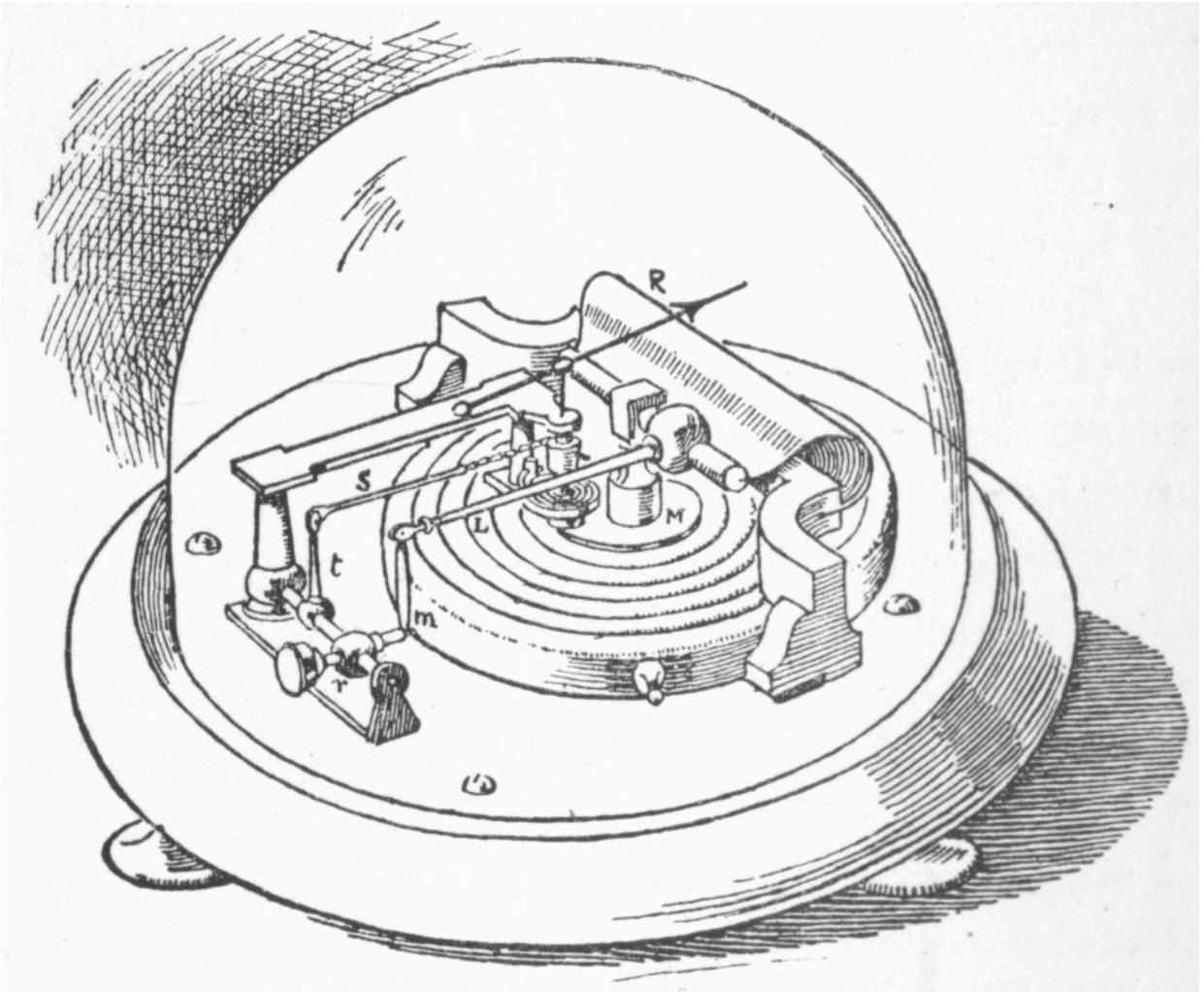
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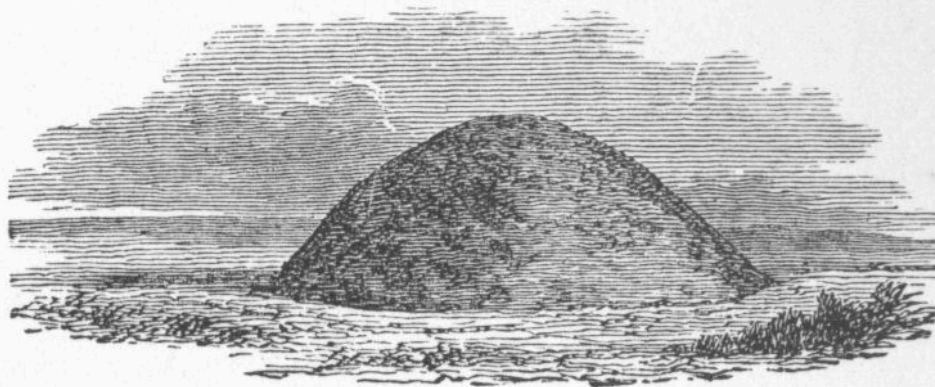
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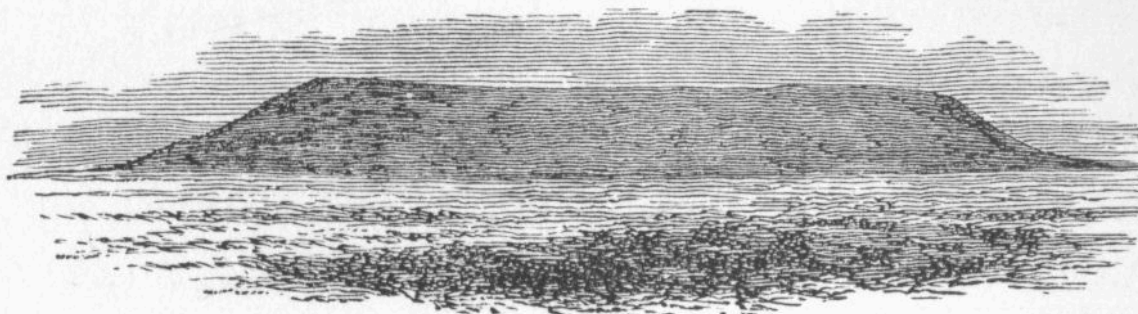
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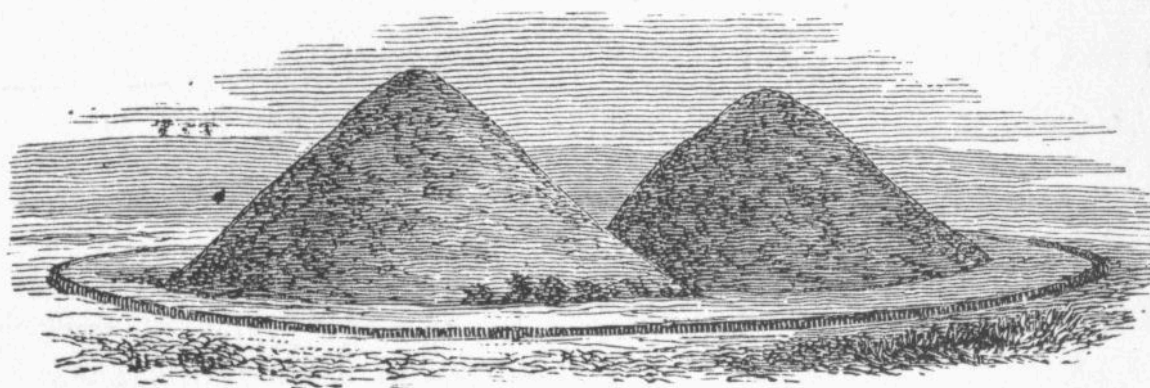
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Bowl Barrow

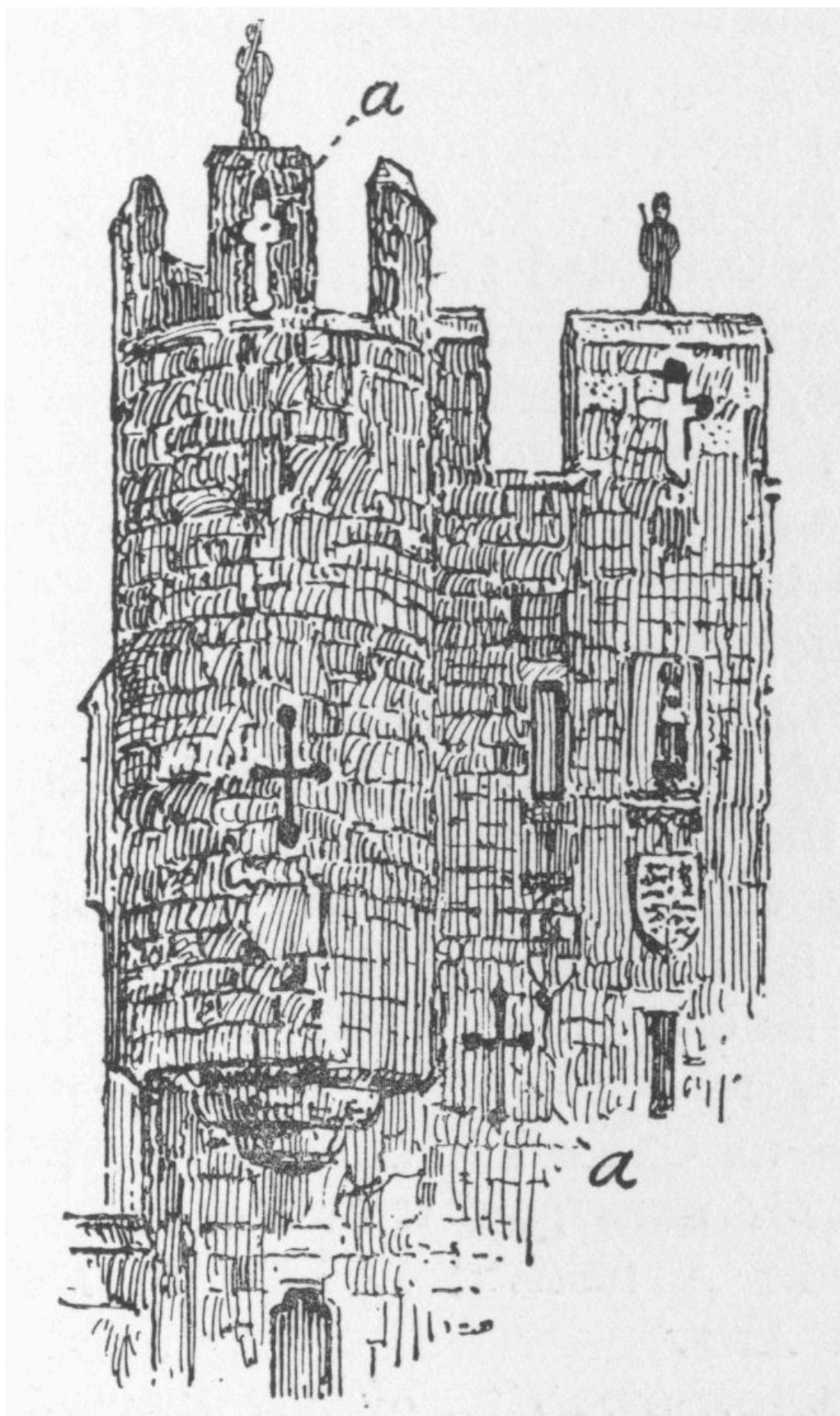


Long Barrow

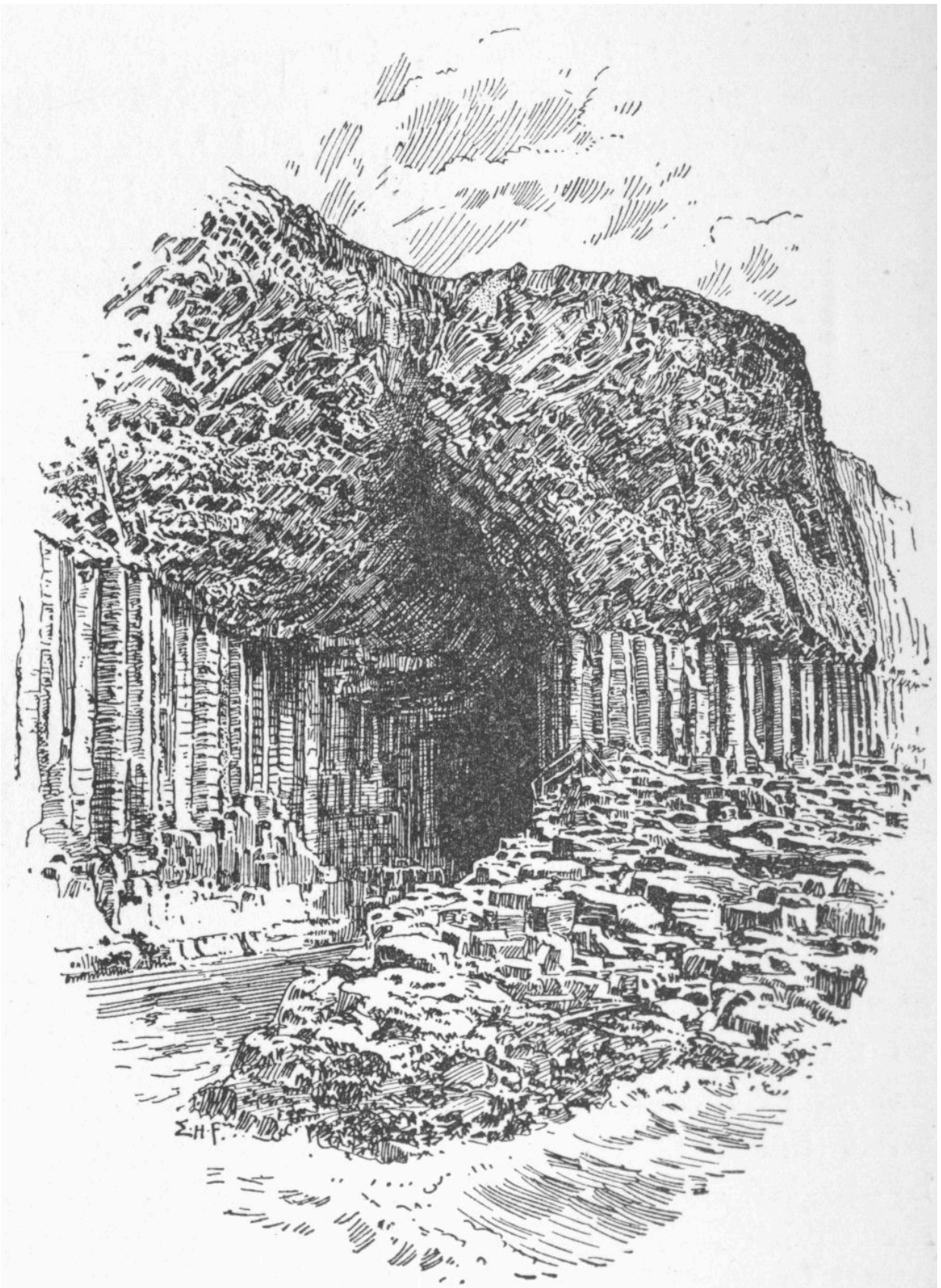


Twin Barrow

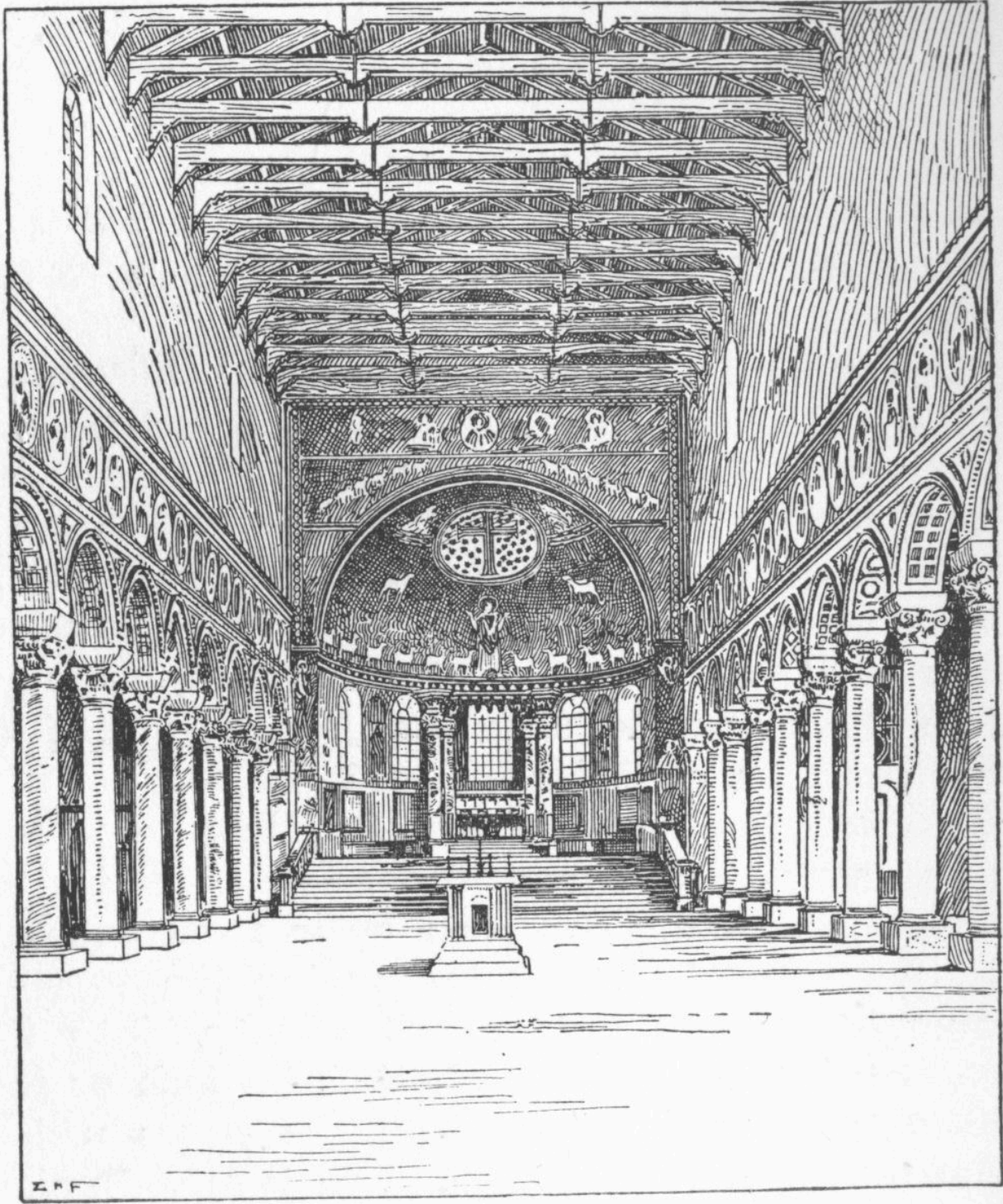
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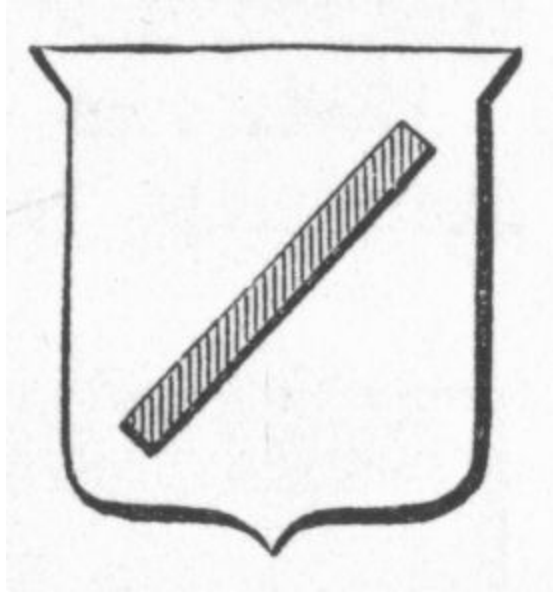
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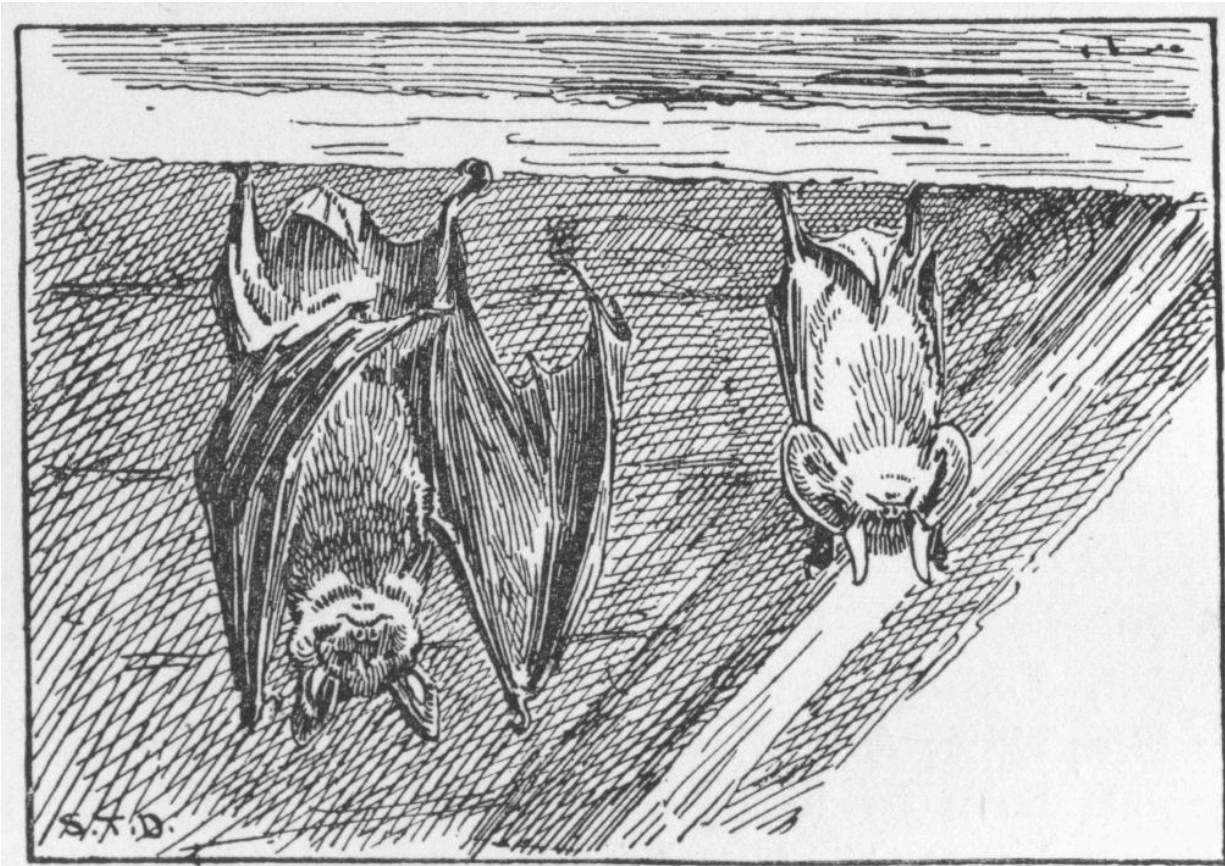
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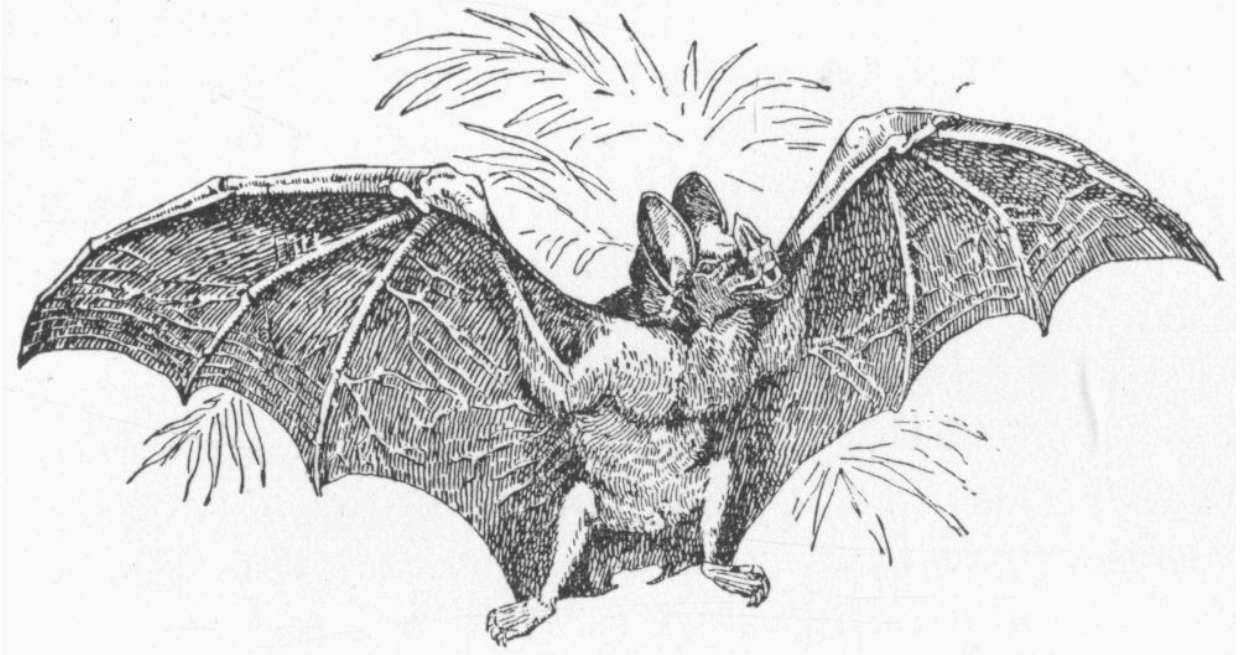
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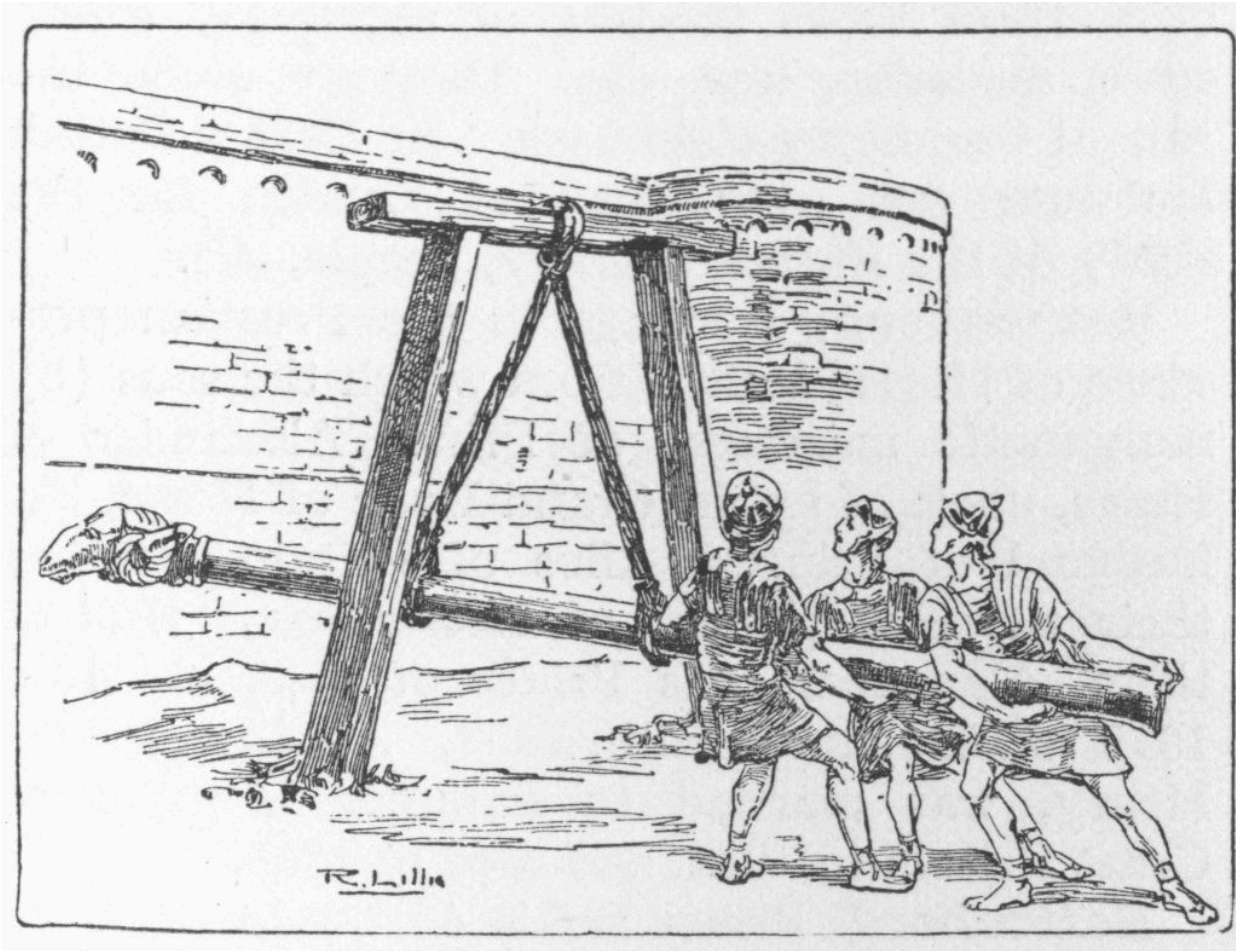
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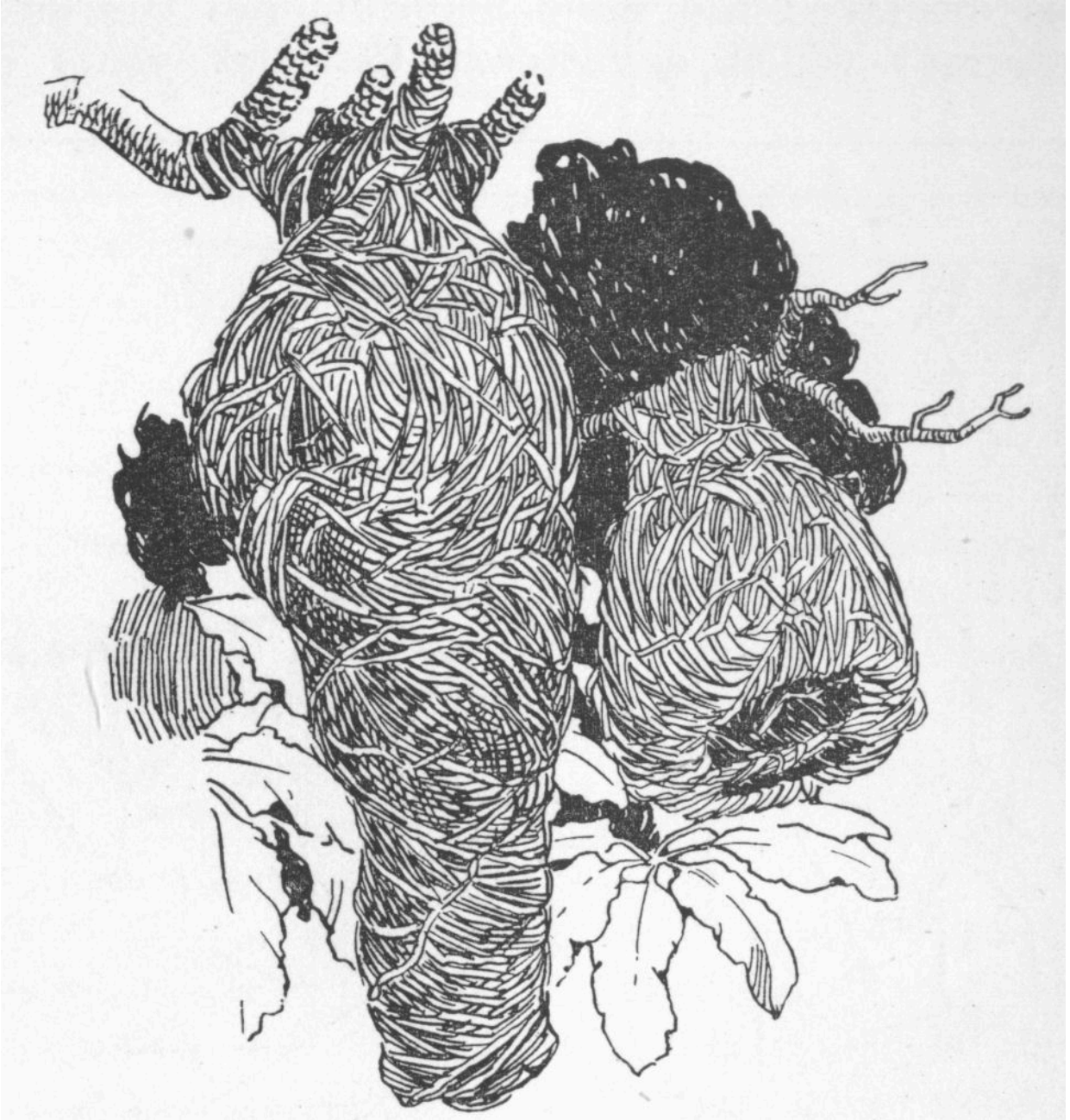
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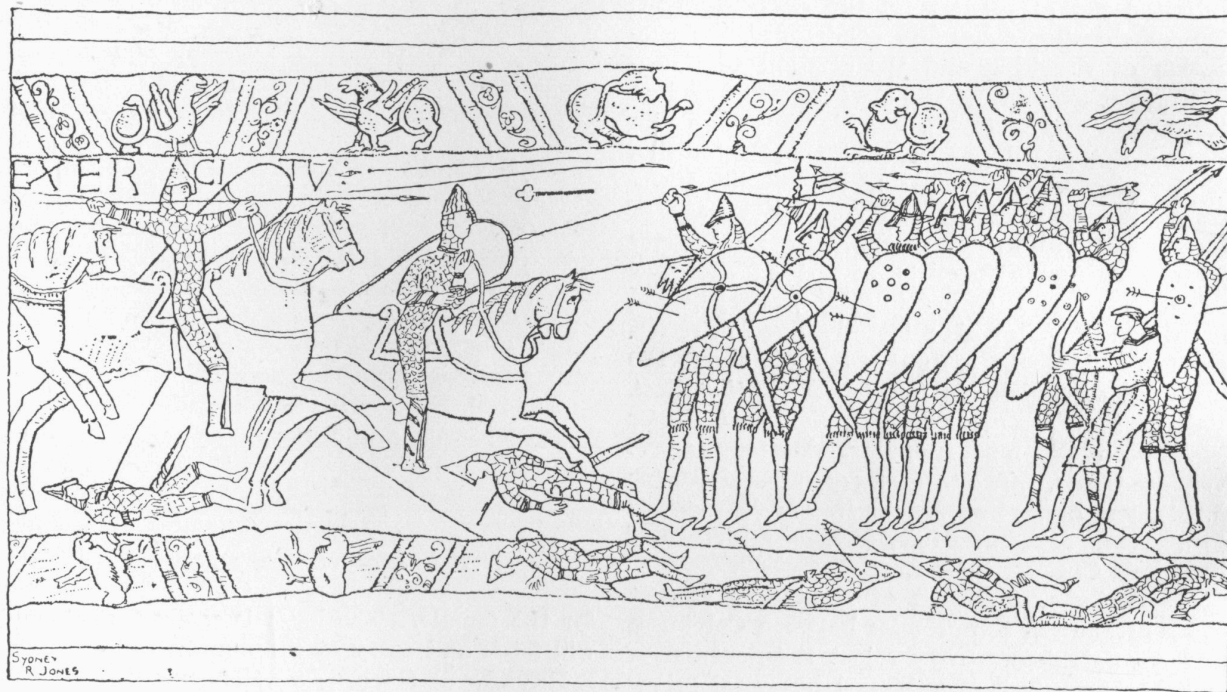
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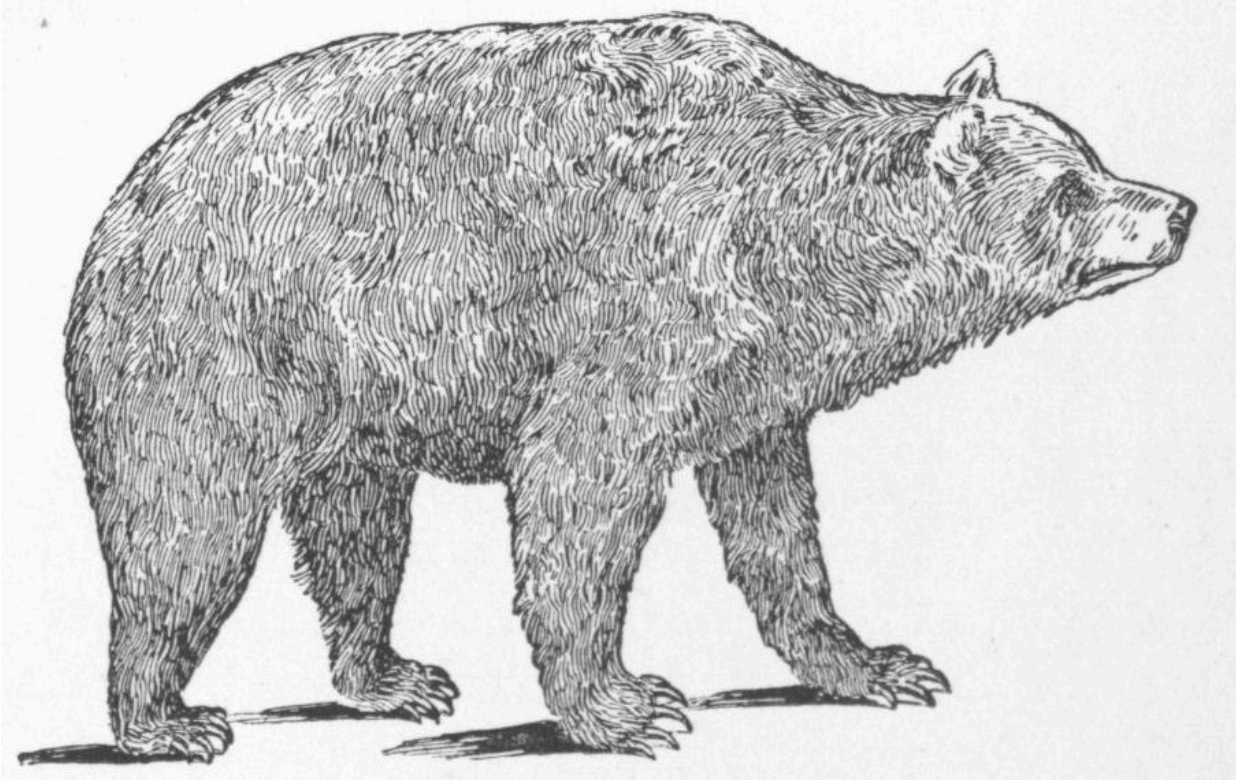
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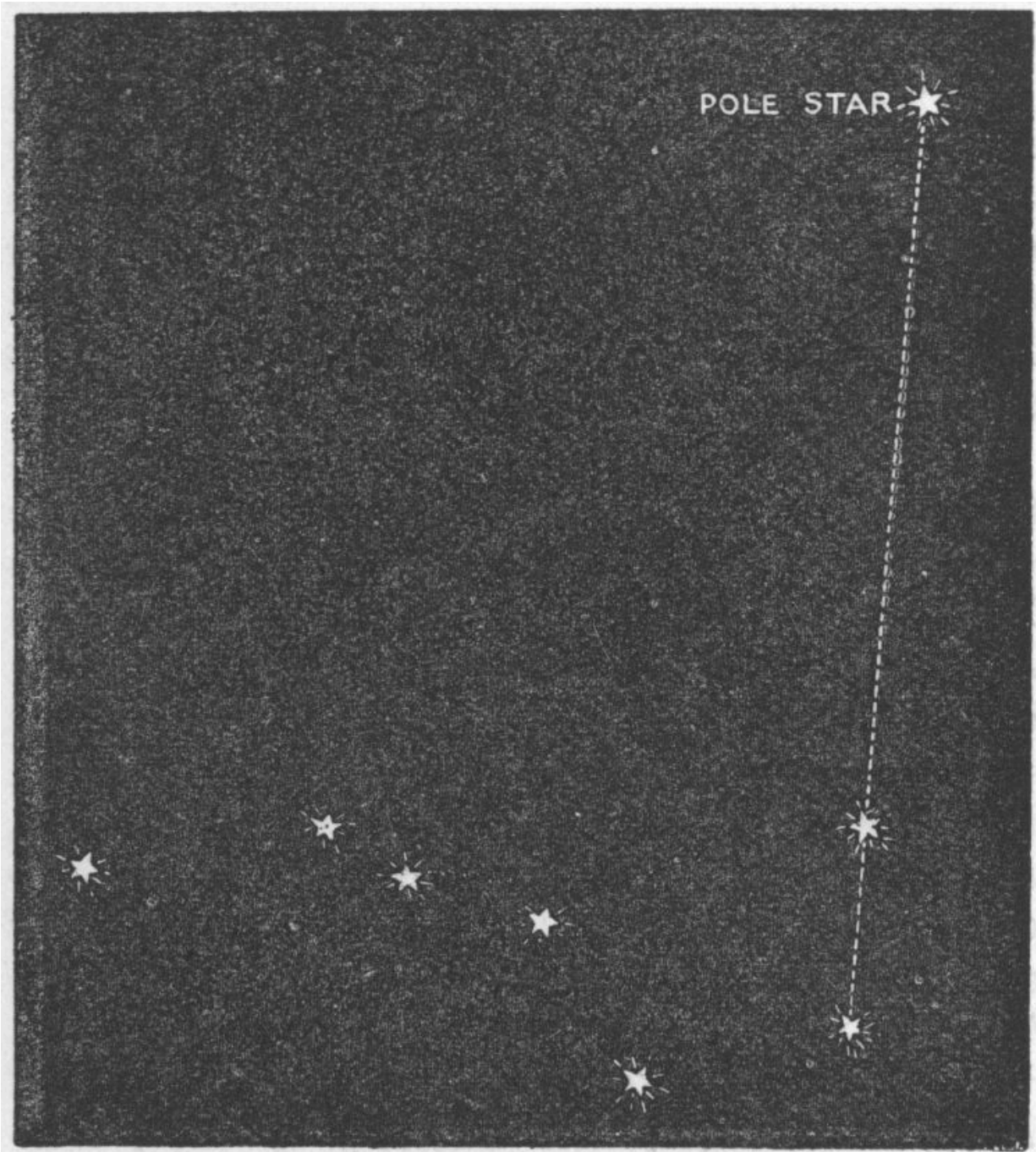
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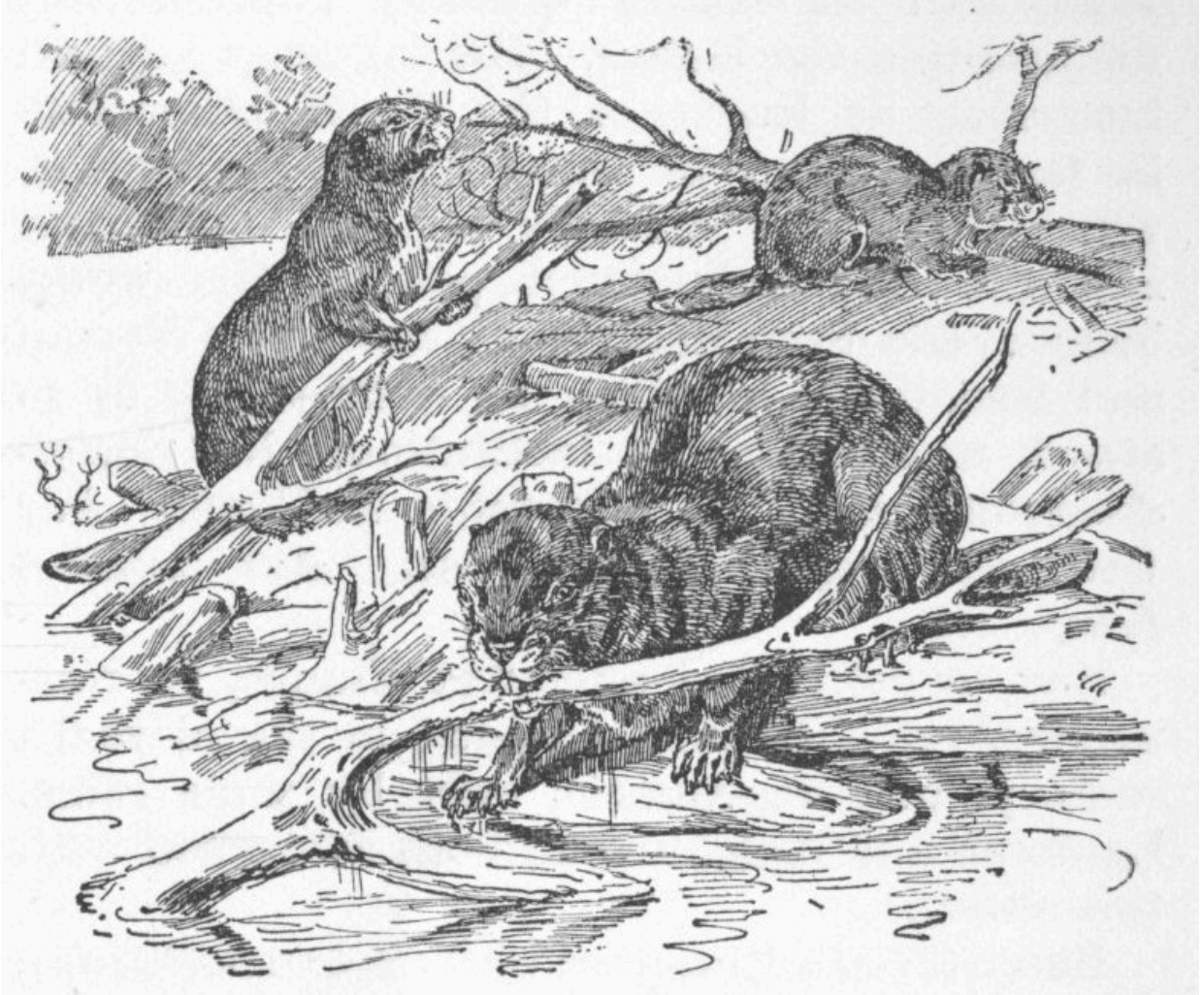
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